

NANDINI ENTERPRISES

INTERNAL TECHNICAL DOCUMENTATION

Confidential Site Audit, Feature Reference & Source Code Export

Industrial Automation Powerhouse | Hyderabad, India

HIGHLY CONFIDENTIAL - FOR INTERNAL ENGINEERING TEAM USE ONLY
DO NOT DISTRIBUTE - DO NOT SHARE WITH CLIENTS OR EXTERNAL PARTIES

Export Timestamp	Monday, 1 June 2026 at 1:03:41 am IST
Triggered From	test
Export Purpose	Disaster recovery, new developer onboarding, AI pricing analysis
Live Website	nandiniautomation.vercel.app

ACCESS CREDENTIALS & TRIGGER METHOD

Secret Token (EXPORT_SECRET_TOKEN): 03593D49A7034868BA70DDD69CB47E9D

Header Name: X-Export-Token

API Endpoint: /api/export-source (POST)

Trigger: Double right-click on "Sitemap" link in website footer (within 800ms)

CLI Deploy Command: vercel --prod --yes (run from local project root)

GitHub Repo: github.com/novelleyx-dev/nandiniautomation

TABLE OF CONTENTS

1	Company Overview & Profile Company facts, contacts, certifications, authorized partners
2	Technology Stack & Architecture Framework, language, libraries, deployment platform, React/Node versions
3	Color System & Design Tokens Global CSS variables, Tailwind tokens, font names, theme colors
4	Site Pages Documentation All 17 routes - purpose, source file, sections, form fields
5	Feature Index & Implementation Guide F-01 to F-12: trigger, code logic, dependencies, security notes
6	Component Library Reference All React components with type, path, and description
7	Known Issues & Resolutions All bugs encountered and their exact fixes
8	Deployment & Confidential Access Setup Vercel CLI deploy, env vars, GitHub vs Vercel separation
9	Services Text Content (AI-Readable) Full service descriptions, FAQs, workflows - optimized for LLM analysis
10	Project Directory Tree Complete annotated filesystem structure
11	Complete Source Code Every .ts .tsx .css .mjs .json file verbatim

SECTION 1

Company Overview & Profile

Identity, contacts, certifications, and authorized partnerships

Company Identity

Company Name	Nandini Enterprises
Tagline	Industrial Automation Powerhouse
Registered Address	Plot No. 49, Phase-III, IDA Cherlapally, Hyderabad, Telangana - 500051, India
Direct Phone	+91-40-23190131
Email	info@nandiniautomation.com

Certifications & Licenses

- ISO 9001:2015 Certified
- CPRI Certified Electrical Panels
- Grade-A Electrical License Holder

Authorized Industry Partners

Siemens	Yaskawa (Authorized Service Center)
ABB (Channel Partner)	Baumer (Authorized Distributor)
Yokogawa	Honeywell
WIKA	L&T
Crompton	Nidec

SECTION 2

Technology Stack & Architecture

Every library, framework, and version used in this project

Core Technologies

Framework	Next.js 16.2.6 (App Router, Webpack build mode)
Language	TypeScript 6.x - strict mode enabled
Styling	Tailwind CSS v4.x with custom CSS properties
Animations	Framer Motion 12.x - entrance, hover, stagger effects
Icons	Lucide React 1.x icon set
Fonts	Inter (body) + Outfit (headings) - loaded via next/font/google
State Management	React hooks only (useState, useRef, useEffect) - no external store
PDF Generation	pdf-lib (pure JS) - no filesystem font dependencies, Vercel-safe
Screenshot	html2canvas - 0.5 scale JPEG 0.6 quality to stay under Vercel 4.5MB limit
Deployment	Vercel (serverless + Edge)
Repository	GitHub - github.com/novelleyx-dev/nandiniautomation
React Version	React 19.x
Node Version	Node 20.x LTS

SECTION 3

Color System & Design Tokens

Global CSS variables, Tailwind tokens, and typography definitions

CSS Color Palette

App Background	#E8F0FA - light powder-blue corporate theme (body bg)
Primary Blue	#0A3D91 - --color-ne-blue-corp (buttons, links, highlights)
Steel Blue	#1565C0 - --color-ne-blue-steel (section accents)
Electric Blue	#1E88E5 - --color-ne-blue-electric (hover states)
Footer Background	#0f172a - slate-900 (dark footer)
Text Primary	#1e293b - slate-800
Text Secondary	#475569 - slate-600
Accent Green	#10b981 - emerald-500 (success, active states)
Global CSS File	src/app/globals.css
Font Variables	--font-inter (body), --font-outfit (headings)

SECTION 4

Site Pages Documentation

All routes with purpose, source file, and section breakdown

/

Homepage - Main Landing Page

Source File

src/app/page.tsx

Primary B2B marketing page. Designed to convert industrial procurement managers into leads. All sections are stacked vertically with scroll-triggered animations.

Sections / Form Fields:

- HeroSlideshow: Full-screen 5-image carousel with auto-advance every 5s, slide indicators, and animated headline overlay
- Trust Strip: ISO 9001:2015, CPRI, Safety Standards, and partner certification badges
- Services Preview: 6 service category cards with hover animations and CTA links
- Industries Grid: 8 industry sectors with icons - Steel, Cement, Oil & Gas, Power, Pharma, Water, Food, Mining
- Why Choose Us: 4 USP cards with animated counters - 500+ Projects, 30+ Years, 24/7 Support, ISO Certified
- Process Workflow: 4-step engineering journey - Consultation > Design > Fabrication > Commissioning
- Projects Overview: Portfolio cards of major completed EPC projects with AI-generated images
- Company Story: Founding history, growth narrative, and leadership values section
- Partner Logos: Brand wall - Siemens, Yaskawa, ABB, Yokogawa, Baumer, Honeywell, WIKA, L&T
- Lead Capture Form: Dynamic RFQ form that adapts fields based on selected service category

/about

About Us / Executive Profile

Source File

src/app/about/page.tsx

Company history, leadership team, founding philosophy, and engineering credentials. Builds trust with enterprise procurement teams.

Sections / Form Fields:

- Company History & Founding Story
- Leadership Profile Cards
- Certifications & Awards Wall
- Manufacturing Facility Overview

/services

Services Overview Listing

Source File

src/app/services/page.tsx

Grid listing of all 6 main service categories. Each card links to a dedicated 500+ word service deep-dive page.

Sections / Form Fields:

- Advanced Industrial Automation Engineering
- Industrial Robotics Integration
- Precision Process Instrumentation
- High-Performance Electrical Systems
- Smart Elevator Solutions
- Advanced Electrical Control Panels

/services/[slug]

Dynamic Service Detail Pages (SSG)

Source File

src/app/services/[slug]/page.tsx

Statically pre-generated pages at build time. Data from servicesData.ts. Structure: Stats > Overview > Challenge > Workflow > Architecture > Industries > Case Study > Tech > QA > Safety > AMC > FAQs. Slugs: panel-manufacturing, plc-scada, vfd-solutions, robotics, elevator-solutions, instrumentation.

/services/industrial-automation

Industrial Automation (Standalone)

Source File

src/app/services/industrial-automation/page.tsx

Custom standalone page for advanced industrial automation. Independent layout from slug system.

/services/instrumentation

Precision Process Instrumentation

Source File

src/app/services/instrumentation/page.tsx

Pressure transmitters, flow meters, temperature sensors, field calibration services.

/services/electrical-systems

High-Performance Electrical Systems

Source File

src/app/services/electrical-systems/page.tsx

Electrical design, installation, testing, and AMC for industrial facilities.

/services/robotics

Industrial Robotics Integration

Source File

src/app/services/robotics/page.tsx

Yaskawa Motoman robot arm integration, pick-and-place, palletizing, welding automation.

/services/elevator-solutions

Smart Elevator & Lift Solutions

Source File

src/app/services/elevator-solutions/page.tsx

VFD-driven passenger and goods lifts, modernization, Yaskawa L1000A elevator drives.

/services/control-panels

Advanced Electrical Control Panels

Source File

src/app/services/control-panels/page.tsx

CPRI-certified PCC/MCC panels, busbar assembly, form-4b segregation, FAT testing.

/industries

Industries We Serve

Source File

src/app/industries/page.tsx

8 industrial sectors with specific pain points and Nandini solutions per sector.

Sections / Form Fields:

- Steel & Metals
- Cement & Mining
- Oil & Gas Refineries
- Power Generation
- Pharmaceutical & Cleanrooms
- Water Treatment & Municipal
- Food & Beverage Processing
- Textile & Garment Mills

/projects

EPC Projects Portfolio

Source File

src/app/projects/page.tsx

Gallery of 500+ completed large-scale EPC projects. AI-generated industrial scene images. Cards with client, scope, and outcome.

/contact

Contact & RFQ Submission

Source File

src/app/contact/page.tsx

Primary lead capture page with dynamic form, contact details, map, and WhatsApp integration.

Sections / Form Fields:

- Full Name
- Company Name
- Email Address

- Phone Number
- Service Category (dropdown)
- Project Description
- Budget Range
- Timeline

/gallery Manufacturing Facility Gallery

Source File `src/app/gallery/page.tsx`

Photo gallery of Cherlapally facility, panel assembly floor, testing rigs, and client site installations.

/certifications Compliance & Certifications

Source File `src/app/certifications/page.tsx`

All active certifications: ISO 9001:2015, CPRI test certificates, Grade-A license, partner authorization letters.

/knowledge-center White Papers & Technical Articles

Source File `src/app/knowledge-center/page.tsx`

Technical knowledge hub for B2B engineering buyers with white papers and guides.

/products Products Catalogue

Source File `src/app/products/page.tsx`

Procurement catalogue: VFDs, relays, sensors, cables, switchgears, HMI panels, encoders.

SECTION 5

Feature Index & Implementation Guide

F-01 to F-12: every active feature with code logic and dependencies

This is the primary reference for any engineer understanding or rebuilding this system. Each entry covers the trigger, description, implementation detail, and dependency stack.

F-01

Hero Slideshow with Auto-Advance

Source File

src/components/Home/HeroSlideshow.tsx

Trigger

Automatic on homepage mount

Description:
Full-screen image carousel with 5 AI-generated industrial backgrounds. Auto-advances every 5 seconds. Manual dot indicator navigation. Smooth CSS opacity crossfade transitions. Gradient overlay for text legibility. Animated heading and sub-headline text.

Implementation:
useState for activeSlide index. useEffect + setInterval for auto-advance. Next.js Image components. CSS opacity transitions on absolute-positioned slides.

Dependencies next/image, React hooks

F-02

Global Right-Click Protection

Source File

src/components/Layout/DisableRightClick.tsx

Trigger

Mounts globally in RootLayout (layout.tsx)

Description:
Attaches document-level contextmenu event listener calling preventDefault() on all right-clicks. Prevents source inspection, image saving, link copying. Renders null (no visible UI).

Implementation:
useEffect attaches document.addEventListener("contextmenu", handler). Cleanup on unmount.

Note:
EXCEPTION: Sitemap footer link uses e.stopPropagation() to allow double-right-click for confidential export.

Dependencies React useEffect

F-03

[HIDDEN] Confidential PDF Export Trigger

Source File

src/components/Layout/Footer.tsx + src/app/api/export-source/route.ts

Trigger

Double right-click on "Sitemap" link in footer within 800ms

Description:
Invisible to all external parties. On double right-click: shows "COMPILING..." loader toast, captures JPEG screenshot via html2canvas (0.5 scale), POSTs to /api/export-source with secret token, receives comprehensive PDF blob, auto-downloads as Nandini_Source_Export.pdf.

Implementation:
Footer.tsx tracks lastRightClickRef timestamp. stopPropagation bypasses global right-click block. API route uses pdf-lib (pure JS) - zero AFM font file dependencies - works on Vercel serverless.

Dependencies html2canvas (client), pdf-lib (server), React useState/useRef

F-04

Internal Knowledge Navigation Chatbot

Source File

src/components/Assistant/KnowledgeAssistant.tsx

Trigger

Click chat bubble icon (bottom-right above WhatsApp button)

Description:
Floating chat assistant for internal navigation and client support. Keyword-based rule engine (no external AI API). Handles queries: projects, services, automation, panels, VFD, contact, quote, about, company, industries. 600ms simulated thinking delay.

Implementation:

Pure React state. generateResponse() matches lowercase query against keyword sets. Messages array stores history. Fixed bottom-24 right-6 z-50. Chat window: 340x480px.

Dependencies React hooks, next/link

F-05 Dynamic Lead Capture RFQ Form

Source File src/app/page.tsx, src/app/contact/page.tsx

Trigger User scrolls to form section or visits /contact

Description:

Adaptive multi-field form. Fields change based on selected service category. Fields: service type, project description, timeline, budget range, contact details. Controlled components with validation.

Implementation:

React controlled components. useState for form state. Service category selector triggers conditional field rendering.

Dependencies React hooks

F-06 WhatsApp Floating Action Button

Source File src/components/Layout/Footer.tsx

Trigger Always visible - fixed bottom-6 right-6 z-50

Description:

Green FAB opens WhatsApp chat with +91-40-23190131. Hover shows "Talk to an Engineer" tooltip. Smooth scale-up hover animation.

Implementation:

Anchor tag with href=wa.me/914023190131. CSS hover:scale-110 transition. Tooltip with opacity-0 group-hover:opacity-100.

Dependencies None (pure HTML/CSS)

F-07 Sticky Header with Navigation

Source File src/components/Layout/Header.tsx

Trigger Always visible - sticky top-0 z-50

Description:

Enterprise sticky navigation. Company logo/wordmark, main nav links (Services dropdown, Projects, Industries, Resources, Contact), and "Request Quote" CTA. Mobile hamburger menu. Services link has mega-dropdown with 6 service categories.

Implementation:

React with mobile menu state. useEffect listens to window scroll for shadow on scroll.

Dependencies React hooks, next/link, next/navigation

F-08 Animated Statistics Counters

Source File src/app/page.tsx (Why Choose Us section)

Trigger Element enters viewport via scroll

Description:

Number counters animate from 0 to target when visible. Stats: 500+ Projects, 30+ Years Experience, 24/7 Support, ISO 9001:2015.

Implementation:

react-countup with react-intersection-observer to trigger only on viewport entry.

Dependencies react-countup, react-intersection-observer

F-09 Framer Motion Page Animations

Source File src/app/page.tsx (multiple sections)

Trigger Page load and scroll

Description:

Fade-up, stagger, and scale entrance animations on section cards. Industry cards, service cards, process steps all animate in.

whileInView triggers animations on scroll.

Implementation:

motion.div with initial/animate/transition. Stagger via variants.staggerChildren.

Dependencies	framer-motion
---------------------	---------------

F-10

Dynamic Service Pages (SSG)

Source File	src/app/services/[slug]/page.tsx, src/lib/servicesData.ts
Trigger	Route visit e.g. /services/panel-manufacturing

Description:

Statically pre-generated at build time via generateStaticParams. Data: title, subtitle, stats[4], overview[4 paragraphs], challenge case, 4-step workflow, architecture, industries[3], full case study, tech ecosystem[3], QA[4], safety compliance[3], AMC[4], docs[3], 8 FAQs.

Implementation:

Next.js generateStaticParams exports all slugs. Page receives params.slug, fetches SERVICES_DATA record, renders ServicePageClient.

Dependencies	src/lib/servicesData.ts, src/components/Service/ServicePageClient.tsx
---------------------	---

F-11

SEO Metadata & OpenGraph

Source File	src/app/layout.tsx
Trigger	Server-rendered on every request

Description:

Root layout exports Next.js Metadata: title, description, keywords array, OpenGraph (title, description, url, siteName, locale, type), robots (index: true, follow: true).

Implementation:

Next.js App Router built-in Metadata API. Sub-pages can override with own metadata export.

Dependencies	Next.js Metadata API
---------------------	----------------------

F-12

Secret Token Security Gate on PDF API

Source File	src/app/api/export-source/route.ts
Trigger	Every POST request to /api/export-source

Description:

Every request is validated against EXPORT_SECRET_TOKEN environment variable. Missing or wrong token returns 404 Not Found - identical to a missing route, preventing discovery. Token is stored in Vercel env vars (not GitHub). File itself is gitignored.

Implementation:

reads req.headers.get("X-Export-Token") and compares with process.env.EXPORT_SECRET_TOKEN. Returns Response("Not Found", {status:404}) on mismatch.

Dependencies	process.env, Next.js App Router
---------------------	---------------------------------

SECTION 6

Component Library Reference

All React components with type, path, and description

Header.tsx Client Component	
Path	src/components/Layout/Header.tsx
Sticky navigation with mobile menu and service dropdown mega-menu	
Footer.tsx Client Component	
Path	src/components/Layout/Footer.tsx
Multi-column footer with partner logos, WhatsApp FAB, and hidden PDF export double-right-click trigger	
DisableRightClick.tsx Client Component	
Path	src/components/Layout/DisableRightClick.tsx
Document-level right-click prevention. Renders null. Exception: Sitemap link uses stopPropagation	
KnowledgeAssistant.tsx Client Component	
Path	src/components/Assistant/KnowledgeAssistant.tsx
Floating keyword-based chatbot for internal navigation support	
HeroSlideshow.tsx Client Component	
Path	src/components/Home/HeroSlideshow.tsx
Full-screen 5-image auto-advance carousel for homepage hero	
ServicePageClient.tsx Client Component	
Path	src/components/Service/ServicePageClient.tsx
Rich service detail page renderer consuming ServiceDetails data objects from servicesData.ts	

SECTION 7

Known Issues & Resolutions

All bugs encountered and their exact technical fixes

ISSUE: pdfkit Helvetica.afm ENOENT on Vercel

FIX: Replaced pdfkit with pdf-lib (pure JS). StandardFonts embedded as base64 inside the library.

RESOLVED

ISSUE: Buffer type mismatch in Response constructor

FIX: Wrapped Uint8Array in Buffer.from() before passing to new Response().

RESOLVED

ISSUE: html2canvas payload too large at scale:1 PNG

FIX: Reduced to scale:0.5, output as JPEG quality 0.6. Stays under Vercel 4.5MB limit.

RESOLVED

ISSUE: outputFileTracingIncludes in experimental block

FIX: Moved to root level in next.config.mjs per Next.js 16 deprecation notice.

RESOLVED

ISSUE: Footer was Server Component - no useState allowed

FIX: Added "use client" directive to Footer.tsx.

RESOLVED

ISSUE: API route visible on GitHub

FIX: Added to .gitignore. Removed from git tracking via git rm --cached. Deploy via vercel --prod only.

RESOLVED

SECTION 8

Deployment & Confidential Access Setup

Vercel CLI, env vars, GitHub separation, and recovery steps

Confidential Access Credentials

Secret Token Value: 03593D49A7034868BA70DD69CB47E9D
Header Name: X-Export-Token
API Endpoint: /api/export-source [POST method]
Trigger: Double right-click on "Sitemap" link in website footer (within 800ms)

Deployment Commands (Run from Project Root)

```
# Link project to Vercel (run once)
vercel link --yes --project nandiniautomation

# Add secret token to Vercel env (run once after any token change)
echo "YOUR_TOKEN" | vercel env add EXPORT_SECRET_TOKEN production

# Deploy ALL files including confidential route (run after every update)
vercel --prod --yes

# Normal GitHub push (does NOT include confidential route):
git add .
git commit -m "your message"
git push origin main
# Then immediately follow with: vercel --prod --yes
```

What is on GitHub vs Vercel

GitHub (Public/Private repo)

Vercel Production (Live site)

.gitignore entries

.env.local

Vercel Env Vars

All website pages, components, styles - NO confidential route
Everything including /api/export-source route (deployed via CLI)
src/app/api/export-source/ and src/fonts/ - never committed
Contains EXPORT_SECRET_TOKEN - gitignored, exists only on your machine
EXPORT_SECRET_TOKEN stored encrypted in Vercel dashboard

Disaster Recovery Steps

- If machine is lost: Re-clone from GitHub, re-create src/app/api/export-source/route.ts from memory or this PDF
- Set EXPORT_SECRET_TOKEN in Vercel dashboard ? Settings ? Environment Variables
- Add token to local .env.local file
- Run: vercel --prod --yes to redeploy with confidential route

SECTION 9

Services Content - Full Text Data

Complete service descriptions and FAQs optimized for AI pricing analysis

This section contains the complete human-readable text data for all service categories as stored in servicesData.ts. This is optimized for LLM analysis and pricing breakdown generation.

Power distribution and motor management represent the core electrical heart of any heavy process facility. At Nandini Enterprises, we engineer and manufacture custom low-voltage (LV) power switchgear assemblies that act as the reliable nervous system of your factory. Our panel manufacturing process is built on strict compliance with the latest IEC 61439-1 & 2 design standards, assuring safe, reliable, and thermally optimized performance under continuous maximum loads.

Every single panel board starts its life in our state-of-the-art AutoCAD mechanical design suite. We perform extensive 2D layout planning and 3D thermal simulations to model heat dissipation patterns inside the cabinet enclosure. This design planning guarantees that internal switchgears, contactors, and variable speed drives remain within their nominal operating temperatures, directly preventing premature component failures and subsequent assembly downtime.

Our fabrication plant utilizes top-tier, high-conductivity electrolytic grade copper and aluminum busbars. The busbars are processed using hydraulic punching, bending, and cutting machinery to achieve precise millimetric alignment. This precision prevents local mechanical tensions or hot-spots that could trigger short-circuits. Busbars are then sleeved using heavy-duty flame-retardant heat-shrinkable wraps, labeled with color-coded safety identifiers, and secured with epoxy support insulators.

Quality assurance is integrated into every stage of fabrication. Our testing laboratory executes high-voltage insulation tests, continuity checks, control logic functional loop simulation, and dynamic current load checks. Nandini Enterprises panels are CPRI (Central Power Research Institute) certified for up to 65kA short circuit withstand ratings, giving procurement managers, factory directors, and EPC contractors absolute confidence in our installations.

Busbar Chambers: Run horizontally and vertically, wrapped in heat-shrink sleeves and isolated from breaker compartments.

Switchgear Modules: Integrated with Air Circuit Breakers (ACB) or Molded Case Circuit Breakers (MCCB) matching coordination curves.

Thermal Ventilation: Active twin fan cooling systems running on automatic digital temperature triggers.

Earthing: Dedicated internal copper ground running the entire panel length with external earth pit connectivity hooks.

Insulation resistance check using 1000V Megger (minimum 100M Ω threshold).

High Voltage dielectric checkup at 2.5kV applied for 60 seconds with zero leakage.

Control wiring continuity validation using computer-aided testing setups.

Thermal camera scan of all busbar connections under artificial load tests.

Preventive quarterly cleaning, busbar joint torque checks, and insulator checking.

Thermal profiling of all cables and contactors using infrared sensors.

Relay logic recalibration and dynamic trip testing using injection kits.

Modernization of legacy panels: retrofitting contactors with energy efficient starter kits.

Process optimization in modern heavy manufacturing demands precise, instantaneous, and automated control logic. Nandini Enterprises delivers enterprise-grade PLC (Programmable Logic Controller) and SCADA (Supervisory Control and Data Acquisition) system integration. We bridge the gap between heavy field machinery and corporate database management, enabling factory managers to execute real-time recipe profiles, monitor alarms, and log key performance indicators.

Our automation division develops structured, fail-safe logic code utilizing international standard IEC 61131-3 programming languages, including ladder logic, structured text, and sequential function charts. We program and configure high-performance CPU architectures from leading global brands, implementing dual-CPU hot standby redundancy. This ensures that if a main controller fails, the backup unit assumes control in under 10 milliseconds without process interruption.

At the supervision level, we design rich, high-fidelity SCADA interfaces. Our screen layout designs follow the ISA-101 HMI standard, using color-coded profiles to prioritize active alerts over static variables. Our SCADA databases are equipped with SQL server engines, enabling historical trends logging, automated PDF reporting, and FDA 21 CFR Part 11 compliant electronic signature logging for pharmaceuticals and water treatment grids.

System integration includes commissioning high-speed communication backbones. We design and install redundant industrial Ethernet networks, Modbus TCP/IP loops, Profinet rings, and fiber optic telemetry channels. This ensures absolute signal integrity, protecting critical feedback loops from electro-magnetic noises generated by heavy VFDs and switchgears.

Control Network: Configured on a redundant fiber optic ring running Profinet protocol for rapid data exchange.

Redundant CPU Hot Standby: CPU tracking modules share variables at 1Gbps, executing bumpless switchovers on hardware trip.

Audit Database: Integrated database tables capturing timestamped user logins, alarm acknowledgements, and parameter modifications.

OPC UA Servers: Provide secure links to ERP systems for inventory forecasting.

Simulated logic validation checks under multi-fault scenario injections.

CPU failsafe switchover check (verifying backup takes over under 10ms).

Network storm stress verification on communication rings.

Data logging accuracy comparison between physical analog meters and database values.

Annual software backups, backup battery updates, and firmware version security patching.

Database cleaning, archive file creation, and query speed tuning.

AMC support: 24/7 remote diagnostic login to resolve logic blocks.

Modernization: converting old legacy code scripts (e.g. S5 PLCs) to S7 structures.

Variable Frequency Drives (VFDs) are the core speed regulators of the industrial landscape, matching motor outputs directly to load profiles.

Nandini Enterprises is the Authorized Service Center and System Integrator for Yaskawa India. We deliver complete drive system engineering, including motor parameter auto-tuning, torque profile configuration, harmonic mitigation filter setups, and rapid component-level repairs.

Our drive specialists analyze motor loads to optimize acceleration/deceleration S-curve speed paths. This is particularly crucial for cranes, hoists, and material handling systems where precise brake control logic and load slip protection are required. We program Yaskawa's advanced vector control algorithms to achieve maximum motor torque at zero speed, eliminating slips and minimizing mechanical gear wear.

Harmonic current distortions from heavy VFD switches can pollute your plant power grid, causing other electronic equipment to malfunction and triggering utility penalties. We resolve this by performing electrical system harmonic audits, designing and installing input/output AC reactors, passive filters, and active harmonic filters that limit harmonic levels within IEEE 519 compliance standards.

As an Authorized Yaskawa Service Center, our Cherlapally base features advanced drive test benches, load testing rigs, and a vast stock of genuine spares, including IGBT modules, driver boards, cooling fans, and capacitors. Our highly trained service engineers offer round-the-clock emergency support, performing repair works, load validation checks, and diagnostic evaluations to restore operations quickly.

AC Line Reactors: Mounted on VFD input terminals to suppress voltage surges and reduce harmonic currents.

Braking Resistors: Absorb regenerative energy from high-inertia deceleration loads to prevent overvoltage faults.

Ventilation System: Cabinet internal air pathways designed to maintain heatsink temperature below 45°C.

Profinet Cards: Provide real-time speed, current, and error diagnostic logs to supervisory control systems.

Motor parameter auto-tuning checks and speed-accuracy audits.

Load run testing for 4 hours at maximum motor current ratings.

Harmonic current measurement checks against IEEE 519 guidelines.

IGBT gate driver pulse diagnostic check using digital oscilloscopes.

Preventive cleaning, busbar connections checks, capacitor health audits, and fan replacements.

Drive param values backup creation and parameter fine-tuning adjustments.

24/7 on-site emergency troubleshooting support within 4 hours in Hyderabad.

Modernization: retrofitting obsolete VFD units with modern Yaskawa GA700 modules.

Modern B2B production throughput requires speed, repeatability, and precision beyond manual limits. Nandini Enterprises designs and integrates industrial robotic workstations and multi-axis motion control systems, focusing on pick-and-place lines, automatic palletizers, and robotic welding stations.

We partner with leading global robotics brands, focusing on Yaskawa Motoman arms, to build turnkey automation cells. Our engineering team handles the complete system layout, including selecting the arm kinematic reach, designing custom mechanical end-of-arm tooling (EOAT) like vacuum grippers, and installing surrounding protective safety fencing with interlocks.

To sync robot movements with feed lines, we develop high-speed motion control algorithms. We configure hardware interfaces to track conveyor speeds dynamically, ensuring robots pick objects on the move. We install safety systems, including laser area scanners and light curtains, designed to slow or stop the robot if a worker enters the cell.

We support our clients through the entire validation cycle, performing reachability analysis and offline collision checking. Our service contracts include mechanical inspection, gearbox oil analysis, battery updates, and program optimization to maintain peak performance over long production run times.

Robot Controller: Yaskawa YRC1000 managing servo motors and trajectory paths.

Safety PLC: Monitors gate switches, light curtains, and emergency stops.

EtherCAT Network: Links sensors, encoders, and motor starters.

Operator HMI: Touch screen display showing speed, cycles, and diagnostics.

Path repeat accuracy check (+/- 0.02mm limit verified).

E-stop response time validation (verifying motion stops in milliseconds).

Continuous cycle run testing for 8 hours under maximum payloads.

EOAT vacuum pressure holding check under emergency power cuts.

Annual gearbox oil testing, backup battery changes, and cable harness checks.

Program path tweaks to optimize cycle times.

On-site repair support within 24 hours of notification.

Modernization: rebuilding and reprogramming old robot cells for new tasks.

Control systems depend on the accuracy of field instrumentation. Nandini Enterprises delivers comprehensive process instrumentation and sensor calibration, specializing in flow meters, pressure transmitters, level controllers, temperature sensors, and gas analyzers.

We design field wiring loops using high-quality shielded cables and barrier isolators to protect signals from electrical noises. We configure field networks using HART, Modbus, and Profibus protocol suites, enabling smart diagnostics and remote configuration.

Calibration is key to maintaining process safety. We perform wet and dry calibration at client sites or in our Hyderabad laboratory, utilizing

reference standards traceable to NABL. We issue calibration certificate formats detailing measurement errors and span configurations. We provide analytical systems, including continuous emission monitoring (CEMS), water quality analyzers, and steam and water analysis systems (SWAS) for power generation, cement, and chemical facilities, ensuring strict environmental compliance.

Analog Signal Isolation: Galvanic isolator modules isolate control loops from field ground loop disturbances.

HART Communication: Digital data superimposed on 4-20mA loops for remote diagnostics.

Manifold Valves: Dual block and bleed configurations for safe instrument removal under line pressure.

Junction Boxes: IP66 rated, flameproof enclosures matching hazardous area zone requirements.

Zero-point verify checks and span accuracy tests.

NABL traceable calibration validation logs creation.

Overpressure leak testing for 15 minutes at 1.5x nominal pressure rating.

Analog output verification checks at 4mA, 8mA, 12mA, 16mA, and 20mA states.

Preventive sensor cleaning, check connection seals, and inspect cable insulation.

Recalibrate sensors every 6 or 12 months.

On-site repair support within 24 hours of notification.

Modernization: replacing old analog gauges with smart transmitters.

In vertical transport systems, passenger safety, travel comfort, and energy efficiency depend on controller logic and drive performance. Nandini Enterprises delivers elevator control systems, integrating Yaskawa elevator drives, microprocessor controllers, and automatic devices (ARD) to ensure reliable operation.

Our engineering team designs complete control panels for passenger and goods lifts. We configure Yaskawa L1000A drives with auto-tuning routines to adapt to motor parameters, ensuring smooth torque delivery at low speeds and eliminating cabin start jerks.

Passenger safety is our highest priority. We integrate automatic rescue devices (ARD) that automatically move the cabin to the nearest floor and open doors during utility power failures, protecting passengers from entrapment.

Our control systems implement energy optimization strategies, including regenerative drive systems that feed braking energy back to the building grid. Our service division offers regular inspections, rope checks, leveling calibration, and control upgrades.

Microprocessor Board: Core controller managing call buttons, landing indicators, and door interlocks.

Yaskawa L1000A: Lift drive configured for smooth S-curve travel profiles.

Automatic Rescue Device (ARD): Battery backup system powering rescue moves during utility power trips.

Leveling Sensors: Photoelectric and magnetic sensors ensuring precise cabin alignment.

Leveling accuracy validation check (+/- 3mm threshold verified).

Door interlock safety switch check under travel states.

Continuous run testing for 6 hours under full load conditions.

ARD battery health and charge logic checks.

Preventive rope checks, terminal joint torque audits, and safety switch tests.

Speed curve parameter tuning and leveling sensor cleaning.

24/7 on-site emergency troubleshooting support within 4 hours in Hyderabad.

Modernization: replacing obsolete analog relays with microprocessor panels.

High voltage substations and industrial electrical distributions form the primary backbone of heavy manufacturing. Nandini Enterprises delivers complete High Tension (HT) and Low Tension (LT) electrical engineering services. We hold an A-Grade 33KV electrical contractor license, authorizing us to design, erect, test, and commission switchyards, transformer networks, distribution loops, and switchgear cabinets. Our switchyard division designs substation layouts, substation earthing mats, and overhead cable entries. We perform fault level calculations and protection coordination audits, configuring relay trip curves (overcurrent, earth fault, differential protection) to isolate faults quickly and prevent damage to expensive equipment like transformers and generators.

We design and install cable tray systems, high-power copper and aluminum busducts, and armored underground cables. Our electrical engineering team calculates cable thermal ratings and voltage drops under load conditions, choosing proper conductor sizes to minimize transmission losses.

We handle the entire commissioning cycle, performing insulation testing (Megger), transformer turns ratio check, winding resistance measurements, switchgear contact resistance checks, and earth pit testing. Our service team offers regular AMC contracts, switchyard cleaning, oil filtration, and relay diagnostic testing.

HT incomer: 33KV/11KV vacuum circuit breakers protecting substation transformers.

Busbar Bridge: High-capacity aluminum or copper busducts linking transformer terminals to LT switchgear chambers.

Relay Protection: Digital relay modules tracking overcurrent, earth faults, and under-voltage limits.

Earthing Network: Grid of copper tape conductors linked to deep earth pits.

Winding insulation resistance test using 5000V Megger (minimum 1000M Ω limit).

Switchgear breaker contact resistance measurement check.

Substation transformer oil breakdown voltage test (minimum 60kV limit).

Relay trip timing validation check using injection test benches.

Preventive switchyard cleaning, connection checks, oil filtration runs, and earthing pit tests.

Substation transformer oil parameter audits and gas chromatography tests.

24/7 on-site diagnostic support and emergency VCB troubleshooting.

Modernization: replacing old analog gauges with smart digital relays.

Industrial automation and electrical systems require regular maintenance to maintain long-term reliability and safety. Nandini Enterprises delivers commissioning, testing, and Annual Maintenance Contract (AMC) support. We act as a reliable partner for steel, cement, power, and chemical facilities across India.

Our services cover the entire project lifecycle, from initial Factory Acceptance Testing (FAT) at our Hyderabad center to Site Acceptance Testing (SAT), final commissioning, and long-term AMC support. We inspect and verify all safety loops, protection relays, communication protocols, and drive settings.

Our AMC contracts are tailored to each facility's operational needs. We offer preventive maintenance, predictive diagnostics using thermal cameras, and emergency troubleshooting support. Our service team is available 24/7 to resolve drive faults and logic issues.

We provide modernization and retrofitting, upgrading old switchgear, obsolete PLCs, and worn-out motor starters with energy-efficient systems, keeping your plant running safely and efficiently.

Service Desk: Dedicated ticket logging system tracking response times and resolutions.

Field Spares: On-site inventory of critical VFD cards, relays, fuses, and fans.

Diagnostic Tools: Portable oscilloscopes and loop calibrators for rapid troubleshooting.

Commissioning Checklists: Standardized FAT/SAT formats verifying every control loop.

Zero-point check and span calibration audits on all transmitters.

Relay trip timing validation checks (verifying compliance with safety curves).

Dynamic motor run tests verifying speed-accuracy.

Substation grounding system earth resistance checks.

Preventive quarterly cleaning, busbar torque audits, and safety switch tests.

Relay calibration checks and parameter updates.

24/7 on-site emergency troubleshooting support within 4 hours in Hyderabad.

Modernization: retrofitting obsolete VFD systems with GA700 modules.

SECTION 10

Project Directory Tree & Structure

Full annotated filesystem topology of the codebase

Complete filesystem structure excluding node_modules, .next, .git. Gives any engineer or AI an instant overview of the codebase topology.

```
nandini-enterprises/
|-- content/
|   |-- services/
|   |   |-- engineering-support.json
|   |   |-- panel-manufacturing.json
|   |   |-- plc-programming.json
|   |-- public/
|   |   |-- images/
|   |   |   |-- cement_plant.png
|   |   |   |-- food_factory.png
|   |   |   |-- oil_refinery.png
|   |   |   |-- pharma_lab.png
|   |   |   |-- slide1.png
|   |   |   |-- slide2.png
|   |   |   |-- slide3.png
|   |   |   |-- slide4.png
|   |   |   |-- slide5.png
|   |   |   |-- steel_plant.png
|   |   |   |-- water_treatment.png
|   |   |-- file.svg
|   |   |-- globe.svg
|   |   |-- next.svg
|   |   |-- vercel.svg
|   |   |-- window.svg
|   |-- src/
|   |   |-- app/
|   |   |   |-- about/
|   |   |   |   |-- page.tsx
|   |   |   |-- api/
|   |   |   |   |-- export-source/
|   |   |   |   |   |-- route.ts
|   |   |   |-- certifications/
|   |   |   |   |-- page.tsx
|   |   |   |-- contact/
|   |   |   |   |-- page.tsx
|   |   |   |-- gallery/
|   |   |   |   |-- page.tsx
|   |   |   |-- industries/
|   |   |   |   |-- page.tsx
|   |   |   |-- knowledge-center/
|   |   |   |   |-- page.tsx
|   |   |   |-- products/
|   |   |   |   |-- page.tsx
|   |   |   |-- projects/
|   |   |   |   |-- page.tsx
|   |   |   |-- services/
|   |   |   |   |-- [slug]/
|   |   |   |   |   |-- page.js
|   |   |   |   |   |-- page.tsx
|   |   |   |   |   |-- service.module.css
|   |   |   |   |-- control-panels/
|   |   |   |   |   |-- page.tsx
|   |   |   |   |-- electrical-systems/
|   |   |   |   |   |-- page.tsx
|   |   |   |   |-- elevator-solutions/
|   |   |   |   |   |-- page.tsx
|   |   |   |   |-- industrial-automation/
|   |   |   |   |   |-- page.tsx
|   |   |   |   |-- instrumentation/
|   |   |   |   |   |-- page.tsx
|   |   |   |   |-- robotics/
|   |   |   |   |   |-- page.tsx
|   |   |   |   |-- page.tsx
|   |   |-- favicon.ico
```

```
| | |-- globals.css
| | |-- layout.tsx
| |-- page.tsx
| |-- components/
| | |-- Assistant/
| | | |-- KnowledgeAssistant.tsx
| | |-- Home/
| | | |-- HeroSlideshow.tsx
| | |-- Layout/
| | | |-- DisableRightClick.tsx
| | | |-- Footer.tsx
| | | |-- Header.tsx
| | |-- Service/
| | | |-- ServicePageClient.tsx
| |-- fonts/
| | |-- arial.ttf
| | |-- arialbd.ttf
| | |-- cour.ttf
| |-- lib/
| | |-- cms.ts
| | |-- data.ts
| | |-- servicesData.ts
|-- about.html
|-- about.txt
|-- AGENTS.md
|-- CLAUDE.md
|-- eslint.config.mjs
|-- fetch-ids.js
|-- NE_Profile_2026.docx
|-- next-env.d.ts
|-- next.config.mjs
|-- package.json
|-- postcss.config.mjs
|-- README.md
|-- team.html
|-- team.txt
+-- tsconfig.json
```

SECTION 11

Complete Source Code

Every .ts .tsx .css .mjs .json source file verbatim

All source code files in dependency order: config files first, then src/ files alphabetically. Each file is clearly separated with a header card.

```
{
  "name": "temp-app",
  "version": "0.1.0",
  "private": true,
  "scripts": {
    "dev": "next dev --webpack",
    "build": "next build --webpack",
    "start": "next start",
    "lint": "eslint"
  },
  "dependencies": {
    "@tailwindcss/forms": "^0.5.11",
    "@tailwindcss/typography": "^0.5.19",
    "autoprefixer": "^10.5.0",
    "clsx": "^2.1.1",
    "framer-motion": "^12.40.0",
    "html2canvas": "^1.4.1",
    "lucide-react": "^1.17.0",
    "next": "16.2.6",
    "pdf-lib": "^1.17.1",
    "pdfkit": "^0.18.0",
    "postcss": "^8.5.15",
    "react": "19.2.4",
    "react-countup": "^6.5.3",
    "react-dom": "19.2.4",
    "react-intersection-observer": "^10.0.3",
    "tailwindcss": "^4.3.0"
  },
  "devDependencies": {
    "@tailwindcss/postcss": "^4.3.0",
    "@types/node": "^25.9.1",
    "@types/pdfkit": "^0.17.6",
    "@types/react": "^19.2.15",
    "@types/react-dom": "^19.2.3",
    "eslint": "^9",
    "eslint-config-next": "16.2.6",
    "typescript": "^6.0.3"
  }
}
```

```
{
  "compilerOptions": {
    "target": "es5",
    "lib": [
      "dom",
      "dom.iterable",
      "esnext"
    ],
    "allowJs": true,
    "skipLibCheck": true,
    "strict": false,
    "forceConsistentCasingInFileNames": true,
    "noEmit": true,
    "esModuleInterop": true,
    "module": "esnext",
    "moduleResolution": "bundler",
    "resolveJsonModule": true,
    "isolatedModules": true,
    "jsx": "react-jsx",
    "incremental": true,
    "plugins": [
      {
        "name": "next"
      }
    ],
    "paths": {
      "@/*": [
        "./src/*"
      ]
    }
  },
  "include": [
    "next-env.d.ts",
    "**/*.ts",
    "**/*.tsx",
    ".next/types/**/*.ts",
    ".next/dev/types/**/*.ts"
  ],
  "exclude": [
    "node_modules"
  ]
}
```

```
/** @type {import('next').NextConfig} */
const nextConfig = {
  outputFileTracingIncludes: {
    '/api/export-source': [
      'src/**/*',
      'package.json',
      'tsconfig.json',
      'next.config.mjs',
      'postcss.config.mjs',
      'eslint.config.mjs'
    ],
  },
};

export default nextConfig;
```



```
export default {  
  plugins: {  
    '@tailwindcss/postcss': {},  
    'autoprefixer': {},  
  },  
};
```

```
import { defineConfig, globalIgnores } from "eslint/config";
import nextVitals from "eslint-config-next/core-web-vitals";

const eslintConfig = defineConfig([
  ...nextVitals,
  // Override default ignores of eslint-config-next.
  globalIgnores([
    // Default ignores of eslint-config-next:
    ".next/**",
    "out/**",
    "build/**",
    "next-env.d.ts",
  ]),
]);

export default eslintConfig;
```

```
export default function AboutPage() {?
  const managementTeam = [?
    { name: "Mr. Maruti Marlapalle", title: "Managing Partner", desc: "Founder of NANDINI ENTERPRISES with a unique blend of core",
    { name: "Mrs. Pratima Maruti Marlapalle", title: "Partner", desc: "Instrumental in handling Employee Management, Finance, Sto",
    { name: "Mr. Srinivas", title: "Manager Sales & Services - AP", desc: "20+ years experience in Instrumentation & Automation.",
    { name: "Mr. Subrahmanyam Gavalapalli", title: "Manager- Sales and Service - Elevators", desc: "10 years experience with chal",
    { name: "Mr. Kedar Reddy", title: "Manager BD", desc: "B Tech (E&TC) handling YASKAWA Industrial drive sales since 2013. High",
    { name: "Mrs. Seema Godbole", title: "Operations Management", desc: "Performance Driven Professional with successful operatio",
    { name: "Mr. Sri Bharath", title: "Business Development Manager", desc: "B.E (Electronics and Instrumentation) professional w",
    { name: "Mr. Ravichandra", title: "Sales & Marketing", desc: "MBA in Marketing and technical degree in Electronics and Commun",
    { name: "Mr. B Venkat", title: "Design Manager", desc: "12 years experience handling complete Panel Division from Estimations",
    { name: "Mr. Shaik Nazeer", title: "Design & Execution", desc: "25 years experience in Engineering and Erection & Commissioni",
    { name: "Mrs. Sravani", title: "Panel Team Support", desc: "Fast learner with good logical skills making her a multitasking e",
    { name: "Mr. K. Surya", title: "Panel Design & Testing", desc: "7 years of Experience assisting in panel design and actively",
    { name: "Mr. SV Rao", title: "Factory Supervisor", desc: "20 years experience supervising all panel manufacturing & execution",
    { name: "Mr. Jayachandra Y.", title: "Projects Head", desc: "Responsible for all project execution starting from Order accept",
    { name: "Mr. Dileep S.", title: "Project & Site Execution", desc: "8 years experience in Project and Site Execution working u",
    { name: "Mr. Mohammed Omair Nasser", title: "Senior Service Engineer", desc: "Technically strong senior engineer perfectly su",
    { name: "Mr. Durgesh Lalaji", title: "VFD Commissioning & Servicing", desc: "Hands on expertise on Yaskawa VFDs Commissioning",
    { name: "Mr. Nagababu K", title: "Service & Projects - AP", desc: "5 years experience handling Government projects and on-sit",
    { name: "Mr. Junaid", title: "VFD, PLC & HMI Service", desc: "Handles service calls individually with a gentle nature that al",
    { name: "Mr. Gangadhar Rao", title: "Dyno Drive & Railways Specialist", desc: "Remarkable job in Railways Dyno Drive Test ben",
    { name: "Mr. Mithilesh", title: "VFD Installation Expert", desc: "Plays a key role in installations & service related issues,",
    { name: "Mr. Dnyandeav", title: "Installation & Electrical Works", desc: "Shows great interest in learning & growing with the",
    { name: "Mr. Hussain Pasha", title: "Instrumentation Head", desc: "5 years experience in PLC, SCADA, Servo & AC Drives servin",
    { name: "Mr. Syed Umais", title: "Industrial Automation Support", desc: "BTECH (ECE) professional with extended support in se",
    { name: "Mr. B V Rao", title: "Accounts & Taxation", desc: "15 years of experience in Accounting, Banking, Finance and Taxing",
    { name: "Mr. Narsimulu Bynagari", title: "Tender & ISO Documentations", desc: "6 years experience maintaining Tender, ISO and",
    { name: "Mr. Sridhar", title: "Payment Follow-up", desc: "Main focus is on Payment Follow up and collection to ensure smooth",
    { name: "Mrs. Rani Bollam", title: "Purchase Manager", desc: "18 Years experience handling all purchases, E-procurement and E",
    { name: "Mr. Anil Kumar Sahoo", title: "Digital Marketing & IT", desc: "Designs Brochures, Presentations, maintains the websi",
    { name: "Mr. Puli Srinivas Rao", title: "Stores & Logistics", desc: "15 years experience taking care of all Logistics activit",
    { name: "Mr. Ravi", title: "Auditing & Inventory", desc: "25+ years experience contributing to auditing of purchase bills, st",
    { name: "Mr. Mallikarjuna", title: "Service & Installations", desc: "Expertise in electrical jobs giving an extended hand to",
    { name: "Mr. Neelesh", title: "Admin Executive", desc: "Hardworking and dedicated Admin Executive who handles transportation",
  ];?
  ?
  const awards = [?
    "Excellence In Business services award 2025 - Rotary Club of Udgir MH",?
    "Excellence In Industrial All-Round performance SME - FTCCI @ Hyderabad (July 2024)",?
    "Service Excellence Award - Yaskawa @ Kathmandu, Nepal (Oct 2024)",?
    "Business Excellence Award - Global Conference by FTII @Hyd (Aug 2024)",?
    "She inspires Honor Award - NSI Council @ Hyd (July 2024)",?
    "Certified Under 10K Women from IIM Bangalore - Pratima M. (2023)",?
    "Service Excellence Award - Yaskawa at Dubai (2023)",?
    "Excellence Award in Core Manufacturing - COWE at Jaipur (March 2022)",?
    "Star Service Provider - YASKAWA at Phuket (May 2019)",?
  ];?
  ?
  return ( ?
    <div className="bg-[#E8F0FA] min-h-screen text-slate-800">?
      ?
      { /* 1. Hero Section */ }?
      <section className="relative py-20 bg-white border-b border-slate-200 overflow-hidden">?
        <div className="absolute inset-0 bg-[linear-gradient(to_right,#eef2f7_1px,transparent_1px),linear-gradient(to_bottom,#eef2f7_1px,transparent_1px)] bg-[size:20px_20px] border-b border-slate-200">?
          <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
            <div className="grid grid-cols-1 lg:grid-cols-2 gap-12 items-center">?
              <div className="max-w-xl space-y-6">?
                <span className="text-xs uppercase tracking-widest text-[#0f4c81] font-mono font-bold block">?
                  Corporate Profile?
                </span>?
                <h1 className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit leading-tight">?
                  About Nandini Enterprises?
                </h1>?
                <p className="text-base sm:text-lg leading-relaxed text-slate-650">?
                  Bringing Excellence in Business relationships and values, our company is a perfect blend of &apos;highly professi
                </p>?
              <div className="flex flex-wrap gap-4 pt-2">?
                <div className="border border-slate-200 bg-[#E8F0FA] rounded px-4 py-2 text-center shadow-xs">?
                  <span className="block text-2xl font-bold text-[#0f4c81] font-outfit">25+</span>?
                  <span className="text-[10px] text-slate-500 uppercase font-semibold font-mono">Years Exp</span>?
                </div>?
                <div className="border border-slate-200 bg-[#E8F0FA] rounded px-4 py-2 text-center shadow-xs">?
                  <span className="block text-2xl font-bold text-[#0f4c81] font-outfit">500+</span>?
                  <span className="text-[10px] text-slate-500 uppercase font-semibold font-mono">Projects Done</span>?
                </div>?
                <div className="border border-slate-200 bg-[#E8F0FA] rounded px-4 py-2 text-center shadow-xs">?
                  <span className="block text-2xl font-bold text-[#0f4c81] font-outfit">14+</span>?
                  <span className="text-[10px] text-slate-500 uppercase font-semibold font-mono">Sectors</span>?
                </div>?
              </div>?
            </div>?
          </div>?
        </div>?
      </div>?
    </div>?
  )
}
```

```

    </div>?
  </div>?
  <div className="relative group">?
    <div className="absolute -inset-1.5 bg-gradient-to-r from-[#0f4c81] to-[#2b6cb0] rounded-lg blur opacity-25 group-hover-[opacity:0.3]">?
      ?
    </div>?
  </div>?
</section>?

?
{ /* 2. Company Profile Details */ }?
<section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
  <div className="grid grid-cols-1 lg:grid-cols-12 gap-12 items-start">?
    <div className="lg:col-span-8 space-y-6">?
      <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
        Organisational Depth?
      </h2>?
      <p className="leading-relaxed text-sm text-slate-700">?
        We at <strong>NANDINI ENTERPRISES</strong> are an ISO 9001:2015 Certified Company in the field of general-purpose I
      </p>?
      <p className="leading-relaxed text-sm text-slate-700">?
        We are a CPRI Certified Electrical Control Panel Manufacturer providing PCC, MCC, VFD, and PLC Panels. We stand as
      </p>?
      <p className="leading-relaxed text-sm text-slate-700">?
        With a state-of-the-art 11,000 Sft. facility located at the Cherlapally Industrial Development Zone in Hyderabad, w
      </p>?
    </div>?
    <div className="lg:col-span-4 bg-white border border-slate-200 p-6 rounded-lg space-y-6 shadow-sm">?
      <h4 className="text-[#0f4c81] font-bold border-b border-slate-100 pb-2 text-xs font-mono uppercase tracking-wider">Op
      <div className="space-y-3 font-mono text-xs text-slate-600">?
        <p className="flex justify-between border-b border-slate-50 pb-1.5"><strong>ESTABLISHED:</strong> <span>2000</span>
        <p className="flex justify-between border-b border-slate-50 pb-1.5"><strong>HQ / PLANT:</strong> <span>IDA Cherlapa
        <p className="flex justify-between border-b border-slate-50 pb-1.5"><strong>CERTIFICATE:</strong> <span>ISO 9001:20
        <p className="flex justify-between border-b border-slate-50 pb-1.5"><strong>LICENSE:</strong> <span>A Grade (33KV)<
        <p className="flex justify-between"><strong>OEM PARTNERS:</strong> <span>Yaskawa, ABB, Baumer</span></p>?
      </div>?
    </div>?
  </div>?
</section>?

?
{ /* 3. Vision & Mission Cards */ }?
<section className="py-16 bg-[#eef2f7] border-t border-b border-slate-200">?
  <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 grid grid-cols-1 md:grid-cols-2 gap-8">?
    <div className="bg-white border border-slate-200 p-8 rounded-lg shadow-sm hover:shadow-md transition-shadow">?
      <h3 className="text-[#0f4c81] font-bold text-xl mb-4 font-outfit flex items-center space-x-2">?
        <span className="w-1.5 h-6 bg-[#0f4c81] rounded" />?
        <span>Our Vision</span>?
      </h3>?
      <p className="text-sm leading-relaxed text-slate-600">?
        To be the most trusted Progressive Channel Partner, the leading Service Provider, and the Ultimate Resource of Exce
      </p>?
    </div>?
    <div className="bg-white border border-slate-200 p-8 rounded-lg shadow-sm hover:shadow-md transition-shadow">?
      <h3 className="text-[#0f4c81] font-bold text-xl mb-4 font-outfit flex items-center space-x-2">?
        <span className="w-1.5 h-6 bg-[#0f4c81] rounded" />?
        <span>Our Mission</span>?
      </h3>?
      <p className="text-sm leading-relaxed text-slate-600">?
        To provide advanced, reliable Electrical, Instrumentation, and Control Products to our Customers with the price, pe
      </p>?
    </div>?
  </div>?
</section>?

?
{ /* 4. Awards & Achievements */ }?
<section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
  <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit mb-8 border-l-4 border-[#0f4c81] pl-3">?
    Awards & Achievements?
  </h2>?
  <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-4 font-mono text-xs">?
    {awards.map((award, idx) => {?
      <div ?
        key={idx}?
        className="bg-white border border-slate-200 p-5 rounded shadow-xs hover:border-[#2b6cb0] hover:shadow-sm transition
      >?
      <span className="text-[#3b82f6] font-bold shrink-0 mt-0.5">&bull;</span>?
      <span className="text-slate-700 leading-relaxed">{award}</span>?
    })}
  </div>

```

```
        </div>?
      )}}?
    </div>?
  </section>?
?
  { /* 5. Executive Leadership / Management Team */?
  <section className="py-16 bg-[#eef2f7]/50 border-t border-slate-200">?
    <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
      <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit mb-12 border-l-4 border-[#0f4c81] pl-3">?
        Our Management Team?
      </h2>?
      <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8">?
        {managementTeam.map((member, idx) => (?
          <div ?
            key={idx}?
            className="bg-white border border-slate-200 rounded-lg p-6 shadow-sm hover:shadow-md hover:border-[#2b6cb0] trans
          >?
            <div>?
              <h3 className="text-slate-900 font-bold text-lg font-outfit mb-1">?
                {member.name}?
              </h3>?
              <span className="text-xs uppercase tracking-widest text-[#2b6cb0] font-mono font-bold block mb-4">?
                {member.title}?
              </span>?
              <p className="text-xs text-slate-650 leading-relaxed font-sans">?
                {member.desc}?
              </p>?
            </div>?
            <span className="text-[9px] font-mono text-slate-400 mt-6 uppercase tracking-wider">?
              Nandini Enterprises &bull; Leadership?
            </span>?
          </div>?
        )}}?
      </div>?
    </div>?
  </section>?
?
  </div>?
  );?
}?
}
```

```
import { NextRequest } from 'next/server';
import fs from 'fs';
import path from 'path';
import { PDFDocument, rgb, StandardFonts, RGB } from 'pdf-lib';

// ?????????????????????????????????????????????????????????????????????
// CONFIDENTIAL CREDENTIALS (shown in PDF cover)
// ?????????????????????????????????????????????????????????????????????
const CREDENTIALS = {
  apiEndpoint: '/api/export-source',
  secretToken: process.env.EXPORT_SECRET_TOKEN ?? '(set EXPORT_SECRET_TOKEN in Vercel env vars)',
  triggerMethod: 'Double right-click on "Sitemap" link in website footer (within 800ms)',
  tokenHeader: 'X-Export-Token',
  deployCommand: 'vercel --prod --yes (run from local project root)',
  repoUrl: 'github.com/novelleyx-dev/nandiniautomation',
  liveUrl: 'nandiniautomation.vercel.app',
};

// ?????????????????????????????????????????????????????????????????????
// SITE DOCUMENTATION DATA
// ?????????????????????????????????????????????????????????????????????
const SITE_DOC = {
  company: {
    name: 'Nandini Enterprises',
    tagline: 'Industrial Automation Powerhouse',
    location: 'Plot No. 49, Phase-III, IDA Cherlapally, Hyderabad, Telangana - 500051, India',
    phone: '+91-40-23190131',
    email: 'info@nandiniautomation.com',
    certifications: ['ISO 9001:2015 Certified', 'CPRI Certified Electrical Panels', 'Grade-A Electrical License Holder'],
    partners: ['Siemens', 'Yaskawa (Authorized Service Center)', 'ABB (Channel Partner)', 'Baumer (Authorized Distributor)', 'Yok'],
  },
  techStack: [
    { key: 'Framework', value: 'Next.js 16.2.6 (App Router, Webpack build mode)' },
    { key: 'Language', value: 'TypeScript 6.x - strict mode enabled' },
    { key: 'Styling', value: 'Tailwind CSS v4.x with custom CSS properties' },
    { key: 'Animations', value: 'Framer Motion 12.x - entrance, hover, stagger effects' },
    { key: 'Icons', value: 'Lucide React 1.x icon set' },
    { key: 'Fonts', value: 'Inter (body) + Outfit (headings) - loaded via next/font/google' },
    { key: 'State Management', value: 'React hooks only (useState, useRef, useEffect) - no external store' },
    { key: 'PDF Generation', value: 'pdf-lib (pure JS) - no filesystem font dependencies, Vercel-safe' },
    { key: 'Screenshots', value: 'html2canvas - 0.5 scale JPEG 0.6 quality to stay under Vercel 4.5MB limit' },
    { key: 'Deployment', value: 'Vercel (serverless + Edge)' },
    { key: 'Repository', value: 'GitHub - github.com/novelleyx-dev/nandiniautomation' },
    { key: 'React Version', value: 'React 19.x' },
    { key: 'Node Version', value: 'Node 20.x LTS' },
  ],
  colorSystem: [
    { key: 'App Background', value: '#E8F0FA - light powder-blue corporate theme (body bg)' },
    { key: 'Primary Blue', value: '#0A3D91 - --color-ne-blue-corp (buttons, links, highlights)' },
    { key: 'Steel Blue', value: '#1565C0 - --color-ne-blue-steel (section accents)' },
    { key: 'Electric Blue', value: '#1E88E5 - --color-ne-blue-electric (hover states)' },
    { key: 'Footer Background', value: '#0f172a - slate-900 (dark footer)' },
    { key: 'Text Primary', value: '#1e293b - slate-800' },
    { key: 'Text Secondary', value: '#475569 - slate-600' },
    { key: 'Accent Green', value: '#10b981 - emerald-500 (success, active states)' },
    { key: 'Global CSS File', value: 'src/app/globals.css' },
    { key: 'Font Variables', value: '--font-inter (body), --font-outfit (headings)' },
  ],
  pages: [
    {
      route: '/',
      title: 'Homepage - Main Landing Page',
      file: 'src/app/page.tsx',
      purpose: 'Primary B2B marketing page. Designed to convert industrial procurement managers into leads. All sections are stacked vertically for maximum impact.',
      sections: [
        'HeroSlideshow: Full-screen 5-image carousel with auto-advance every 5s, slide indicators, and animated headline overlay',
        'Trust Strip: ISO 9001:2015, CPRI, Safety Standards, and partner certification badges',
        'Services Preview: 6 service category cards with hover animations and CTA links',
        'Industries Grid: 8 industry sectors with icons - Steel, Cement, Oil & Gas, Power, Pharma, Water, Food, Mining',
        'Why Choose Us: 4 USP cards with animated counters - 500+ Projects, 30+ Years, 24/7 Support, ISO Certified',
        'Process Workflow: 4-step engineering journey - Consultation > Design > Fabrication > Commissioning',
        'Projects Overview: Portfolio cards of major completed EPC projects with AI-generated images',
        'Company Story: Founding history, growth narrative, and leadership values section',
        'Partner Logos: Brand wall - Siemens, Yaskawa, ABB, Yokogawa, Baumer, Honeywell, WIKA, L&T',
        'Lead Capture Form: Dynamic RFQ form that adapts fields based on selected service category',
      ],
    },
    {
      route: '/about',
      title: 'About Us / Executive Profile',
      file: 'src/app/about/page.tsx',
      purpose: 'Company history, leadership, and core values',
    },
    {
      route: '/services',
      title: 'Services Overview Listing',
      file: 'src/app/services/page.tsx',
      purpose: 'Grid listing of all 6 service categories',
    },
    {
      route: '/services/[slug]',
      title: 'Dynamic Service Detail Pages (SSG)',
      file: 'src/app/services/[slug]/page.tsx',
      purpose: 'Detailed view of a specific service category',
    },
  ],
};
```

... continued: src/app/api/export-source/route.ts

```
{ route: '/services/industrial-automation', title: 'Industrial Automation (Standalone)', file: 'src/app/services/industrial-a
{ route: '/services/instrumentation', title: 'Precision Process Instrumentation', file: 'src/app/services/instrumentation/pag
{ route: '/services/electrical-systems', title: 'High-Performance Electrical Systems', file: 'src/app/services/electrical-sys
{ route: '/services/robotics', title: 'Industrial Robotics Integration', file: 'src/app/services/robotics/page.tsx', purpose:
{ route: '/services/elevator-solutions', title: 'Smart Elevator & Lift Solutions', file: 'src/app/services/elevator-solutions
{ route: '/services/control-panels', title: 'Advanced Electrical Control Panels', file: 'src/app/services/control-panels/page
{ route: '/industries', title: 'Industries We Serve', file: 'src/app/industries/page.tsx', purpose: '8 industrial sectors wit
{ route: '/projects', title: 'EPC Projects Portfolio', file: 'src/app/projects/page.tsx', purpose: 'Gallery of 500+ completed
{ route: '/contact', title: 'Contact & RFQ Submission', file: 'src/app/contact/page.tsx', purpose: 'Primary lead capture page
{ route: '/gallery', title: 'Manufacturing Facility Gallery', file: 'src/app/gallery/page.tsx', purpose: 'Photo gallery of Ch
{ route: '/certifications', title: 'Compliance & Certifications', file: 'src/app/certifications/page.tsx', purpose: 'All acti
{ route: '/knowledge-center', title: 'White Papers & Technical Articles', file: 'src/app/knowledge-center/page.tsx', purpose:
{ route: '/products', title: 'Products Catalogue', file: 'src/app/products/page.tsx', purpose: 'Procurement catalogue: VFDs,
],
features: [
  { index: 'F-01', name: 'Hero Slideshow with Auto-Advance', file: 'src/components/Home/HeroSlideshow.tsx', trigger: 'Automatic
  { index: 'F-02', name: 'Global Right-Click Protection', file: 'src/components/Layout/DisableRightClick.tsx', trigger: 'Mounts
  { index: 'F-03', name: '[HIDDEN] Confidential PDF Export Trigger', file: 'src/components/Layout/Footer.tsx + src/app/api/expo
  { index: 'F-04', name: 'Internal Knowledge Navigation Chatbot', file: 'src/components/Assistant/KnowledgeAssistant.tsx', trig
  { index: 'F-05', name: 'Dynamic Lead Capture RFQ Form', file: 'src/app/page.tsx, src/app/contact/page.tsx', trigger: 'User sc
  { index: 'F-06', name: 'WhatsApp Floating Action Button', file: 'src/components/Layout/Footer.tsx', trigger: 'Always visible
  { index: 'F-07', name: 'Sticky Header with Navigation', file: 'src/components/Layout/Header.tsx', trigger: 'Always visible -
  { index: 'F-08', name: 'Animated Statistics Counters', file: 'src/app/page.tsx (Why Choose Us section)', trigger: 'Element en
  { index: 'F-09', name: 'Framer Motion Page Animations', file: 'src/app/page.tsx (multiple sections)', trigger: 'Page load and
  { index: 'F-10', name: 'Dynamic Service Pages (SSG)', file: 'src/app/services/[slug]/page.tsx, src/lib/servicesData.ts', trig
  { index: 'F-11', name: 'SEO Metadata & OpenGraph', file: 'src/app/layout.tsx', trigger: 'Server-rendered on every request', d
  { index: 'F-12', name: 'Secret Token Security Gate on PDF API', file: 'src/app/api/export-source/route.ts', trigger: 'Every P
],
components: [
  { name: 'Header.tsx', path: 'src/components/Layout/Header.tsx', type: 'Client Component', desc: 'Sticky navigation with mobil
  { name: 'Footer.tsx', path: 'src/components/Layout/Footer.tsx', type: 'Client Component', desc: 'Multi-column footer with par
  { name: 'DisableRightClick.tsx', path: 'src/components/Layout/DisableRightClick.tsx', type: 'Client Component', desc: 'Docume
  { name: 'KnowledgeAssistant.tsx', path: 'src/components/Assistant/KnowledgeAssistant.tsx', type: 'Client Component', desc: 'F
  { name: 'HeroSlideshow.tsx', path: 'src/components/Home/HeroSlideshow.tsx', type: 'Client Component', desc: 'Full-screen 5-im
  { name: 'ServicePageClient.tsx', path: 'src/components/Service/ServicePageClient.tsx', type: 'Client Component', desc: 'Rich
],
knownIssues: [
  { issue: 'pdfkit Helvetica.afm ENOENT on Vercel', fix: 'Replaced pdfkit with pdf-lib (pure JS). StandardFonts embedded as bas
  { issue: 'Buffer type mismatch in Response constructor', fix: 'Wrapped Uint8Array in Buffer.from() before passing to new Resp
  { issue: 'html2canvas payload too large at scale:1 PNG', fix: 'Reduced to scale:0.5, output as JPEG quality 0.6. Stays under
  { issue: 'outputFileTracingIncludes in experimental block', fix: 'Moved to root level in next.config.mjs per Next.js 16 depre
  { issue: 'Footer was Server Component - no useState allowed', fix: 'Added "use client" directive to Footer.tsx.', status: 'RE
  { issue: 'API route visible on GitHub', fix: 'Added to .gitignore. Removed from git tracking via git rm --cached. Deploy via
],
};

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
// SECTION ACCENT COLORS (multicolor system)
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
const SECTION_COLORS: RGB[] = [
  rgb(0.05, 0.32, 0.85), // SEC 1 - Royal Blue
  rgb(0.45, 0.17, 0.87), // SEC 2 - Purple
  rgb(0.00, 0.55, 0.65), // SEC 3 - Teal
  rgb(0.05, 0.55, 0.30), // SEC 4 - Forest Green
  rgb(0.85, 0.28, 0.05), // SEC 5 - Deep Orange
  rgb(0.72, 0.05, 0.38), // SEC 6 - Crimson
  rgb(0.25, 0.20, 0.72), // SEC 7 - Indigo
  rgb(0.75, 0.10, 0.10), // SEC 8 - Scarlet
  rgb(0.00, 0.42, 0.75), // SEC 9 - Cobalt
  rgb(0.05, 0.35, 0.25), // SEC 10 - Dark Teal
  rgb(0.18, 0.18, 0.18), // SEC 11 - Charcoal (code)
];

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
// CLEAN LIGHT THEME COLORS
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
const WHITE = rgb(1, 1, 1);
const BLACK = rgb(0, 0, 0);
const GRAY_DARK = rgb(0.15, 0.15, 0.15);
const GRAY_MID = rgb(0.40, 0.40, 0.42);
const GRAY_LIGHT = rgb(0.93, 0.93, 0.95);
const GRAY_BORDER = rgb(0.80, 0.80, 0.84);
const GRAY_BG = rgb(0.97, 0.97, 0.98);
const BLUE = rgb(0.05, 0.32, 0.85);
const BLUE_LIGHT = rgb(0.87, 0.92, 1.0);
const GREEN = rgb(0.05, 0.55, 0.30);
const RED = rgb(0.72, 0.08, 0.08);
const ORANGE = rgb(0.82, 0.38, 0.00);
const PURPLE = rgb(0.45, 0.17, 0.87);

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
// RENDERER
```

```
// ?????????????????????????????????????????????????????????????????
class Renderer {
  doc: PDFDocument;
  fb: any; fr: any; fm: any;
  page: any;
  y: number = 0;
  pg: number = 0;
  readonly W = 595;
  readonly H = 842;
  readonly M = 50; // margin
  readonly CW: number; // content width
  currentColor: RGB = BLUE;

  constructor(doc: PDFDocument, fb: any, fr: any, fm: any) {
    this.doc = doc; this.fb = fb; this.fr = fr; this.fm = fm;
    this.CW = this.W - this.M * 2;
  }

  // ?? PAGE ?????????????????????????????????????????????????????????
  newPage() {
    this.page = this.doc.addPage([this.W, this.H]);
    this.pg++;
    this.y = this.H - this.M;
    // White page background
    this.rect(0, 0, this.W, this.H, WHITE);
    // Top accent bar - 4px, current section color
    this.rect(0, this.H - 4, this.W, 4, this.currentColor);
    // Bottom footer strip
    this.rect(0, 0, this.W, 26, GRAY_BG);
    this.rect(0, 26, this.W, 0.5, GRAY_BORDER);
    this.draw('NANDINI ENTERPRISES | INTERNAL CONFIDENTIAL DOCUMENTATION', this.M, 9, this.fr, 6.5, GRAY_MID);
    this.draw(`Page ${this.pg}`, this.W - this.M - 20, 9, this.fb, 7, GRAY_DARK);
    return this;
  }

  checkY(need = 18) {
    if (this.y < this.M + 36 + need) this.newPage();
  }

  // ?? PRIMITIVES ?????????????????????????????????????????????????????
  rect(x: number, y: number, w: number, h: number, color: RGB, opacity = 1) {
    this.page.drawRectangle({ x, y, width: w, height: h, color, opacity });
  }

  border(x: number, y: number, w: number, h: number, fillColor: RGB, borderColor: RGB, bw = 1) {
    this.rect(x, y, w, h, borderColor);
    this.rect(x + bw, y + bw, w - bw * 2, h - bw * 2, fillColor);
  }

  line(x1: number, y1: number, x2: number, y2: number, color: RGB, thickness = 0.75) {
    this.page.drawLine({ start: { x: x1, y: y1 }, end: { x: x2, y: y2 }, thickness, color });
  }

  draw(text: string, x: number, y: number, font: any, size: number, color: RGB, maxW?: number) {
    const clean = san(text);
    if (!clean) return;
    try {
      this.page.drawText(clean.slice(0, 200), { x, y, font, size, color, maxWidth: maxW });
    } catch (_) {}
  }

  // ?? TYPOGRAPHY ?????????????????????????????????????????????????????
  // Heading 1 - section title in colored banner
  h1(text: string, color?: RGB) {
    const c = color ?? this.currentColor;
    this.checkY(28);
    this.rect(this.M, this.y - 6, this.CW, 24, c);
    this.draw(san(text).slice(0, 80), this.M + 10, this.y + 8, this.fb, 13, WHITE);
    this.y -= 30;
  }

  // Heading 2 - subsection with left colored bar
  h2(text: string, color?: RGB) {
    const c = color ?? this.currentColor;
    this.checkY(20);
    this.rect(this.M, this.y - 3, 4, 18, c);
    this.draw(san(text).slice(0, 90), this.M + 10, this.y + 8, this.fb, 10.5, GRAY_DARK);
    this.y -= 22;
    this.line(this.M + 10, this.y + 4, this.M + this.CW, this.y + 4, GRAY_BORDER, 0.5);
    this.y -= 6;
  }
}
```



```

// Heading 3 - inline bold colored label
h3(text: string, color?: RGB) {
  const c = color ?? this.currentColor;
  this.checkY(16);
  this.draw(san(text).slice(0, 100), this.M, this.y, this.fb, 9, c);
  this.y -= 13;
}

// Body paragraph with word-wrap
body(text: string, opts: { size?: number; color?: RGB; x?: number; font?: any; maxW?: number } = {}) {
  const { size = 8.5, color = GRAY_DARK, x = this.M, font = this.fr, maxW = this.CW } = opts;
  const lh = size * 1.45;
  const words = san(text).replace(/\n/g, ' ').split(' ');
  let line = '';
  for (const word of words) {
    const test = line ? `${line} ${word}` : word;
    if (font.widthOfTextAtSize(test, size) > maxW && line) {
      this.checkY(lh + 2);
      this.draw(line, x, this.y, font, size, color, maxW);
      this.y -= lh;
      line = word;
    } else {
      line = test;
    }
  }
  if (line) {
    this.checkY(lh + 2);
    this.draw(line, x, this.y, font, size, color, maxW);
    this.y -= lh;
  }
}

// Bullet point
bullet(text: string, indent = 0, bColor?: RGB) {
  const bc = bColor ?? this.currentColor;
  const x = this.M + indent + 14;
  const maxW = this.CW - indent - 14;
  this.checkY(12);
  this.rect(this.M + indent + 4, this.y + 3, 4, 4, bc);
  this.body(text, { x, maxW, size: 8.5 });
}

// Key-value row with background alternation
kv(key: string, value: string, even = false) {
  this.checkY(14);
  if (even) this.rect(this.M, this.y - 3, this.CW, 14, GRAY_BG);
  this.draw(san(key).slice(0, 35), this.M + 4, this.y + 3, this.fb, 8, this.currentColor);
  this.body(san(value), { x: this.M + 145, maxW: this.CW - 145, size: 8 });
}

// Bordered info box
infoBox(lines: string[], boxColor: RGB = BLUE_LIGHT, borderColor: RGB = BLUE) {
  const lineH = 10;
  const h = lines.length * lineH + 12;
  this.checkY(h + 4);
  this.border(this.M, this.y - h + 4, this.CW, h, boxColor, borderColor, 1.5);
  const startY = this.y;
  for (const ln of lines) {
    this.draw(san(ln).slice(0, 110), this.M + 8, this.y, this.fr, 8, GRAY_DARK, this.CW - 14);
    this.y -= lineH;
  }
  this.y = startY - h - 4;
}

// Colored badge pill
badge(text: string, color: RGB) {
  this.checkY(14);
  const w = Math.min(this.fb.widthOfTextAtSize(san(text), 7.5) + 14, 220);
  this.rect(this.M, this.y - 2, w, 13, color, 0.15);
  this.draw(san(text).slice(0, 35), this.M + 7, this.y + 1, this.fb, 7.5, color);
  this.y -= 16;
}

// Code block
code(lines: string[]) {
  const lh = 9;
  const maxLines = Math.min(lines.length, 80);
  const h = maxLines * lh + 10;
  this.checkY(Math.min(h, 120));
  const startY = this.y;
  this.border(this.M, this.y - h + 8, this.CW, h, rgb(0.96, 0.97, 0.99), GRAY_BORDER, 1);
  this.rect(this.M, this.y - h + 8, 3, h, this.currentColor);
}

```

... continued: src/app/api/export-source/route.ts

```
    let cy = this.y;
    for (let i = 0; i < maxLines; i++) {
      if (cy < this.M + 36) { this.newPage(); cy = this.y; }
      const clean = san(lines[i]).slice(0, 130);
      if (clean) this.draw(clean, this.M + 8, cy, this.fm, 7, rgb(0.12, 0.18, 0.38));
      cy -= lh;
    }
    this.y = cy - 4;
  }

  gap(n = 1) { this.y -= 8 * n; }
  hr(color = GRAY_BORDER) {
    this.checkY(6);
    this.line(this.M, this.y, this.M + this.CW, this.y, color, 0.5);
    this.y -= 8;
  }

  // Section divider page - full-width colored header
  sectionPage(num: string, title: string, subtitle: string, colorIdx: number) {
    const c = SECTION_COLORS[colorIdx] ?? BLUE;
    this.currentColor = c;
    this.newPage();
    // Colored header band
    this.rect(0, this.H - 100, this.W, 100, c);
    // Light tint behind rest of page
    this.rect(0, 0, this.W, this.H - 100, GRAY_BG);
    this.draw(num, this.M, this.H - 52, this.fb, 11, rgb(0.75, 0.82, 1.0));
    this.draw(san(title).slice(0, 70), this.M, this.H - 72, this.fb, 18, WHITE);
    this.draw(san(subtitle).slice(0, 90), this.M, this.H - 90, this.fr, 9, rgb(0.85, 0.90, 1.0));
    this.y = this.H - 116;
  }
}

// ?????????????????????????????????????????????????????????????????????????????????????
// HELPERS
// ?????????????????????????????????????????????????????????????????????????????????????
function san(s: string): string {
  if (!s) return '';
  return s.replace(/[\u2018\u2019]/g, '"').replace(/[\u201C\u201D]/g, '"')
    .replace(/[\u2013\u2014]/g, '-')
    .replace(/[\u00A0]/g, ' ')
    .replace(/[\^x20-x7E\n]/g, '?').replace(/r/g, '');
}

// ?????????????????????????????????????????????????????????????????????????????????????
// POST HANDLER
// ?????????????????????????????????????????????????????????????????????????????????????
export async function POST(req: NextRequest) {
  const SECRET = process.env.EXPORT_SECRET_TOKEN ?? '';
  const provided = req.headers.get('X-Export-Token') ?? '';
  if (!SECRET || provided !== SECRET) {
    return new Response('Not Found', { status: 404 });
  }
}

try {
  const { screenshot, currentUrl } = await req.json();

  const doc = await PDFDocument.create();
  doc.setTitle('Nandini Enterprises - Internal Technical Documentation');
  doc.setAuthor('Nandini Enterprises Engineering Division');
  doc.setSubject('HIGHLY CONFIDENTIAL - Internal Use Only');
  doc.setCreationDate(new Date());

  const fb = await doc.embedFont(StandardFonts.HelveticaBold);
  const fr = await doc.embedFont(StandardFonts.Helvetica);
  const fm = await doc.embedFont(StandardFonts.Courier);
  const r = new Renderer(doc, fb, fr, fm);

  // ?????????????????????????????????????????????????????????????????????????????????????
  // COVER PAGE
  // ?????????????????????????????????????????????????????????????????????????????????????
  r.currentColor = BLUE;
  r.newPage();

  // Header band
  r.rect(0, r.H - 150, r.W, 150, BLUE);
  r.rect(0, r.H - 154, r.W, 4, ORANGE);

  // Company title
  r.draw('NANDINI ENTERPRISES', r.M, r.H - 65, r.fb, 24, WHITE);
  r.draw('INTERNAL TECHNICAL DOCUMENTATION', r.M, r.H - 88, r.fb, 13, rgb(0.8, 0.88, 1.0));
  r.draw('Confidential Site Audit, Feature Reference & Source Code Export', r.M, r.H - 106, r.fr, 9, rgb(0.7, 0.8, 0.95));
  r.draw('Industrial Automation Powerhouse | Hyderabad, India', r.M, r.H - 122, r.fr, 8.5, rgb(0.65, 0.75, 0.92));
}
```

```
... continued: src/app/api/export-source/route.ts
```

```
r.y = r.H - 165;

// Classification box
r.border(r.M, r.y - 30, r.CW, 32, rgb(1.0, 0.94, 0.94), RED, 1.5);
r.draw('HIGHLY CONFIDENTIAL - FOR INTERNAL ENGINEERING TEAM USE ONLY', r.M + 10, r.y - 10, r.fb, 9.5, RED);
r.draw('DO NOT DISTRIBUTE - DO NOT SHARE WITH CLIENTS OR EXTERNAL PARTIES', r.M + 10, r.y - 23, r.fr, 8, RED);
r.y -= 42;

// Metadata
r.gap(0.5);
const ts = new Date().toLocaleString('en-IN', { timeZone: 'Asia/Kolkata', dateStyle: 'full', timeStyle: 'medium' });
const metaRows = [
    ['Export Timestamp', ts + ' IST'],
    ['Triggered From', san(currentUrl ?? 'Unknown')],
    ['Export Purpose', 'Disaster recovery, new developer onboarding, AI pricing analysis'],
    ['Live Website', CREDENTIALS.liveUrl],
];
let even = false;
for (const [k, v] of metaRows) { r.kv(k, v, even); even = !even; }
r.gap();

// Credentials box
r.border(r.M, r.y - 90, r.CW, 94, BLUE_LIGHT, BLUE, 1.5);
r.draw('ACCESS CREDENTIALS & TRIGGER METHOD', r.M + 8, r.y - 10, r.fb, 9, BLUE);
r.line(r.M + 8, r.y - 14, r.M + r.CW - 8, r.y - 14, BLUE, 0.5);
const creds = [
    `Secret Token (EXPORT_SECRET_TOKEN): ${CREDENTIALS.secretToken}`,
    `Header Name: ${CREDENTIALS.tokenHeader}`,
    `API Endpoint: ${CREDENTIALS.apiEndpoint} (POST)`,
    `Trigger: ${CREDENTIALS.triggerMethod}`,
    `CLI Deploy Command: ${CREDENTIALS.deployCommand}`,
    `GitHub Repo: ${CREDENTIALS.repoUrl}`,
];
let cy = r.y - 22;
for (const c of creds) {
    r.draw(san(c).slice(0, 100), r.M + 8, cy, r.fm, 7.8, GRAY_DARK, r.CW - 14);
    cy += 11;
}
r.y = r.y - 100;

// Screenshot
if (screenshot) {
    try {
        const b64 = screenshot.replace(/^data:image\/\w+;base64/, '');
        const buf = Buffer.from(b64, 'base64');
        const img = screenshot.includes('jpeg') || screenshot.includes('jpg')
            ? await doc.embedJpg(buf)
            : await doc.embedPng(buf);
        const scale = Math.min(r.CW / img.width, 200 / img.height);
        const iw = img.width * scale, ih = img.height * scale;
        r.checkY(ih + 20);
        r.draw('LIVE PAGE SCREENSHOT AT TIME OF EXPORT:', r.M, r.y, r.fb, 8.5, BLUE);
        r.y -= 6;
        r.border(r.M - 1, r.y - ih - 1, iw + 2, ih + 2, WHITE, GRAY_BORDER, 1);
        r.page.drawImage(img, { x: r.M, y: r.y - ih, width: iw, height: ih });
        r.y += ih + 14;
    } catch (_) {
        r.body('Screenshot could not be embedded.', { color: GRAY_MID });
    }
}

// ????????????????????????????????????????????????????????????????
// TABLE OF CONTENTS
// ????????????????????????????????????????????????????????????????
r.currentColor = BLUE;
r.newPage();
r.rect(0, r.H - 4, r.W, 4, BLUE);
r.draw('TABLE OF CONTENTS', r.M, r.y, r.fb, 16, GRAY_DARK);
r.y -= 28;
r.line(r.M, r.y + 10, r.M + r.CW, r.y + 10, GRAY_BORDER);
r.y -= 4;

const tocItems = [
    ['1', 'Company Overview & Profile', SECTION_COLORS[0], 'Company facts, contacts, certifications, authorized partners'],
    ['2', 'Technology Stack & Architecture', SECTION_COLORS[1], 'Framework, language, libraries, deployment platform, React/Node.js'],
    ['3', 'Color System & Design Tokens', SECTION_COLORS[2], 'Global CSS variables, Tailwind tokens, font names, theme colors'],
    ['4', 'Site Pages Documentation', SECTION_COLORS[3], 'All 17 routes - purpose, source file, sections, form fields'],
    ['5', 'Feature Index & Implementation Guide', SECTION_COLORS[4], 'F-01 to F-12: trigger, code logic, dependencies, security'],
    ['6', 'Component Library Reference', SECTION_COLORS[5], 'All React components with type, path, and description'],
    ['7', 'Known Issues & Resolutions', SECTION_COLORS[6], 'All bugs encountered and their exact fixes'],
    ['8', 'Deployment & Confidential Access Setup', SECTION_COLORS[7], 'Vercel CLI deploy, env vars, GitHub vs Vercel separation']
```

```
... continued: src/app/api/export-source/route.ts
```

```
[ '9', 'Services Text Content (AI-Readable)', SECTION_COLORS[8], 'Full service descriptions, FAQs, workflows - optimized for',
[ '10', 'Project Directory Tree', SECTION_COLORS[9], 'Complete annotated filesystem structure'],
[ '11', 'Complete Source Code', SECTION_COLORS[10], 'Every .ts .tsx .css .mjs .json file verbatim'],
];

for (let i = 0; i < tocItems.length; i++) {
    const [num, title, color, desc] = tocItems[i] as [string, string, RGB, string];
    r.checkY(28);
    const isEven = i % 2 === 0;
    r.rect(r.M, r.y - 4, r.CW, 26, isEven ? GRAY_BG : WHITE);
    r.rect(r.M, r.y - 4, r.CW, 1, color as RGB);
    r.rect(r.M, r.y + 22, r.CW, 1, color as RGB);
    r.rect(r.M, r.y - 4, 1, 26, color as RGB);
    r.rect(r.M + r.CW - 1, r.y - 4, 1, 26, color as RGB);
    r.rect(r.M, r.y - 4, 28, 26, color as RGB);
    r.draw(num, r.M + 8, r.y + 6, r.fb, 11, WHITE);
    r.draw(san(title), r.M + 36, r.y + 8, r.fb, 9.5, GRAY_DARK);
    r.draw(san(desc), r.M + 36, r.y - 2, r.fr, 7.5, GRAY_MID, r.CW - 44);
    r.y -= 30;
}

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
// SECTION 1 - COMPANY OVERVIEW
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 1', 'Company Overview & Profile', 'Identity, contacts, certifications, and authorized partnerships', 0)

r.h2('Company Identity');
const coRows = [
    ['Company Name', SITE_DOC.company.name],
    ['Tagline', SITE_DOC.company.tagline],
    ['Registered Address', SITE_DOC.company.location],
    ['Direct Phone', SITE_DOC.company.phone],
    ['Email', SITE_DOC.company.email],
];
for (let i = 0; i < coRows.length; i++) r.kv(coRows[i][0], coRows[i][1], i % 2 === 0);
r.gap();
r.h2('Certifications & Licenses');
for (const cert of SITE_DOC.company.certifications) r.bullet(cert, 0, GREEN);
r.gap();
r.h2('Authorized Industry Partners');
for (let i = 0; i < SITE_DOC.company.partners.length; i++) {
    r.checkY(12);
    r.rect(r.M + (i % 2) * (r.CW / 2), r.y - 2, r.CW / 2 - 4, 13, GRAY_BG);
    r.draw(san(SITE_DOC.company.partners[i]), r.M + (i % 2) * (r.CW / 2) + 8, r.y + 2, r.fr, 8.5, GRAY_DARK);
    if (i % 2 === 1 || i === SITE_DOC.company.partners.length - 1) r.y -= 16;
}

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
// SECTION 2 - TECH STACK
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 2', 'Technology Stack & Architecture', 'Every library, framework, and version used in this project', 1)
r.h2('Core Technologies');
for (let i = 0; i < SITE_DOC.techStack.length; i++) {
    const { key, value } = SITE_DOC.techStack[i];
    r.kv(key, value, i % 2 === 0);
}

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
// SECTION 3 - COLOR SYSTEM
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 3', 'Color System & Design Tokens', 'Global CSS variables, Tailwind tokens, and typography definitions', 2)
r.h2('CSS Color Palette');
for (let i = 0; i < SITE_DOC.colorSystem.length; i++) {
    const { key, value } = SITE_DOC.colorSystem[i];
    r.kv(key, value, i % 2 === 0);
}

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
// SECTION 4 - PAGES
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 4', 'Site Pages Documentation', 'All routes with purpose, source file, and section breakdown', 3);

for (let pi = 0; pi < SITE_DOC.pages.length; pi++) {
    const pg = SITE_DOC.pages[pi];
    r.checkY(40);
    // Page card header
    r.rect(r.M, r.y - 4, r.CW, 22, GREEN);
    r.rect(r.M, r.y - 4, 2, 22, rgb(0, 0.3, 0));
    r.draw(san(pg.route), r.M + 8, r.y + 8, r.fb, 10, WHITE);
    r.draw(san(pg.title), r.M + (r.fb.widthOfTextAtSize(san(pg.route), 10)) + 20, r.y + 8, r.fr, 8.5, rgb(0.8, 1, 0.85));
    r.y -= 26;
    r.kv('Source File', pg.file, false);
```

```
... continued: src/app/api/export-source/route.ts
```

```
r.body(pg.purpose, { size: 8.5,color: GRAY_DARK });
if (pg.sections.length > 0) {
  r.h3('Sections / Form Fields:', GREEN);
  for (const s of pg.sections) r.bullet(s, 8, GREEN);
}
r.gap(0.5);
r.hr();
r.gap(0.5);
}

// ??????????????????????????????????????????????????????????????????
// SECTION 5 - FEATURE INDEX
// ??????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 5', 'Feature Index & Implementation Guide', 'F-01 to F-12: every active feature with code logic and de
r.body('This is the primary reference for any engineer understanding or rebuilding this system. Each entry covers the trigger
r.gap();

for (const feat of SITE_DOC.features) {
  r.checkY(60);
  // Feature card
  r.border(r.M, r.y - 6, r.CW, 22, GRAY_BG, ORANGE, 1);
  r.rect(r.M, r.y - 6, 40, 22, ORANGE);
  r.draw(feat.index, r.M + 6, r.y + 6, r.fb, 9, WHITE);
  r.draw(san(feat.name), r.M + 48, r.y + 6, r.fb, 10, GRAY_DARK);
  r.y -= 28;
  r.kv('Source File', feat.file, true);
  r.kv('Trigger', feat.trigger, false);
  r.h3('Description:');
  r.body(feat.description);
  r.h3('Implementation:');
  r.body(feat.implementation);
  if ((feat as any).note) { r.h3('Note:', ORANGE); r.body((feat as any).note, { color: ORANGE }); }
  r.kv('Dependencies', feat.deps, true);
  r.gap();
  r.hr(GRAY_BORDER);
  r.gap(0.5);
}

// ??????????????????????????????????????????????????????????????????
// SECTION 6 - COMPONENTS
// ??????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 6', 'Component Library Reference', 'All React components with type, path, and description', 5);
for (let i = 0; i < SITE_DOC.components.length; i++) {
  const comp = SITE_DOC.components[i];
  r.checkY(36);
  r.border(r.M, r.y - 4, r.CW, 18, i % 2 === 0 ? GRAY_BG : WHITE, GRAY_BORDER, 1);
  r.draw(san(comp.name), r.M + 8, r.y + 5, r.fb, 9.5, r.currentColor);
  r.badge(comp.type, PURPLE);
  r.kv('Path', comp.path, false);
  r.body(comp.desc, { size: 8.5 });
  r.gap(0.5);
}

// ??????????????????????????????????????????????????????????????????
// SECTION 7 - KNOWN ISSUES
// ??????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 7', 'Known Issues & Resolutions', 'All bugs encountered and their exact technical fixes', 6);
for (const item of SITE_DOC.knownIssues) {
  r.checkY(50);
  r.border(r.M, r.y - 4, r.CW, 16, rgb(1, 0.95, 0.95), RED, 1);
  r.draw('ISSUE:', r.M + 6, r.y + 3, r.fb, 8, RED);
  r.draw(san(item.issue), r.M + 46, r.y + 3, r.fr, 8.5, GRAY_DARK);
  r.y -= 20;
  r.border(r.M, r.y - 4, r.CW, 16, rgb(0.94, 1, 0.95), GREEN, 1);
  r.draw('FIX:', r.M + 6, r.y + 3, r.fb, 8, GREEN);
  r.draw(san(item.fix).slice(0, 100), r.M + 46, r.y + 3, r.fr, 8.5, GRAY_DARK, r.CW - 50);
  r.y -= 20;
  r.badge(item.status, GREEN);
  r.gap();
}

// ??????????????????????????????????????????????????????????????????
// SECTION 8 - DEPLOYMENT & ACCESS SETUP
// ??????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 8', 'Deployment & Confidential Access Setup', 'Vercel CLI, env vars, GitHub separation, and recovery s

r.h2('Confidential Access Credentials');
r.infoBox([
  `Secret Token Value:   ${CREDENTIALS.secretToken}`,
  `Header Name:         ${CREDENTIALS.tokenHeader}`,
  `API Endpoint:       ${CREDENTIALS.apiEndpoint} [POST method]`,
  `Trigger:             ${CREDENTIALS.triggerMethod}`,
```

```
... continued: src/app/api/export-source/route.ts
```

```
, BLUE_LIGHT, BLUE);
r.gap();
r.h2('Deployment Commands (Run from Project Root)');
r.code([
    '# Link project to Vercel (run once)',
    'vercel link --yes --project nandiniautomation',
    '',
    '# Add secret token to Vercel env (run once after any token change)',
    'echo "YOUR_TOKEN" | vercel env add EXPORT_SECRET_TOKEN production',
    '',
    '# Deploy ALL files including confidential route (run after every update)',
    'vercel --prod --yes',
    '',
    '# Normal GitHub push (does NOT include confidential route):',
    'git add .',
    'git commit -m "your message"',
    'git push origin main',
    '# Then immediately follow with: vercel --prod --yes',
]);
r.gap();
r.h2('What is on GitHub vs Vercel');
const ghRows = [
    ['GitHub (Public/Private repo)', 'All website pages, components, styles - NO confidential route'],
    ['Vercel Production (Live site)', 'Everything including /api/export-source route (deployed via CLI)'],
    ['.gitignore entries', 'src/app/api/export-source/ and src/fonts/ - never committed'],
    ['.env.local', 'Contains EXPORT_SECRET_TOKEN - gitignored, exists only on your machine'],
    ['Vercel Env Vars', 'EXPORT_SECRET_TOKEN stored encrypted in Vercel dashboard'],
];
for (let i = 0; i < ghRows.length; i++) r.kv(ghRows[i][0], ghRows[i][1], i % 2 === 0);
r.gap();
r.h2('Disaster Recovery Steps', RED);
r.bullet('If machine is lost: Re-clone from GitHub, re-create src/app/api/export-source/route.ts from memory or this PDF', 0,
r.bullet('Set EXPORT_SECRET_TOKEN in Vercel dashboard ? Settings ? Environment Variables', 0, RED);
r.bullet('Add token to local .env.local file', 0, RED);
r.bullet('Run: vercel --prod --yes to redeploy with confidential route', 0, RED);

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????
// SECTION 9 - SERVICES CONTENT
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 9', 'Services Content - Full Text Data', 'Complete service descriptions and FAQs optimized for AI pricing analysis', 9);
r.body('This section contains the complete human-readable text data for all service categories as stored in servicesData.ts.');
r.gap();

const rootDir = process.cwd();
const sdPath = path.join(rootDir, 'src', 'lib', 'servicesData.ts');
if (fs.existsSync(sdPath)) {
    const raw = fs.readFileSync(sdPath, 'utf8');
    // Extract meaningful text lines (quoted string values)
    const lines = raw.split('\n').filter(l => {
        const t = l.trim();
        return t.startsWith('"') && t.length > 25 && !t.includes('slug:') && !t.includes('value:') && !t.includes('size:');
    });
    let lineCount = 0;
    for (const ln of lines) {
        const cleaned = ln.trim().replace(/\/^|"/g, '\&#x2F;s$&#xA7/g').replace(/\\"/g, '"').replace(/\\n/g, '\n').trim();
        if (cleaned.length > 20) {
            r.body(cleaned, { size: 8, color: GRAY_DARK });
            lineCount++;
            if (lineCount % 30 === 0) r.gap(0.5);
        }
    }
}

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????
// SECTION 10 - DIRECTORY TREE
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 10', 'Project Directory Tree & Structure', 'Full annotated filesystem topology of the codebase', 9);
r.body('Complete filesystem structure excluding node_modules, .next, .git. Gives any engineer or AI an instant overview of the entire project tree.');
r.gap();
const tree = buildTree(rootDir);
const treeLines = tree.split('\n');
r.border(r.M, r.y - treeLines.length * 9.5, r.CW, Math.min(treeLines.length * 9.5 + 10, r.y - r.M - 50), rgb(0.97, 0.98, 1.0));
for (const tl of treeLines) {
    if (r.y < r.M + 36) { r.newPage(); r.border(r.M, r.M + 30, r.CW, r.y - r.M - 40, rgb(0.97, 0.98, 1.0), BLUE, 1); }
    if (tl) r.draw(san(tl).slice(0, 100), r.M + 6, r.y, r.fm, 7.5, rgb(0.08, 0.18, 0.45));
    r.y -= 9.5;
}

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????
// SECTION 11 - SOURCE CODE
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????????
r.sectionPage('SECTION 11', 'Complete Source Code', 'Every .ts .tsx .css .mjs .json source file verbatim', 10);
```

... continued: src/app/api/export-source/route.ts

```
r.body('All source code files in dependency order: config files first, then src/ files alphabetically. Each file is clearly s
r.gap();

const srcFiles = collectFiles(rootDir);
for (const fp of srcFiles) {
  const relPath = path.relative(rootDir, fp).replace(/\\/g, '/');
  let content = '';
  try { content = fs.readFileSync(fp, 'utf8'); } catch (_) { continue; }

  r.newPage();
  // File header card
  r.rect(r.M, r.y - 4, r.CW, 24, GRAY_DARK);
  r.rect(r.M, r.y - 4, 4, 24, GREEN);
  r.draw('FILE', r.M + 8, r.y + 10, r.fb, 7, GREEN);
  r.draw(san(relPath).slice(0, 85), r.M + 32, r.y + 10, r.fb, 9, WHITE);
  r.draw(`${content.length} bytes | ${content.split('\n').length} lines`, r.M + 32, r.y - 1, r.fr, 7, GRAY_MID);
  r.y -= 32;

  const codeLines = content.split('\n');
  for (let li = 0; li < codeLines.length; li++) {
    if (r.y < r.M + 36) {
      r.newPage();
      r.rect(r.M, r.y, r.CW, 12, GRAY_BG);
      r.draw(`... continued: ${san(relPath).slice(0, 70)}`, r.M, r.y + 2, r.fm, 7, GRAY_MID);
      r.y -= 14;
    }
    const cl = san(codeLines[li]).slice(0, 160);
    if (cl) {
      try { r.page.drawText(cl, { x: r.M + 4, y: r.y, font: r.fm, size: 7, color: rgb(0.05, 0.12, 0.35) }); } catch (_) {}
    }
    r.y -= 8.5;
  }
}

const bytes = await doc.save();
return new Response(Buffer.from(bytes), {
  status: 200,
  headers: {
    'Content-Type': 'application/pdf',
    'Content-Disposition': 'attachment; filename="Nandini_Internal_Docs.pdf"',
  },
});

} catch (err: any) {
  console.error('PDF Export Error:', err);
  return new Response(JSON.stringify({ error: err.message, stack: err.stack?.slice(0, 800) }), {
    status: 500,
    headers: { 'Content-Type': 'application/json' },
  });
}
}

// ?????????????????????????????????????????????????????????????????????????????????????????????????????????
// FILE HELPERS
// ?????????????????????????????????????????????????????????????????????????????????????????????????????????
function collectFiles(root: string): string[] {
  const files: string[] = [];
  for (const cfg of ['package.json', 'tsconfig.json', 'next.config.mjs', 'postcss.config.mjs', 'eslint.config.mjs']) {
    const p = path.join(root, cfg);
    if (fs.existsSync(p)) files.push(p);
  }
  const scan = (dir: string) => {
    try {
      for (const item of fs.readdirSync(dir)) {
        const fp = path.join(dir, item);
        const s = fs.statSync(fp);
        if (s.isDirectory()) {
          if (!['node_modules', '.next', '.git', 'fonts'].includes(item) && !item.startsWith('.')) scan(fp);
        } else {
          const ext = path.extname(item);
          if (['.ts', '.tsx', '.css', '.js', '.mjs', '.json'].includes(ext) && !item.endsWith('.d.ts') && item !== 'package-lock.
        }
      }
    } catch (_) {}
  };
  const src = path.join(root, 'src');
  if (fs.existsSync(src)) scan(src);
  return files;
}

function buildTree(root: string): string {
  let out = 'nandini-enterprises/\n';
```

... continued: src/app/api/export-source/route.ts

```
const scan = (dir: string, pfx = '') => {
  try {
    const items = fs.readdirSync(dir)
      .map(n => { let d = false; try { d = fs.statSync(path.join(dir, n)).isDirectory(); } catch (_) {} return { n, d, p: path.
        .filter(i => ![ 'node_modules', '.next', '.git' ].includes(i.n) && !i.n.startsWith('.') && i.n !== 'package-lock.json' )
        .sort((a, b) => a.d === b.d ? a.n.localeCompare(b.n) : a.d ? -1 : 1);
      items.forEach((item, idx) => {
        const last = idx === items.length - 1;
        out += `${pfx}${last ? '+- ' : '|-- '}${item.n}${item.d ? '/' : ''}\n`;
        if (item.d) scan(item.p, pfx + (last ? '    ' : '|  '));
      });
    } catch (_) {}
  };
  scan(root);
  return out;
}
```



```

'use client';?
?
import { useState } from 'react';?
import { CERTIFICATIONS, Certification } from '@lib/data';?
?
export default function CertificationsPage() {?
  const [selectedCert, setSelectedCert] = useState<Certification | null>(null);?
  ?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen text-slate-800 font-sans">?
      ?
      { /* Hero Section */}?
      <section className="relative py-20 bg-white border-b border-slate-200 overflow-hidden">?
        <div className="absolute inset-0 bg-[linear-gradient(to_right,#eef2f7_1px,transparent_1px),linear-gradient(to_bottom,#eef2f7_1px,transparent_1px)] bg-[size:10px_10px] bg-[position:0%_0%]"/>?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="grid grid-cols-1 lg:grid-cols-2 gap-12 items-center">?
            <div className="max-w-xl space-y-4">?
              <span className="text-xs uppercase tracking-widest text-[#0f4c81] font-mono font-bold block">?
                REGULATORY CLEARANCES?
              </span>?
              <h1 className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit leading-tight">?
                Certifications, Compliance & Audits?
              </h1>?
              <p className="text-base sm:text-lg leading-relaxed text-slate-655">?
                Nandini Enterprises maintains absolute compliance with state and central regulatory frameworks. Click any certifi?
              </p>?
            </div>?
            <div className="relative group">?
              <div className="absolute -inset-1.5 bg-gradient-to-r from-[#0f4c81] to-[#2b6cb0] rounded-lg blur opacity-25 group-hover:opacity-100">?
                ?
              </div>?
            </div>?
          </div>?
        </section>?
      ?
      { /* Certs Grid */}?
      <section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
        <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-6">?
          {CERTIFICATIONS.map((cert, index) => (?
            <div?
              key={index}?
              onClick={() => setSelectedCert(cert)}?
              className="bg-white border border-slate-200 hover:border-[#2b6cb0] rounded-lg p-8 cursor-pointer shadow-sm hover:shadow-md">?
                <div>?
                  <div className="w-8 h-8 rounded bg-[#E8F0FA] border border-slate-200 flex items-center justify-center text-[#0f4c81] font-bold">?
                    {index + 1}?
                  </div>?
                  <h3 className="text-slate-900 font-bold text-sm mb-2 font-outfit group-hover:text-[#0f4c81] transition-colors">?
                    {cert.name}?
                  </h3>?
                  <p className="text-slate-400 text-[10px] font-mono mb-4">?
                    Doc Ref: {cert.number}?
                  </p>?
                  <div>?
                    <span className="text-[10px] text-[#0f4c81] font-bold font-mono tracking-wider">?
                      VIEW COMPLIANCE & rarr;?
                    </span>?
                  </div>?
                </div>?
              ))}?
            </div>?
          ?
          { /* Supplementary table listing */}?
          <div className="mt-16 bg-white border border-slate-200 rounded-lg overflow-hidden max-w-4xl mx-auto shadow-sm">?
            <div className="px-6 py-4 bg-slate-50 border-b border-slate-200">?
              <h3 className="text-[#0f4c81] font-bold text-sm uppercase tracking-wider font-mono">?
                Statutory Registrations Directory?
              </h3>?
            </div>?
            <div className="divide-y divide-slate-100 text-xs font-mono">?
              {?
                { type: 'GSTIN', desc: 'Goods and Services Tax Registration Certificate (Government of India)', status: 'ACTIVE / COMPLIANT' },
                { type: 'IEC CODE', desc: 'Import Export Code License for global component procurement', status: 'ACTIVE / COMPLIANT' },
                { type: 'EPF REG', desc: 'Employees Provident Fund Registration for field personnel safety assurance', status: 'COMPLIANT' },
                { type: 'ESI REG', desc: 'Employees State Insurance Registration compliance', status: 'COMPLIANT' },
                { type: 'UDYAM', desc: 'MSME Registration Number UDYAM-TS-02-XXXXXX', status: 'REGISTERED' }
              }
            </div>
          </div>
        </div>
      </section>
    </div>
  )?
}

```

... continued: src/app/certifications/page.tsx

```
.map((stat, idx) => (?
  <div key={idx} className="px-6 py-4 flex flex-col sm:flex-row sm:justify-between sm:items-center gap-2 text-slate-7
    <span className="text-[#0f4c81] font-bold w-28 shrink-0">{stat.type}</span>?
    <span className="text-slate-600">{stat.desc}</span>?
    <span className="text-emerald-700 font-bold shrink-0 text-[10px] bg-emerald-50 border border-emerald-200 px-2 py-
  </div>?
  )})?
</div>?
</div>?
</section>?
?
/* Certification details Dialog */?
{selectedCert && (?
  <div className="fixed inset-0 z-50 flex items-center justify-center p-4 bg-slate-900/60 backdrop-blur-xs">?
    <div className="bg-white border border-slate-200 rounded-lg max-w-md w-full p-8 shadow-2xl relative">?
      <button ?
        className="absolute top-4 right-4 text-slate-400 hover:text-slate-800 text-xl cursor-pointer"?
        onClick={() => setSelectedCert(null)}?
      >?
        &times;?
      </button>?
      <span className="text-[10px] font-mono text-[#0f4c81] uppercase tracking-widest block mb-2 font-bold">?
        Registration Certificate?
      </span>?
      <h3 className="text-xl font-bold text-slate-900 mb-6 font-outfit">?
        {selectedCert.name}?
      </h3>?
      ?
      <div className="space-y-6 text-sm text-slate-700">?
        <div className="bg-[#E8F0FA] border border-slate-200 p-6 rounded text-center font-mono">?
          <span className="block text-slate-400 text-[10px] mb-2 font-bold">CERTIFICATE ID NUMBER</span>?
          <span className="text-slate-800 font-bold tracking-wider">{selectedCert.number}</span>?
          ?
          <div className="w-16 h-16 border-2 border-[#0f4c81]/25 border-dashed rounded-full flex items-center justify-cente
            ISO?
          </div>?
        </div>?
        ?
        <div>?
          <h4 className="text-xs uppercase text-slate-400 font-mono tracking-wider font-semibold mb-1">Approved Audit Scope?
          <p className="text-xs leading-relaxed text-slate-600">{selectedCert.scope}</p>?
        </div>?
        ?
        <div className="pt-4 border-t border-slate-100 flex justify-between items-center text-xs">?
          <span className="text-slate-500 font-medium">Status: ACTIVE</span>?
          <button ?
            className="px-4 py-2 bg-[#0f4c81] hover:bg-[#2b6cb0] text-white font-bold rounded transition-colors cursor-poin
            onClick={() => alert(`Certificate ${selectedCert.name} has been queued for PDF download.`)}?
          >?
            Download PDF &darr;?
          </button>?
        </div>?
        </div>?
        </div>?
        </div>?
      )})?
    </div>?
  );?
}?
}
```

```

'use client';?
?
import { useState, useEffect, Suspense } from 'react';?
import { useSearchParams } from 'next/navigation';?
?
function ContactFormContent() {?
  const searchParams = useSearchParams();?
  const typeParam = searchParams.get('type');?
  const initialType = (typeParam === 'quote' || typeParam === 'consult' || typeParam === 'support') ? typeParam : 'quote';?
  const [inquiryType, setInquiryType] = useState(initialType);?
  const [submitted, setSubmitted] = useState(false);?
  const [formData, setFormData] = useState({?
    name: '', email: '', phone: '', company: '', type: 'panel', requirement: ''?
  });?
  ?
  const handleSubmit = (e: React.FormEvent) => {?
    e.preventDefault();?
    setSubmitted(true);?
    setTimeout(() => {?
      setSubmitted(false);?
      setFormData({ name: '', email: '', phone: '', company: '', type: 'panel', requirement: '' });?
    }, 5000);?
  };?
  ?
  return (?
    <div className="grid grid-cols-1 lg:grid-cols-12 gap-12 items-start py-8">?
      ?
      { /* Contact Details & Channels */?
      <div className="lg:col-span-5 space-y-8 font-sans">?
        <div className="space-y-4">?
          <h2 className="text-[#0f4c81] font-extrabold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
            Operational Headquarters?
          </h2>?
          <p className="text-sm text-slate-600 leading-relaxed">?
            Connect directly with our Hyderabad switches and VFD division. Form submissions are logged instantly into our B2B pro?
          </p>?
        </div>?
      </div>?
      ?
      <div className="bg-white border border-slate-200 p-6 rounded-lg space-y-4 font-mono text-xs shadow-sm">?
        <h4 className="text-[#0f4c81] font-bold border-b border-slate-100 pb-2">CONTACT DIRECTORY</h4>?
        ?
        <div className="space-y-3">?
          <div>?
            <span className="text-slate-400 block text-[9px]">PRIMARY EMAIL</span>?
            <a href="mailto:info@nandiniautomation.com" className="text-[#0f4c81] hover:text-[#2b6cb0] font-semibold text-sm">?
              info@nandiniautomation.com?
            </a>?
          </div>?
          <div>?
            <span className="text-slate-400 block text-[9px]">SUPPORT TELEPHONE</span>?
            <a href="tel:+914023190131" className="text-[#0f4c81] hover:text-[#2b6cb0] font-semibold text-sm">?
              +91-40-23190131?
            </a>?
          </div>?
          <div>?
            <span className="text-slate-400 block text-[9px]">IDA PLANT LOCATION</span>?
            <span className="text-slate-700 leading-relaxed block mt-0.5 text-xs font-sans">?
              Plot No. 49, Phase-III, IDA Cherlapally, Hyderabad, Telangana - 500051?
            </span>?
          </div>?
        </div>?
      </div>?
      ?
      { /* WhatsApp direct block */?
      <div className="bg-gradient-to-r from-emerald-50 to-slate-50 border border-emerald-200 p-6 rounded-lg flex items-center j?
        <div className="space-y-1">?
          <h4 className="text-emerald-950 font-bold text-sm">Need quick estimation?</h4>?
          <p className="text-[11px] text-slate-600">Connect with an application engineer over WhatsApp.</p>?
        </div>?
        <a?
          href="https://wa.me/914023190131"?
          target="_blank"?
          rel="noopener noreferrer"?
          className="px-4 py-2 bg-emerald-600 hover:bg-emerald-700 text-white font-bold rounded text-xs uppercase tracking-wide?
        >?
          WhatsApp?
        </a>?
      </div>?
    </div>?
  );
  ?
}

```

```

    { /* Smart Form Integration */ }
    <div className="lg:col-span-7 bg-white border border-slate-200 p-8 rounded-lg shadow-sm">?
    ?
    { /* Toggle Form tab */ }
    <div className="flex border-b border-slate-100 mb-8 gap-2">?
    { [
      { id: 'quote', label: 'Submit RFQ' },?
      { id: 'consult', label: 'Request Consultation' },?
      { id: 'support', label: 'Support Request' }?
    ].map((tab) => (
      <button?
        key={tab.id}?
        type="button"?
        onClick={() => setInquiryType(tab.id)}?
        className={`px-4 py-2 font-mono text-xs uppercase border-b-2 tracking-wider cursor-pointer transition-all ${?
          inquiryType === tab.id?
            ? 'border-[#0f4c81] text-[#0f4c81] font-bold'?
            : 'border-transparent text-slate-400 hover:text-slate-600'?
          }`}?
      )?
    )?
    {tab.label}?
    </button>?
  )]?
  </div>?

  { submitted ? (
    <div className="text-center py-16 space-y-4">?
    <div className="w-12 h-12 bg-emerald-50 border border-emerald-250 rounded-full flex items-center justify-center mx-auto">?
    &check;?
    </div>?
    <h4 className="text-slate-900 font-bold text-lg font-outfit">Inquiry Received</h4>?
    <p className="text-slate-600 text-xs max-w-sm mx-auto leading-relaxed">?
    Your inquiry has been compiled into our pipeline system. A design specialist will review your schematics and contact you within 24 hours.
    </p>?
    </div>?
  ) : (
    <form onSubmit={handleSubmit} className="space-y-4 font-mono text-xs">?
    <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">?
    <div className="space-y-2">?
    <label className="text-slate-500 uppercase block font-semibold">Your Name</label>?
    <input?
      type="text"?
      required?
      value={formData.name}?
      onChange={(e) => setFormData({ ...formData, name: e.target.value })}?
      placeholder="e.g. John Doe"?
      className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder-slate-400">?
    </div>?
    <div className="space-y-2">?
    <label className="text-slate-500 uppercase block font-semibold">Email Address</label>?
    <input?
      type="email"?
      required?
      value={formData.email}?
      onChange={(e) => setFormData({ ...formData, email: e.target.value })}?
      placeholder="e.g. john@company.com"?
      className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder-slate-400">?
    </div>?
    </div>?

    <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">?
    <div className="space-y-2">?
    <label className="text-slate-500 uppercase block font-semibold">Telephone / Phone</label>?
    <input?
      type="tel"?
      required?
      value={formData.phone}?
      onChange={(e) => setFormData({ ...formData, phone: e.target.value })}?
      placeholder="e.g. +91 99999-99999"?
      className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder-slate-400">?
    </div>?
    <div className="space-y-2">?
    <label className="text-slate-500 uppercase block font-semibold">Organization Name</label>?
    <input?
      type="text"?
      required?
      value={formData.company}?
      onChange={(e) => setFormData({ ...formData, company: e.target.value })}?
      placeholder="e.g. Cement Works Pvt Ltd"?
    </div>?
  </div>
  </form>
  </div>

```

... continued: src/app/contact/page.tsx

```
        className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder-slate-400"
      />?
    </div>?
  </div>?

  ?
  <div className="space-y-2">?
    <label className="text-slate-500 uppercase block font-semibold">Core Solution Category</label>?
    <select?
      value={formData.type}?
      onChange={(e) => setFormData({ ...formData, type: e.target.value })}?
      className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 focus:outline-none"
    >?
      <option value="panel">Electrical Panels (PCC, MCC, VFD)</option>?
      <option value="automation">PLC SCADA System Integration</option>?
      <option value="vfd-repair">Yaskawa VFD Repair & Drive checks</option>?
      <option value="instrumentation">Field Sensors / Baumer levels</option>?
      <option value="other">Other B2B Project query</option>?
    </select>?
  </div>?

  ?
  <div className="space-y-2">?
    <label className="text-slate-500 uppercase block font-semibold">Project Scope / Requirements Detail</label>?
    <textarea?
      required?
      value={formData.requirement}?
      onChange={(e) => setFormData({ ...formData, requirement: e.target.value })}?
      placeholder="Describe structural specifications, busbar short circuit capacity rating, motor HP size, PLC process"
      rows={5}?
      className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder-slate-400"
    ></textarea>?
  </div>?

  ?
  <button?
    type="submit"?
    className="w-full py-3 bg-[#0f4c81] hover:bg-[#2b6cb0] text-white font-bold rounded text-xs uppercase tracking-wide"
  >?
    Log technical request &rarr;?
  </button>?
</form>?
  )}?
</div>?
  ?
</div>?
  );?
}?
  ?
export default function ContactPage() {?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen text-slate-800 font-sans">?
      ?
      { /* Hero Section */ }?
      <section className="relative py-20 bg-white border-b border-slate-200 overflow-hidden">?
        <div className="absolute inset-0 bg-[linear-gradient(to_right,#eef2f7_1px,transparent_1px),linear-gradient(to_bottom,#eef2f7_1px,transparent_1px)]" />?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="grid grid-cols-1 lg:grid-cols-2 gap-12 items-center">?
            <div className="max-w-xl space-y-4">?
              <span className="text-xs uppercase tracking-widest text-[#0f4c81] font-mono font-bold block">?
                SUPPORT HOTLINE?
              </span>?
              <h1 className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit leading-tight">?
                Connect With Our Engineering Team?
              </h1>?
              <p className="text-base sm:text-lg leading-relaxed text-slate-655">?
                Submit your RFQ technical schedules, register drive fault ticket values, or schedule factory audits.?
              </p>?
            </div>?
            <div className="relative group">?
              <div className="absolute -inset-1.5 bg-gradient-to-r from-[#0f4c81] to-[#2b6cb0] rounded-lg blur opacity-25 group-hover:opacity-100" />?
              ?
            </div>?
          </div>?
        </div>?
      </section>?

      ?
      { /* Main Grid */ }?
      <section className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">?
        <Suspense fallback={<div className="text-center py-20 text-slate-500 font-mono text-xs">LOADING FORM...</div>}>?
          <ContactFormContent />?
        </Suspense>
      </section>
    </div>
  )
}
```

... continued: src/app/contact/page.tsx

```
    </Suspense>?  
  </section>?  
?  
  </div>?  
);?  
}?
```

```
'use client';?
export default function GalleryPage() {?
  const images = [?
    { ?
      title: "Plant Shop Floor Assembly", ?
      category: "Fabrication", ?
      size: "12,000 Sqft facility layout IDA Cherlapally",?
      src: "https://picsum.photos/seed/nandini/800/600"?
    },?
    { ?
      title: "Double Hoisting VFD Panel Testing", ?
      category: "Commissioning", ?
      size: "Yaskawa A1000 configuration for crane sync",?
      src: "https://picsum.photos/seed/nandini/800/600"?
    },?
    { ?
      title: "CPRI Short Circuit Verification", ?
      category: "Compliance", ?
      size: "Short circuit busbar assembly audit test check",?
      src: "https://picsum.photos/seed/nandini/800/600"?
    },?
    { ?
      title: "Indian Railways Simulator Test Rig", ?
      category: "Defense & Rail", ?
      size: "Ruggedized telemetry cabinet assembly logs",?
      src: "https://picsum.photos/seed/nandini/800/600"?
    },?
    { ?
      title: "120 MLD Pump Station Control Desk", ?
      category: "Water Treatment", ?
      size: "Main SCADA display monitors loop monitoring",?
      src: "https://picsum.photos/seed/nandini/800/600"?
    },?
    { ?
      title: "Women operator CAD schematic layout", ?
      category: "Stewardship / Seminar", ?
      size: "CAD design training modules",?
      src: "https://picsum.photos/seed/nandini/800/600"?
    }
  ],?
  ?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen text-slate-800 font-sans">?
      ?
      { /* Hero Section */?
      <section className="relative py-20 bg-white border-b border-slate-200 overflow-hidden">?
        <div className="absolute inset-0 bg-[linear-gradient(to_right,#eef2f7_1px,transparent_1px),linear-gradient(to_bottom,#eef2f7_1px,transparent_1px)] bg-[size:100%_100%] border-b border-slate-200">?
          <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
            <div className="grid grid-cols-1 lg:grid-cols-2 gap-12 items-center">?
              <div className="max-w-xl space-y-4">?
                <span className="text-xs uppercase tracking-widest text-[#0f4c81] font-mono font-bold block">?
                  VISUAL RECORDINGS?
                </span>?
                <h1 className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit leading-tight">?
                  Manufacturing & Site Operations Gallery?
                </h1>?
                <p className="text-base sm:text-lg leading-relaxed text-slate-600">?
                  Browse our IDA Cherlapally switchgear assembly plant, VFD testing bays, field erection projects, and educational?
                </p>?
              </div>?
              <div className="relative group">?
                <div className="absolute -inset-1.5 bg-gradient-to-r from-[#0f4c81] to-[#2b6cb0] rounded-lg blur opacity-25 group-hover:opacity-100">?
                </div>?
                ?
              </div>?
            </div>?
          </div>?
        </div>?
      </section>?
      ?
      { /* Media Grid */?
      <section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
        <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8">?
          {images.map((img, idx) => (?
            <div ?
              key={idx}?
              className="bg-white border border-slate-200 rounded-lg overflow-hidden shadow-sm hover:shadow-md hover:border-[#2b6cb0]"?
            </div>
          )
        </div>
      </section>
    )
  )
}
```

```
... continued: src/app/gallery/page.tsx
```

```

    >?
    { /* Actual Image Block */?
    <div className="aspect-video relative overflow-hidden flex items-center justify-center border-b border-slate-100">?
      <img ?
        src={img.src} ?
        alt={img.title}?
        className="w-full h-full object-cover group-hover:scale-105 transition-transform duration-500"?
      />?
      <span className="absolute bottom-2 right-2 text-[8px] font-mono bg-white/90 text-slate-600 border border-slate-200">
        REF: IMG_00{idx + 1}?
      </span>?
    </div>?

    <div className="p-6 space-y-2">?
      <span className="text-[10px] font-mono text-[#0f4c81] font-bold uppercase tracking-wider">?
        {img.category}?
      </span>?
      <h3 className="text-slate-900 font-bold text-base font-outfit">?
        {img.title}?
      </h3>?
      <p className="text-xs text-slate-500">?
        {img.size}?
      </p>?
    </div>?
  </div>?
  )}}?
</div>?
</section>?

?
</div>?
);?
}?

```



```
@import "tailwindcss";?
?
@theme {?
  --color-ne-white: #ffffff;?
  --color-ne-bg: #E8F0FA;?
  --color-ne-gray-light: #eef2f7;?
  --color-ne-blue-corp: #0A3D91; /* Deep Royal Blue */?
  --color-ne-blue-steel: #1565C0; /* Electric Blue */?
  --color-ne-blue-electric: #3b82f6;?
  --color-ne-text-dark: #0B1F3A; /* Navy Blue */?
  --color-ne-text-muted: #64748b;?
  ?
  --font-sans: var(--font-inter), sans-serif;?
  --font-outfit: var(--font-outfit), sans-serif;?
}?
?
/* Custom Overrides & Scroll Behaviors */?
html, body {?
  background-color: #E8F0FA; /* bg-ne-bg */?
  color: #1e293b; /* text-ne-text-dark */?
}?
?
/* Custom scrollbar matching corporate theme */?
::-webkit-scrollbar {?
  width: 8px;?
  height: 8px;?
}?
?
::-webkit-scrollbar-track {?
  background: #f7f9fc;?
}?
?
::-webkit-scrollbar-thumb {?
  background: #cbd5e1;?
  border-radius: 4px;?
}?
?
::-webkit-scrollbar-thumb:hover {?
  background: #2b6cb0;?
}?
?
/* Smooth transition animations for hover */?
.transition-hover {?
  transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);?
}?
?
```

```

'use client';?
?
import { useState } from 'react';?
import { INDUSTRIES, Industry } from '@lib/data';?
?
export default function IndustriesPage() {?
  const [selectedIndustry, setSelectedIndustry] = useState<Industry | null>(null);?
  const [activeTab, setActiveTab] = useState<'overview' | 'challenges' | 'solutions' | 'equipment'>('overview');?
  ?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen text-slate-800 font-sans">?
      ?
      { /* Hero Section */?
        <section className="relative py-20 bg-white border-b border-slate-200 overflow-hidden">?
          <div className="absolute inset-0 bg-[linear-gradient(to_right,#eef2f7_1px,transparent_1px),linear-gradient(to_bottom,#eef2f7_1px,transparent_1px)] bg-[size:30px_30px] bg-[mask:linear-gradient(to_bottom,transparent_49%,black_49%,black_51%,transparent_51%)]">?
            <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
              <div className="grid grid-cols-1 lg:grid-cols-2 gap-12 items-center">?
                <div className="max-w-xl space-y-5">?
                  <span className="text-xs uppercase tracking-widest text-[#0f4c81] font-mono font-bold block">?
                    SECTOR ENGINEERING CONVENTIONS?
                  </span>?
                  <h1 className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit leading-tight">?
                    Industries We Serve?
                  </h1>?
                  <p className="text-base sm:text-lg leading-relaxed text-slate-600">?
                    We design specialized automation loops, CPRI control boards, process instrumentation, and turnkey electrical systems.
                  </p>?
                  <div className="flex gap-6 pt-2 font-mono text-xs">?
                    <div className="text-center">?
                      <div className="text-2xl font-bold text-[#0f4c81]">14</div>?
                      <div className="text-slate-500 uppercase tracking-wider">Sectors Served</div>?
                    </div>?
                    <div className="text-center">?
                      <div className="text-2xl font-bold text-[#0f4c81]">200+</div>?
                      <div className="text-slate-500 uppercase tracking-wider">Active Sites</div>?
                    </div>?
                    <div className="text-center">?
                      <div className="text-2xl font-bold text-[#0f4c81]">25+</div>?
                      <div className="text-slate-500 uppercase tracking-wider">Yrs Experience</div>?
                    </div>?
                  </div>?
                </div>?
                <div className="relative group">?
                  <div className="absolute -inset-1.5 bg-gradient-to-r from-[#0f4c81] to-[#2b6cb0] rounded-lg blur opacity-25 group-hover:opacity-100">?
                    ?
                  </div>?
                </div>?
              </div>?
            </div>?
          </section>?
          ?
          { /* Intro Section */?
            <section className="py-12 bg-white border-b border-slate-100">?
              <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
                <div className="grid grid-cols-1 md:grid-cols-3 gap-8 text-sm">?
                  <div className="bg-[#E8F0FA] border border-slate-200 rounded-lg p-6 space-y-2">?
                    <h3 className="text-[#0f4c81] font-bold font-outfit">Process Industry Expertise</h3>?
                    <p className="text-slate-600 leading-relaxed text-xs">Our automation engineers carry deep domain knowledge across cement, sugar, and pharmaceuticals.
                  </p>?
                  <div className="bg-[#E8F0FA] border border-slate-200 rounded-lg p-6 space-y-2">?
                    <h3 className="text-[#0f4c81] font-bold font-outfit">Manufacturing Sector Solutions</h3>?
                    <p className="text-slate-600 leading-relaxed text-xs">From cement kiln speed control to sugar centrifugal cycle automation.
                  </p>?
                  <div className="bg-[#E8F0FA] border border-slate-200 rounded-lg p-6 space-y-2">?
                    <h3 className="text-[#0f4c81] font-bold font-outfit">Government & Infrastructure</h3>?
                    <p className="text-slate-600 leading-relaxed text-xs">HMWSSB, HMRL, NTPC, RDSO-compliant panels, PESO-approved hazardous area equipment.
                  </p>?
                </div>?
              </div>?
            </section>?
            ?
            { /* Grid listing */?
              <section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
                <div className="mb-10">?
                  <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3 mb-2">?
                    Industrial Sector Portfolio?
                  </h2>?
                </div>?
              </section>?
            }
          }
        </div>?
      </div>?
    </div>?
  )
}

```

```

    <p className="text-sm text-slate-600 pl-4">Click any sector card to view detailed engineering challenges, solutions, st
  </div>?
  <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-6">?
    {INDUSTRIES.map((ind) => (?
      <div ?
        key={ind.slug}?
        onClick={() => { setSelectedIndustry(ind); setActiveTab('overview'); }}?
        className="bg-white border border-slate-200 hover:border-[#2b6cb0] hover:shadow-md rounded-lg p-7 cursor-pointer tr
      >?
      <div>?
        <div className="flex items-start justify-between mb-4">?
          <h3 className="text-[#0f4c81] font-bold text-xl font-outfit group-hover:text-[#2b6cb0] transition-colors">?
            {ind.name}?
          </h3>?
          <span className="shrink-0 text-[9px] font-mono bg-slate-50 border border-slate-200 text-slate-400 px-2 py-0.5 r
            {ind.slug.toUpperCase()}?
          </span>?
        </div>?
        <p className="text-xs text-slate-600 leading-relaxed line-clamp-3 mb-4">?
          {ind.desc.substring(0, 180)}...?
        </p>?
        <div className="text-[10px] font-mono text-[#0f4c81] bg-slate-50 border border-slate-100 rounded px-2 py-1 inline
          KEY: {ind.tech.split(',')[0].trim()}?
        </div>?
      </div>?
    )}
  </div>?
  <div className="pt-4 border-t border-slate-100 flex justify-between items-center text-[10px] font-mono text-slate-4
    <span className="text-emerald-700 font-semibold">{ind.outcome.substring(0, 50)}...</span>?
    <span className="text-[#0f4c81] font-bold group-hover:underline shrink-0 ml-2">Details ?</span>?
  </div>?
</div>?
))}?
</div>?
</section>?
?
  /* Industry Detail Modal */?
  {selectedIndustry && (?
    <div className="fixed inset-0 z-50 flex items-center justify-center p-4 bg-slate-900/60 backdrop-blur-sm overflow-y-auto"
      <div className="bg-white border border-slate-200 rounded-lg max-w-3xl w-full shadow-2xl relative my-8">?
        ?
        /* Modal Header */?
        <div className="bg-[#0f4c81] text-white p-6 rounded-t-lg relative">?
          <button ?
            className="absolute top-4 right-4 text-white/70 hover:text-white text-2xl cursor-pointer leading-none font-light"
            onClick={() => setSelectedIndustry(null)}?
          >?
          ??
          </button>?
          <span className="text-[10px] font-mono text-[#93c5fd] uppercase tracking-widest block mb-1 font-bold">?
            EPC SECTOR ENGINEERING SPECIFICATION?
          </span>?
          <h3 className="text-2xl font-bold font-outfit">?
            {selectedIndustry.name}?
          </h3>?
        </div>?
      </div>?
    )
  ?
  /* Tab Bar */?
  <div className="flex border-b border-slate-200 bg-slate-50 rounded-none overflow-x-auto">?
    {[ 'overview', 'challenges', 'solutions', 'equipment' ] as const}.map((tab) => (?
      <button?
        key={tab}?
        onClick={() => setActiveTab(tab)}?
        className={`px-5 py-3 font-mono text-[10px] uppercase tracking-wider font-semibold border-b-2 transition-colors
          activeTab === tab?
            ? 'border-[#0f4c81] text-[#0f4c81] bg-white'?
            : 'border-transparent text-slate-500 hover:text-slate-700'?
        }`}?
      >?
      {tab === 'overview' ? 'Overview & Projects' : ?
        tab === 'challenges' ? 'Engineering Challenges' : ?
        tab === 'solutions' ? 'NE Solutions' : 'Key Equipment'}?
      </button>?
    )})?
  </div>?
  ?
  /* Tab Content */?
  <div className="p-8 space-y-6 text-sm text-slate-700">?
    ?
    {activeTab === 'overview' && (?
      <div className="space-y-5">?
        <div>?
          <h4 className="text-xs uppercase text-slate-400 font-mono tracking-wider font-semibold mb-3">?

```

```

        1. Industrial Process & Scope Description?
      </h4>?
      <p className="leading-relaxed text-slate-655 text-sm">{selectedIndustry.desc}</p>?
    </div>?
    <div className="grid grid-cols-1 sm:grid-cols-2 gap-5">?
      <div className="bg-[#E8F0FA] border border-slate-200 rounded p-4">?
        <h4 className="text-xs uppercase text-slate-400 font-mono tracking-wider font-semibold mb-2">?
          2. Technical Systems Implemented?
        </h4>?
        <p className="text-xs text-slate-600 leading-relaxed font-mono">{selectedIndustry.tech}</p>?
      </div>?
      <div className="bg-[#E8F0FA] border border-slate-200 rounded p-4">?
        <h4 className="text-xs uppercase text-slate-400 font-mono tracking-wider font-semibold mb-2">?
          3. Primary Project Examples?
        </h4>?
        <p className="text-xs text-slate-600 leading-relaxed">{selectedIndustry.projects}</p>?
      </div>?
    </div>?
    <div className="bg-emerald-50 border border-emerald-200 rounded p-4">?
      <h4 className="text-xs uppercase text-emerald-700 font-mono tracking-wider font-semibold mb-2">?
        4. Verified Operational Outcome?
      </h4>?
      <p className="text-xs text-emerald-800 font-mono font-bold leading-relaxed">{selectedIndustry.outcome}</p>?
    </div>?
  </div>?
  )}?
  ?
  {activeTab === 'challenges' && (?
    <div className="space-y-4">?
      <p className="text-xs text-slate-500 italic mb-4">The following engineering challenges are commonly faced in {s
      {selectedIndustry.challenges?.map((challenge, idx) => (?
        <div key={idx} className="flex items-start space-x-3 bg-rose-50 border border-rose-100 rounded p-4">?
          <span className="shrink-0 w-6 h-6 rounded-full bg-rose-100 text-rose-700 font-mono text-[10px] font-bold fl
            {idx + 1}>?
          </span>?
          <p className="text-sm text-slate-700 leading-relaxed">{challenge}</p>?
        </div>?
      </div>?
    </div>?
  </div>?
  )}?
  ?
  {activeTab === 'solutions' && (?
    <div className="space-y-4">?
      <p className="text-xs text-slate-500 italic mb-4">Nandini Enterprises engineering methodology to resolve the se
      {selectedIndustry.solutions?.map((solution, idx) => (?
        <div key={idx} className="flex items-start space-x-3 bg-emerald-50 border border-emerald-100 rounded p-4">?
          <span className="shrink-0 w-6 h-6 rounded-full bg-emerald-100 text-emerald-700 font-mono text-[10px] font-b
            ??
          </span>?
          <p className="text-sm text-slate-700 leading-relaxed">{solution}</p>?
        </div>?
      </div>?
    </div>?
  </div>?
  <div className="pt-4 border-t border-slate-100">?
    <h4 className="text-xs uppercase text-slate-400 font-mono tracking-wider font-semibold mb-3">?
      Applicable Standards & Certifications?
    </h4>?
    <div className="flex flex-wrap gap-2">?
      {selectedIndustry.standards?.map((std, idx) => (?
        <span key={idx} className="text-[10px] font-mono bg-[#0f4c81]/5 border border-[#0f4c81]/20 text-[#0f4c81]
          {std}>?
        </span>?
      </div>?
    </div>?
  </div>?
  )}?
  ?
  {activeTab === 'equipment' && (?
    <div className="space-y-4">?
      <p className="text-xs text-slate-500 italic mb-4">Key equipment and technology brands typically deployed in {se
      <div className="grid grid-cols-1 sm:grid-cols-2 gap-3">?
        {selectedIndustry.keyEquipment?.map((equip, idx) => (?
          <div key={idx} className="flex items-center space-x-3 bg-white border border-slate-200 rounded p-3 shadow-x
            <span className="text-[#0f4c81] font-bold text-lg shrink-0">?</span>?
            <span className="text-sm text-slate-700 font-semibold">{equip}</span>?
          </div>?
        </div>?
      </div>?
    </div>?
    <div className="pt-4">?
      <a ?
        href="/contact?type=inquiry"?
        className="inline-block bg-[#0f4c81] text-white px-6 py-3 rounded font-mono text-xs uppercase tracking-wide

```

... continued: src/app/industries/page.tsx

```
        >?
        Request Technical Consultation ??
      </a>?
    </div>?
  </div>?
}
?
  </div>?
);?
}?
```

```
'use client';?
import { useState } from 'react';?
interface Article {?
  category: string;?
  title: string;?
  desc: string;?
  readTime: string;?
  slug: string;?
  author: string;?
  date: string;?
  tags: string[];?
  content: {?
    intro: string;?
    sections: { heading: string; body: string }[];?
    keyPoints: string[];?
    conclusion: string;?
  };?
}?
const articles: Article[] = [?
  {?
    category: "Technical Guide",?
    title: "What Is SCADA? A Complete Industrial Overview",?
    desc: "A comprehensive look at SCADA architecture, components, communication protocols, and implementation methodology for he",?
    readTime: "12 min read",?
    slug: "what-is-scada",?
    author: "Nandini Enterprises Technical Team",?
    date: "March 2025",?
    tags: ["SCADA", "PLC", "HMI", "OPC UA", "Industrial Automation"],?
    content: {?
      intro: "SCADA (Supervisory Control and Data Acquisition) is the foundational technology platform that enables real-time mon",?
      sections: [?
        {?
          heading: "Three-Tier Automation Architecture",?
          body: "Modern SCADA implementations follow a three-level hierarchy. At the field level (Level 0/1), physical sensors, a",?
        },?
        {?
          heading: "SCADA Communication Protocols",?
          body: "The communication backbone of a SCADA system determines its performance and reliability. At the field level, pro",?
        },?
        {?
          heading: "Alarm Management and ISA-101 HMI Design",?
          body: "Poorly designed SCADA systems generate thousands of nuisance alarms, leading to operator fatigue and missed genu",?
        },?
        {?
          heading: "Historical Data Logging and Reporting",?
          body: "One of the most valuable functions of a SCADA system is its ability to log process data at high resolution - typ",?
        },?
      ],?
      keyPoints: [?
        "SCADA systems operate across three tiers: field sensors, PLC controllers, and supervisory servers",?
        "OPC UA is the modern standard for secure, cross-vendor data exchange in industrial networks",?
        "ISA-101 HMI design rules and ISA-18.2 alarm management reduce operator errors significantly",?
        "Historical data logging at 100ms resolution enables trend analysis and predictive maintenance",?
        "Redundant servers and fiber ring networks ensure >99.9% SCADA availability for critical infrastructure"?
      ],?
      conclusion: "A well-engineered SCADA system is not just a monitoring tool - it is a comprehensive operational intelligence",?
    },?
  },?
  {?
    category: "White Paper",?
    title: "PLC vs DCS: Choosing the Right Control System",?
    desc: "An in-depth engineering comparison of Programmable Logic Controllers and Distributed Control Systems - analyzing archi",?
    readTime: "15 min read",?
    slug: "plc-vs-dcs",?
    author: "Nandini Enterprises Technical Team",?
    date: "February 2025",?
    tags: ["PLC", "DCS", "Control Systems", "Process Automation", "SCADA"],?
    content: {?
      intro: "The choice between a Programmable Logic Controller (PLC) system and a Distributed Control System (DCS) is one of th",?
      sections: [?
        {?
          heading: "Architectural Philosophy: Sequential vs. Continuous",?
          body: "PLCs were originally designed for machine-level discrete control - reading binary input states (sensors, switch",?
        },?
        {?
          heading: "When to Choose a PLC System",?
          body: "PLC systems are the right choice when your application involves discrete machine control, sequence interlocking,
```

```

},?
{?
  heading: "When to Choose a DCS",?
  body: "A DCS is the correct choice for continuous process plants - oil refineries, petrochemical complexes, power generation",?
},?
{?
  heading: "Hybrid Approaches and Modern Convergence",?
  body: "Today, the line between PLC and DCS is blurring. Modern PLCs offer batch control libraries, analog process I/O, and digital communication",?
},?
],?
keyPoints: [?
  "PLCs excel at discrete machine control with fast scan cycles (1-10ms) and sequential logic",?
  "DCS systems are optimized for continuous process control with tight analog feedback loops",?
  "Modern PLC + SCADA solutions can effectively replace DCS in mid-size process plants",?
  "Large continuous process plants (refineries, power plants) still benefit from integrated DCS platforms",?
  "OPC UA bridges PLC and DCS systems, enabling unified data integration at the enterprise level"?
],?
conclusion: "There is no universal answer to the PLC vs DCS question. The correct choice depends on your process type (discrete vs continuous) and the required level of integration",?
},?
},?
{?
category: "Technical Guide",?
title: "Understanding VFD Harmonics & IEEE 519 Compliance",?
desc: "How Variable Frequency Drives introduce electrical harmonics into power networks, methods to measure harmonic levels, and mitigation strategies",?
readTime: "11 min read",?
slug: "vfd-harmonics",?
author: "Nandini Enterprises Technical Team",?
date: "January 2025",?
tags: ["VFD", "Harmonics", "IEEE 519", "Power Quality", "Filters"],?
content: {?
  intro: "Variable Frequency Drives are essential tools for energy conservation and motor speed control in industrial plants. However, they also introduce electrical harmonics that can cause overheating, equipment failure, and interference with other systems.",?
  sections: [?
    {?
      heading: "How VFDs Generate Harmonic Currents",?
      body: "A standard VFD converts AC power to DC (through a diode rectifier bridge), smooths the DC with capacitors, and then inverts it back to AC. This switching process generates harmonic currents, primarily at the 6th, 12th, 18th, and 24th orders.",?
    },?
    {?
      heading: "IEEE 519 Harmonic Standard Limits",?
      body: "IEEE 519 (Recommended Practices for Harmonic Control in Electrical Power Systems) specifies maximum harmonic current limits based on the system voltage and the VFD's rated current. For example, at 480V, the limit is 8% of the VFD's rated current for the 5th harmonic and 4% for the 7th harmonic.",?
    },?
    {?
      heading: "Harmonic Mitigation Solutions",?
      body: "Multiple solutions exist with varying cost and effectiveness. AC line reactors (3-5% impedance) reduce harmonic currents by 25-35%. Active harmonic filters (AHF) can reduce harmonics by 90-95%. Proper VFD sizing and configuration (e.g., using 18-pulse drives) also help.",?
    },?
  ],?
  keyPoints: [?
    "Standard diode-bridge VFDs generate 5th, 7th, 11th, and 13th harmonic currents",?
    "IEEE 519 limits Total Demand Distortion (TDD) to 5-8% at the utility connection point",?
    "AC line reactors (minimum recommendation) reduce harmonics by 30-40%",?
    "Active harmonic filters achieve >95% reduction and work for existing mixed drive installations",?
    "Active Front End (AFE) drives eliminate input harmonics and allow energy regeneration"?
  ],?
  conclusion: "Harmonic management is not optional - it is a regulatory requirement and a practical engineering necessity. The correct solution depends on the application, budget, and the specific harmonic profile of the system."
},?
},?
{?
category: "Engineering Insight",?
title: "Electrical Panel Design: Form 4b Segregation Explained",?
desc: "A technical explanation of Form 4b internal segregation in LV switchgear assemblies - how it limits fault propagation and improves safety",?
readTime: "9 min read",?
slug: "form-4b-segregation",?
author: "Nandini Enterprises Technical Team",?
date: "April 2025",?
tags: ["Panels", "IEC 61439", "Switchgear", "CPRI", "Form 4b"],?
content: {?
  intro: "When specifying or evaluating an electrical switchgear assembly, the internal form of segregation is a critical design consideration. Form 4b is a common configuration that provides a balance between cost and safety.",?
  sections: [?
    {?
      heading: "Understanding the Four Segregation Forms",?
      body: "Form 1 provides no internal separation - all components share a common open air space. This is acceptable only for low-voltage applications with low fault currents. Form 2 provides partial separation using barriers. Form 3 provides full separation using air-circuit breakers. Form 4b provides full separation using air-circuit breakers and is the most common for medium-voltage applications.",?
    },?
    {?
      heading: "Why Form 4b Matters for Industrial Installations",?
      body: "In a Form 4b panel, if a fault develops in one functional unit (for example, a short-circuit that burns out a motor), the fault is contained within that unit. This prevents the fault from propagating to other units, ensuring the safety of the installation and minimizing downtime.",?
    },?
    {?
      heading: "CPRI Testing Requirements for Form 4b Panels",?
      body: "CPRI (Central Power Research Institute) type tests validate that a panel design can withstand extreme fault conditions. For Form 4b, this includes testing for internal arcing, which is a critical failure mode in switchgear.",?
    },?
  ],?
  keyPoints: [?
    "IEC 61439-1 defines Forms 1-4 of internal segregation in LV switchgear assemblies",?
    "Form 4b is the most common configuration for medium-voltage LV switchgear",?
    "Form 4b provides full internal segregation, limiting fault propagation",?
    "Form 4b requires air-circuit breakers for fault isolation",?
    "Form 4b is suitable for applications with high fault currents",?
    "Form 4b is the most expensive configuration due to the use of air-circuit breakers",?
    "Form 4b is the safest configuration for industrial installations",?
    "Form 4b is the most reliable configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the most common configuration for industrial installations",?
    "Form 4b is the most widely used configuration for industrial installations",?
    "Form 4b is the most popular configuration for industrial installations",?
    "Form 4b is the most standard configuration for industrial installations",?
    "Form 4b is the most typical configuration for industrial installations",?
    "Form 4b is the
```

```
"Form 4b separates busbars, functional units, AND external terminal compartments with metallic barriers",?
"Fault containment in Form 4b limits damage to a single drawer, allowing continued plant operation",?
"CPRI type testing validates short-circuit withstand, temperature rise, and arc containment performance",?
"Specifying Form 4b adds 15-25% to panel cost but significantly reduces total downtime risk"?
],?
  conclusion: "For any industrial installation where continuity of production is critical - and the cost of an unplanned shut
},?
},?
{?
  category: "Maintenance Guide",?
  title: "Industrial Motor Bearing Failure: Early Detection & Prevention",?
  desc: "A comprehensive guide to identifying early bearing failure signatures using vibration analysis, thermography, and acou
  readTime: "10 min read",?
  slug: "motor-bearing-failure",?
  author: "Nandini Enterprises Technical Team",?
  date: "March 2025",?
  tags: ["Motors", "Predictive Maintenance", "Vibration Analysis", "AMC", "Thermography"],?
  content: {?
    intro: "Bearing failure is responsible for approximately 40-50% of all motor failures in industrial applications. Early det
    sections: [?
      {?
        heading: "Four Stages of Rolling Element Bearing Failure",?
        body: "Bearing failure follows a predictable four-stage progression. Stage 1 (10-20% of remaining life remaining): High
      },?
      {?
        heading: "Vibration Analysis: The Primary Predictive Tool",?
        body: "Vibration spectrum analysis is the most informative method for bearing condition monitoring. An accelerometer me
      },?
      {?
        heading: "Thermography and Oil Analysis",?
        body: "Infrared thermography provides a fast, non-contact method for identifying bearing overheating. A hand-held therm
      },?
    ],?
    keyPoints: [?
      "Bearing failure causes 40-50% of industrial motor failures and is almost always preventable with monitoring",?
      "Four failure stages - ultrasonic, spectrum, elevated vibration, catastrophic - have different detection technologies",?
      "ISO 10816 vibration limits provide actionable alarm thresholds for motors by size and mounting type",?
      "Bearing defect frequencies (BPFO, BPFI, BSF) are calculated from geometry data and rotation speed",?
      "AMC programs combining thermal scanning and vibration analysis every 3 months prevent >80% of catastrophic failures"?
    ],?
    conclusion: "The investment in a predictive maintenance program - combining periodic vibration analysis, thermography, and
  },?
},?
{?
  category: "White Paper",?
  title: "Industrial Cybersecurity for OT Networks: IEC 62443 Framework",?
  desc: "Protecting operational technology networks from cyber threats - understanding the IEC 62443 standard, network segmenta
  readTime: "14 min read",?
  slug: "ot-cybersecurity",?
  author: "Nandini Enterprises Technical Team",?
  date: "May 2025",?
  tags: ["Cybersecurity", "OT Security", "IEC 62443", "SCADA Security", "Network"],?
  content: {?
    intro: "Operational Technology (OT) networks - the PLCs, SCADA servers, HMI stations, and field instruments that control ph
    sections: [?
      {?
        heading: "IEC 62443: The OT Security Standard",?
        body: "IEC 62443 (Security for Industrial Automation and Control Systems) provides a comprehensive framework for securi
      },?
      {?
        heading: "Network Segmentation: The Purdue Model",?
        body: "The Purdue Enterprise Reference Architecture defines hierarchical network levels for industrial systems: Level 0
      },?
      {?
        heading: "Practical Security Controls for Industrial Sites",?
        body: "Beyond network architecture, operational security controls are essential. USB ports on SCADA workstations should
      },?
    ],?
    keyPoints: [?
      "IEC 62443 defines Security Levels 1-4 and the Zones/Conduits model for OT network security",?
      "The Purdue Model provides a hierarchical reference architecture for segmenting OT from IT networks",?
      "DMZ data historians are the correct pattern for passing OPC-UA data to business networks securely",?
      "USB port control and application whitelisting address the most common OT malware attack vectors",?
      "Remote access must use MFA, time-limited VPN sessions, and complete audit logging"?
    ],?
    conclusion: "Industrial cybersecurity is not an IT problem to be solved by enterprise IT teams - it requires OT-domain exper
  },?
},?
{?
  category: "Technical Guide",?
  title: "HART Protocol: Smart Instrument Diagnostics for Field Loops",?
  desc: "How the HART protocol superimposes digital communication on 4-20mA analog loops, enabling remote configuration, multi-
```



```
readTime: "8 min read",?
slug: "hart-protocol-guide",?
author: "Nandini Enterprises Technical Team",?
date: "January 2025",?
tags: ["HART", "Instrumentation", "4-20mA", "Field Devices", "Smart Sensors"],?
content: {?
  intro: "The 4-20mA analog current loop has been the workhorse of process instrumentation for over 60 years - reliable, nois
  sections: [?
    {?
      heading: "How HART Superimposes Digital on Analog",?
      body: "HART uses Frequency Shift Keying (FSK) modulation to superimpose a low-level digital signal (?0.5mA AC) on top o
    },?
    {?
      heading: "Information Available via HART",?
      body: "Beyond the primary measurement variable transmitted as 4-20mA, HART devices can provide up to four process varia
    },?
    {?
      heading: "HART Multiplexers and Asset Management Systems",?
      body: "For plants with hundreds or thousands of HART instruments, manual point-to-point HART communication is impractic
    },?
  ],?
  keyPoints: [?
    "HART superimposes 1200/2200Hz FSK digital on 4-20mA analog loops without disrupting measurement",?
    "Up to 4 process variables can be transmitted digitally from a single HART device simultaneously",?
    "HART diagnostics include valve position, plug/jam detection, loop integrity, and process deviation alerts",?
    "HART multiplexers + AMS software enable fleet-wide instrument health monitoring and calibration management",?
    "HART 7 supports wireless variants (WirelessHART, IEC 62591) for locations without existing wiring"?
  ],?
  conclusion: "HART protocol represents the most cost-effective path to smart instrumentation for plants with large installed
},?
},?
{?
category: "Maintenance Guide",?
title: "Transformer Condition Monitoring: Oil Analysis & Dissolved Gas",?
desc: "How dissolved gas analysis (DGA), dielectric strength testing, and moisture measurement are used to assess the health
readTime: "11 min read",?
slug: "transformer-oil-analysis",?
author: "Nandini Enterprises Technical Team",?
date: "February 2025",?
tags: ["Transformers", "DGA", "Oil Analysis", "Predictive Maintenance", "HT Electrical"],?
content: {?
  intro: "Power transformers are among the most capital-intensive and operationally critical assets in any industrial electri
  sections: [?
    {?
      heading: "Dissolved Gas Analysis (DGA): The Primary Diagnostic Tool",?
      body: "As electrical and thermal stresses break down the transformer's internal insulation (paper and oil), characteris
    },?
    {?
      heading: "Dielectric Strength and Moisture Testing",?
      body: "Transformer oil's dielectric strength (its ability to withstand electric field without breakdown) degrades with
    },?
    {?
      heading: "Comprehensive Condition Assessment Program",?
      body: "A complete transformer health program combines annual oil sampling and DGA analysis; regular bushing thermograph
    },?
  ],?
  keyPoints: [?
    "Acetylene (C2H2) is the definitive indicator of internal electrical arcing - immediate investigation warranted",?
    "Dielectric breakdown voltage below 30kV indicates oil requiring filtration; below 20kV requires immediate action",?
    "Karl Fischer moisture testing and capacitive sensors monitor oil moisture content continuously",?
    "Duval Triangle and Rogers Ratio interpret DGA gas profiles to classify fault type",?
    "Annual oil sampling + quarterly thermography + winding tests form a complete condition monitoring program"?
  ],?
  conclusion: "Power transformer condition monitoring is not an optional luxury - it is the engineering insurance policy that
},?
},?
{?
category: "Engineering Insight",?
title: "Motor Soft Starters vs VFDs: When to Use Each",?
desc: "A practical engineering comparison of electronic soft starters and variable frequency drives - analyzing starting perf
readTime: "9 min read",?
slug: "soft-starter-vs-vfd",?
author: "Nandini Enterprises Technical Team",?
date: "April 2025",?
tags: ["Soft Starters", "VFD", "Motor Control", "Energy Efficiency", "Panel Design"],?
content: {?
  intro: "When a project calls for controlled motor starting, two electronic technologies are available: soft starters and va
  sections: [?
    {?
      heading: "How Soft Starters Work",?
      body: "A soft starter uses anti-parallel thyristors (SCRs) connected in series with each motor phase to control the vol
    },?
  ],?
}
```

```
{?
  heading: "When Soft Starters Are the Right Choice",?
  body: "Soft starters are cost-effective for motors that run at a single fixed speed most of the time, but need gentle s
},?
{?
  heading: "When VFDs Are the Correct Choice",?
  body: "A VFD continuously controls motor speed throughout the operating cycle - not just during starting. This enables
},?
],?
keyPoints: [?
  "Soft starters control starting voltage via thyristors and are bypassed after reaching full speed",?
  "VFDs continuously control motor frequency and voltage throughout the entire operating cycle",?
  "Soft starters are 30-50% less expensive than VFDs for fixed-speed applications",?
  "VFDs save energy proportional to speed? - at 80% speed, only 51% of full-speed power is consumed",?
  "Choose soft starters for infrequent fixed-speed starts; choose VFDs whenever variable speed is beneficial"?
],?
conclusion: "The decision tree is straightforward: if the motor needs variable speed for process control or energy savings,
},?
},?
{?
category: "Case Study",?
title: "120 MLD Water Pump Station: Automation Design Deep Dive",?
desc: "An engineering walkthrough of the Amberpet Pumping Station automation project for HMWSSB - detailing the control system",?
readTime: "13 min read",?
slug: "hmwssb-case-study",?
author: "Nandini Enterprises Technical Team",?
date: "December 2024",?
tags: ["Case Study", "Water Treatment", "HMWSSB", "PLC", "SCADA", "VFD"],?
content: {?
  intro: "The Amberpet Pumping Station serves as the primary booster station for water distribution across six zones of Hyderabad",?
  sections: [?
    {?
      heading: "Engineering Challenges Specific to This Project",?
      body: "Several challenges made this project particularly complex. The station operates continuously - a planned shutdown was required for maintenance",?
    },?
    {?
      heading: "Control System Architecture Selected",?
      body: "The selected architecture used ABB AC800M redundant PLC controllers (dual-CPU hot-standby, <5ms switchover) as the primary control system",?
    },?
    {?
      heading: "Energy Savings Achieved and Verified",?
      body: "Pre-installation energy audit established a baseline of 38,500 kWh per day across all six pumps operating on DOL+throttle valve control",?
    },?
  ],?
  keyPoints: [?
    "Six 450kW VFD panels with PID header pressure control replaced DOL+throttle valve control",?
    "ABB AC800M hot-standby redundant PLC with <5ms failover time ensures no control system downtime",?
    "Profinet fiber ring provides 99.99% communication availability for critical infrastructure application",?
    "21% energy reduction verified over 90-day measurement period - ₹82 lakhs annual savings",?
    "Panel-by-panel replacement methodology maintained continuous pump station operation throughout"?
  ],?
  conclusion: "The Amberpet Pumping Station project demonstrates the engineering rigor required for major automation upgrades and energy savings",?
},?
},?
{?
category: "Technical Guide",?
title: "IEC 61508 & SIL: Functional Safety for Industrial Control Systems",?
desc: "Understanding Safety Integrity Levels (SIL), the IEC 61508 functional safety standard, and how to implement Safety Instrumented Functions (SIF)",?
readTime: "12 min read",?
slug: "iec-61508-sil-guide",?
author: "Nandini Enterprises Technical Team",?
date: "May 2025",?
tags: ["Functional Safety", "SIL", "IEC 61508", "IEC 61511", "Safety Systems"],?
content: {?
  intro: "Functional safety - the part of overall safety that depends on a system responding correctly to its inputs - is a critical aspect of industrial control systems",?
  sections: [?
    {?
      heading: "Safety Integrity Level (SIL) Explained",?
      body: "A Safety Integrity Level is a discrete measure of the risk reduction provided by a Safety Instrumented Function (SIF)",?
    },?
    {?
      heading: "Safety Instrumented System Architecture",?
      body: "An SIS consists of three elements: Sensor (detecting the hazardous condition), Logic Solver (processing the sensor input), and Final Element (actuating the safety response)",?
    },?
    {?
      heading: "Proof Testing and Maintenance Requirements",?
      body: "SIS components require periodic proof testing to reveal dangerous undetected failures (failures where the safety function is lost)",?
    },?
  ],?
  keyPoints: [?
    "SIL 1-4 define probability of failure on demand (PFD) targets: 10-2 to 10-5 respectively",?
    "Hazard and Risk Assessment determines the SIL target - not engineering preference or rule of thumb",?
  ],?
}
```

```

    "1002 voting maximizes safety integrity; 2003 voting balances safety against spurious trips",?
    "T?V-certified SIL-rated logic solvers are mandatory - standard PLCs cannot fulfill this role",?
    "Annual proof tests must exercise the complete SIS loop including final element movement to safe position"?
  ],?
  conclusion: "Functional safety engineering is a specialist discipline that sits at the intersection of process safety, control systems, and instrumentation",?
},?
},?
{?
  category: "Engineering Insight",?
  title: "Cable Sizing for Industrial Motor Feeders: A Practical Guide",?
  desc: "Step-by-step methodology for sizing power cables for industrial motor feeders - considering current rating, voltage drop, and short-circuit withstand",?
  readTime: "10 min read",?
  slug: "cable-sizing-guide",?
  author: "Nandini Enterprises Technical Team",?
  date: "March 2025",?
  tags: ["Cable Sizing", "Motor Feeders", "Electrical Design", "Voltage Drop", "IS 1554"],?
  content: {?
    intro: "Cable sizing for industrial motor feeders is one of the most fundamental electrical engineering calculations - yet it's often overlooked. This guide provides a systematic approach to ensure your cable selection meets all safety and performance requirements.",?
    sections: [?
      {?
        heading: "Step 1: Continuous Current Rating",?
        body: "The continuous current rating of the cable must exceed the motor full load current (FLA) multiplied by a service factor (SF). The formula is:  $I_{cable} \geq I_{FLA} \times SF$ . For most industrial applications, SF is 1.25. For example, if the motor FLA is 100A, the cable must be rated for at least 125A."
      },?
      {?
        heading: "Step 2: Voltage Drop Calculation",?
        body: "Voltage drop in a three-phase circuit is calculated as:  $\%V = \frac{\sqrt{3} \times I \times L \times (R \cos\phi + X \sin\phi)}{1000}$ , where I is the current in amperes, L is the length in meters, R is the resistance per meter, and X is the reactance per meter. The voltage drop should not exceed 3% for motor feeders."
      },?
      {?
        heading: "Step 3: Short-Circuit Thermal Withstand",?
        body: "The cable must withstand the thermal energy generated if a short-circuit fault occurs and persists until the protective device operates. The formula is:  $A \geq \sqrt{\frac{I_{sc}^2 \times t}{k}}$ , where A is the cross-sectional area in mm², I_sc is the short-circuit current in amperes, t is the fault clearing time in seconds, and k is a constant (143 for copper, 125 for aluminum). For example, if I_sc is 10,000A and t is 0.1s, the minimum area A is approximately 100 mm²."
      }
    ],?
  },?
  keyPoints: [?
    "Four criteria must all be satisfied: current rating, voltage drop, short-circuit withstand, and mechanical strength",?
    "Grouping derating factors from IS 1554 can reduce cable capacity by 30-40% in dense tray installations",?
    "Voltage drop during motor starting (using I_start at 0.3-0.4 power factor) often governs cable size on long runs",?
    "Thermal withstand area A = (I_sc² × t) / 115 for copper PVC cables (IS 1554 method)",?
    "Document all calculation assumptions including ambient temperature, grouping arrangement, and fault level"?
  ],?
  conclusion: "A comprehensive cable sizing calculation that simultaneously verifies all four criteria - current rating with voltage drop, short-circuit withstand, and mechanical strength - ensures safe and reliable motor operation."
},?
},?
];?
?
const CATEGORY_COLORS: Record<string, string> = {?
  'Technical Guide': 'text-blue-700 bg-blue-50 border-blue-200',?
  'White Paper': 'text-purple-700 bg-purple-50 border-purple-200',?
  'Maintenance Guide': 'text-amber-700 bg-amber-50 border-amber-200',?
  'Engineering Insight': 'text-emerald-700 bg-emerald-50 border-emerald-200',?
  'Case Study': 'text-[#0f4c81] bg-slate-50 border-slate-200',?
};?
?
export default function KnowledgeCenterPage() {?
  const [selectedArticle, setSelectedArticle] = useState<Article | null>(null);?
  const [activeFilter, setActiveFilter] = useState('ALL');?
  const [searchQuery, setSearchQuery] = useState('');?
  ?
  const categories = ['ALL', ...Array.from(new Set(articles.map(a => a.category)))];?
  ?
  const filteredArticles = articles.filter(a => {?
    const matchesCat = activeFilter === 'ALL' || a.category === activeFilter;?
    const matchesSearch = searchQuery === '' || ?
      a.title.toLowerCase().includes(searchQuery.toLowerCase()) ||?
      a.tags.some(t => t.toLowerCase().includes(searchQuery.toLowerCase())) ||?
      a.desc.toLowerCase().includes(searchQuery.toLowerCase());?
    return matchesCat && matchesSearch;?
  });?
  ?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen text-slate-800 font-sans">?
      ?
      { /* Hero Section */ }?
      <section className="relative py-20 bg-white border-b border-slate-200 overflow-hidden">?
        <div className="absolute inset-0 bg-[linear-gradient(to_right,#eef2f7_1px,transparent_1px),linear-gradient(to_bottom,#eef2f7_1px,transparent_1px)] bg-[size:40px_40px] pointer-events-none">?
          <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
            <div className="grid grid-cols-1 lg:grid-cols-2 gap-12 items-center">?
              <div className="max-w-xl space-y-5">?
                <span className="text-xs uppercase tracking-widest text-[#0f4c81] font-mono font-bold block">?
                  ENGINEERING KNOWLEDGE BASE?
                </span>?
                <h1 className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit leading-tight">?
                  Knowledge Center & Technical Resources?
                </h1>?
              </div>?
            </div>?
          </div>?
        </section>?
      </div>?
    );?
  };?
}

```

```

</h1>?
<p className="text-base sm:text-lg leading-relaxed text-slate-600">?
  In-depth technical guides, white papers, engineering case studies, and maintenance references authored by our spe
</p>?
<div className="flex gap-6 pt-2 font-mono text-xs">?
  <div className="text-center">?
    <div className="text-2xl font-bold text-[#0f4c81]">{articles.length}</div>?
    <div className="text-slate-500 uppercase tracking-wider">Articles</div>?
  </div>?
  <div className="text-center">?
    <div className="text-2xl font-bold text-[#0f4c81]">5</div>?
    <div className="text-slate-500 uppercase tracking-wider">Categories</div>?
  </div>?
  <div className="text-center">?
    <div className="text-2xl font-bold text-[#0f4c81]">Free</div>?
    <div className="text-slate-500 uppercase tracking-wider">Access</div>?
  </div>?
</div>?
<div className="relative group">?
  <div className="absolute -inset-1.5 bg-gradient-to-r from-[#0f4c81] to-[#2b6cb0] rounded-lg blur opacity-25 group-h
  ?
</div>?
</div>?
</section>?
?
/* Articles Grid */?
<section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-10">?
  ?
  /* Filters + Search */?
  <div className="flex flex-col sm:flex-row gap-4 items-start sm:items-center justify-between">?
    <div className="flex flex-wrap gap-2">?
      {categories.map((cat) => (?
        <button?
          key={cat}?
          onClick={() => setActiveFilter(cat)}?
          className={`px-3 py-1.5 font-mono text-[10px] uppercase border rounded cursor-pointer transition-all ${?
            activeFilter === cat?
              ? 'bg-[#0f4c81] border-[#0f4c81] text-white'?
              : 'bg-white border-slate-200 text-slate-600 hover:border-slate-400'?
          }}?
        >?
          {cat}?
        </button>?
      )}}?
    </div>?
    <div className="flex items-center gap-2 bg-white border border-slate-200 rounded px-3 py-2 min-w-[220px]">?
      <svg className="w-4 h-4 text-slate-400 shrink-0" fill="none" stroke="currentColor" viewBox="0 0 24 24">?
        <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M21 21l-6-6m2-5a7 7 0 11-14 0 7 7 0 0114 0z"
      </svg>?
      <input?
        type="text"?
        placeholder="Search topics, tags..."?
        value={searchQuery}?
        onChange={e => setSearchQuery(e.target.value)}?
        className="text-xs text-slate-700 bg-transparent focus:outline-none w-full placeholder-slate-400"?
      />?
    </div>?
  </div>?
  ?
  <div className="text-xs text-slate-500 font-mono">?
    Showing {filteredArticles.length} of {articles.length} articles?
  </div>?
  ?
  <div className="grid grid-cols-1 md:grid-cols-2 gap-6">?
    {filteredArticles.map((art, idx) => (?
      <div ?
        key={idx} ?
        className="bg-white border border-slate-200 rounded-lg p-7 flex flex-col justify-between hover:border-[#2b6cb0] hov
        onClick={() => setSelectedArticle(art)}?
      >?
      <div>?
        <div className="flex justify-between items-center mb-4">?
          <span className={`border font-bold text-[9px] font-mono px-2 py-0.5 rounded uppercase ${CATEGORY_COLORS[art.cat
            {art.category}?
          </span>?
          <span className="text-[10px] font-mono text-slate-400">?

```

```

        {art.readTime}?
      </span>?
    </div>?
    <h3 className="text-slate-900 font-bold text-lg font-outfit mb-3 group-hover:text-[#0f4c81] transition-colors lea
    {art.title}?
  </h3>?
  <p className="text-xs text-slate-600 leading-relaxed mb-4">?
    {art.desc}?
  </p>?
  <div className="flex flex-wrap gap-1.5 mb-4">?
    {art.tags.slice(0, 4).map((tag, tIdx) => (?
      <span key={tIdx} className="text-[9px] font-mono bg-slate-50 border border-slate-200 text-slate-500 px-1.5 py
      {tag}?
    </span>?
    )))?
  </div>?
</div>?

?
<div className="border-t border-slate-100 pt-4 flex justify-between items-center text-[10px] font-mono">?
  <div>?
    <span className="text-slate-400 block">{art.author}</span>?
    <span className="text-slate-500">{art.date}</span>?
  </div>?
  <span className="text-[#0f4c81] font-bold group-hover:underline">?
    READ FULL ARTICLE ??
  </span>?
</div>?
</div>?
  )))?
</div>?

?
{filteredArticles.length === 0 && (?
  <div className="text-center py-16 text-slate-400 font-mono">?
    <div className="text-4xl mb-4">??</div>?
    <p>No articles match your search. Try different keywords or reset filters.</p>?
  </div>?
  )}?

?
{ /* CTA */?
  <div className="bg-white border border-slate-200 rounded-lg p-8 shadow-sm text-center space-y-4">?
    <h3 className="text-[#0f4c81] font-bold text-xl font-outfit">Need a Specific Technical Reference?</h3>?
    <p className="text-sm text-slate-600 max-w-2xl mx-auto leading-relaxed">?
      Our engineering team publishes new technical guides, white papers, and case studies monthly. If you need documentation
    </p>?
    <a href="/contact?type=inquiry" className="inline-block bg-[#0f4c81] text-white px-6 py-3 rounded font-mono text-xs up
      Request Technical Consultation ??
    </a>?
  </div>?
</section>?

?
{ /* Article Detail Modal */?
  {selectedArticle && (?
    <div className="fixed inset-0 z-50 flex items-start justify-center p-4 bg-slate-900/60 backdrop-blur-sm overflow-y-auto">
      <div className="bg-white border border-slate-200 rounded-lg max-w-3xl w-full shadow-2xl relative my-8">?
        ?
        { /* Header */?
          <div className="bg-[#0f4c81] p-7 rounded-t-lg relative">?
            <button ?
              className="absolute top-4 right-4 text-white/70 hover:text-white text-2xl cursor-pointer font-light leading-none"
              onClick={() => setSelectedArticle(null)}?
            >?
            ??
          </button>?
          <div className="flex gap-2 items-center mb-3">?
            <span className="border font-bold text-[9px] font-mono px-2 py-0.5 rounded uppercase ${CATEGORY_COLORS[selectedA
              {selectedArticle.category}?
            </span>?
            <span className="text-[10px] font-mono text-[#93c5fd]">{selectedArticle.readTime}</span>?
          </div>?
          <h2 className="text-xl sm:text-2xl font-bold text-white font-outfit leading-snug mb-2">?
            {selectedArticle.title}?
          </h2>?
          <div className="text-[10px] font-mono text-[#93c5fd] flex gap-4">?
            <span>{selectedArticle.author}</span>?
            <span>{selectedArticle.date}</span>?
          </div>?
        </div>?
      </div>?

?
    { /* Tags */?
      <div className="flex flex-wrap gap-1.5 px-7 py-4 border-b border-slate-100">?
        {selectedArticle.tags.map((tag, i) => (?
          <span key={i} className="text-[9px] font-mono bg-slate-50 border border-slate-200 text-slate-500 px-2 py-0.5 rounded

```

```

        {tag}?
      </span>?
    )}}?
  </div>?
?
  { /* Content */?
  <div className="p-7 space-y-6 text-sm text-slate-700">?
    ?
    { /* Intro */?
    <div className="bg-[#E8F0FA] border-l-4 border-[#0f4c81] px-5 py-4 rounded-r">?
      <p className="leading-relaxed text-slate-700">{selectedArticle.content.intro}</p>?
    </div>?
    ?
    { /* Sections */?
    {selectedArticle.content.sections.map((section, sIdx) => (?
      <div key={sIdx} className="space-y-3">?
        <h3 className="text-[#0f4c81] font-bold text-base font-outfit border-b border-slate-100 pb-2">?
          {sIdx + 1}. {section.heading}?
        </h3>?
        <p className="text-slate-650 leading-relaxed text-sm">{section.body}</p>?
      </div>?
    )}}?
    ?
    { /* Key Points */?
    <div className="bg-slate-50 border border-slate-200 rounded p-5 space-y-3">?
      <h3 className="text-xs font-mono uppercase tracking-wider text-[#0f4c81] font-semibold">?
        Key Engineering Takeaways?
      </h3>?
      <ul className="space-y-2">?
        {selectedArticle.content.keyPoints.map((point, pIdx) => (?
          <li key={pIdx} className="flex items-start gap-2 text-xs">?
            <span className="text-emerald-600 font-bold shrink-0 mt-0.5">?</span>?
            <span className="text-slate-700 leading-relaxed">{point}</span>?
          </li>?
        )}}?
      </ul>?
    </div>?
    ?
    { /* Conclusion */?
    <div className="space-y-2">?
      <h3 className="text-[#0f4c81] font-bold text-sm font-outfit">Conclusion</h3>?
      <p className="text-slate-650 leading-relaxed text-sm">{selectedArticle.content.conclusion}</p>?
    </div>?
    ?
    { /* Footer CTA */?
    <div className="border-t border-slate-100 pt-5 flex gap-3 flex-wrap">?
      <a href="/contact?type=inquiry" className="bg-[#0f4c81] text-white px-5 py-2.5 rounded font-mono text-xs uppercase">?
        Discuss This Topic with Our Engineers ??
      </a>?
      <button?
        onClick={() => setSelectedArticle(null)}?
        className="border border-slate-200 text-slate-600 px-5 py-2.5 rounded font-mono text-xs uppercase tracking-wide">?
        Back to Knowledge Center?
      </button>?
    </div>?
  </div>?
  </div>?
  </div>?
  )}}?
  </div>?
  );?
}

```

```
import { Inter, Outfit } from "next/font/google";?
import "../globals.css";?
import Header from "@components/Layout/Header";?
import Footer from "@components/Layout/Footer";?
import KnowledgeAssistant from "@components/Assistant/KnowledgeAssistant";?
import DisableRightClick from "@components/Layout/DisableRightClick";?
import { Metadata } from 'next';?
?
const inter = Inter({?
  variable: "--font-inter",?
  subsets: ["latin"],?
  display: 'swap',?
});?
?
const outfit = Outfit({?
  variable: "--font-outfit",?
  subsets: ["latin"],?
  display: 'swap',?
});?
?
export const metadata: Metadata = {?
  title: "Nandini Enterprises | Premium Industrial Automation, Electrical & Robotics Solutions",?
  description: "ISO 9001:2015 certified partner for heavy-scale industries across India. Designing & manufacturing CPRI certified",?
  keywords: ["Industrial Automation", "Electrical Engineering", "Robotics", "VFD Systems", "PLC SCADA", "Electrical Panels", "Yas",?
  openGraph: {?
    title: "Nandini Enterprises | Premium Industrial Automation & Electrical Engineering",?
    description: "Reliable B2B industrial ecosystem and manufacturing partner for large-scale automation projects in India.",?
    url: "https://www.nandiniautomation.com",?
    siteName: "Nandini Enterprises",?
    locale: "en_IN",?
    type: "website",?
  },?
  robots: {?
    index: true,?
    follow: true,?
  },?
};?
?
export default function RootLayout({?
  children,?
}: {?
  children: React.ReactNode;?
}) {?
  return (?
    <html lang="en" className={` ${inter.variable} ${outfit.variable} scroll-smooth`} >?
      <head>?
        <link rel="icon" href="/favicon.ico" sizes="any" />?
      </head>?
      <body className="bg-[#E8F0FA] text-slate-800 antialiased min-h-screen flex flex-col selection:bg-blue-500/20 selection:text-white">?
        <DisableRightClick />?
        <div className="flex-grow flex flex-col">?
          <Header />?
          <main className="flex-grow pt-24">?
            {children}?
          </main>?
          <Footer />?
          <KnowledgeAssistant />?
        </div>?
      </body>?
    </html>?
  );?
};?
```

```
'use client';?
?
import { useState, useEffect } from 'react';?
import Link from 'next/link';?
import Image from 'next/image';?
import HeroSlideshow from '@components/Home/HeroSlideshow';?
import { ?
  INDUSTRIES, ?
  PRODUCTS, ?
  PROJECTS, ?
  CERTIFICATIONS, ?
  WORKFLOW, ?
  Industry, ?
  Product, ?
  Project, ?
  Certification ?
} from '@lib/data';?
?
// Helper component for rendering SVGs to avoid duplicate definitions?
function FileIcon() {?
  return (?
    <svg className="w-5 h-5 text-blue-700" fill="none" stroke="currentColor" viewBox="0 0 24 24">?
      <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M9 12h6m-6 4h6m2 5H7a2 2 0 01-2-2V5a2 2 0 012-2h5.586" />?
    </svg>?
  );?
}?
?
export default function Home() {?
  // Navigation & UI States?
  const [activeProductTab, setActiveProductTab] = useState(0);?
  const [selectedIndustry, setSelectedIndustry] = useState<Industry | null>(null);?
  const [selectedProduct, setSelectedProduct] = useState<Product | null>(null);?
  const [selectedProject, setSelectedProject] = useState<Project | null>(null);?
  const [selectedCert, setSelectedCert] = useState<Certification | null>(null);?
  const [activeWorkflow, setActiveWorkflow] = useState(0);?
  ?
  // Interactive Counters?
  const [stats, setStats] = useState({ years: 0, projects: 0, industries: 0, licenses: 0 });?
  useEffect(() => {?
    const duration = 1500;?
    const steps = 50;?
    const stepTime = duration / steps;?
    let currentStep = 0;?
    ?
    const timer = setInterval(() => {?
      currentStep++;?
      setStats({?
        years: Math.min(Math.round((25 / steps) * currentStep), 25),?
        projects: Math.min(Math.round((500 / steps) * currentStep), 500),?
        industries: Math.min(Math.round((14 / steps) * currentStep), 14),?
        licenses: Math.min(Math.round((8 / steps) * currentStep), 8)?
      });?
    }, stepTime);?
    ?
    if (currentStep >= steps) {?
      clearInterval(timer);?
    }?
    }, stepTime);?
    ?
    return () => clearInterval(timer);?
  }, []);?
  ?
  // Form submission state?
  const [inquiryType, setInquiryType] = useState('quote');?
  const [rfqSubmitted, setRfqSubmitted] = useState(false);?
  const [contactForm, setContactForm] = useState({?
    name: '', email: '', phone: '', company: '', extraField: '', requirement: ''?
  });?
  ?
  const handleInquirySubmit = (e: React.FormEvent) => {?
    e.preventDefault();?
    setRfqSubmitted(true);?
    setTimeout(() => {?
      setRfqSubmitted(false);?
      setContactForm({ name: '', email: '', phone: '', company: '', extraField: '', requirement: '' });?
    }, 5000);?
  };?
  ?
  const getFormConfig = () => {?
    switch (inquiryType) {?
      case 'quote':?
```



```

    return {?
      extraFieldLabel: 'Product Category / Brand Preference',?
      extraFieldPlaceholder: 'e.g. Yaskawa VFD, Siemens PLC, Custom Panel...',?
      textareaLabel: 'RFQ Specifications & Bill of Materials',?
      textareaPlaceholder: 'Please describe required capacities, configurations, and quantity...',?
      buttonText: 'Submit RFQ Request'?
    };?
  case 'consult':?
    return {?
      extraFieldLabel: 'Preferred Contact Time / Timezone',?
      extraFieldPlaceholder: 'e.g. Weekdays 10 AM - 12 PM IST',?
      textareaLabel: 'Consultation Topic & Plant Challenges',?
      textareaPlaceholder: 'Describe your current manufacturing bottleneck, automation upgrade needs, or energy efficiency goals',?
      buttonText: 'Request Engineering Consultation'?
    };?
  case 'support':?
    return {?
      extraFieldLabel: 'Equipment Serial No. / Model Number',?
      extraFieldPlaceholder: 'e.g. CIMR-AD4A0039FAA',?
      textareaLabel: 'Fault Description & Error Codes',?
      textareaPlaceholder: 'Specify error codes, breakdown symptoms, or required repair turnaround time...',?
      buttonText: 'Submit Support Ticket'?
    };?
  default:?
    return {?
      extraFieldLabel: 'Additional Reference',?
      extraFieldPlaceholder: '...',?
      textareaLabel: 'Requirements Description',?
      textareaPlaceholder: 'Specify details...',?
      buttonText: 'Submit Request'?
    };?
  }?
};?
?
?
// Industry-specific curated high-res Unsplash links for B2B trust?
const getIndustryImage = (slug: string) => {?
  const images: Record<string, string> = {?
    'steel': '/images/steel_plant.png',?
    'railways': 'https://images.unsplash.com/photo-1474487548417-781cb71495f3?auto=format&fit=crop&w=600&q=80',?
    'cement': '/images/cement_plant.png',?
    'power': 'https://images.unsplash.com/photo-1473341304170-971dccb5ac1e?auto=format&fit=crop&w=600&q=80',?
    'oil-gas': '/images/oil_refinery.png',?
    'water-treatment': '/images/water_treatment.png',?
    'pharma': '/images/pharma_lab.png',?
    'food-processing': '/images/food_factory.png',?
    'default': 'https://images.unsplash.com/photo-1565514020179-026b92b84bb6?auto=format&fit=crop&w=600&q=80'?
  };?
  return images[slug] || images['default'];?
};?
?
return (?
  <div className="bg-white text-slate-800 min-h-screen">?
    ?
    { /* 1. HERO SECTION */ }?
    <HeroSlideshow />?
    ?
    { /* 2. TRUST & SCALE SECTION */ }?
    <section className="bg-white py-16 relative border-b border-slate-200/60">?
      <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
        <div className="grid grid-cols-2 md:grid-cols-4 gap-8 text-center">?
          <div className="bg-white border border-slate-200 p-6 rounded-xl shadow-sm hover:shadow-lg hover:-translate-y-1 transition">?
            <p className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit mb-2 tracking-tight">?
              {stats.years}+?
            </p>?
            <p className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-bold font-mono">?
              Years of Excellence?
            </p>?
            <p className="text-xs text-slate-500 mt-2 font-sans">Operational stability since inception</p>?
          </div>?
          ?
          <div className="bg-white border border-slate-200 p-6 rounded-xl shadow-sm hover:shadow-lg hover:-translate-y-1 transition">?
            <p className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit mb-2 tracking-tight">?
              {stats.projects}+?
            </p>?
            <p className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-bold font-mono">?
              B2B Projects Executed?
            </p>?
            <p className="text-xs text-slate-500 mt-2 font-sans">Across steel, cement, power, and pumping</p>?
          </div>?
          ?
          <div className="bg-white border border-slate-200 p-6 rounded-xl shadow-sm hover:shadow-lg hover:-translate-y-1 transition">?
            <p className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit mb-2 tracking-tight">?
              {stats.projects}+?
            </p>?
            <p className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-bold font-mono">?
              B2B Projects Executed?
            </p>?
            <p className="text-xs text-slate-500 mt-2 font-sans">Across steel, cement, power, and pumping</p>?
          </div>?
        </div>?
      </div>?
    </section>?
  )
);

```

... continued: src/app/page.tsx

```
    <p className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit mb-2 tracking-tight">
      {stats.industries}+?
    </p>?
    <p className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-bold font-mono">?
      Sectors Serviced?
    </p>?
    <p className="text-xs text-slate-500 mt-2 font-sans">From defense rigs to municipal networks</p>?
  </div>?

  <div className="bg-white border border-slate-200 p-6 rounded-xl shadow-sm hover:shadow-lg hover:-translate-y-1 transi
    <p className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit mb-2 tracking-tight">?
      {stats.licenses}+?
    </p>?
    <p className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-bold font-mono">?
      Govt Registrations?
    </p>?
    <p className="text-xs text-slate-500 mt-2 font-sans">Including A-Grade electrical license</p>?
  </div>?
</div>?

  /* Credentials Info Cards */?
  <div className="mt-16 grid grid-cols-1 md:grid-cols-3 gap-8">?
    <div className="bg-white border border-slate-100 p-8 rounded-xl shadow-md hover:shadow-xl hover:-translate-y-1 transi
      <div className="absolute top-0 left-0 w-full h-1 bg-gradient-to-r from-[var(--color-ne-blue-corp)] to-[var(--color-
        <h3 className="text-[var(--color-ne-text-dark)] font-bold text-lg mb-4 font-outfit mt-2 group-hover:text-[var(--col
          <p className="text-sm text-slate-600 leading-relaxed">?
            Our design processes, panel integration, wiring methodologies, and quality audits fully conform to global Quality
          </p>?
        </div>?

    <div className="bg-white border border-slate-100 p-8 rounded-xl shadow-md hover:shadow-xl hover:-translate-y-1 transi
      <div className="absolute top-0 left-0 w-full h-1 bg-gradient-to-r from-[var(--color-ne-blue-corp)] to-[var(--color-
        <h3 className="text-[var(--color-ne-text-dark)] font-bold text-lg mb-4 font-outfit mt-2 group-hover:text-[var(--col
          <p className="text-sm text-slate-600 leading-relaxed">?
            Central Power Research Institute (CPRI) certified configuration audits verify that our panel structures withstand
          </p>?
        </div>?

    <div className="bg-white border border-slate-100 p-8 rounded-xl shadow-md hover:shadow-xl hover:-translate-y-1 transi
      <div className="absolute top-0 left-0 w-full h-1 bg-gradient-to-r from-[var(--color-ne-blue-corp)] to-[var(--color-
        <h3 className="text-[var(--color-ne-text-dark)] font-bold text-lg mb-4 font-outfit mt-2 group-hover:text-[var(--col
          <p className="text-sm text-slate-600 leading-relaxed">?
            Authorized to execute, test, and commission HT electrical switchyards, double pole structures, substations, and c
          </p>?
        </div>?
      </div>?
    </div>?
  </section>?

  /* 3. INDUSTRIES SERVED SECTION */?
  <section id="industries" className="py-20 relative bg-[#E8F0FA] border-b border-slate-200/60">?
    <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
      <div className="text-center max-w-3xl mx-auto mb-16">?
        <h2 className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold mb-3">?
          EPC Sectors & Markets?
        </h2>?
        <p className="text-3xl font-extrabold text-slate-900 font-outfit">?
          Custom Automation Solutions Across Diverse Industrial Landscapes?
        </p>?
        <p className="text-slate-500 text-sm mt-4">?
          Click any industry card below to drill down into our functional process designs, technical architectures, and real-
        </p>?
      </div>?

    <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-6">?
      {INDUSTRIES.slice(0, 8).map((industry) => (?
        <div ?
          key={industry.slug}?
          onClick={() => setSelectedIndustry(industry)}?
          className="bg-white border border-slate-100 rounded-xl overflow-hidden cursor-pointer shadow-sm hover:shadow-xl h
        >?
          /* Real industrial imagery integration with soft color-graded overlay */?
          <div className="h-40 relative bg-slate-100 overflow-hidden">?
            <img ?
              src={getIndustryImage(industry.slug)} ?
              alt={industry.name} ?
              className="w-full h-full object-cover transition-transform duration-700 group-hover:scale-110"?
            >?
            <div className="absolute inset-0 bg-gradient-to-t from-[var(--color-ne-text-dark)]/90 via-[var(--color-ne-text-
              <span className="absolute bottom-3 left-3 text-white font-bold text-base font-outfit drop-shadow">?
                {industry.name}?
              </span>?
            </div>?
          </div>?
        </div>?
      )}
    </div>
  </section>
```

```

    </div>?
    ?
    <div className="p-5 flex-grow flex flex-col justify-between">?
      <p className="text-xs text-slate-500 line-clamp-3 mb-6 leading-relaxed">?
        {industry.desc}?
      </p>?
      ?
      <div className="flex justify-between items-center text-[10px] font-mono text-slate-400 pt-3 border-t border-slate-200">?
        <span>SYSTEM SCENARIO</span>?
        <span className="text-[var(--color-ne-blue-corp)] font-bold group-hover:text-[var(--color-ne-blue-steel)]">tra
      </div>?
    </div>?
  </div>?
  )))?
</div>?
</div>?
</section>?
?
/* 4. PRODUCTS & SOLUTIONS ECOSYSTEM */?
<section id="products" className="py-20 relative bg-white border-b border-slate-200/60">?
  <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
    <div className="text-center max-w-3xl mx-auto mb-12">?
      <h2 className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold mb-3">?
        Production Portfolio?
      </h2>?
      <p className="text-3xl font-extrabold text-slate-900 font-outfit">?
        Advanced Electrical Panels, Automation Systems & Instrumentation?
      </p>?
      <p className="text-slate-550 text-sm mt-4">?
        Select category to review technical build sheets, working logic, and compliance standards. Click card to review eng
      </p>?
    </div>?
  ?
  /* Product Tabs */?
  <div className="flex flex-wrap justify-center border-b border-slate-200 mb-12 gap-2">?
    {PRODUCTS.map((category, index) => (?
      <button?
        key={index}?
        onClick={() => setActiveProductTab(index)}?
        className={`px-6 py-3.5 font-bold text-xs tracking-wider uppercase font-mono border-b-2 transition-all ${?
          activeProductTab === index?
            ? 'border-[var(--color-ne-blue-corp)] text-[var(--color-ne-blue-corp)] bg-slate-50'?
            : 'border-transparent text-slate-400 hover:text-slate-600'?
          }`}?
      >?
        {category.title}?
      </button>?
    )}}?
  </div>?
  ?
  /* Product Grid */?
  <div className="grid grid-cols-1 md:grid-cols-2 gap-8">?
    {PRODUCTS[activeProductTab].items.map((product, idx) => (?
      <div ?
        key={idx}?
        onClick={() => setSelectedProduct(product)}?
        className="bg-[#E8F0FA] hover:bg-white border border-slate-200 hover:border-blue-400 rounded-lg p-8 cursor-pointer">?
      >?
        /* Technical grid blueprint lines */?
        <div className="absolute inset-0 bg-[linear-gradient(to_right,#e2e8f0_1px,transparent_1px),linear-gradient(to_bottom,#e2e8f0_1px,transparent_1px)]">?
        ?
        <div className="relative z-10">?
          <div className="flex justify-between items-start mb-4">?
            <h3 className="text-slate-900 font-bold text-lg font-outfit group-hover:text-blue-700 transition-colors">?
              {product.name}?
            </h3>?
            <span className="text-[9px] uppercase font-mono text-[var(--color-ne-blue-corp)] bg-blue-50 px-2.5 py-1 rounded">?
              SPECS SHEET?
            </span>?
          </div>?
          <p className="text-xs text-slate-600 leading-relaxed mb-6">?
            {product.principle}?
          </p>?
          ?
          /* Highlights preview */?
          <ul className="space-y-2 mb-6 font-mono text-[10px] text-slate-550">?
            {product.specs.slice(0, 3).map((spec, sIdx) => (?
              <li key={sIdx} className="flex items-center space-x-2">?
                <span className="w-1.5 h-1.5 bg-[var(--color-ne-blue-corp)] rounded-full" />?
                <span className="truncate">{spec}</span>?
              </li>?
            )}}?
          </ul>?
        </div>?
      </div>?
    )}}?
  </div>?

```

```

        </ul>?
      </div>?
    ?
    <div className="border-t border-slate-200 pt-4 flex justify-between items-center text-[10px] font-mono text-slate-900">
      <span>STANDARD CONFORMITY</span>?
      <span className="text-[var(--color-ne-blue-corp)] font-bold group-hover:underline">Open Data sheet &arr;</span>
    </div>?
  </div>?
)??
</div>?
</div>?
</section>?
?
{ /* 5. MAJOR PROJECTS SHOWCASE */?
<section id="projects" className="py-20 relative bg-[#E8F0FA] border-b border-slate-200/60">?
  <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
    <div className="text-center max-w-3xl mx-auto mb-16">?
      <h2 className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold mb-3">?
        Project Execution Log?
      </h2>?
      <p className="text-3xl font-extrabold text-slate-900 font-outfit">?
        500+ Large-Scale Turnkey EPC Projects Commissioned Nationwide?
      </p>?
      <p className="text-slate-500 text-sm mt-4">?
        Review our industrial project execution reports. Click card to load engineering strategy and operational outcome.?
      </p>?
    </div>?
  ?
  <div className="grid grid-cols-1 md:grid-cols-2 gap-8">?
    {PROJECTS.slice(0, 4).map((project, idx) => (?
      <div ?
        key={idx}?
        onClick={() => setSelectedProject(project)}?
        className="bg-white border border-slate-200 rounded-lg p-8 cursor-pointer shadow-sm hover:shadow-md transition-all">?
      >?
        <div className="absolute top-0 left-0 w-[4px] h-full bg-[#cbd5e1] group-hover:bg-[var(--color-ne-blue-corp)] transition-colors">?
        ?
        <div>?
          <span className="text-[10px] font-mono text-blue-700 uppercase tracking-widest block mb-2 font-bold">?
            {project.industry} &bull; {project.location}?
          </span>?
          <h3 className="text-slate-900 font-bold text-xl mb-4 font-outfit group-hover:text-blue-700 transition-colors">?
            {project.title}?
          </h3>?
          <p className="text-xs text-slate-600 line-clamp-2 leading-relaxed mb-6">?
            <strong>Scope:</strong> {project.scope}?
          </p>?
        </div>?
        ?
        <div className="border-t border-slate-100 pt-4 flex justify-between items-center text-[10px] font-mono">?
          <span className="text-slate-450 uppercase font-bold">Client: {project.client}</span>?
          <span className="text-[var(--color-ne-blue-corp)] font-bold group-hover:underline">CASE REPORT &arr;</span>?
        </div>?
      </div>?
    )??
  </div>?
  ?
  <div className="mt-12 text-center">?
    <Link ?
      href="/projects" ?
      className="inline-block px-8 py-4 bg-[var(--color-ne-blue-corp)] hover:bg-[var(--color-ne-blue-steel)] text-white font-outfit">?
    >?
      View All 500+ Projects & Proof of Execution &arr;?
    </Link>?
  </div>?
</div>?
</section>?
?
{ /* 6. ENGINEERING WORKFLOW (PIPELINE UI) */?
<section className="py-20 bg-white border-b border-slate-200/60 relative">?
  <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
    <div className="text-center max-w-3xl mx-auto mb-16">?
      <h2 className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold mb-3">?
        Standard Operating Workflow?
      </h2>?
      <p className="text-3xl font-extrabold text-slate-900 font-outfit">?
        From Schematic Blueprints to Lifetime Technical Support?
      </p>?
      <p className="text-slate-500 text-sm mt-4">?
        Click the pipeline phases below to trace our quality-assured project lifecycle methodology.?
      </p>?
    </div>?
  </div>?

```

?

```

    { /* Workflow Pipeline Progress Bar */ }
    <div className="relative border border-slate-200 rounded-lg p-6 bg-slate-50 mb-8">
      <div className="absolute top-1/2 left-0 right-0 h-0.5 bg-slate-200 -translate-y-1/2 hidden md:block" />
      <div ?
        className="absolute top-1/2 left-0 h-0.5 bg-[var(--color-ne-blue-corp)] -translate-y-1/2 transition-all duration-500"
        style={{ width: `${(activeWorkflow / (WORKFLOW.length - 1)) * 100}%` }}?
      />

```

?

```

    <div className="grid grid-cols-2 md:grid-cols-6 gap-4 relative z-10">
      {WORKFLOW.map((item, idx) => ( ?
        <button?
          key={idx}?
          onClick={() => setActiveWorkflow(idx)}?
          className={`flex flex-col items-center text-center p-3 rounded transition-all focus:outline-none ${?
            activeWorkflow === idx?
              ? 'bg-white border border-[var(--color-ne-blue-corp)]/30 shadow-sm'?
              : 'bg-transparent border border-transparent hover:border-slate-200'?
            }`}?
          >?
            <span className={`w-8 h-8 rounded-full flex items-center justify-center font-mono text-xs font-bold mb-2 transi
              activeWorkflow === idx?
                ? 'bg-[var(--color-ne-blue-corp)] text-white'?
                : 'bg-slate-200 text-slate-500'?
              }`}>?
              0{idx + 1}?
            </span>?
            <span className={`text-[10px] uppercase font-mono tracking-wider font-bold ${?
              activeWorkflow === idx ? 'text-[var(--color-ne-blue-corp)]' : 'text-slate-400'?
            }`}>?
              {item.step}?
            </span>?
          </button>?
        )})?
      </div>?
    </div>?

```

?

```

    { /* Workflow Detail Display Card */ }
    <div className="bg-[#E8F0FA] border border-slate-200 p-8 rounded-lg flex flex-col md:flex-row md:items-center justify-b
      <div className="space-y-4 max-w-xl">?
        <span className="text-[10px] font-mono text-[var(--color-ne-blue-corp)] uppercase tracking-widest border border-bl
          PHASE 0{activeWorkflow + 1} METRIC?
        </span>?
        <h3 className="text-slate-900 font-extrabold text-2xl font-outfit">?
          {WORKFLOW[activeWorkflow].title}?
        </h3>?
        <p className="text-sm text-slate-600 leading-relaxed font-sans">?
          {WORKFLOW[activeWorkflow].desc}?
        </p>?
      </div>?
      <div className="flex-shrink-0">?
        <svg className="w-32 h-32 text-blue-700/10" viewBox="0 0 100 100" fill="none" stroke="currentColor" strokeWidth="1.
          <rect x="20" y="20" width="60" height="60" rx="4" />?
          <path d="M20,40 H80 M20,60 H80 M40,20 V80 M60,20 V80" strokeDasharray="1,2" />?
          <circle cx="50" cy="50" r="8" className="fill-[var(--color-ne-blue-corp)] stroke-blue-400" />?
        </svg>?
      </div>?
    </div>?
  </div>?
</section>?

```

?

```

    { /* 7. SERVICE SUPPORT ECOSYSTEM */ }
    <section className="py-20 relative bg-[#E8F0FA] border-b border-slate-200/60">?
      <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
        <div className="grid grid-cols-1 lg:grid-cols-12 gap-12 items-center">?
          ?
          <div className="lg:col-span-6 space-y-6">?
            <div className="inline-block bg-blue-50 border border-blue-200 px-3 py-1 rounded font-mono text-xs text-[var(--colo
              Support Infrastructure?
            </div>?
            <h2 className="text-3xl font-extrabold text-slate-900 font-outfit">?
              24/7 Field Service Engineering & Authorized VFD Repair Center?
            </h2>?
            <p className="text-sm text-slate-600 leading-relaxed font-sans">?
              Nandini Enterprises offers a high-performance backup maintenance ecosystem. ?
              We are equipped with advanced testing devices and custom diagnostic equipment ?
              to analyze, tune, and test heavy industrial hardware.?
            </p>?

```

?

```

      <div className="space-y-4">?
        {[?
          { title: "24-Hour Technical Hotline", desc: "Direct channel connection with service coordinator to mobilize eng

```

... continued: src/app/page.tsx

```
{ title: "Authorized Yaskawa Repair Base", desc: "OEM component repairs, drive load testing rigs, and immediate
{ title: "Comprehensive AMCs", desc: "Preventive calendar health evaluations, parameter diagnostic profiling, a
].map((item, idx) => (?
  <div key={idx} className="flex items-start space-x-3">?
    <span className="w-2.5 h-2.5 bg-[var(--color-ne-blue-corp)] rounded-full mt-1.5 shrink-0" />?
    <div>?
      <h4 className="text-slate-900 font-bold text-sm font-sans">{item.title}</h4>?
      <p className="text-xs text-slate-500 mt-0.5 font-sans">{item.desc}</p>?
    </div>?
  </div>?
)}?
</div>?
</div>?
?
/* Support Center Visual Infographic Card */?
<div className="lg:col-span-6">?
  <div className="bg-white border border-slate-200 rounded-lg p-8 space-y-6 font-mono text-xs shadow-md">?
    <div className="flex justify-between items-center border-b border-slate-100 pb-4">?
      <span className="text-[var(--color-ne-blue-corp)] font-bold uppercase tracking-wider">EMERGENCY RESPONSE PROTOC
      <span className="text-slate-400 text-[10px]">VER - 4.1</span>?
    </div>?
    <div className="space-y-3">?
      <div className="bg-[#E8F0FA] p-3 rounded border border-slate-100">?
        <span className="block text-slate-450 text-[9px] font-bold">STEP 01: FAULT REPORT</span>?
        <span className="text-slate-700 font-sans">Client log call & rarr; Diagnostic ticket issued in under 15 mins.</span>?
      </div>?
      <div className="bg-[#E8F0FA] p-3 rounded border border-slate-100">?
        <span className="block text-slate-450 text-[9px] font-bold">STEP 02: TELE-ASSESSMENT</span>?
        <span className="text-slate-700 font-sans">Drive expert attempts remote parameter correction over line.</span>?
      </div>?
      <div className="bg-[#E8F0FA] p-3 rounded border border-slate-100">?
        <span className="block text-slate-455 text-[9px] font-bold">STEP 03: ENGINEER DISPATCH</span>?
        <span className="text-slate-700 font-sans">On-site deployment of service team with standard diagnostic tools.</span>?
      </div>?
    </div>?
    <div className="flex flex-col sm:flex-row justify-between items-center pt-4 border-t border-slate-100 gap-4 text-
      <div>?
        <span className="block text-slate-400 text-[9px]">RESPONSE WINDOW</span>?
        <span className="text-slate-800 font-bold text-sm">??????: &lt; 4 Hrs &bull; India: &lt; 24 Hrs</span>?
      </div>?
      <a?
        href="tel:+914023190131"?
        className="w-full sm:w-auto px-4 py-2 bg-red-600 hover:bg-red-500 text-white font-bold rounded text-center tr
      >?
        Call Support Hotline?
      </a>?
    </div>?
  </div>?
</div>?
</div>?
</div>?
</section>?
?
/* 8. PARTNERS & ASSOCIATIONS */?
<section className="py-16 bg-white border-b border-slate-200/60">?
  <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 text-center">?
    <h2 className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold mb-10">?
      AUTHORIZED ASSOCIATIONS & OEM CHANNELS?
    </h2>?
    <div className="grid grid-cols-2 sm:grid-cols-3 lg:grid-cols-5 gap-6">?
      {[?
        { name: 'YASKAWA', desc: 'Authorized Service Center for Variable Frequency Drives (VFDs)', url: 'https://www.yaskaw
        { name: 'ABB', desc: 'Authorized Channel Partner for Drives & Automation panels', url: 'https://new.abb.com/indian-
        { name: 'BAUMER', desc: 'Authorized System Integrator for Process Sensors', url: 'https://www.baumer.com/' },?
        { name: 'SIEMENS', desc: 'Industrial System Partner for PLC, SCADA, & Switchgears', url: 'https://www.siemens.com/'
        { name: 'YOKOGAWA', desc: 'Process Control Integration & Instrument Config partner', url: 'https://www.yokogawa.com
      ].map((partner, idx) => (?
        <a ?
          key={idx}?
          href={partner.url}?
          target="_blank"?
          rel="noopener noreferrer"?
          className="bg-[#E8F0FA] border border-slate-200 p-6 rounded flex flex-col justify-between hover:border-blue-300 s
        >?
          <span className="text-[var(--color-ne-blue-corp)] font-extrabold text-lg tracking-widest font-mono group-hover:te
            {partner.name}?
          </span>?
          <span className="text-[10px] text-slate-500 mt-4 leading-normal font-sans font-medium">?
            {partner.desc}?
          </span>?
        </a>?
      )?
    </div>?
  </div>?
</section>
```

```

    </a>?
  )}}?
</div>?
</div>?
</section>?
?
{ /* 9. CERTIFICATIONS & COMPLIANCE SECTION */?
<section id="certifications" className="py-20 relative bg-[#E8F0FA] border-b border-slate-200/60">?
  <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
    <div className="text-center max-w-3xl mx-auto mb-16">?
      <h2 className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold mb-3">?
        Standard Compliance?
      </h2>?
      <p className="text-3xl font-extrabold text-slate-900 font-outfit">?
        Statutory Registration, Compliance & Quality Licenses?
      </p>?
      <p className="text-slate-500 text-sm mt-4">?
        We maintain absolute compliance with state and central regulatory frameworks. Click any certificate card below to v
      </p>?
    </div>?
  ?
  <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-6">?
    {CERTIFICATIONS.map((cert, index) => (?
      <div?
        key={index}?
        onClick={() => setSelectedCert(cert)}?
        className="bg-white border border-slate-200 hover:border-blue-300 rounded shadow-sm hover:shadow-md p-6 cursor-po
      >?
      <div>?
        <div className="w-8 h-8 rounded bg-blue-50 border border-blue-200 flex items-center justify-center text-[var(--
          0{index + 1}?
        </div>?
        <h3 className="text-slate-900 font-bold text-sm mb-2 font-outfit group-hover:text-blue-700 transition-colors">?
          {cert.name}?
        </h3>?
        <p className="text-slate-450 text-[10px] font-mono mb-4">?
          Doc Ref: {cert.number}?
        </p>?
      </div>?
      <span className="text-[10px] text-[var(--color-ne-blue-corp)] font-bold font-mono tracking-wider">?
        VIEW COMPLIANCE & rarr;?
      </span>?
    </div>?
  )}}?
</div>?
?
{ /* Detailed supplementary listing */?
<div className="mt-12 bg-white border border-slate-200 p-6 rounded text-center max-w-4xl mx-auto shadow-sm">?
  <p className="text-xs text-slate-500 leading-relaxed font-mono font-medium">?
    Additional Registrations: GST Certificate &bull; Import Export Code (IEC) &bull; Provident Fund (EPF) Registration
  <br />?
  <span className="text-[10px] text-slate-400 block mt-2">All credential verification documents can be requested from
  </span>?
</div>?
</div>?
</section>?
?
{ /* 10. ABOUT COMPANY SECTION */?
<section id="about" className="py-20 relative bg-white border-b border-slate-200/60">?
  <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
    <div className="grid grid-cols-1 lg:grid-cols-12 gap-12 items-center">?
      { /* Manufacturing capabilities */?
      <div className="lg:col-span-6 space-y-6">?
        <span className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold">?
          Company Infrastructure?
        </span>?
        <h2 className="text-3xl sm:text-4xl font-extrabold text-slate-900 font-outfit">?
          India&apos;s Reliable Automation & Heavy Electrical Switchgear Assembler?
        </h2>?
        <p className="text-sm text-slate-655 leading-relaxed font-sans">?
          Nandini Enterprises has operated as an industrial integration leader since 2000. ?
          Based out of IDA Cherlapally, Hyderabad&apos;s premier industrial corridor, our facilities ?
          span a state-of-the-art sheet metal assembly plant, busbar fabrication machinery, ?
          and dedicated test bays.?
        </p>?
      ?
      <div className="grid grid-cols-2 gap-4 font-mono text-xs">?
        <div className="bg-[#E8F0FA] p-4 rounded border border-slate-200">?
          <span className="block text-[var(--color-ne-blue-corp)] font-bold mb-1">CHERLAPALLY PLANT</span>?
          <span className="text-slate-500 text-[10px] font-sans">Over 12,000 sq.ft custom shop floor for mechanical and w
        </div>?
        <div className="bg-[#E8F0FA] p-4 rounded border border-slate-200">?

```


... continued: src/app/page.tsx

```
        <span className="block text-[var(--color-ne-blue-corp)] font-bold mb-1">ENGINEERING TEAM</span>?
        <span className="text-slate-500 text-[10px] font-sans">Highly certified design engineers and automation experts
      </div>?
    </div>?
  </div>?
?
  { /* Vision / Mission / Timelines */?
  <div className="lg:col-span-6">?
    <div className="space-y-6 bg-[#E8F0FA] border border-slate-200 rounded-lg p-8 shadow-sm">?
      <div>?
        <h3 className="text-[var(--color-ne-blue-corp)] font-bold text-sm uppercase tracking-wider mb-2 font-mono">?
          OUR CORE MISSION?
        </h3>?
        <p className="text-xs text-slate-600 leading-relaxed font-sans">?
          To deliver premium, safe, and CPRI-certified industrial power control systems and smart automation workflows
        </p>?
      </div>?
      <div className="border-t border-slate-200 pt-6">?
        <h3 className="text-[var(--color-ne-blue-corp)] font-bold text-sm uppercase tracking-wider mb-2 font-mono">?
          OUR VISION?
        </h3>?
        <p className="text-xs text-slate-600 leading-relaxed font-sans">?
          To remain India's premier B2B system integrator, recognized by leading global OEMs as the preferred high
        </p>?
      </div>?
    </div>?
  </div>?
  </div>?
  </div>?
  </section>?
?
  { /* 11. CSR & SEMINARS / EXPOS SECTION */?
  <section className="py-20 bg-[#E8F0FA] border-b border-slate-200/60">?
    <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10 text-center">?
      <div className="max-w-3xl mx-auto mb-12">?
        <h2 className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold mb-3">?
          Corporate Stewardship & Expos?
        </h2>?
        <p className="text-2xl font-extrabold text-slate-900 font-outfit">?
          Nurturing Talent & Driving Industry-Academia Linkages?
        </p>?
      </div>?
    </div>?
    <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-6 font-sans text-left">?
      {[?
        { title: "Industrial Training Seminars", desc: "Hosting structured automation programming bootcamps for engineering
        { title: "Women in Engineering Projects", desc: "Targeted hiring and skill training initiatives enabling female ope
        { title: "Industrial Expos & Summits", desc: "Regularly participating in Automation Expo India and national electri
      ].map((activity, idx) => (?
        <div key={idx} className="bg-white border border-slate-200 p-6 rounded-lg flex flex-col justify-between shadow-sm h
        <div>?
          <h4 className="text-slate-900 font-bold text-base mb-3 font-outfit">{activity.title}</h4>?
          <p className="text-xs text-slate-500 leading-relaxed">{activity.desc}</p>?
        </div>?
        <span className="text-[10px] text-slate-400 font-mono mt-6 font-semibold">COMMUNITY IMPACT PROFILES</span>?
      </div>?
    )]}?
  </div>?
  </div>?
  </section>?
?
  { /* 12. ADVANCED INTERACTIVE MEDIA CARD PREVIEWS */?
  <section className="py-20 bg-white border-b border-slate-200/60">?
    <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
      <div className="text-center max-w-3xl mx-auto mb-16">?
        <h2 className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold mb-3">?
          Interactive Media Logs?
        </h2>?
        <p className="text-2xl font-extrabold text-slate-900 font-outfit">?
          Engineering Media & Visual Documentation Drawer?
        </p>?
        <p className="text-slate-500 text-sm mt-4">?
          Expand visual cards below to browse high-definition diagnostic schematics, assembly operations, and project deliver
        </p>?
      </div>?
    </div>?
    <div className="grid grid-cols-1 md:grid-cols-3 gap-8 text-left font-mono text-xs">?
      <div className="bg-[#E8F0FA] border border-slate-200 rounded-lg p-6 hover:border-blue-300 shadow-sm transition-all">?
        <span className="text-[9px] text-slate-450 block mb-1">MEDIA ID: V-401</span>?
        <h4 className="text-slate-900 font-bold mb-4 font-outfit text-sm">VFD Load Testing Bay Operations</h4>?
        <p className="text-xs text-slate-500 font-sans mb-6">?
          Demonstrates high-power load runs on a newly assembled Yaskawa VFD control panel before dispatch to JSW Steel.?
```



```

        </p>?
        <span className="text-[var(--color-ne-blue-corp)] font-bold cursor-pointer" onClick={() => alert("Loading mock video
          Watch Video Demonstration &arr;?
        </span>?
      </div>?
?
      <div className="bg-[#E8F0FA] border border-slate-200 rounded-lg p-6 hover:border-blue-300 shadow-sm transition-all">?
        <span className="text-[9px] text-slate-450 block mb-1">MEDIA ID: S-102</span>?
        <h4 className="text-slate-900 font-bold mb-4 font-outfit text-sm">120 MLD PLC Logic Diagrams</h4>?
        <p className="text-xs text-slate-500 font-sans mb-6">?
          Detailed functional control sequence logic flowchart matching HMWSSB&apos;s main water pumping loops.?
        </p>?
        <span className="text-[var(--color-ne-blue-corp)] font-bold cursor-pointer" onClick={() => alert("Loading layout sc
          Open Engineering Drawing &arr;?
        </span>?
      </div>?
?
      <div className="bg-[#E8F0FA] border border-slate-200 rounded-lg p-6 hover:border-blue-300 shadow-sm transition-all">?
        <span className="text-[9px] text-slate-455 block mb-1">MEDIA ID: D-88</span>?
        <h4 className="text-slate-900 font-bold mb-4 font-outfit text-sm">Company B2B Product Brochure</h4>?
        <p className="text-xs text-slate-500 font-sans mb-6">?
          Download the complete executive brochure containing technical specifications for all CPRI panels and drive system
        </p>?
        <a ?
          href="/NE_Profile_2026.pdf" ?
          className="text-[var(--color-ne-blue-corp)] font-bold hover:underline block"?
          onClick={(e) => {?
            e.preventDefault();?
            alert("Converting the profile profile_2026 to PDF download stream. Completed!");?
          }}?
        >?
          Download PDF Brochure &darr;?
        </a>?
      </div>?
    </div>?
  </div>?
</section>?
?
  </* 13. LEAD GENERATION SYSTEM (SMART FORMS) */?
  <section id="rfq-section" className="py-20 relative bg-[#E8F0FA]">?
    <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
      <div className="grid grid-cols-1 lg:grid-cols-12 gap-12 items-center">?
        ?
        </* Quote details */?
        <div className="lg:col-span-5 space-y-6">?
          <div className="inline-block bg-blue-50 border border-blue-200 px-3 py-1 rounded font-mono text-xs text-[var(--color-ne-blue-corp)] font-outfit">
            Lead Capture Center?
          </div>?
          <h2 className="text-3xl font-extrabold text-slate-900 font-outfit">?
            Ready to Optimize Your Plant Operations??
          </h2>?
          <p className="text-sm text-slate-600 leading-relaxed font-sans">?
            Connect with our Hyderabad-based technical experts. Submit details to launch formal system engineering assessment
          </p>?
?
          <div className="bg-white border border-slate-200 rounded-lg p-6 font-mono text-xs space-y-4 shadow-sm">?
            <span className="block text-slate-800 font-bold border-b border-slate-100 pb-2 uppercase tracking-wide">Technical
              <div className="flex items-center space-x-3 text-slate-600">?
                <span className="text-[var(--color-ne-blue-corp)] font-bold w-24">Sales RFQ:</span>?
                <span>info@nandiniautomation.com</span>?
              </div>?
              <div className="flex items-center space-x-3 text-slate-600">?
                <span className="text-[var(--color-ne-blue-corp)] font-bold w-24">Drive Desk:</span>?
                <span>+91-40-23190131</span>?
              </div>?
              <div className="flex items-center space-x-3 text-slate-600">?
                <span className="text-[var(--color-ne-blue-corp)] font-bold w-24">Location:</span>?
                <span>Cherlapally, Hyd - 500051</span>?
              </div>?
            </div>?
          </div>?
        </div>?
      </div>?
    </div>?
  </* Interactive smart B2B form */?
  <div className="lg:col-span-7 bg-white border border-slate-200 p-8 rounded-lg shadow-md">?
    <div className="flex border-b border-slate-200 mb-6 gap-2">?
      {[?
        { id: 'quote', label: 'Request RFQ' },?
        { id: 'consult', label: 'Consultation' },?
        { id: 'support', label: 'Technical Support' }?
      ].map((tab) => (?
        <button?
          key={tab.id}?

```

... continued: src/app/page.tsx

```
        onClick={() => setInquiryType(tab.id)}?
        className={`px-4 py-2 font-mono text-xs uppercase border-b-2 tracking-wider ${?
            inquiryType === tab.id?
                ? 'border-[var(--color-ne-blue-corp)] text-[var(--color-ne-blue-corp)] font-bold'?
                : 'border-transparent text-slate-400'?
            }`}?
    >?
    {tab.label}?
  </button>?
)}?
</div>?

?

{rfqSubmitted ? (?
  <div className="text-center py-12 space-y-4">?
    <div className="w-12 h-12 bg-green-50 border border-green-500 rounded-full flex items-center justify-center mx-?
      &check;?
    </div>?
    <h4 className="text-slate-905 font-bold text-lg font-outfit">RFQ Inquiry Logged Successfully</h4>?
    <p className="text-slate-500 text-xs max-w-md mx-auto">?
      Your details have been registered into our B2B lead portal. A senior service coordinator will contact you sho
    </p>?
  </div>?
) : (?
  <form onSubmit={handleInquirySubmit} className="space-y-4">?
    <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">?
      <div className="space-y-2">?
        <label className="text-[10px] font-mono text-slate-500 uppercase">Contact Name</label>?
        <input?
          type="text"?
          required?
          value={contactForm.name}?
          onChange={(e) => setContactForm({ ...contactForm, name: e.target.value })}?
          placeholder="John Doe"?
          className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder?
        </div>?
      <div className="space-y-2">?
        <label className="text-[10px] font-mono text-slate-500 uppercase">Email Address</label>?
        <input?
          type="email"?
          required?
          value={contactForm.email}?
          onChange={(e) => setContactForm({ ...contactForm, email: e.target.value })}?
          placeholder="john@company.com"?
          className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder?
        </div>?
      </div>?
    </div>?

    ?

    <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">?
      <div className="space-y-2">?
        <label className="text-[10px] font-mono text-slate-500 uppercase">Telephone / Phone</label>?
        <input?
          type="tel"?
          required?
          value={contactForm.phone}?
          onChange={(e) => setContactForm({ ...contactForm, phone: e.target.value })}?
          placeholder="+91-XXXXX-XXXXX"?
          className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder?
        </div>?
      <div className="space-y-2">?
        <label className="text-[10px] font-mono text-slate-500 uppercase">Company Name</label>?
        <input?
          type="text"?
          required?
          value={contactForm.company}?
          onChange={(e) => setContactForm({ ...contactForm, company: e.target.value })}?
          placeholder="Steel Corp Ltd"?
          className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder?
        </div>?
      </div>?

    ?

    <div className="space-y-2">?
      <label className="text-[10px] font-mono text-slate-500 uppercase">{getFormConfig().extraFieldLabel}</label>?
      <input?
        type="text"?
        value={contactForm.extraField}?
        onChange={(e) => setContactForm({ ...contactForm, extraField: e.target.value })}?
        placeholder={getFormConfig().extraFieldPlaceholder}?
        className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder?
```

```

    />?
  </div>?

  ?
    <div className="space-y-2">?
      <label className="text-[10px] font-mono text-slate-500 uppercase">{getFormConfig().textareaLabel}</label>?
      <textarea?
        required?
        value={contactForm.requirement}?
        onChange={(e) => setContactForm({ ...contactForm, requirement: e.target.value })}?
        placeholder={getFormConfig().textareaPlaceholder}?
        rows={4}?
        className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-3 py-2 text-sm text-slate-800 placeholder"
      ></textarea>?
    </div>?

  ?
    /* Formspree submission hooks simulated */?
    <button?
      type="submit"?
      className="w-full py-3 bg-gradient-to-r from-[var(--color-ne-blue-corp)] to-blue-700 hover:from-blue-700 hover"
    >?
      {getFormConfig().buttonText}?
    </button>?
  </form>?
  )}?
</div>?
</div>?
</div>?
</section>?

  ?
  /* ===== */?
  /* 14. MODAL POPUPS (DETAILED DATA VIEWER) */?
  /* ===== */?
  ?
  /* Industry Detail Modal */?
  {selectedIndustry && (?
    <div className="fixed inset-0 z-100 flex items-center justify-center p-4 bg-slate-900/60 backdrop-blur-sm">?
      <div className="bg-white border border-slate-200 rounded-lg max-w-2xl w-full p-8 shadow-2xl relative text-slate-700 font"
        <button ?
          className="absolute top-4 right-4 text-slate-450 hover:text-slate-800 text-xl"?
          onClick={() => setSelectedIndustry(null)}?
        >?
          &times;?
        </button>?
        <span className="text-[10px] font-mono text-[var(--color-ne-blue-corp)] uppercase tracking-widest block mb-2 font-bol"
          EPC Industry Detail?
        </span>?
        <h3 className="text-2xl font-bold text-slate-900 mb-6 font-outfit">?
          {selectedIndustry.name}?
        </h3>?

      ?
      <div className="space-y-6 text-sm text-slate-600">?
        <div>?
          <h4 className="text-xs uppercase text-slate-500 font-mono tracking-wider font-bold mb-2">1. Functional Process De
          <p className="leading-relaxed">{selectedIndustry.desc}</p>?
        </div>?
        <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">?
          <div>?
            <h4 className="text-xs uppercase text-slate-500 font-mono tracking-wider font-bold mb-2">2. Technical Systems I
            <p className="text-xs text-slate-550">{selectedIndustry.tech}</p>?
          </div>?
          <div>?
            <h4 className="text-xs uppercase text-slate-500 font-mono tracking-wider font-bold mb-2">3. Primary Project Sce
            <p className="text-xs text-slate-550">{selectedIndustry.projects}</p>?
          </div>?
        </div>?
        <div>?
          <h4 className="text-xs uppercase text-slate-500 font-mono tracking-wider font-bold mb-2">4. Operational Case Outc
          <p className="text-xs text-green-700 font-mono font-semibold">{selectedIndustry.outcome}</p>?
        </div>?
      </div>?
    </div>?
  )}?

  ?
  /* Product Data sheet Drawer */?
  {selectedProduct && (?
    <div className="fixed inset-0 z-100 flex items-center justify-center p-4 bg-slate-900/60 backdrop-blur-sm">?
      <div className="bg-white border border-slate-200 rounded-lg max-w-2xl w-full p-8 shadow-2xl relative text-slate-700 font"
        <button ?
          className="absolute top-4 right-4 text-slate-450 hover:text-slate-800 text-xl"?
          onClick={() => setSelectedProduct(null)}?
        >?

```

```
... continued: src/app/page.tsx
```

?

?

?

... continued: src/app/page.tsx

```

    <p className="text-xs font-mono text-[var(--color-ne-blue-corp)] font-bold">{selectedProject.tech}</p>?
  </div>?
?
  <div>?
    <h4 className="text-xs uppercase text-slate-500 font-mono tracking-wider font-bold mb-2">3. Verified Process Impr
    <p className="text-xs text-green-700 font-mono font-bold">{selectedProject.outcome}</p>?
  </div>?
</div>?
</div>?
)}?
?
{ /* Certification zoom dialog */?
{selectedCert && (?
  <div className="fixed inset-0 z-100 flex items-center justify-center p-4 bg-slate-900/60 backdrop-blur-sm">?
    <div className="bg-white border border-slate-200 rounded-lg max-w-md w-full p-8 shadow-2xl relative text-slate-700 font
      <button ?
        className="absolute top-4 right-4 text-slate-450 hover:text-slate-800 text-xl"?
        onClick={() => setSelectedCert(null)}?
      >?
        &times;?
      </button>?
      <span className="text-[10px] font-mono text-[var(--color-ne-blue-corp)] uppercase tracking-widest block mb-2 font-bol
        Registration Certificate?
      </span>?
      <h3 className="text-xl font-bold text-slate-900 mb-6 font-outfit">?
        {selectedCert.name}?
      </h3>?
      ?
      <div className="space-y-6 text-sm text-slate-600">?
        <div className="bg-[#E8F0FA] border border-slate-200 p-6 rounded text-center font-mono">?
          <span className="block text-slate-450 text-[10px] mb-2 font-bold">CERTIFICATE ID NUMBER</span>?
          <span className="text-slate-800 font-bold tracking-wider">{selectedCert.number}</span>?
          ?
          <div className="w-16 h-16 border-2 border-blue-200 border-dashed rounded-full flex items-center justify-center mx
            ISO?
          </div>?
        </div>?
      </div>?
      <div>?
        <h4 className="text-xs uppercase text-slate-500 font-mono tracking-wider font-bold mb-1">Approved Audit Scope:</h
        <p className="text-xs leading-relaxed">{selectedCert.scope}</p>?
      </div>?
      ?
      <div className="pt-4 border-t border-slate-200 flex justify-between items-center text-xs">?
        <span className="text-slate-500 font-semibold">Status: ACTIVE</span>?
        <button ?
          className="px-3.5 py-2 bg-slate-100 hover:bg-slate-200 text-[var(--color-ne-blue-corp)] rounded font-bold trans
          onClick={() => alert(`Certificate ${selectedCert.name} has been queued for PDF download.`)}?
        >?
          Download PDF &darr;?
        </button>?
      </div>?
    </div>?
  </div>?
)}?
?
</div>?
);?
}?
}
```

```
'use client';?
import { useState } from 'react';?
interface BrandSpec {?
  parameter: string;?
  value: string;?
}?
interface BrandProduct {?
  name: string;?
  desc: string;?
  specs: BrandSpec[];?
  applications: string[];?
  advantage: string;?
}?
interface Brand {?
  name: string;?
  tagline: string;?
  desc: string;?
  partnership: string;?
  products: BrandProduct[];?
  brands_note?: string;?
}?
const brands: Brand[] = [?
  {?
    name: "NE Manufacturing",?
    tagline: "In-House Engineered Control Panel Systems",?
    desc: "Nandini Enterprises' own line of custom-engineered electrical control panels and automation skids, manufactured at our",?
    partnership: "OEM Manufacturer - Internal Brand",?
    products: [?
      {?
        name: "Power Control Centers (PCC)",?
        desc: "Centralized power distribution switchgear assemblies for main incoming and outgoing feeder management.",?
        specs: [?
          { parameter: "Busbar Rating", value: "400A to 6300A (copper or aluminum)" },?
          { parameter: "Short-Circuit Rating", value: "Up to 65kA / 1 second (CPRI certified)" },?
          { parameter: "Incomer Type", value: "ACB with electronic trip unit (LSI/LSIG)" },?
          { parameter: "Form Segregation", value: "Form 4b - full internal isolation" },?
          { parameter: "Enclosure Protection", value: "IP41 standard, IP55/65 optional" },?
          { parameter: "Design Standard", value: "IEC/IS 61439-1 & 2" },?
        ],?
        applications: ["Steel rolling mills", "Cement plants", "Chemical facilities", "Commercial complexes", "Power plant auxiliaries"],?
        advantage: "CPRI-tested architecture ensures bus fault containment; active ventilation maintains internal temperatures 18°C or lower",?
      },?
      {?
        name: "Motor Control Centers (MCC)",?
        desc: "Centralized motor starting, protection, and monitoring systems with optional smart bus integration.",?
        specs: [?
          { parameter: "Configuration", value: "Fixed or drawout modular drawers" },?
          { parameter: "Starter Type", value: "DOL, Star-Delta, Soft Starter, VFD" },?
          { parameter: "Protection Class", value: "Type-1 or Type-2 coordination" },?
          { parameter: "Communication", value: "Profibus-DP, Modbus RTU/TCP" },?
          { parameter: "Monitoring", value: "Integrated multifunction relays" },?
          { parameter: "Enclosure", value: "IP54 standard; IP65 outdoor version" },?
        ],?
        applications: ["Pumping stations", "Process plants with 10-200 motors", "HVAC distribution centers", "Conveyor motor groups"],?
        advantage: "Intelligent MCC with per-drawer motor diagnostics - running hours, start count, earth fault status - accessible from inside",?
      },?
      {?
        name: "VFD Drive Panels",?
        desc: "Custom-built enclosures housing variable frequency drives with integrated protection, filtration, and communication",?
        specs: [?
          { parameter: "Drive Range", value: "0.75kW to 1000kW (LV); up to 10MW (MV)" },?
          { parameter: "Input Filtering", value: "AC line reactor + harmonic filter" },?
          { parameter: "Braking", value: "Dynamic braking resistors available" },?
          { parameter: "Bypass", value: "Manual or automatic DOL bypass" },?
          { parameter: "Communication", value: "Profinet, Modbus TCP, EtherNet-IP" },?
          { parameter: "Cooling", value: "Forced ventilation with thermostat control" },?
        ],?
        applications: ["Pumps and fans", "Crane/hoist systems", "Elevators", "Conveyors", "Centrifuges"],?
        advantage: "Integrated bypass switchgear allows motor to continue operating on DOL if drive trips - critical for continuous operation",?
      },?
      {?
        name: "Elevator Controller Panels",?
        desc: "Microprocessor-based lift controller assemblies for passenger, goods, and service elevator applications.",?
        specs: [?
          { parameter: "Controller Type", value: "32-bit microprocessor with dual redundancy" },?
        ],?
      },?
    ],?
  },?
],?
};
```

```
... continued: src/app/products/page.tsx
```

```

    {
      parameter: "Drive Integration", value: "Yaskawa L1000A vector drive" },?
    {
      parameter: "Speed Range", value: "0.5 m/s to 2.5 m/s" },?
    {
      parameter: "Rescue Device", value: "ARD battery backup, auto-level to floor" },?
    {
      parameter: "Leveling Accuracy", value: "?3mm with magnetic/optical sensors" },?
    {
      parameter: "Standards", value: "IS 14665, EN 81-20, ASME A17.1" },?
  },?
  applications: ["High-rise residential buildings", "Hospitals and healthcare", "Industrial goods elevators", "Commercial c
  advantage: "ARD (Automatic Rescue Device) included as standard - moves cabin to nearest floor and opens doors within 15 s
},?
},?
},?
{?
  name: "YASKAWA",?
  tagline: "World's Largest Motion Control Manufacturer",?
  desc: "Yaskawa Electric Corporation (Japan, founded 1915) is the global leader in variable frequency drives, servo systems, a
  partnership: "Authorized Service Center & System Integrator - Yaskawa India Ltd.",?
  products: [?
    {?
      name: "GA700 / A1000 General Purpose VFDs",?
      desc: "High-performance vector control drives for demanding industrial applications requiring precision torque and speed
      specs: [?
        {
          parameter: "Power Range", value: "0.4kW to 630kW (200V/400V class)" },?
        {
          parameter: "Control Mode", value: "V/f, Open-loop vector, Closed-loop vector" },?
        {
          parameter: "Torque at 0Hz", value: "150% (open-loop), 200% (closed-loop)" },?
        {
          parameter: "Speed Accuracy", value: "?0.2% (open-loop), ?0.02% (closed-loop)" },?
        {
          parameter: "Protection", value: "IP20 standard, IP55 option" },?
        {
          parameter: "Communication", value: "Profinet, EtherNet-IP, Modbus TCP" },?
      ],?
      applications: ["Pumps and fans", "Compressors", "Conveyors", "Machine tools", "Mixers and agitators"],?
      advantage: "Auto-tuning wizard configures motor parameters in 5 minutes without load - eliminating manual parameter entry
    },?
    {?
      name: "L1000A Elevator Drive",?
      desc: "Dedicated elevator drive with integral passenger comfort features, safety functions, and rescue operation capabili
      specs: [?
        {
          parameter: "Power Range", value: "5.5kW to 110kW" },?
        {
          parameter: "S-Curve Profiles", value: "5-parameter jerk control" },?
        {
          parameter: "Leveling Speed", value: "Configurable 0.01-0.5 m/s" },?
        {
          parameter: "Safety Functions", value: "STO (Safe Torque Off), SS1, SS2" },?
        {
          parameter: "Rescue Operation", value: "ARD battery input standard" },?
        {
          parameter: "Standards", value: "EN 81-20/50, IEC 61800-5-2" },?
      ],?
      applications: ["Passenger elevators", "Goods elevators", "Hospital lifts", "Escalators"],?
      advantage: "Dedicated elevator firmware with pre-programmed leveling, brake control, and rescue sequences - reduces commi
    },?
    {?
      name: "Motoman Industrial Robots",?
      desc: "High-speed, high-accuracy articulated arm robots for welding, assembly, pick-and-place, and palletizing applicatio
      specs: [?
        {
          parameter: "Payload Range", value: "3kg (GP4) to 600kg (MH600)" },?
        {
          parameter: "Reach", value: "550mm to 3159mm depending on model" },?
        {
          parameter: "Repeatability", value: "?0.02mm to ?0.07mm" },?
        {
          parameter: "Axes", value: "4, 6, or 7 axes depending on model" },?
        {
          parameter: "Controller", value: "YRC1000 - EtherCAT, Profinet ready" },?
        {
          parameter: "IP Rating", value: "IP67 standard (washdown versions)" },?
      ],?
      applications: ["Pick-and-place assembly", "Arc/spot welding", "Palletizing", "Machine tending", "Quality inspection"],?
      advantage: "YRC1000 controller runs EtherCAT at 250?s cycle time - enabling real-time force control for precision assembl
    },?
  },?
},?
},?
{?
  name: "ABB",?
  tagline: "Technology Leader in Electrification & Automation",?
  desc: "ABB Ltd. (Switzerland, founded 1988 from ASEA/BBC merger) is a world-leading power and automation technology company.
  partnership: "Authorized Supplier & Integrator - ABB India Ltd.",?
  products: [?
    {?
      name: "ACS880 Industrial Drives",?
      desc: "Direct torque control (DTC) drives for the most demanding process industry motor control requirements.",?
      specs: [?
        {
          parameter: "Power Range", value: "0.37kW to 5600kW (modular frames)" },?
        {
          parameter: "Control Algorithm", value: "Direct Torque Control (DTC)" },?
        {
          parameter: "Torque Response", value: "<2ms full torque step response" },?
        {
          parameter: "Speed Accuracy", value: "?0.1% without encoder" },?
        {
          parameter: "Energy Optimizer", value: "Automatic flux optimization" },?
        {
          parameter: "Communication", value: "PROFINET, EtherNet/IP, Modbus TCP" },?
      ],?
      applications: ["Pumps and fans (energy optimizer)", "Compressors and blowers", "Cranes and hoists", "Centrifuges", "Marin
      advantage: "Direct Torque Control delivers 2ms torque response - 10? faster than standard V/f control - critical for cran
    },?
  },?
},?
},?
{?
  name: "ABB",?
  tagline: "Technology Leader in Electrification & Automation",?
  desc: "ABB Ltd. (Switzerland, founded 1988 from ASEA/BBC merger) is a world-leading power and automation technology company.
  partnership: "Authorized Supplier & Integrator - ABB India Ltd.",?
  products: [?
    {?
      name: "ACS880 Industrial Drives",?
      desc: "Direct torque control (DTC) drives for the most demanding process industry motor control requirements.",?
      specs: [?
        {
          parameter: "Power Range", value: "0.37kW to 5600kW (modular frames)" },?
        {
          parameter: "Control Algorithm", value: "Direct Torque Control (DTC)" },?
        {
          parameter: "Torque Response", value: "<2ms full torque step response" },?
        {
          parameter: "Speed Accuracy", value: "?0.1% without encoder" },?
        {
          parameter: "Energy Optimizer", value: "Automatic flux optimization" },?
        {
          parameter: "Communication", value: "PROFINET, EtherNet/IP, Modbus TCP" },?
      ],?
      applications: ["Pumps and fans (energy optimizer)", "Compressors and blowers", "Cranes and hoists", "Centrifuges", "Marin
      advantage: "Direct Torque Control delivers 2ms torque response - 10? faster than standard V/f control - critical for cran
    },?
  },?
},?
},?
{?
  name: "ABB",?
  tagline: "Technology Leader in Electrification & Automation",?
  desc: "ABB Ltd. (Switzerland, founded 1988 from ASEA/BBC merger) is a world-leading power and automation technology company.
  partnership: "Authorized Supplier & Integrator - ABB India Ltd.",?
  products: [?
    {?
      name: "ACS880 Industrial Drives",?
      desc: "Direct torque control (DTC) drives for the most demanding process industry motor control requirements.",?
      specs: [?
        {
          parameter: "Power Range", value: "0.37kW to 5600kW (modular frames)" },?
        {
          parameter: "Control Algorithm", value: "Direct Torque Control (DTC)" },?
        {
          parameter: "Torque Response", value: "<2ms full torque step response" },?
        {
          parameter: "Speed Accuracy", value: "?0.1% without encoder" },?
        {
          parameter: "Energy Optimizer", value: "Automatic flux optimization" },?
        {
          parameter: "Communication", value: "PROFINET, EtherNet/IP, Modbus TCP" },?
      ],?
      applications: ["Pumps and fans (energy optimizer)", "Compressors and blowers", "Cranes and hoists", "Centrifuges", "Marin
      advantage: "Direct Torque Control delivers 2ms torque response - 10? faster than standard V/f control - critical for cran
    },?
  },?
},?
},?
{?
  name: "ABB",?
  tagline: "Technology Leader in Electrification & Automation",?
  desc: "ABB Ltd. (Switzerland, founded 1988 from ASEA/BBC merger) is a world-leading power and automation technology company.
  partnership: "Authorized Supplier & Integrator - ABB India Ltd.",?
  products: [?
    {?
      name: "ACS880 Industrial Drives",?
      desc: "Direct torque control (DTC) drives for the most demanding process industry motor control requirements.",?
      specs: [?
        {
          parameter: "Power Range", value: "0.37kW to 5600kW (modular frames)" },?
        {
          parameter: "Control Algorithm", value: "Direct Torque Control (DTC)" },?
        {
          parameter: "Torque Response", value: "<2ms full torque step response" },?
        {
          parameter: "Speed Accuracy", value: "?0.1% without encoder" },?
        {
          parameter: "Energy Optimizer", value: "Automatic flux optimization" },?
        {
          parameter: "Communication", value: "PROFINET, EtherNet/IP, Modbus TCP" },?
      ],?
      applications: ["Pumps and fans (energy optimizer)", "Compressors and blowers", "Cranes and hoists", "Centrifuges", "Marin
      advantage: "Direct Torque Control delivers 2ms torque response - 10? faster than standard V/f control - critical for cran
    },?
  },?
},?
},?
{?
  name: "ABB",?
  tagline: "Technology Leader in Electrification & Automation",?
  desc: "ABB Ltd. (Switzerland, founded 1988 from ASEA/BBC merger) is a world-leading power and automation technology company.
  partnership: "Authorized Supplier & Integrator - ABB India Ltd.",?
  products: [?
    {?
      name: "ACS880 Industrial Drives",?
      desc: "Direct torque control (DTC) drives for the most demanding process industry motor control requirements.",?
      specs: [?
        {
          parameter: "Power Range", value: "0.37kW to 5600kW (modular frames)" },?
        {
          parameter: "Control Algorithm", value: "Direct Torque Control (DTC)" },?
        {
          parameter: "Torque Response", value: "<2ms full torque step response" },?
        {
          parameter: "Speed Accuracy", value: "?0.1% without encoder" },?
        {
          parameter: "Energy Optimizer", value: "Automatic flux optimization" },?
        {
          parameter: "Communication", value: "PROFINET, EtherNet/IP, Modbus TCP" },?
      ],?
      applications: ["Pumps and fans (energy optimizer)", "Compressors and blowers", "Cranes and hoists", "Centrifuges", "Marin
      advantage: "Direct Torque Control delivers 2ms torque response - 10? faster than standard V/f control - critical for cran
    },?
  },?
},?
},?
{?
  name: "ABB",?
  tagline: "Technology Leader in Electrification & Automation",?
  desc: "ABB Ltd. (Switzerland, founded 1988 from ASEA/BBC merger) is a world-leading power and automation technology company.
  partnership: "Authorized Supplier & Integrator - ABB India Ltd.",?
  products: [?
    {?
      name: "ACS880 Industrial Drives",?
      desc: "Direct torque control (DTC) drives for the most demanding process industry motor control requirements.",?
      specs: [?
        {
          parameter: "Power Range", value: "0.37kW to 5600kW (modular frames)" },?
        {
          parameter: "Control Algorithm", value: "Direct Torque Control (DTC)" },?
        {
          parameter: "Torque Response", value: "<2ms full torque step response" },?
        {
          parameter: "Speed Accuracy", value: "?0.1% without encoder" },?
        {
          parameter: "Energy Optimizer", value: "Automatic flux optimization" },?
        {
          parameter: "Communication", value: "PROFINET, EtherNet/IP, Modbus TCP" },?
      ],?
      applications: ["Pumps and fans (energy optimizer)", "Compressors and blowers", "Cranes and hoists", "Centrifuges", "Marin
      advantage: "Direct Torque Control delivers 2ms torque response - 10? faster than standard V/f control - critical for cran
    },?
  },?
},?
},?
{?
  name: "ABB",?
  tagline: "Technology Leader in Electrification & Automation",?
  desc: "ABB Ltd. (Switzerland, founded 1988 from ASEA/BBC merger) is a world-leading power and automation technology company.
  partnership: "Authorized Supplier & Integrator - ABB India Ltd.",?
  products: [?
    {?
      name: "ACS880 Industrial Drives",?
      desc: "Direct torque control (DTC) drives for the most demanding process industry motor control requirements.",?
      specs: [?
        {
          parameter: "Power Range", value: "0.37kW to 5600kW (modular frames)" },?
        {
          parameter: "Control Algorithm", value: "Direct Torque Control (DTC)" },?
        {
          parameter: "Torque Response", value: "<2ms full torque step response" },?
        {
          parameter: "Speed Accuracy", value: "?0.1% without encoder" },?
        {
          parameter: "Energy Optimizer", value: "Automatic flux optimization" },?
        {
          parameter: "Communication", value: "PROFINET, EtherNet/IP, Modbus TCP" },?
      ],?
      applications: ["Pumps and fans (energy optimizer)", "Compressors and blowers", "Cranes and hoists", "Centrifuges", "Marin
      advantage: "Direct Torque Control delivers 2ms torque response - 10? faster than standard V/f control - critical for cran
    },?
  },?
},?
},?
{?
  name: "ABB",?
  tagline: "Technology Leader in Electrification & Automation",?
  desc: "ABB Ltd. (Switzerland, founded 1988 from ASEA/BBC merger) is a world-leading power and automation technology company.
  partnership: "Authorized Supplier & Integrator - ABB India Ltd.",?
  products: [?
    {?
      name: "ACS880 Industrial Drives",?
      desc: "Direct torque control (DTC) drives for the most demanding process industry motor control requirements.",?
      specs: [?
        {
          parameter: "Power Range", value: "0.37kW to 5600kW (modular frames)" },?
        {
          parameter: "Control Algorithm", value: "Direct Torque Control (DTC)" },?
        {
          parameter: "Torque Response", value: "<2ms full torque step response" },?
        {
          parameter: "Speed Accuracy", value: "?0.1% without encoder" },?
        {
          parameter: "Energy Optimizer", value: "Automatic flux optimization" },?
        {
          parameter:
```



```
name: "HXR / AMI High-Efficiency Motors",?
desc: "Premium efficiency IE3/IE4 motors designed for continuous heavy industrial operation.",?
specs: [?
  { parameter: "Power Range", value: "75kW to 1400kW (HXR cast iron frame)" },?
  { parameter: "Efficiency Class", value: "IE3 (Premium), IE4 (Super Premium)" },?
  { parameter: "Insulation Class", value: "Class F (155°C) rated, Class B rise" },?
  { parameter: "Bearing Type", value: "Anti-friction, grease/oil lubricated" },?
  { parameter: "Frame Material", value: "Cast iron (HXR), aluminum (AMI)" },?
  { parameter: "Standards", value: "IEC 60034-1, ATEX Zone 2 versions" },?
],?
applications: ["Compressors", "Pumps above 90kW", "Fans and blowers", "Mixers", "Mill drives"],?
advantage: "IE4 Super Premium motors reduce energy consumption 2-4% vs IE3 - on a 200kW motor running 6000 hours/year, th
  },?
},?
{?
  name: "YOKOGAWA",?
  tagline: "Precision Measurement & Process Control Excellence",?
  desc: "Yokogawa Electric Corporation (Japan, founded 1915) specializes in industrial automation, control instrumentation, and
  partnership: "Authorized Supplier & Service Partner - Yokogawa India Ltd.",?
  products: [?
    {?
      name: "EJA/EJX Transmitters",?
      desc: "Differential pressure, gauge pressure, and absolute pressure transmitters for flow, level, and pressure measurement",?
      specs: [?
        { parameter: "Accuracy", value: "±0.04% (EJX890A flagship model)" },?
        { parameter: "Turndown Ratio", value: "Up to 200:1 (EJX series)" },?
        { parameter: "Response Time", value: "90ms (standard), 10ms (fast response)" },?
        { parameter: "Communication", value: "HART 7, FOUNDATION Fieldbus, PROFIBUS-PA" },?
        { parameter: "Wetted Material", value: "316SS, Hastelloy C276, Tantalum" },?
        { parameter: "Approvals", value: "ATEX Zone 1/2, IECEx, SIL-2 capable" },?
      ],?
      applications: ["Flow measurement via orifice/venturi", "Tank level monitoring", "Boiler drum level", "Filter differential pressure"],?
      advantage: "EJX890A multi-sensing transmitter measures both differential and static pressure simultaneously - replacing t
    },?
    {?
      name: "AXF Electromagnetic Flow Meters",?
      desc: "Magnetic flow meters for conductive liquid flow measurement from small process lines to large municipal mains.",?
      specs: [?
        { parameter: "Line Size", value: "DN10 to DN2400" },?
        { parameter: "Accuracy", value: "±0.35% of reading" },?
        { parameter: "Liner Material", value: "PTFE, Hard Rubber, PFA" },?
        { parameter: "Electrode Material", value: "316SS, Hastelloy, Platinum" },?
        { parameter: "Communication", value: "HART, PROFIBUS-PA, Modbus" },?
        { parameter: "Standards", value: "MID certified, OIML R49" },?
      ],?
      applications: ["Water distribution mains", "Wastewater treatment", "Chemical dosing", "Pulp and paper", "Food and beverage"],?
      advantage: "Dual-frequency excitation (AXF models) eliminates measurement noise in slurry and low-conductivity applications"
    },?
  ],?
},?
{?
  name: "CENTUM VP Distributed Control System",?
  desc: "Enterprise DCS platform integrating process control, safety, HMI, and plant information management.",?
  specs: [?
    { parameter: "I/O Capacity", value: "Up to 100,000 I/O points" },?
    { parameter: "Control Network", value: "Vnet/IP (1Gbps, redundant)" },?
    { parameter: "Scan Cycle", value: "100ms to 1000ms per control station" },?
    { parameter: "Redundancy", value: "Dual-redundant controllers, networks, and HMIs" },?
    { parameter: "Integration", value: "OPC DA/UA, Modbus, PROFIBUS" },?
    { parameter: "History", value: "Exaquantum plant historian" },?
  ],?
  applications: ["Oil refineries", "Chemical plants", "Power generation", "LNG terminals", "Pharmaceutical manufacturing"],?
  advantage: "FCS (Field Control Station) self-diagnosing capability detects I/O card faults automatically, reducing mean t
  },?
},?
{?
  name: "BAUMER",?
  tagline: "Innovative Sensor Technology & Precision Encoders",?
  desc: "Baumer Group (Switzerland, founded 1952) develops and manufactures innovative sensor solutions for industrial automation",?
  partnership: "Authorized Distributor & Application Partner - Baumer India Pvt. Ltd.",?
  products: [?
    {?
      name: "Heavy Duty Industrial Encoders",?
      desc: "Rotary encoders for speed, position, and angle measurement in heavy industrial drive applications.",?
      specs: [?
        { parameter: "Type", value: "Incremental & Absolute (single/multi-turn)" },?
        { parameter: "Resolution", value: "Up to 65,536 steps/revolution (16-bit)" },?
        { parameter: "Max Speed", value: "6000 RPM continuous" },?
        { parameter: "Shaft Loading", value: "Up to 400N radial, 200N axial" },?
        { parameter: "Protection", value: "IP67, IP69K (washdown)" },?
        { parameter: "Interface", value: "HTL, TTL, SSI, PROFIBUS, EtherCAT" },?
      ],?
    },
  ],?
}
```



```

applications: ["Crane rope drum speed", "Conveyor belt speed", "Robot joint angles", "Wind turbine rotor position", "P
advantage: "Magnetic encoder technology operates without optical components - immune to dust, condensation, and mechanical
},?
{?
name: "OADM Radar & Ultrasonic Level Sensors",?
desc: "Non-contact level measurement for liquids, pastes, and bulk solids in storage vessels and process tanks.",?
specs: [?
{ parameter: "Technology", value: "80GHz FMCW radar or 40kHz ultrasonic" },?
{ parameter: "Range", value: "Radar: 0.05m to 70m; Ultrasonic: 0.3m to 15m" },?
{ parameter: "Accuracy", value: "?3mm (radar at full range)" },?
{ parameter: "Process Temp", value: "Up to 200°C (high-temp radar)" },?
{ parameter: "Output", value: "4-20mA + HART, IO-Link, Profibus" },?
{ parameter: "Approvals", value: "ATEX Zone 0/1, FDA compliant" },?
],?
applications: ["Chemical storage tanks", "Water reservoirs", "Silo level (bulk solids)", "Pharmaceutical mixing vessels",
advantage: "80GHz radar beam angle of 3° (vs 10-15° for older 26GHz technology) - eliminates false echoes from internal t
},?
],?
},?
{?
name: "HONEYWELL",?
tagline: "Advanced Industrial Process Control & Safety Systems",?
desc: "Honeywell Process Solutions (USA, fortune 100 company) provides DCS, safety systems, advanced control software, and pr
partnership: "Authorized Partner - Honeywell India",?
products: [?
{?
name: "Experion PKS / C300 DCS",?
desc: "Enterprise-grade DCS platform for large continuous process facilities with integrated safety and advanced control.
specs: [?
{ parameter: "Controller", value: "C300 with dual-redundant CPUs" },?
{ parameter: "I/O Capacity", value: "Up to 60,000 I/O points per system" },?
{ parameter: "Control Network", value: "Fault-Tolerant Ethernet (FTE) - 1Gbps" },?
{ parameter: "Scan Rate", value: "50ms to 500ms configurable" },?
{ parameter: "History", value: "Uniformance Plant Historian (SQL)" },?
{ parameter: "Certifications", value: "ISA-99, IEC 62443 cybersecurity" },?
],?
applications: ["Oil refineries", "Gas processing plants", "Chemical complexes", "Pharmaceutical batch plants"],?
advantage: "Experion Advanced Process Control (APC) integration reduces process variability by 30-50% and increases throug
},?
{?
name: "HC900 Hybrid Controller",?
desc: "Mid-range process controller combining PLC discrete logic with DCS-quality analog loop control in a single compact
specs: [?
{ parameter: "I/O Capacity", value: "Up to 256 mixed I/O points" },?
{ parameter: "Control Loops", value: "Up to 64 PID loops" },?
{ parameter: "Redundancy", value: "Dual CPU hot-standby available" },?
{ parameter: "HMI", value: "Built-in web server + Honeywell Vista HMI" },?
{ parameter: "Communication", value: "Modbus TCP/RTU, EtherNet/IP" },?
{ parameter: "Approvals", value: "UL listed, CE, FM" },?
],?
applications: ["Standalone process units", "HVAC building controls", "Batch reactors <20 vessels", "Test bench automation
advantage: "Web server-based remote monitoring without VPN infrastructure - ideal for multi-site monitoring from a centra
},?
},?
},?
{?
name: "WIKA",?
tagline: "Global Standard in Pressure & Temperature Measurement",?
desc: "WIKA Alexander Wiegand SE & Co. KG (Germany, founded 1946) is the world market leader in pressure and temperature meas
partnership: "Authorized Distributor - WIKA Instruments India Pvt. Ltd.",?
products: [?
{?
name: "S-10 / A-10 Pressure Transmitters",?
desc: "Industrial pressure transmitters for demanding process applications requiring high accuracy and long-term stabilit
specs: [?
{ parameter: "Accuracy", value: "?0.5% (standard), ?0.1% (precision)" },?
{ parameter: "Pressure Ranges", value: "-1 Bar vacuum to 1600 Bar" },?
{ parameter: "Output Signal", value: "4-20mA + HART; 0-10V; IO-Link" },?
{ parameter: "Process Connection", value: "G1/4 to G1, NPT, flanged" },?
{ parameter: "Wetted Material", value: "316L SS, Hastelloy, titanium" },?
{ parameter: "Approvals", value: "ATEX Zone 1/2, IECEx, SIL-2 capable" },?
],?
applications: ["Hydraulic systems", "Compressed air monitoring", "Process piping pressure", "Pump differential monitoring
advantage: "Flush diaphragm versions eliminate clogging risk in viscous, crystallizing, or particle-laden media - a commo
},?
},?
{?
name: "Pressure Gauges (Bourdon Type)",?
desc: "Mechanical pressure gauges for local indication without power supply requirement.",?
specs: [?
{ parameter: "Accuracy Class", value: "0.1%, 0.25%, 0.5%, 1.0%" },?
{ parameter: "Dial Size", value: "63mm, 100mm, 160mm standard" },?
{ parameter: "Pressure Range", value: "-1 Bar to 1000 Bar" },?

```

... continued: src/app/products/page.tsx

```
    { parameter: "Case Material", value: "Stainless steel or phenol resin" },?
    { parameter: "Fill Liquid", value: "Glycerin or silicone for vibration" },?
    { parameter: "Standards", value: "EN 837-1, DIN 16001, ASME B40.100" },?
  ],?
  applications: ["Process plant field indication", "Pump suction/discharge monitoring", "Compressor pressure", "Test & cali
  advantage: "Glycerin-filled gauges dampen needle oscillation in pulsating systems - extending pointer and movement life 3
  ]?
  }?
  };?
  ?
  export default function ProductsPage() {?
    const [searchTerm, setSearchTerm] = useState('');?
    const [activeBrand, setActiveBrand] = useState<number | null>(null);?
    const [activeProduct, setActiveProduct] = useState<number>(0);?
    ?
    const filteredBrands = searchTerm === '' ? brands : brands.filter(b => ?
      b.name.toLowerCase().includes(searchTerm.toLowerCase()) ||?
      b.products.some(p => ?
        p.name.toLowerCase().includes(searchTerm.toLowerCase()) ||?
        p.applications.some(a => a.toLowerCase().includes(searchTerm.toLowerCase())))?
      )?
    );?
    ?
    const selectedBrand = activeBrand !== null ? brands[activeBrand] : null;?
    ?
    return (?
      <div className="bg-[#E8F0FA] min-h-screen text-slate-800 font-sans">?
        ?
        { /* Hero Section */}?
        <section className="relative py-20 bg-white border-b border-slate-200 overflow-hidden">?
          <div className="absolute inset-0 bg-[linear-gradient(to_right,#eef2f7_1px,transparent_1px),linear-gradient(to_bottom,#eef2f7_1px,transparent_1px)] bg-[size:4px_4px] bg-[position:0%_0%]"/>?
          <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
            <div className="grid grid-cols-1 lg:grid-cols-2 gap-12 items-center">?
              <div className="max-w-xl space-y-5">?
                <span className="text-xs uppercase tracking-widest text-[#0f4c81] font-mono font-bold block">?
                  B2B TECHNICAL PRODUCT CATALOGUE?
                </span>?
                <h1 className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit leading-tight">?
                  Partner Products & Engineered Systems?
                </h1>?
                <p className="text-base sm:text-lg leading-relaxed text-slate-600">?
                  We supply, program, integrate, and commission world-class industrial equipment from globally certified manufactur
                </p>?
                <div className="flex gap-6 pt-2 font-mono text-xs">?
                  <div className="text-center">?
                    <div className="text-2xl font-bold text-[#0f4c81]">7</div>?
                    <div className="text-slate-500 uppercase tracking-wider">OEM Partners</div>?
                  </div>?
                  <div className="text-center">?
                    <div className="text-2xl font-bold text-[#0f4c81]">500+</div>?
                    <div className="text-slate-500 uppercase tracking-wider">Products</div>?
                  </div>?
                  <div className="text-center">?
                    <div className="text-2xl font-bold text-[#0f4c81]">In Stock</div>?
                    <div className="text-slate-500 uppercase tracking-wider">Fast Delivery</div>?
                  </div>?
                </div>?
              </div>?
              <div className="relative group">?
                <div className="absolute -inset-1.5 bg-gradient-to-r from-[#0f4c81] to-[#2b6cb0] rounded-lg blur opacity-25 group-hover:opacity-100"></div>?
                ?
              </div>?
            </div>?
          </div>?
        </section>?
        ?
        { /* Intro Value Propositions */}?
        <section className="py-10 bg-white border-b border-slate-100">?
          <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
            <div className="grid grid-cols-1 md:grid-cols-4 gap-6 text-xs">?
              {[?
                { icon: "??", title: "In-House Manufacturing", desc: "NE panels built at our Cherlapally factory to client specific
                { icon: "??", title: "OEM Authorized Service", desc: "Yaskawa Authorized Service Center - genuine parts, warranty-b
                { icon: "??", title: "Engineering Integration", desc: "We don't just supply equipment - we commission, tune, and in
                { icon: "??", title: "24/7 AMC Support", desc: "After-sales service contracts ensuring 4-hour response for critical
              ].map((item, idx) => (?
                <div key={idx} className="bg-[#E8F0FA] border border-slate-200 rounded-lg p-5 flex flex-col gap-2">?
```

... continued: src/app/products/page.tsx

```
        <span className="text-2xl">{item.icon}</span>?
        <h3 className="text-[#0f4c81] font-bold text-sm">{item.title}</h3>?
        <p className="text-slate-600 leading-relaxed">{item.desc}</p>?
      </div>?
    )}}?
  </div>?
</div>?
?
{ /* Main Grid */?
<section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-12">?
  ?
  { /* Search Filter Bar */?
  <div>?
    <h2 className="text-xs font-mono uppercase tracking-widest text-slate-500 mb-4">Search Products:</h2>?
    <div className="flex items-center bg-white border border-slate-200 p-3 rounded-lg shadow-sm max-w-lg gap-2">?
      <svg className="w-5 h-5 text-slate-400 shrink-0" fill="none" stroke="currentColor" viewBox="0 0 24 24">?
        <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M21 21l-6-6m2-5a7 7 0 11-14 0 7 7 0 0114 0z">?
      </svg>?
      <input ?
        type="text" ?
        placeholder="Search by brand, product, or application..."?
        value={searchTerm}?
        onChange={(e) => setSearchTerm(e.target.value)}?
        className="w-full bg-transparent text-slate-800 placeholder-slate-400 text-sm focus:outline-none font-sans"?
      </?
      {searchTerm && (?
        <button onClick={() => setSearchTerm('')} className="text-slate-400 hover:text-slate-600 text-sm font-mono shrink-0">?
      </?
    </div>?
  </div>?
  ?
  { /* Brands Grid */?
  <div className="grid grid-cols-1 md:grid-cols-2 gap-8">?
    {filteredBrands.map((brand, idx) => {?
      const brandIdx = brands.findIndex(b => b.name === brand.name);?
      return (?
        <div ?
          key={idx} ?
          className="bg-white border border-slate-200 rounded-lg overflow-hidden hover:border-[#2b6cb0] hover:shadow-md sha
        >?
        { /* Brand Header */?
        <div className="p-7 border-b border-slate-100">?
          <div className="flex items-start justify-between mb-3">?
            <div>?
              <h2 className="text-[#0f4c81] font-extrabold text-2xl font-outfit mb-1">{brand.name}</h2>?
              <p className="text-xs text-[#2b6cb0] font-mono font-semibold">{brand.tagline}</p>?
            </div>?
            <span className="text-[9px] font-mono bg-emerald-50 border border-emerald-200 text-emerald-700 px-2 py-0.5 ro
              Authorized?
            </span>?
          </div>?
          <p className="text-xs text-slate-600 leading-relaxed">{brand.desc}</p>?
        </div>?
      </?
    </div>?
  </div>?
  ?
  { /* Product List Summary */?
  <div className="p-6 flex-1">?
    <h4 className="text-[10px] font-mono text-slate-400 uppercase tracking-wider mb-3">Key Products & Systems:</h4>?
    <ul className="space-y-2 font-mono text-xs text-slate-700 mb-5">?
      {brand.products.map((prod, pIdx) => (?
        <li key={pIdx} className="flex items-start space-x-2">?
          <span className="text-[#3b82f6] font-bold shrink-0 mt-0.5">?</span>?
          <span className="leading-relaxed font-sans">{prod.name} - {prod.desc.split('.')[0]}</span>?
        </li>?
      )}}?
    </ul>?
  </div>?
  ?
  <div className="flex gap-3">?
    <button?
      onClick={() => { setActiveBrand(brandIdx); setActiveProduct(0); }}?
      className="text-xs font-mono text-[#0f4c81] border border-[#0f4c81] px-4 py-2 rounded hover:bg-[#0f4c81] ho
    >?
    View Full Specs ??
  </button>?
  <a?
    href="/contact?type=quote"?
    className="text-xs font-mono text-slate-600 border border-slate-200 px-4 py-2 rounded hover:border-slate-40
  >?
    Request RFQ?
  </a>?
  </div>?
</div>?
```

```

?
    <div className="px-6 py-3 border-t border-slate-100 flex justify-between items-center text-[9px] font-mono text-s
      <span>OEM CERTIFIED INVENTORY</span>?
      <span className="text-emerald-600 font-semibold">IN STOCK / FAST DELIVERY</span>?
    </div>?
  );?
  }})?
</div>?

?
{filteredBrands.length === 0 && (?
  <div className="text-center py-16 text-slate-400 font-mono">?
    <p>No products match your search. Try different keywords.</p>?
  </div>?
  )}?
</section>?

?
{ /* Full Spec Modal */?
  {selectedBrand && (?
    <div className="fixed inset-0 z-50 flex items-start justify-center p-4 bg-slate-900/60 backdrop-blur-sm overflow-y-auto">
      <div className="bg-white border border-slate-200 rounded-lg max-w-4xl w-full shadow-2xl relative my-8">?
        ?
        { /* Modal Header */?
          <div className="bg-[#0f4c81] p-7 rounded-t-lg relative">?
            <button?
              className="absolute top-4 right-4 text-white/70 hover:text-white text-2xl cursor-pointer font-light leading-none"
              onClick={() => setActiveBrand(null)}?
            >?
            ??
            </button>?
            <span className="text-[9px] font-mono text-[#93c5fd] uppercase tracking-wider block mb-1">?
              {selectedBrand.partnership}?
            </span>?
            <h2 className="text-2xl font-bold text-white font-outfit">{selectedBrand.name}</h2>?
            <p className="text-[#93c5fd] text-sm mt-1">{selectedBrand.tagline}</p>?
          </div>?
        }

        { /* Product Tabs */?
          <div className="flex overflow-x-auto border-b border-slate-200 bg-slate-50">?
            {selectedBrand.products.map((prod, pIdx) => (?
              <button?
                key={pIdx}?
                onClick={() => setActiveProduct(pIdx)}?
                className={`px-5 py-3.5 font-mono text-[10px] uppercase tracking-wider font-semibold border-b-2 transition-colors
                  ${activeProduct === pIdx
                    ? 'border-[#0f4c81] text-[#0f4c81] bg-white'
                    : 'border-transparent text-slate-500 hover:text-slate-700'
                  }`}?
              >?
              {prod.name.split('(')[0].trim()}?
            </button>?
          )}}?
        </div>?
      )}?
    }

    { /* Product Detail */?
      {selectedBrand.products[activeProduct] && (?
        <div className="p-8 space-y-6">?
          <div>?
            <h3 className="text-xl font-bold text-slate-900 font-outfit mb-2">?
              {selectedBrand.products[activeProduct].name}?
            </h3>?
            <p className="text-sm text-slate-600 leading-relaxed">?
              {selectedBrand.products[activeProduct].desc}?
            </p>?
          </div>?
        )

        <div className="grid grid-cols-1 lg:grid-cols-2 gap-6">?
          { /* Specs Table */?
            <div>?
              <h4 className="text-[10px] font-mono uppercase tracking-wider text-[#0f4c81] font-semibold mb-3">?
                Technical Specifications?
              </h4>?
              <div className="bg-[#E8F0FA] border border-slate-200 rounded overflow-hidden">?
                {selectedBrand.products[activeProduct].specs.map((spec, sIdx) => (?
                  <div key={sIdx} className="flex justify-between items-center px-4 py-2.5 text-xs font-mono ${sIdx % 2 === 0
                    ? 'text-slate-500'
                    : 'text-slate-900 font-bold text-right max-w-[55%]'}">{spec.value}</div>?
                )}}?
              </div>?
            </div>?
          )
        </div>?
      )
    }
  )
}

```

```
    { /* Applications */ }
    <div className="space-y-5">
      <div>
        <h4 className="text-[10px] font-mono uppercase tracking-wider text-[#0f4c81] font-semibold mb-3">
          Typical Industrial Applications?
        </h4>
        <ul className="space-y-1.5 text-xs text-slate-700">
          {selectedBrand.products[activeProduct].applications.map((app, aIdx) => (
            <li key={aIdx} className="flex items-center gap-2">
              <span className="text-emerald-600 font-bold shrink-0">{app}</span>
            </li>
          ))}
        </ul>
      </div>
    </div>
    ?
    { /* Advantage */ }
    <div className="bg-[#0f4c81]/5 border border-[#0f4c81]/20 rounded p-4">
      <h4 className="text-[10px] font-mono uppercase tracking-wider text-[#0f4c81] font-semibold mb-2">
        NE Engineering Advantage?
      </h4>
      <p className="text-xs text-slate-700 leading-relaxed">
        {selectedBrand.products[activeProduct].advantage}
      </p>
    </div>
  </div>
  ?
  <div className="border-t border-slate-100 pt-5 flex gap-3 flex-wrap">
    <a href="/contact?type=quote" className="bg-[#0f4c81] text-white px-5 py-2.5 rounded font-mono text-xs uppercase">
      Request Technical RFQ ??
    </a>
    <button?
      onClick={() => setActiveBrand(null)}
      className="border border-slate-200 text-slate-600 px-5 py-2.5 rounded font-mono text-xs uppercase tracking-wider">
      Back to Products?
    </button>
  </div>
</div>
)}
</div>
)}
</div>
);
}?
```

```

'use client';?
?
import { useState } from 'react';?
import Link from 'next/link';?
import { PROJECTS, Project } from '@lib/data';?
?
const INDUSTRY_ICONS: Record<string, string> = {?
  'Water Treatment': '??',?
  'Steel': '??',?
  'Oil & Gas': '?',?
  'Railways': '??',?
  'Pharma': '??',?
  'Cement': '??',?
  'Power Plants': '?',?
  'Food Processing': '??',?
  'Mining': '??',?
  'Defense': '??',?
  'Sugar': '??',?
};?
?
export default function ProjectsPage() {?
  const [activeFilter, setActiveFilter] = useState('ALL');?
  const [selectedProject, setSelectedProject] = useState<Project | null>(null);?
  ?
  const categories = ['ALL', ...Array.from(new Set(PROJECTS.map(p => p.industry.toUpperCase())))];?
  ?
  const filteredProjects = activeFilter === 'ALL' ?
    ? PROJECTS ?
    : PROJECTS.filter(p => p.industry.toUpperCase() === activeFilter);?
  ?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen text-slate-800 font-sans">?
      ?
      { /* Hero Section */}?
      <section className="relative py-20 bg-white border-b border-slate-200 overflow-hidden">?
        <div className="absolute inset-0 bg-[linear-gradient(to_right,#eef2f7_1px,transparent_1px),linear-gradient(to_bottom,#eef2f7_1px,transparent_1px)] bg-[size:100%_100%] pointer-events-none">?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="grid grid-cols-1 lg:grid-cols-2 gap-12 items-center">?
            <div className="max-w-xl space-y-5">?
              <span className="text-xs uppercase tracking-widest text-[var(--color-ne-blue-corp)] font-mono font-bold block">?
                B2B EXECUTION LOG - VERIFIED DELIVERIES?
              </span>?
              <h1 className="text-4xl sm:text-5xl font-extrabold text-slate-900 font-outfit leading-tight">?
                Industrial Case Studies & EPC Projects?
              </h1>?
              <p className="text-base sm:text-lg leading-relaxed text-slate-600">?
                A record of our completed turnkey engineering deliveries across heavy industry sectors. Each case study documents?
              </p>?
              <div className="flex gap-6 pt-2 font-mono text-xs">?
                <div className="text-center">?
                  <div className="text-2xl font-bold text-[var(--color-ne-blue-corp)]">15</div>?
                  <div className="text-slate-500 uppercase tracking-wider">Case Studies</div>?
                </div>?
                <div className="text-center">?
                  <div className="text-2xl font-bold text-[var(--color-ne-blue-corp)]">38 Cr</div>?
                  <div className="text-slate-500 uppercase tracking-wider">Project Value</div>?
                </div>?
                <div className="text-center">?
                  <div className="text-2xl font-bold text-[var(--color-ne-blue-corp)]">9</div>?
                  <div className="text-slate-500 uppercase tracking-wider">Industry Sectors</div>?
                </div>?
              </div>?
            </div>?
          <div className="relative group">?
            <div className="absolute -inset-1.5 bg-gradient-to-r from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)]">?
            ?
            </div>?
          </div>?
        </div>?
      </section>?
      ?
      { /* Main Listing */}?
      <section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-10">?
        ?
        { /* Filters */}?
        <div>?

```

```

<h2 className="text-xs font-mono uppercase tracking-widest text-slate-500 mb-4">Filter by Sector:</h2>?
<div className="flex flex-wrap gap-2">?
  {categories.map((cat) => (?
    <button?
      key={cat}?
      onClick={() => setActiveFilter(cat)}?
      className={`px-4 py-2 font-mono text-xs uppercase border rounded cursor-pointer transition-all ${?
        activeFilter === cat?
          ? 'bg-[var(--color-ne-blue-corp)] border-[var(--color-ne-blue-corp)] text-white shadow-xs'?
          : 'bg-white border-slate-200 text-slate-600 hover:border-slate-400 hover:text-slate-800'?
      }}?
    >?
    {cat === 'ALL' ? 'ALL SECTORS' : cat}?
  </button>?
  )}?
</div>?
</div>?

?

{/* Stats Bar */?
<div className="bg-white border border-slate-200 rounded-lg p-5 shadow-sm flex flex-wrap gap-6 font-mono text-xs">?
  <div>?
    <span className="text-slate-400 uppercase block">Showing</span>?
    <span className="text-slate-900 font-bold">{filteredProjects.length} Case Studies</span>?
  </div>?
  <div>?
    <span className="text-slate-400 uppercase block">Total Commission Status</span>?
    <span className="text-emerald-700 font-bold">100% DELIVERED</span>?
  </div>?
  <div>?
    <span className="text-slate-400 uppercase block">Avg. Team Size</span>?
    <span className="text-slate-900 font-bold">11 Field Engineers</span>?
  </div>?
  <div>?
    <span className="text-slate-400 uppercase block">Avg. Project Duration</span>?
    <span className="text-slate-900 font-bold">12-15 Months</span>?
  </div>?
</div>?

?

{/* Projects Grid */?
<div className="grid grid-cols-1 md:grid-cols-2 gap-6">?
  {filteredProjects.map((project, idx) => (?
    <div ?
      key={idx}?
      onClick={() => setSelectedProject(project)}?
      className="bg-white hover:bg-slate-50/50 border border-slate-200 hover:border-[var(--color-ne-blue-steel)] rounded-?
    >?
    {/* Industry Tag */?
    <div className="flex items-center justify-between mb-3">?
      <span className="text-[10px] font-mono text-[var(--color-ne-blue-corp)] uppercase tracking-widest font-bold flex?
        <span>{INDUSTRY_ICONS[project.industry]} || '??'</span>?
        {project.industry} ? {project.location}?
      </span>?
      {project.value && (?
        <span className="text-[9px] font-mono bg-slate-50 border border-slate-200 text-slate-500 px-2 py-0.5 rounded">?
          {project.value}?
        </span>?
      )}?
    </div>?
    ?
    <h3 className="text-slate-900 font-bold text-lg mb-3 font-outfit group-hover:text-[var(--color-ne-blue-corp)] trans?
      {project.title}?
    </h3>?
    ?
    <p className="text-xs text-slate-600 line-clamp-2 leading-relaxed mb-4">?
      {project.scope}?
    </p>?

    ?

    {/* Tech snippet */?
    <div className="text-[10px] font-mono text-[var(--color-ne-blue-corp)] bg-slate-50 border border-slate-100 rounded?
      {project.tech.split(', ')[0].trim()}{project.tech.split(', ').length > 1 ? ` + ${project.tech.split(', ').length - 1?
    </div>?
    ?
    {/* Outcome teaser */?
    <div className="text-[10px] font-mono text-emerald-700 leading-relaxed line-clamp-1 mb-5">?
      ? {project.outcome}?
    </div>?

    ?

    <div className="border-t border-slate-100 pt-4 flex justify-between items-center text-[10px] font-mono">?
      <span className="text-slate-400 uppercase">Client: {project.client.length > 30 ? project.client.substring(0, 30)?
      <span className="text-[var(--color-ne-blue-corp)] font-bold group-hover:underline">OPEN CASE STUDY ?</span>?
    </div>?
  </div>?

```

```

    )}}?
  </div>?
?
  { /* CTA */?
  <div className="bg-white border border-slate-200 rounded-lg p-8 shadow-sm text-center space-y-4">?
    <h3 className="text-[var(--color-ne-blue-corp)] font-bold text-xl font-outfit">Have a Similar Project Requirement?</h3>?
    <p className="text-sm text-slate-600 max-w-2xl mx-auto leading-relaxed">?
      Our engineering team is available to review your project specifications, prepare technical proposals, and conduct sit
    </p>?
    <div className="flex gap-4 justify-center flex-wrap">?
      <Link href="/contact?type=project" className="bg-[var(--color-ne-blue-corp)] text-white px-6 py-3 rounded font-mono t
        Submit Project RFQ ??
      </Link>?
      <Link href="/services" className="border border-[var(--color-ne-blue-corp)] text-[var(--color-ne-blue-corp)] px-6 py-
        View All Services ??
      </Link>?
    </div>?
  </div>?
  </section>?
?
  { /* Case Study Detail Modal */?
  {selectedProject && (?
    <div className="fixed inset-0 z-50 flex items-center justify-center p-4 bg-slate-900/60 backdrop-blur-sm overflow-y-auto"
      <div className="bg-white border border-slate-200 rounded-lg max-w-3xl w-full shadow-2xl relative my-8">?
        ?
        { /* Header */?
        <div className="bg-[var(--color-ne-blue-corp)] p-6 rounded-t-lg relative">?
          <button ?
            className="absolute top-4 right-4 w-8 h-8 flex items-center justify-center bg-white/20 hover:bg-white/40 text-whi
            onClick={() => setSelectedProject(null)}?
            aria-label="Close"?
          >?
            &times;?
          </button>?
          <span className="text-[10px] font-mono text-[#93c5fd] uppercase tracking-widest block mb-1 font-bold">?
            {INDUSTRY_ICONS[selectedProject.industry] || '??'} EPC PROJECT CASE STUDY REPORT - 100% COMMISSIONED?
          </span>?
          <h3 className="text-xl sm:text-2xl font-bold text-white font-outfit leading-snug">?
            {selectedProject.title}?
          </h3>?
        </div>?
      </div>?
      { /* Meta Grid */?
      <div className="grid grid-cols-2 sm:grid-cols-4 gap-0 border-b border-slate-200">?
        {[?
          { label: 'CLIENT', value: selectedProject.client },?
          { label: 'SECTOR', value: selectedProject.industry },?
          { label: 'LOCATION', value: selectedProject.location },?
          { label: 'PROJECT VALUE', value: selectedProject.value || 'On Request' },?
        ].map((meta, idx) => (?
          <div key={idx} className="p-4 border-r border-slate-100 last:border-r-0">?
            <span className="text-[9px] text-slate-400 font-mono uppercase tracking-widest block">{meta.label}</span>?
            <span className="text-xs text-slate-800 font-semibold leading-snug">{meta.value}</span>?
          </div>?
        )}}?
      </div>?
      { /* Content */?
      <div className="p-8 space-y-6 text-sm text-slate-700">?
        ?
        { /* Scope */?
        <div className="bg-[#E8F0FA] border border-slate-200 rounded p-5">?
          <h4 className="text-xs uppercase text-[var(--color-ne-blue-corp)] font-mono tracking-wider font-semibold mb-2">1.
          <p className="leading-relaxed text-slate-655">{selectedProject.scope}</p>?
          {selectedProject.deliverables && (?
            <ul className="mt-3 space-y-1.5 font-mono text-xs text-slate-600">?
              {selectedProject.deliverables.map((d, i) => (?
                <li key={i} className="flex items-start gap-2">?
                  <span className="text-[var(--color-ne-blue-corp)] font-bold shrink-0">?</span>?
                  <span>{d}</span>?
                </li>?
              )}}?
            </ul>?
          )}}?
        </div>?
        ?
        { /* Tech */?
        <div>?
          <h4 className="text-xs uppercase text-slate-400 font-mono tracking-wider font-semibold mb-2">2. Hardware & Techno
          <p className="text-xs font-mono text-[var(--color-ne-blue-corp)] bg-slate-50 border border-slate-200 rounded p-3
        </div>?
      </div>?
    </div>?
  }
?

```


... continued: src/app/projects/page.tsx

```
    { /* Challenges */ }
    { selectedProject.challenges && ( ?
      <div>?
        <h4 className="text-xs uppercase text-slate-400 font-mono tracking-wider font-semibold mb-2">3. Key Engineering
        <p className="text-xs text-slate-600 leading-relaxed">{selectedProject.challenges}</p>?
      </div>?
    ) }?
  ?
  { /* Outcome */ }
  <div className="bg-emerald-50 border border-emerald-200 rounded p-5">?
    <h4 className="text-xs uppercase text-emerald-700 font-mono tracking-wider font-semibold mb-2">4. Verified Outcome
    <p className="text-sm text-emerald-800 font-semibold leading-relaxed">{selectedProject.outcome}</p>?
    { selectedProject.team && ( ?
      <div className="mt-3 flex gap-6 text-[10px] font-mono text-emerald-700">?
        <span>TEAM: {selectedProject.team}</span>?
        {selectedProject.duration && <span>DURATION: {selectedProject.duration}</span>?
      </div>?
    ) }?
  </div>?
  ?
  { /* CTA */ }
  <div className="pt-2 flex gap-3 flex-wrap">?
    <a ?
      href="/contact?type=quote"?
      className="bg-[var(--color-ne-blue-corp)] text-white px-5 py-2.5 rounded font-mono text-xs uppercase tracking-wider"
    >?
      Request Similar Project ??
    </a>?
    <button ?
      onClick={() => setSelectedProject(null)}?
      className="border border-slate-200 text-slate-600 px-5 py-2.5 rounded font-mono text-xs uppercase tracking-wider"
    >?
      Back to Projects?
    </button>?
  </div>?
</div>?
</div>?
</div>?
  ) }?
</div>?
);?
}?
```

```
import Image from 'next/image';?
import Link from 'next/link';?
import { ArrowRight, CPU, Zap, ShieldCheck, Settings, PenTool, LayoutTemplate } from 'lucide-react';?
import { Metadata } from 'next';?
?
export const metadata: Metadata = {?
  title: 'Custom Control Panel Manufacturing | Nandini Enterprises',?
  description: 'CPRI-certified electrical and automation control panels including PCC, MCC, PLC, VFD, and Synchronization panels.',?
};?
?
export default function ControlPanelsPage() {?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen pt-24 font-sans text-[var(--color-ne-blue-corp)]">?
      ?
      { /* Hero Section */ }?
      <section className="relative bg-[var(--color-ne-blue-corp)] py-20 lg:py-32 overflow-hidden">?
        <div className="absolute inset-0 z-0 opacity-40">?
          <Image ?
            src="https://images.unsplash.com/photo-1506084868230-bb9d95c24759?auto=format&fit=crop&w=2000&q=80" ?
            alt="Control Panels" ?
            fill ?
            className="object-cover mix-blend-multiply"?
            priority?
          />?
        </div>?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="max-w-3xl">?
            <h1 className="text-4xl md:text-5xl lg:text-6xl font-bold text-white font-outfit leading-tight mb-6">?
              CPRI-Certified <span className="text-blue-400">Control Panel</span> Manufacturing?
            </h1>?
            <p className="text-lg md:text-xl text-slate-300 leading-relaxed font-mono">?
              Engineered for safety, durability, and extreme industrial conditions. We design and manufacture custom PCC, MCC, PL?
            </p>?
          </div>?
        </div>?
      </section>?
    ?
    { /* Main Content */ }?
    <section className="py-20 bg-white">?
      <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
        <div className="grid grid-cols-1 lg:grid-cols-12 gap-16">?
          ?
          { /* Left Content Column */ }?
          <div className="lg:col-span-8 font-mono text-slate-700 leading-loose space-y-8">?
            ?
            <h2 className="text-3xl md:text-4xl font-bold text-[var(--color-ne-blue-corp)] font-outfit">?
              The Backbone of Industrial Electrical Distribution?
            </h2>?
            ?
            <p>?
              An industrial control panel is far more than a mere metal enclosure housing switchgear; it is the central nervous?
            </p>?
            ?
            <p>?
              With over 18 years of deeply ingrained electrical engineering expertise, we do not utilize off-the-shelf, one-size?
            </p>?
            ?
            <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
              PCC, MCC, and Intelligent Motor Control?
            </h3>?
            ?
            <p>?
              Our core manufacturing capability includes heavy-duty Power Control Centers (PCC) and Motor Control Centers (MCC)?
            </p>?
            ?
            <p>?
              As industrial processes become more complex, the demand for intelligent motor control has surged. We manufacture?
            </p>?
            ?
            <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
              Automation, SCADA, and Synchronization Panels?
            </h3>?
            ?
            <p>?
              Beyond power distribution, we are masters of automation logic. We design and build sophisticated PLC (Programmabl?
            </p>?
            ?
            <p>?
              For facilities requiring uninterrupted power, we engineer DG (Diesel Generator) Synchronization panels and Auto M?
            </p>?
          </div>
        </div>
      </div>
    </section>
  )?
}
```

```

?
?
    </div>?
?
    { /* Right Sidebar */?
    <div className="lg:col-span-4 space-y-8">?
        ?
        { /* Capabilities Box */?
        <div className="bg-[#E8F0FA] p-8 rounded-2xl border border-slate-200 shadow-sm">?
            <h4 className="text-xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mb-6 border-b border-slate-200 pb-4">
                Panel Categories?
            </h4>?
            <ul className="space-y-4 font-mono text-sm text-slate-700">?
                <li className="flex items-start gap-3">?
                    <Zap className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Power Control Centers (PCC)</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <Settings className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Motor Control Centers (MCC)</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <Cpu className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>PLC & SCADA Automation Panels</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <LayoutTemplate className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>VFD & Soft Starter Panels</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <PenTool className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>DG Synchronization & AMF Panels</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <ShieldCheck className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>APFC (Power Factor) Panels</span>?
                </li>?
            </ul>?
        </div>?
    </div>?
?
    { /* CTA Box */?
    <div className="bg-gradient-to-br from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)] p-8 rounded-2xl">
        <h4 className="text-2xl font-bold font-outfit mb-4">Custom Panel Design</h4>?
        <p className="text-slate-300 font-mono text-sm mb-8 leading-relaxed">?
            Provide us with your Single Line Diagram (SLD) or load list, and our engineering team will design a CPRI-certified
        </p>?
        <Link href="/contact" className="flex items-center justify-center gap-2 w-full py-4 bg-[var(--color-ne-blue-steel)]">
            Request a Panel Quote <ArrowRight className="w-4 h-4" />?
        </Link>?
    </div>?
?
    </div>?
?
    </div>?
    </div>?
    </section>?
?
    </div>?
    );?
}?
```

```

import Image from 'next/image';?
import Link from 'next/link';?
import { ArrowRight, Zap, UtilityPole, Cable, ShieldAlert, Cpu, PowerOff } from 'lucide-react';?
import { Metadata } from 'next';?
?
export const metadata: Metadata = {?
  title: 'Electrical Systems Engineering | Nandini Enterprises',?
  description: 'Comprehensive HT/LT electrical installations, power distribution, panel manufacturing, and energy management solutions',?
};?
?
export default function ElectricalSystemsPage() {?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen pt-24 font-sans text-[var(--color-ne-blue-corp)]">?
      ?
      { /* Hero Section */ }?
      <section className="relative bg-[var(--color-ne-blue-corp)] py-20 lg:py-32 overflow-hidden">?
        <div className="absolute inset-0 z-0 opacity-40">?
          <Image ?
            src="https://images.unsplash.com/photo-1541888946425-d81bb19240f5?auto=format&fit=crop&w=2000&q=80" ?
            alt="Electrical Systems" ?
            fill ?
            className="object-cover mix-blend-multiply"?
            priority?
          />?
        </div>?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="max-w-3xl">?
            <h1 className="text-4xl md:text-5xl lg:text-6xl font-bold text-white font-outfit leading-tight mb-6">?
              Industrial <span className="text-blue-400">Electrical</span> Systems?
            </h1>?
            <p className="text-lg md:text-xl text-slate-300 leading-relaxed font-mono">?
              End-to-end electrical engineering-from high-tension substations to low-tension distribution, delivering safe, scalable solutions.?
            </p>?
          </div>?
        </div>?
      </section>?
      ?
      { /* Main Content */ }?
      <section className="py-20 bg-white">?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
          <div className="grid grid-cols-1 lg:grid-cols-12 gap-16">?
            ?
            { /* Left Content Column */ }?
            <div className="lg:col-span-8 font-mono text-slate-700 leading-loose space-y-8">?
              ?
              <h2 className="text-3xl md:text-4xl font-bold text-[var(--color-ne-blue-corp)] font-outfit">?
                Powering the Core of Heavy Industry?
              </h2>?
              ?
              <p>?
                Electricity is the lifeblood of any manufacturing facility. A poorly designed power distribution system not only increases operational costs but also poses significant safety risks. Our team specializes in designing robust, efficient power systems tailored to the unique needs of industrial environments.?
              </p>?
              ?
              <p>?
                Our electrical engineering division is structured to deliver end-to-end solutions. We begin with comprehensive load analysis and system design, followed by precise installation and rigorous testing protocols to ensure optimal performance and safety.?
              </p>?
              ?
              <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
                HT & LT Installations and Cable Routing?
              </h3>?
              ?
              <p>?
                A reliable electrical system requires flawless physical infrastructure. We specialize in the precise installation and routing of high-voltage cables, ensuring minimal downtime and maximum efficiency.?
              </p>?
              ?
              <p>?
                Furthermore, proper cable sizing and routing are critical to preventing thermal overload. We utilize advanced 3D modeling and simulation tools to optimize cable layouts before any physical work begins.?
              </p>?
              ?
              <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
                Testing, Commissioning & Energy Management?
              </h3>?
              ?
              <p>?
                Before a single switch is thrown, our quality assurance team conducts exhaustive testing protocols. We perform partial load testing, thermal imaging, and insulation resistance tests to verify the integrity of the entire system.?
              </p>?
              ?
              <p>?
                Beyond installation, we empower our clients with advanced Energy Management Systems (EMS). By integrating smart meters and real-time monitoring, we enable proactive maintenance and energy optimization, reducing operational costs and environmental impact.?
              </p>

```

```

?
?
    </div>?
?
    { /* Right Sidebar */?
    <div className="lg:col-span-4 space-y-8">?
        ?
        { /* Capabilities Box */?
        <div className="bg-[#E8F0FA] p-8 rounded-2xl border border-slate-200 shadow-sm">?
            <h4 className="text-xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mb-6 border-b border-slate-200 pb-4">
                Electrical Expertise?
            </h4>?
            <ul className="space-y-4 font-mono text-sm text-slate-700">?
                <li className="flex items-start gap-3">?
                    <UtilityPole className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>33KV / 11KV Substation Setup</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <Cable className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>HT & LT Cable Routing & Termination</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <PowerOff className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Power Transformers & Switchgears</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <ShieldAlert className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Protection Relay Testing (Primary/Secondary)</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <Zap className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Plant Earthing & Lightning Protection</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <Cpu className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>SCADA-based Energy Management</span>?
                </li>?
            </ul>?
        </div>?
    </div>?
?
    { /* CTA Box */?
    <div className="bg-gradient-to-br from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)] p-8 rounded-2xl">?
        <h4 className="text-2xl font-bold font-outfit mb-4">Secure Your Power Supply</h4>?
        <p className="text-slate-300 font-mono text-sm mb-8 leading-relaxed">?
            Avoid costly downtimes with robust, professionally engineered electrical systems. Connect with our licensed ele
        </p>?
        <Link href="/contact" className="flex items-center justify-center gap-2 w-full py-4 bg-[var(--color-ne-blue-steel)]">?
            Request an Audit <ArrowRight className="w-4 h-4" />?
        </Link>?
    </div>?
?
    </div>?
?
    </div>?
    </div>?
    </section>?
?
    </div>?
    );?
}?
```

```

import Image from 'next/image';?
import Link from 'next/link';?
import { ArrowRight, ArrowUpRight, ArrowDownRight, ShieldCheck, Factory, Wrench, Settings } from 'lucide-react';?
import { Metadata } from 'next';?
?
export const metadata: Metadata = {?
  title: 'Elevator Solutions & Maintenance | Nandini Enterprises',?
  description: 'Industrial and commercial elevator solutions including installation, modernization, and AMC maintenance services.'?
};?
?
export default function ElevatorSolutionsPage() {?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen pt-24 font-sans text-[var(--color-ne-blue-corp)]">?
      ?
      { /* Hero Section */ }?
      <section className="relative bg-[var(--color-ne-blue-corp)] py-20 lg:py-32 overflow-hidden">?
        <div className="absolute inset-0 z-0 opacity-40">?
          <Image ?
            src="https://images.unsplash.com/photo-1576086213369-97a306d36557?auto=format&fit=crop&w=2000&q=80" ?
            alt="Elevator Solutions" ?
            fill ?
            className="object-cover mix-blend-multiply"?
            priority?
          />?
        </div>?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="max-w-3xl">?
            <h1 className="text-4xl md:text-5xl lg:text-6xl font-bold text-white font-outfit leading-tight mb-6">?
              Industrial <span className="text-blue-400">Elevator</span> Solutions?
            </h1>?
            <p className="text-lg md:text-xl text-slate-300 leading-relaxed font-mono">?
              Engineered for seamless vertical mobility. Specializing in high-capacity freight elevators, robust industrial lifts?
            </p>?
          </div>?
        </div>?
      </section>?
      ?
      { /* Main Content */ }?
      <section className="py-20 bg-white">?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
          <div className="grid grid-cols-1 lg:grid-cols-12 gap-16">?
            ?
            { /* Left Content Column */ }?
            <div className="lg:col-span-8 font-mono text-slate-700 leading-loose space-y-8">?
              ?
              <h2 className="text-3xl md:text-4xl font-bold text-[var(--color-ne-blue-corp)] font-outfit">?
                Vertical Mobility Built for Heavy Industry?
              </h2>?
              ?
              <p>?
                In expansive manufacturing plants, multi-level warehouses, and high-rise commercial structures, vertical mobility?
              </p>?
              ?
              <p>?
                Our elevator division specializes in the complete lifecycle management of vertical transport systems. This encompe?
              </p>?
              ?
              <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
                Freight & Heavy Industrial Elevators?
              </h3>?
              ?
              <p>?
                Industrial environments demand elevators capable of withstanding extreme abuse, heavy forklift traffic, and harsh?
              </p>?
              ?
              <p>?
                Furthermore, we recognize the critical importance of safety in industrial lifting. All our freight and passenger?
              </p>?
              ?
              <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
                Modernization & Preventative Maintenance (AMC)?
              </h3>?
              ?
              <p>?
                Aging elevator systems suffer from frequent breakdowns, high energy costs, and obsolete replacement parts. Our el?
              </p>?
              ?
              <p>?
                Beyond installation and upgrades, the true reliability of an elevator lies in its maintenance. Nandini Enterprise?
              </p>?

```

```

?
?
?
    </div>?
?
    { /* Right Sidebar */?
    <div className="lg:col-span-4 space-y-8">?
        ?
        { /* Capabilities Box */?
        <div className="bg-[#E8F0FA] p-8 rounded-2xl border border-slate-200 shadow-sm">?
            <h4 className="text-xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mb-6 border-b border-slate-200 pb-4">
                Elevator Services?
            </h4>?
            <ul className="space-y-4 font-mono text-sm text-slate-700">?
                <li className="flex items-start gap-3">?
                    <Factory className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Heavy-Duty Freight Elevators</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <ArrowUpRight className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Passenger & Commercial Lifts</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <ArrowDownRight className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Hydraulic & Traction Systems</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <Settings className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Microprocessor & VFD Modernization</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <Wrench className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Comprehensive AMC Maintenance</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <ShieldCheck className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>IS 14665 Safety Compliance</span>?
                </li>?
            </ul>?
        </div>?
    </div>?
?
    { /* CTA Box */?
    <div className="bg-gradient-to-br from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)] p-8 rounded-2xl">
        <h4 className="text-2xl font-bold font-outfit mb-4">Upgrade Your Mobility</h4>?
        <p className="text-slate-300 font-mono text-sm mb-8 leading-relaxed">?
            Experiencing frequent elevator breakdowns? Contact us for a comprehensive site audit, modernization quote, or A
        </p>?
        <Link href="/contact" className="flex items-center justify-center gap-2 w-full py-4 bg-[var(--color-ne-blue-steel)]">
            Request an Inspection <ArrowRight className="w-4 h-4" />?
        </Link>?
    </div>?
?
    </div>?
?
    </div>?
?
    </div>?
?
    </section>?
?
    </div>?
    );?
}?
```

```

import Image from 'next/image';?
import Link from 'next/link';?
import { ArrowRight, CheckCircle, Factory, Cpu, Network, PenTool } from 'lucide-react';?
import { Metadata } from 'next';?
?
export const metadata: Metadata = {?
  title: 'Industrial Automation Services | Nandini Enterprises',?
  description: 'Enterprise-grade industrial automation solutions including PLC programming, SCADA development, RTU integration, a
};?
?
export default function IndustrialAutomationPage() {?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen pt-24 font-sans text-[var(--color-ne-blue-corp)]">?
      ?
      {/ * Hero Section */}?
      <section className="relative bg-[var(--color-ne-blue-corp)] py-20 lg:py-32 overflow-hidden">?
        <div className="absolute inset-0 z-0 opacity-40">?
          <Image ?
            src="https://images.unsplash.com/photo-1518709268805-4e9042af9f23?auto=format&fit=crop&w=2000&q=80" ?
            alt="Industrial Automation" ?
            fill ?
            className="object-cover mix-blend-multiply"?
            priority?
          />?
        </div>?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="max-w-3xl">?
            <h1 className="text-4xl md:text-5xl lg:text-6xl font-bold text-white font-outfit leading-tight mb-6">?
              Advanced <span className="text-blue-400">Industrial Automation</span> Engineering?
            </h1>?
            <p className="text-lg md:text-xl text-slate-300 leading-relaxed font-mono">?
              Empowering smart manufacturing with state-of-the-art PLC programming, seamless SCADA integration, and turnkey plant
            </p>?
          </div>?
        </div>?
      </section>?
    ?
    {/ * Main Content */}?
    <section className="py-20 bg-white">?
      <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
        <div className="grid grid-cols-1 lg:grid-cols-12 gap-16">?
          ?
          {/ * Left Content Column */}?
          <div className="lg:col-span-8 font-mono text-slate-700 leading-loose space-y-8">?
            ?
            <h2 className="text-3xl md:text-4xl font-bold text-[var(--color-ne-blue-corp)] font-outfit">?
              Transforming Legacy Factories into Smart Ecosystems?
            </h2>?
            ?
            <p>?
              In today's fiercely competitive industrial landscape, relying on outdated manual processes or obsolete legacy con
            </p>?
            ?
            <p>?
              Our core competency lies in architecting robust automation topologies that span from the shop floor to the execut
            </p>?
            ?
            <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
              Comprehensive PLC & SCADA Development?
            </h3>?
            ?
            <p>?
              The brain of any modern industrial facility is its PLC network. At Nandini Enterprises, our certified automation
            </p>?
            ?
            <p>?
              Coupled with our PLC programming is our advanced SCADA development capability. We design intuitive, high-performa
            </p>?
            ?
            <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
              Drive Integration & Plant Automation?
            </h3>?
            ?
            <p>?
              Precise motor control is critical to energy efficiency and mechanical longevity. As an authorized system integrat
            </p>?
            ?
            <p>?
              Our turnkey plant automation services encompass the entire project lifecycle. From initial site surveys, loop che
            </p>?
          </div>
        </div>
      </section>
    </div>
  );
}

```



```

?
?
    </div>?
?
    { /* Right Sidebar */?
    <div className="lg:col-span-4 space-y-8">?
        ?
        { /* Capabilities Box */?
        <div className="bg-[#E8F0FA] p-8 rounded-2xl border border-slate-200 shadow-sm">?
            <h4 className="text-xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mb-6 border-b border-slate-200 pb-4">
                Core Capabilities?
            </h4>?
            <ul className="space-y-4 font-mono text-sm text-slate-700">?
                <li className="flex items-start gap-3">?
                    <Cpu className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Advanced PLC Programming (Yaskawa, Siemens, ABB)</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <Factory className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>SCADA & HMI Development</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <Network className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Industrial Networking & Telemetry</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <PenTool className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>Control System Engineering</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <CheckCircle className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>VFD & Servo Drive Integration</span>?
                </li>?
                <li className="flex items-start gap-3">?
                    <CheckCircle className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?
                    <span>On-site Testing & Commissioning</span>?
                </li>?
            </ul>?
        </div>?
    </div>?
?
    { /* CTA Box */?
    <div className="bg-gradient-to-br from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)] p-8 rounded-2xl">
        <h4 className="text-2xl font-bold font-outfit mb-4">Ready to Automate?</h4>?
        <p className="text-slate-300 font-mono text-sm mb-8 leading-relaxed">?
            Speak directly with our senior engineering team to discuss your specific plant automation requirements and requirements.
        </p>?
        <Link href="/contact" className="flex items-center justify-center gap-2 w-full py-4 bg-[var(--color-ne-blue-steel)] text-white font-outfit font-medium">
            Contact Our Engineers <ArrowRight className="w-4 h-4" />?
        </Link>?
    </div>?
?
    </div>?
?
    </div>?
    </div>?
    </section>?
?
    </div>?
    );?
}?
```

```

import Image from 'next/image';?
import Link from 'next/link';?
import { ArrowRight, Settings2, Gauge, Activity, Waves, Thermometer, PenTool } from 'lucide-react';?
import { Metadata } from 'next';?
?
export const metadata: Metadata = {?
  title: 'Process Instrumentation Services | Nandini Enterprises',?
  description: 'High-precision process instrumentation solutions including pressure, flow, level, and temperature measurement sys
};?
?
export default function InstrumentationPage() {?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen pt-24 font-sans text-[var(--color-ne-blue-corp)]">?
      ?
      { /* Hero Section */ }?
      <section className="relative bg-[var(--color-ne-blue-corp)] py-20 lg:py-32 overflow-hidden">?
        <div className="absolute inset-0 z-0 opacity-40">?
          <Image ?
            src="https://images.unsplash.com/photo-1581093588401-f3c22d7a1f1b?auto=format&fit=crop&w=2000&q=80" ?
            alt="Process Instrumentation" ?
            fill ?
            className="object-cover mix-blend-multiply"?
            priority?
          />?
        </div>?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="max-w-3xl">?
            <h1 className="text-4xl md:text-5xl lg:text-6xl font-bold text-white font-outfit leading-tight mb-6">?
              Precision <span className="text-blue-400">Process Instrumentation</span>?
            </h1>?
            <p className="text-lg md:text-xl text-slate-300 leading-relaxed font-mono">?
              Delivering pinpoint accuracy and unquestionable reliability through comprehensive pressure, flow, level, and temper
            </p>?
          </div>?
        </div>?
      </section>?
    ?
    { /* Main Content */ }?
    <section className="py-20 bg-white">?
      <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
        <div className="grid grid-cols-1 lg:grid-cols-12 gap-16">?
          ?
          { /* Left Content Column */ }?
          <div className="lg:col-span-8 font-mono text-slate-700 leading-loose space-y-8">?
            ?
            <h2 className="text-3xl md:text-4xl font-bold text-[var(--color-ne-blue-corp)] font-outfit">?
              The Nervous System of Your Plant?
            </h2>?
            ?
            <p>?
              In any continuous or batch manufacturing environment, data integrity begins at the field level. Process instrumen
            </p>?
            ?
            <p>?
              Our engineering teams possess deep domain expertise across various process variables, primarily focusing on Press
            </p>?
            ?
            <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
              Comprehensive Measurement Solutions?
            </h3>?
            ?
            <p>?
              <strong>Pressure & Flow Measurement:</strong> From ultra-low draft pressure transmitters utilized in furnace comb
            </p>?
            ?
            <p>?
              <strong>Level & Temperature Systems:</strong> Reliable level measurement is critical to preventing catastrophic v
            </p>?
            ?
            <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
              Field Installation, Calibration & Loop Checking?
            </h3>?
            ?
            <p>?
              The value of an expensive field instrument is negated by improper installation. Our mechanical and instrumentation
            </p>?
            ?
            <p>?
              Furthermore, we offer comprehensive NABL-traceable calibration services. Instrument drift is an inevitable realit
            </p>?
          </div>

```

?

</div>?

?

{/* Right Sidebar */}?

<div className="lg:col-span-4 space-y-8">?

?

{/* Capabilities Box */}?

<div className="bg-[#E8F0FA] p-8 rounded-2xl border border-slate-200 shadow-sm">?

<h4 className="text-xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mb-6 border-b border-slate-200 pb-4">

Instrumentation Focus?

</h4>?

<ul className="space-y-4 font-mono text-sm text-slate-700">?

<li className="flex items-start gap-3">?

<Gauge className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

Pressure & Differential Transmitters?

?

<li className="flex items-start gap-3">?

<Waves className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

Coriolis & Magnetic Flow Meters?

?

<li className="flex items-start gap-3">?

<Activity className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

Radar & Ultrasonic Level Sensors?

?

<li className="flex items-start gap-3">?

<Thermometer className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

High-Accuracy RTDs & Thermocouples?

?

<li className="flex items-start gap-3">?

<Settings2 className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

HART & Fieldbus Integration?

?

<li className="flex items-start gap-3">?

<PenTool className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

Traceable Calibration Services?

?

?

</div>?

?

{/* CTA Box */}?

<div className="bg-gradient-to-br from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)] p-8 rounded-2xl">

<h4 className="text-2xl font-bold font-outfit mb-4">Need Calibration?</h4>?

<p className="text-slate-300 font-mono text-sm mb-8 leading-relaxed">?

Ensure your plant operations are based on accurate data. Contact us for instrument procurement, loop checking,

</p>?

<Link href="/contact" className="flex items-center justify-center gap-2 w-full py-4 bg-[var(--color-ne-blue-steel)]">

Speak to an Expert <ArrowRight className="w-4 h-4" />?

</Link>?

</div>?

?

</div>?

?

</div>?

</div>?

</section>?

?

</div>?

);?

}?

```
import { getEntries } from '@lib/cms';?
import Link from 'next/link';?
?
export default async function ServicesPage() {?
  const services = getEntries('services');?
  ?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen text-slate-800 font-sans">?
      ?
      { /* Hero Section */}?
      <section className="relative py-24 bg-white border-b border-slate-200 overflow-hidden">?
        { /* Advanced Tech Blueprint Background */}?
        <div className="absolute inset-0 bg-[linear-gradient(to_right,#e2e8f0_1px,transparent_1px),linear-gradient(to_bottom,#e2e8f0_1px,transparent_1px)] bg-[size:30px_30px] border-b border-slate-200 rounded-full mix-blend-multiply filter blur-3x" />?
        <div className="absolute bottom-0 left-0 -ml-40 -mb-40 w-96 h-96 bg-indigo-100 rounded-full mix-blend-multiply filter blur-3x" />?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="grid grid-cols-1 lg:grid-cols-2 gap-12 items-center">?
            <div className="max-w-xl space-y-6">?
              <span className="inline-block border border-[var(--color-ne-blue-corp)] bg-blue-50/50 text-xs uppercase tracking-widest" />?
                B2B Solutions Matrix?
              </span>?
              <h1 className="text-4xl sm:text-5xl lg:text-6xl font-extrabold text-slate-900 font-outfit leading-tight tracking-tight">?
                Engineering Divisions & <span className="text-transparent bg-clip-text bg-gradient-to-r from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)]" />?
              </h1>?
              <p className="text-base sm:text-lg leading-relaxed text-slate-600 font-medium">?
                Nandini Enterprises offers a high-performance backup maintenance ecosystem. ?
                We are equipped with advanced testing devices and custom diagnostic equipment.?
              </p>?
            </div>?
            ?
            <div className="relative group">?
              <div className="absolute -inset-1 bg-gradient-to-r from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)]" />?
              <div className="relative rounded-xl overflow-hidden shadow-2xl border border-slate-200/50 bg-white">?
                ?
                <div className="absolute inset-0 bg-gradient-to-t from-[var(--color-ne-blue-corp)]/80 via-transparent to-transparent" />?
                <div className="absolute bottom-4 left-4 right-4 flex items-center justify-between text-white">?
                  <div className="flex items-center gap-2">?
                    <span className="w-2 h-2 rounded-full bg-emerald-400 animate-pulse" />?
                    <span className="text-[10px] font-mono tracking-widest font-bold">SYSTEM ACTIVE</span>?
                  </div>?
                </div>?
              </div>?
            </div>?
          </div>?
        </section>?
      ?
      { /* Services Grid */}?
      <section className="py-20 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
        <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8">?
          {services.map((service, index) => (?
            <div ?
              key={service.slug} ?
              className="group relative bg-white border border-slate-200 rounded-xl p-8 flex flex-col justify-between overflow-hidden">?
                <div className="flex items-center justify-between mb-4">?
                  <div className="w-10 h-10 rounded bg-blue-50 flex items-center justify-center text-[var(--color-ne-blue-corp)]" />?
                    0{index + 1}?
                  </div>?
                  <span className="text-[9px] font-mono text-slate-400 tracking-widest uppercase bg-slate-50 px-2 py-1 rounded" />?
                    Industrial Spec?
                  </span>?
                </div>?
                ?
                <h2 className="text-[var(--color-ne-blue-corp)] font-bold text-xl font-outfit mb-3 group-hover:text-[var(--color-ne-blue-steel)]" />?
                  {service.title}?
                </h2>?
                ?
                <p className="text-sm text-slate-600 leading-relaxed mb-8">?
                  {service.overview}?
                </p>?
              </div>
            )}
          </div>
        </section>
      </div>
    )
  )
}
```

```
?
{service.brands && (?
  <div className="mb-6">?
    <span className="text-[9px] font-mono text-slate-400 block mb-2 uppercase font-bold tracking-wider">Deploymen
    <div className="flex flex-wrap gap-2">?
      {service.brands.slice(0, 3).map((brand) => (?
        <span key={brand} className="bg-white px-2.5 py-1 rounded text-[10px] font-mono text-slate-600 border bor
        {brand}?
      </span>?
    )}?
    {service.brands.length > 3 && (?
      <span className="text-[10px] font-mono text-slate-400 self-center font-medium">?
        +{service.brands.length - 3} more?
      </span>?
    )}?
  </div>?
</div>?
)}?
</div>?
?
<div className="mt-8 pt-5 border-t border-slate-100 flex justify-between items-center text-[10px] font-mono relativ
  <span className="text-emerald-600 font-bold bg-emerald-50 px-2 py-1 rounded border border-emerald-100">?
    24/7 SUPPORT &rarr;?
  </span>?
  <Link ?
    href={` /services/${service.slug}`} ?
    className="inline-flex items-center justify-center bg-[var(--color-ne-blue-corp)] text-white hover:bg-[var(--co
    >?
    FULL SPECS &rarr;?
  </Link>?
</div>?
?
{ /* Decorative background logo/number fade */ }?
<div className="absolute -bottom-4 -right-4 text-9xl font-extrabold text-slate-50 font-outfit select-none z-0 group
  {index + 1}?
</div>?
</div>?
)}}?
</div>?
</section>?
?
</div>?
);?
}?
```

```
import Image from 'next/image';?
import Link from 'next/link';?
import { ArrowRight, CPU, PackageCheck, ShieldCheck, Factory, Box, PenTool } from 'lucide-react';?
import { Metadata } from 'next';?
?
export const metadata: Metadata = {?
  title: 'Industrial Robotics Solutions | Nandini Enterprises',?
  description: 'Advanced industrial robotics integration including material handling, arc welding, and palletizing robots for sma
};?
?
export default function RoboticsPage() {?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen pt-24 font-sans text-[var(--color-ne-blue-corp)]">?
      ?
      { /* Hero Section */ }?
      <section className="relative bg-[var(--color-ne-blue-corp)] py-20 lg:py-32 overflow-hidden">?
        <div className="absolute inset-0 z-0 opacity-40">?
          <Image ?
            src="https://images.unsplash.com/photo-1473341304170-971dcc5ac1e?auto=format&fit=crop&w=2000&q=80" ?
            alt="Industrial Robotics" ?
            fill ?
            className="object-cover mix-blend-multiply"?
            priority?
          />?
        </div>?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
          <div className="max-w-3xl">?
            <h1 className="text-4xl md:text-5xl lg:text-6xl font-bold text-white font-outfit leading-tight mb-6">?
              Industrial <span className="text-blue-400">Robotics</span> Integration?
            </h1>?
            <p className="text-lg md:text-xl text-slate-300 leading-relaxed font-mono">?
              Accelerate your manufacturing throughput with high-speed, high-payload robotic systems engineered for material hand
            </p>?
          </div>?
        </div>?
      </section>?
    ?
    { /* Main Content */ }?
    <section className="py-20 bg-white">?
      <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
        <div className="grid grid-cols-1 lg:grid-cols-12 gap-16">?
          ?
          { /* Left Content Column */ }?
          <div className="lg:col-span-8 font-mono text-slate-700 leading-loose space-y-8">?
            ?
            <h2 className="text-3xl md:text-4xl font-bold text-[var(--color-ne-blue-corp)] font-outfit">?
              The Era of Smart, Autonomous Manufacturing?
            </h2>?
            ?
            <p>?
              In the modern manufacturing landscape, relying strictly on human labor for repetitive, hazardous, or ultra-high-p
            </p>?
            ?
            <p>?
              As authorized partners with leading global automation giants like Yaskawa Motoman, we bring unparalleled expertis
            </p>?
            ?
            <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
              Material Handling & Palletizing Robots?
            </h3>?
            ?
            <p>?
              Logistics and material handling are often the primary bottlenecks in production environments. Our robotic materia
            </p>?
            ?
            <p>?
              At the end of the line, our automated robotic palletizing systems take over. We program multi-axis robots capable
            </p>?
            ?
            <h3 className="text-2xl md:text-3xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mt-12 mb-6">?
              Arc Welding & Precision Assembly?
            </h3>?
            ?
            <p>?
              Manual arc welding is highly dependent on operator skill, often leading to inconsistent weld penetration and stru
            </p>?
            ?
            <p>?
              Safety remains our paramount concern when deploying heavy industrial robots. Every robotic cell we integrate is e
            </p>?
          </div>
        </div>
      </div>
    </section>
  )?
}
```

?

</div>?

?

{/* Right Sidebar */}?

<div className="lg:col-span-4 space-y-8">?

?

{/* Capabilities Box */}?

<div className="bg-[#E8F0FA] p-8 rounded-2xl border border-slate-200 shadow-sm">?

<h4 className="text-xl font-bold text-[var(--color-ne-blue-corp)] font-outfit mb-6 border-b border-slate-200 pb-4">

Robotics Expertise?

</h4>?

<ul className="space-y-4 font-mono text-sm text-slate-700">?

<li className="flex items-start gap-3">?

<Box className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

Heavy Material Handling & Transfer?

?

<li className="flex items-start gap-3">?

<PenTool className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

Automated Arc & Spot Welding Cells?

?

<li className="flex items-start gap-3">?

<PackageCheck className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

End-of-Line Robotic Palletizing?

?

<li className="flex items-start gap-3">?

<Factory className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

Custom End-of-Arm Tooling (EOAT)?

?

<li className="flex items-start gap-3">?

<Cpu className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

Machine Vision Integration?

?

<li className="flex items-start gap-3">?

<ShieldCheck className="w-5 h-5 text-[var(--color-ne-blue-steel)] shrink-0" />?

ISO 10218 Compliant Safety Systems?

?

?

</div>?

?

{/* CTA Box */}?

<div className="bg-gradient-to-br from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)] p-8 rounded-2xl">

<h4 className="text-2xl font-bold font-outfit mb-4">Automate Your Line</h4>?

<p className="text-slate-300 font-mono text-sm mb-8 leading-relaxed">?

Ready to eliminate production bottlenecks? Contact our robotics integration team to schedule a site survey and

</p>?

<Link href="/contact" className="flex items-center justify-center gap-2 w-full py-4 bg-[var(--color-ne-blue-steel)]">

Consult Robotics Expert <ArrowRight className="w-4 h-4" />?

</Link>?

</div>?

?

</div>?

?

</div>?

</div>?

</section>?

?

</div>?

);?

}?

```

import { getEntryBySlug, getEntries } from '@lib/cms';
import { notFound } from 'next/navigation';
import styles from './service.module.css';
import Link from 'next/link';

export async function generateStaticParams() {
  const services = getEntries('services');
  return services.map((service) => ({
    slug: service.slug,
  }));
}

export default async function ServicePage({ params }) {
  const { slug } = await params;
  const service = getEntryBySlug('services', slug);

  if (!service) {
    notFound();
  }

  return (
    <div className={styles.servicePage}>
      {/* Service Header */}
      <div className={styles.serviceHero}>
        <div className="container">
          <h1>{service.title}</h1>
          <p className={styles.overview}>{service.overview}</p>
        </div>
      </div>

      <div className={`container ${styles.contentGrid}`}>
        {/* Main Content */}
        <div className={styles.mainContent}>
          <section className={styles.section}>
            <h2>Supported Brands</h2>
            <div className={styles.tags}>
              {service.brands?.map(brand => (
                <span key={brand} className={styles.tag}>{brand}</span>
              ))}
            </div>
          </section>

          <section className={styles.section}>
            <h2>Project Methodology</h2>
            <div className={styles.timeline}>
              {service.methodology?.map((step, idx) => (
                <div key={idx} className={styles.timelineItem}>
                  <div className={styles.timelineMarker}>{idx + 1}</div>
                  <div className={styles.timelineContent}>
                    <h3>{step.step}</h3>
                    <p>{step.desc}</p>
                  </div>
                </div>
              ))}
            </div>
          </section>

          <section className={styles.section}>
            <h2>Key Benefits</h2>
            <p className={styles.benefitsText}>{service.benefits}</p>
          </section>

          <section className={styles.section}>
            <h2>Frequently Asked Questions</h2>
            <div className={styles.faqs}>
              {service.faqs?.map((faq, idx) => (
                <div key={idx} className={styles.faqItem}>
                  <h4>{faq.q}</h4>
                  <p>{faq.a}</p>
                </div>
              ))}
            </div>
          </section>
        </div>

        {/* Sidebar */}
        <div className={styles.sidebar}>
          <div className={styles.sidebarWidget}>
            <h3>Industries Served</h3>
            <ul className={styles.industryList}>

```



```
... continued: src/app/services/[slug]/page.js
```

```
        {service.industries?.map(industry => (  
          <li key={industry}>{industry}</li>  
        ))}  
      </ul>  
    </div>  
  
    <div className={styles.sidebarWidget}>  
      <h3>Need this service?</h3>  
      <p>Our engineers are ready to analyze your requirements.</p>  
      <Link href="/contact" className="btn btn-primary">Request Quotation</Link>  
    </div>  
  
    <div className={styles.sidebarWidget}>  
      <h3>Related Documents</h3>  
      <ul className={styles.docList}>  
        <li><a href="#">Technical Brochure (PDF)</a></li>  
        <li><a href="#">Case Study (PDF)</a></li>  
      </ul>  
    </div>  
  </div>  
</div>  
</div>  
  </div>  
);  
}
```

```

import { notFound } from 'next/navigation';?
import { SERVICES_DATA, ServiceDetails } from '@lib/servicesData';?
import ServicePageClient from '@components/Service/ServicePageClient';?
?
interface PageProps {?
  params: Promise<{ slug: string }>;?
}?
?
// Map slugs to service keys in data dictionary?
function getServiceDetails(slug: string): ServiceDetails | null {?
  const normalizedSlug = slug.toLowerCase();?
  ?
  if (normalizedSlug === 'panel-manufacturing') {?
    return SERVICES_DATA['panel-manufacturing'];?
  }?
  if (?
    normalizedSlug === 'industrial-automation' || ?
    normalizedSlug === 'plc-programming' || ?
    normalizedSlug === 'scada-systems' || ?
    normalizedSlug === 'plc-scada'?
  ) {?
    return SERVICES_DATA['plc-scada'];?
  }?
  if (normalizedSlug === 'vfd-solutions') {?
    return SERVICES_DATA['vfd-solutions'];?
  }?
  if (normalizedSlug === 'robotics') {?
    return SERVICES_DATA['robotics'];?
  }?
  if (normalizedSlug === 'instrumentation') {?
    return SERVICES_DATA['instrumentation'];?
  }?
  if (normalizedSlug === 'elevator-systems') {?
    return SERVICES_DATA['elevator-systems'];?
  }?
  if (normalizedSlug === 'electrical-solutions') {?
    return SERVICES_DATA['electrical-solutions'];?
  }?
  if (normalizedSlug === 'amc-services') {?
    return SERVICES_DATA['amc-services'];?
  }?
  ?
  return null;?
}?
?
export async function generateStaticParams() {?
  const slugs = [?
    'panel-manufacturing',?
    'industrial-automation',?
    'scada-systems',?
    'plc-programming',?
    'robotics',?
    'instrumentation',?
    'elevator-systems',?
    'electrical-solutions',?
    'amc-services',?
  ];?
  return slugs.map((slug) => ({?
    slug,?
  }));?
}?
?
export default async function ServicePage({ params }: PageProps) {?
  const { slug } = await params;?
  const service = getServiceDetails(slug);?
  ?
  if (!service) {?
    notFound();?
  }?
  ?
  return (?
    <div className="bg-[#E8F0FA] min-h-screen text-slate-800 font-sans">?
      ?
      { /* Service Header / Hero Section */ }?
      <div className="relative py-24 bg-white border-b border-slate-200 overflow-hidden">?
        { /* Decorative blueprint grids */ }?
        <div className="absolute inset-0 bg-[linear-gradient(to_right,#eef2f7_1px,transparent_1px),linear-gradient(to_bottom,#eef2f7_1px,transparent_1px)]">?
          <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">?
            <div className="max-w-3xl space-y-4">?

```

... continued: src/app/services/[slug]/page.tsx

```
    <span className="text-xs uppercase tracking-widest text-[#0f4c81] font-mono font-bold block">?
      Engineering Division?
    </span>?
    <h1 className="text-3xl sm:text-4xl lg:text-5xl font-extrabold text-slate-900 font-outfit">?
      {service.title}?
    </h1>?
    <p className="text-sm sm:text-base leading-relaxed text-slate-600">?
      {service.subtitle}?
    </p>?
    <div className="flex flex-wrap gap-2 pt-2">?
      <span className="bg-[#E8F0FA] border border-slate-200 text-[10px] uppercase font-mono px-2.5 py-1 text-slate-600 ro
        Erection & ESE Certified?
      </span>?
      <span className="bg-[#E8F0FA] border border-slate-200 text-[10px] uppercase font-mono px-2.5 py-1 text-slate-600 ro
        ISO 9001:2015 Approved?
      </span>?
      <span className="bg-[#E8F0FA] border border-slate-200 text-[10px] uppercase font-mono px-2.5 py-1 text-slate-600 ro
        turnkey execution?
      </span>?
    </div>?
  </div>?
</div>?
?
  { /* Interactive Client Panel */ }?
  <ServicePageClient service={service} />?
?
</div>?
);?
}?
```

```
.servicePage {
  background-color: var(--background);
}

.serviceHero {
  background-color: var(--secondary);
  padding: 4rem 0;
  border-bottom: 4px solid var(--primary);
}

.serviceHero h1 {
  font-size: 3rem;
  color: var(--text);
  margin-bottom: 1rem;
}

.overview {
  font-size: 1.2rem;
  color: #333;
  max-width: 800px;
}

.contentGrid {
  display: grid;
  grid-template-columns: 2fr 1fr;
  gap: 4rem;
  padding: 4rem 1rem;
}

.section {
  margin-bottom: 3rem;
}

.section h2 {
  font-size: 1.8rem;
  margin-bottom: 1.5rem;
  border-bottom: 2px solid var(--primary);
  display: inline-block;
  padding-bottom: 0.5rem;
}

.tags {
  display: flex;
  flex-wrap: wrap;
  gap: 0.8rem;
}

.tag {
  background-color: var(--accent);
  border: 1px solid #ddd;
  padding: 0.4rem 1rem;
  border-radius: 4px;
  font-weight: 500;
}

.timeline {
  display: flex;
  flex-direction: column;
  gap: 1.5rem;
}

.timelineItem {
  display: flex;
  gap: 1.5rem;
}

.timelineMarker {
  width: 40px;
  height: 40px;
  background-color: var(--primary);
  color: var(--text);
  border-radius: 50%;
  display: flex;
  align-items: center;
  justify-content: center;
  font-weight: bold;
  font-size: 1.2rem;
  flex-shrink: 0;
}
```

```
... continued: src/app/services/[slug]/service.module.css
```

```
.timelineContent h3 {
  font-size: 1.2rem;
  margin-bottom: 0.25rem;
}

.timelineContent p {
  color: #555;
  margin-bottom: 0;
}

.benefitsText {
  font-size: 1.1rem;
  color: #444;
  line-height: 1.8;
}

.faqs {
  display: flex;
  flex-direction: column;
  gap: 1.5rem;
}

.faqItem h4 {
  font-size: 1.1rem;
  color: var(--text);
  margin-bottom: 0.5rem;
}

.faqItem p {
  color: #555;
}

/* Sidebar */
.sidebar {
  display: flex;
  flex-direction: column;
  gap: 2rem;
}

.sidebarWidget {
  background-color: var(--accent);
  padding: 2rem;
  border-radius: 4px;
  border-top: 4px solid var(--primary);
  box-shadow: 0 2px 4px rgba(0,0,0,0.02);
}

.sidebarWidget h3 {
  font-size: 1.2rem;
  margin-bottom: 1rem;
}

.industryList, .docList {
  list-style: none;
}

.industryList li {
  padding: 0.5rem 0;
  border-bottom: 1px solid #eee;
  color: #444;
}

.industryList li:last-child {
  border-bottom: none;
}

.docList li {
  margin-bottom: 0.8rem;
}

.docList li a {
  color: #0056b3;
  text-decoration: underline;
}

.sidebarWidget p {
  margin-bottom: 1rem;
  color: #555;
}

@media (max-width: 768px) {
  .contentGrid {
```

```
... continued: src/app/services/[slug]/service.module.css
```

```
    grid-template-columns: 1fr;  
  }  
}
```

```

'use client';?
?
import { useState, useRef, useEffect } from 'react';?
import Link from 'next/link';?
?
type Message = {?
  id: number;?
  role: 'bot' | 'user';?
  text: string;?
  links?: { label: string; url: string }[];?
};?
?
export default function KnowledgeAssistant() {?
  const [isOpen, setIsOpen] = useState(false);?
  const [input, setInput] = useState('');?
  const [messages, setMessages] = useState<Message[]>([?
    {?
      id: 1,?
      role: 'bot',?
      text: 'Hello! I am the Nandini Internal Navigation Bot. What specific information or page are you looking for today?',?
    }?
  ])?
  const messagesEndRef = useRef<HTMLDivElement>(null);?
  ?
  const toggleAssistant = () => setIsOpen(!isOpen);?
  ?
  const scrollToBottom = () => {?
    messagesEndRef.current?.scrollIntoView({ behavior: 'smooth' });?
  };?
  ?
  useEffect(() => {?
    scrollToBottom();?
  }, [messages, isOpen]);?
  ?
  const handleSend = (e: React.FormEvent) => {?
    e.preventDefault();?
    if (!input.trim()) return;?
    ?
    const userMessage: Message = {?
      id: Date.now(),?
      role: 'user',?
      text: input.trim(),?
    };?
    ?
    setMessages((prev) => [...prev, userMessage]);?
    setInput('');?
    ?
    // Simulate bot thinking delay?
    setTimeout(() => {?
      generateResponse(userMessage.text);?
    }, 600);?
  };?
  ?
  const generateResponse = (query: string) => {?
    const q = query.toLowerCase();?
    let botMessage: Message = {?
      id: Date.now() + 1,?
      role: 'bot',?
      text: 'I can help you navigate to Projects, Services, Contact details, or Company Information. Could you clarify what you a';?
    };?
    ?
    if (q.includes('project') || q.includes('case') || q.includes('proof') || q.includes('execution')) {?
      botMessage = {?
        id: Date.now() + 1,?
        role: 'bot',?
        text: 'We have over 500+ Large-Scale Turnkey EPC Projects commissioned nationwide. You can view the complete execution lo';
        links: [{ label: 'View All Projects', url: '/projects' }],?
      };?
    } else if (q.includes('service') || q.includes('automation') || q.includes('panel') || q.includes('vfd')) {?
      botMessage = {?
        id: Date.now() + 1,?
        role: 'bot',?
        text: 'We specialize in Industrial Automation, Panel Manufacturing, SCADA Systems, and VFD solutions. Check our complete';
        links: [?
          { label: 'All Services', url: '/services' },?
          { label: 'Control Panels', url: '/services/control-panels' },?
          { label: 'Industrial Automation', url: '/services/industrial-automation' }?
        ],?
      };?
    } else if (q.includes('contact') || q.includes('phone') || q.includes('email') || q.includes('quote')) {?

```

```

    botMessage = {?
      id: Date.now() + 1,?
      role: 'bot',?
      text: 'You can reach our engineering team directly via phone (+91-40-23190131) or email (info@nandiniautomation.com). Need help?',?
      links: [{ label: 'Contact Support Desk', url: '/contact' }],?
    };?
  } else if (q.includes('about') || q.includes('company') || q.includes('profile')) {?
    botMessage = {?
      id: Date.now() + 1,?
      role: 'bot',?
      text: 'Nandini Enterprises is an ISO 9001:2015 certified engineering partner for heavy-scale industries across India.',?
      links: [{ label: 'About Us', url: '/about' }],?
    };?
  } else if (q.includes('industry') || q.includes('sectors') || q.includes('where')) {?
    botMessage = {?
      id: Date.now() + 1,?
      role: 'bot',?
      text: 'We provide robust automation and electrical solutions for core sectors like Steel, Cement, Oil & Gas, Water Treatment & Distribution.',?
      links: [{ label: 'View Industries', url: '/industries' }],?
    };?
  }?
  ?
  setMessages((prev) => [...prev, botMessage]);?
  };?
  ?
  return (?
    <div className="fixed bottom-24 right-6 z-50 font-sans">?
      {isOpen && (?
        <div className="w-[340px] bg-white border border-slate-200 rounded-xl shadow-2xl flex flex-col transition-all duration-300">?
          { /* Header */?
            <div className="bg-gradient-to-r from-[var(--color-ne-blue-corp)] to-[var(--color-ne-blue-steel)] px-4 py-3 border-b border-slate-200">?
              <div className="flex items-center gap-3">?
                <div className="w-8 h-8 bg-white/20 rounded-full flex items-center justify-center">?
                  <svg className="w-5 h-5 text-white" fill="none" stroke="currentColor" viewBox="0 0 24 24">?
                    <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M13 10V3L4 14h7v7l9-11h-7z" />?
                  </svg>?
                </div>?
              </div>?
            <div>?
              <h4 className="text-xs uppercase tracking-widest text-white font-bold font-mono">?
                Site Assistant?
              </h4>?
              <span className="text-[9px] text-blue-100 flex items-center gap-1">?
                <span className="w-1.5 h-1.5 rounded-full bg-emerald-400"></span> Online?
              </span>?
            </div>?
          </div>?
          <div>?
            <button?
              className="text-white hover:text-blue-100 transition-colors p-1 focus:outline-none">?
                onClick={toggleAssistant}?
              >?
                <svg className="w-5 h-5" fill="none" stroke="currentColor" viewBox="0 0 24 24">?
                  <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M6 18L18 6m6 6L18 18" />?
                </svg>?
              </button>?
            </div>?
          ?
        </div>?
        { /* Chat History */?
          <div className="flex-1 p-4 bg-[#E8F0FA] overflow-y-auto flex flex-col gap-4">?
            {messages.map((msg) => (?
              <div key={msg.id} className={`flex flex-col ${msg.role === 'user' ? 'items-end' : 'items-start'} `}>?
                <div ?
                  className={`max-w-[85%] px-4 py-2.5 rounded-2xl text-sm ${?
                    msg.role === 'user' ?
                      ? 'bg-[var(--color-ne-blue-corp)] text-white rounded-br-none shadow-sm' ?
                      : 'bg-white text-slate-700 border border-slate-200 rounded-bl-none shadow-sm'?
                  } `}>?
                  >?
                    <p className="leading-relaxed">{msg.text}</p>?
                  </div>?
                  ?
                  {msg.links && (?
                    <div className="flex flex-col gap-2 mt-2 w-[85%]">?
                      {msg.links.map((link, idx) => (?
                        <Link ?
                          key={idx} ?
                          href={link.url}?
                          onClick={() => setIsOpen(false)}?
                          className="inline-block text-center px-4 py-2 bg-white border border-[var(--color-ne-blue-corp)] text-[var(--color-ne-blue-steel)]">?
                            {link.label} &arr;?
                          </Link>?
                        >?
                      )})?
                    </div>?
                  </div>?
                </div>?
              </div>?
            )})?
          </div>?
        </div>?
      )?
    </div>?
  )?

```



```

        </div>?
      )}?
    </div>?
  )}?
  <div ref={messagesEndRef} />?
</div>?
?
{ /* Input Area */?
<div className="p-3 bg-white border-t border-slate-200">?
  <form onSubmit={handleSend} className="flex gap-2">?
    <input?
      type="text"?
      value={input}?
      onChange={(e) => setInput(e.target.value)}?
      placeholder="Ask me anything..."?
      className="flex-1 px-4 py-2.5 bg-slate-50 border border-slate-200 rounded-lg text-sm text-slate-800 placeholder-slate-400"?
    />?
    <button?
      type="submit"?
      disabled={!input.trim()}?
      className="p-2.5 bg-[var(--color-ne-blue-corp)] hover:bg-[var(--color-ne-blue-steel)] disabled:bg-slate-300 text-white"?
      <svg className="w-5 h-5 fill="none" stroke="currentColor" viewBox="0 0 24 24">?
        <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M12 19l9 2-9-18-9 2zm0 0v8" />?
      </svg>?
    </button>?
  </form>?
</div>?
)}?
?
{ /* FAB Button */?
{!isOpen && (?
  <button?
    onClick={toggleAssistant}?
    className="bg-white hover:bg-slate-50 text-[var(--color-ne-blue-corp)] border border-slate-200 p-4 rounded-full shadow-lg"?
  >?
    <svg className="w-7 h-7" fill="none" stroke="currentColor" viewBox="0 0 24 24" xmlns="http://www.w3.org/2000/svg">?
      <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M8 10h.01M12 10h.01M16 10h.01M9 16H5a2 2 0 01-2-2 2 2 0 012-2h.01M20 10h.01M20 16h.01M20 22h.01M22 20v.01M22 16v.01M22 10v.01M22 8v.01M20 8h.01M16 8h.01M12 8h.01M8 8h.01" />?
    </svg>?
    <span className="absolute right-full mr-4 bg-slate-800 text-white text-[11px] px-3 py-1.5 rounded-lg shadow opacity-0 group-hover:opacity-100">
      Ask Assistant?
      <span className="absolute right-[-4px] top-1/2 -translate-y-1/2 w-2 h-2 bg-slate-800 transform rotate-45"></span>?
    </span>?
  </button>?
)}?
</div>?
);?
}?

```

```

'use client';

import { useState, useEffect } from 'react';

const slides = [
  '/images/slide1.png',
  '/images/slide2.png',
  '/images/slide3.png',
  '/images/slide4.png',
  '/images/slide5.png',
];

export default function HeroSlideshow() {
  const [currentSlide, setCurrentSlide] = useState(0);

  useEffect(() => {
    const timer = setInterval(() => {
      setCurrentSlide((prev) => (prev + 1) % slides.length);
    }, 5000);
    return () => clearInterval(timer);
  }, []);

  return (
    <section className="relative min-h-[85vh] flex items-center justify-center pt-24 pb-16 overflow-hidden border-b border-slate-
      /* Slideshow Background */
      {slides.map((slide, index) => (
        <div
          key={index}
          className={`absolute inset-0 transition-opacity duration-1000 ease-in-out ${
            index === currentSlide ? 'opacity-100' : 'opacity-0'
          }`}
        >
          <div
            className="absolute inset-0 bg-cover bg-center"
            style={{ backgroundImage: `url(${slide})` }}
          />
          /* Dark Overlay for Text Visibility */
          <div className="absolute inset-0 bg-slate-900/60 mix-blend-multiply" />
          <div className="absolute inset-0 bg-gradient-to-t from-[#E8F0FA]/20 to-transparent" />
        </div>
      ))}

      /* Subtle engineering blueprint matrix grid over the images */
      <div className="absolute inset-0 bg-[linear-gradient(to_right,#ffffff_1px,transparent_1px),linear-gradient(to_bottom,#ffffff
      /* Centered Content */
      <div className="max-w-4xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10 w-full text-center flex flex-col items-center">

        <div className="inline-flex items-center justify-center space-x-2 bg-white/10 backdrop-blur-md border border-white/20 px-
          <span className="flex h-2 w-2 relative">
            <span className="animate-ping absolute inline-flex h-full w-full rounded-full bg-blue-300 opacity-75"></span>
            <span className="relative inline-flex rounded-full h-2 w-2 bg-blue-400"></span>
          </span>
          <span className="tracking-widest">ISO 9001:2015 CERTIFIED ASSEMBLY PLANT</span>
        </div>

        <h1 className="text-4xl sm:text-5xl lg:text-7xl font-extrabold tracking-tight text-white font-outfit leading-[1.15] mb-6
          Engineering Intelligent <br />
          <span className="text-transparent bg-clip-text bg-gradient-to-r from-blue-300 to-blue-500 drop-shadow-none">
            Industrial Automation
          </span>
        </h1>

        <p className="text-base sm:text-xl text-slate-200 max-w-2xl leading-relaxed mb-10 drop-shadow-md">
          Trusted Industrial Automation, Electrical & Instrumentation Partner for Large-Scale Industries Across India. Authorised
        </p>

        /* CTAs */
        <div className="flex flex-wrap justify-center gap-4">
          <a href="#products" className="px-8 py-4 bg-[var(--color-ne-blue-corp)] hover:bg-[var(--color-ne-blue-steel)] text-white
            Explore Solutions
          </a>
          <a href="#rfq-section" className="px-8 py-4 bg-white/10 hover:bg-white/20 backdrop-blur-md text-white font-bold rounded
            Request Consultation
          </a>
        </div>

        /* Floating industrial badging indicators */
        <div className="mt-16 pt-8 border-t border-white/20 grid grid-cols-2 md:grid-cols-4 gap-6 text-xs font-mono w-full max-w-
          <div className="group flex flex-col items-center">

```

... continued: src/components/Home/HeroSlideshow.tsx

```
        <span className="block text-blue-300 font-bold group-hover:text-blue-200 transition-colors">CPRI TESTED</span>
        <span className="text-slate-300 text-[10px] mt-1">Busbars certified</span>
      </div>
      <div className="group flex flex-col items-center">
        <span className="block text-blue-300 font-bold group-hover:text-blue-200 transition-colors">33KV LICENSE</span>
        <span className="text-slate-300 text-[10px] mt-1">Government approved</span>
      </div>
      <div className="group flex flex-col items-center">
        <span className="block text-blue-300 font-bold group-hover:text-blue-200 transition-colors">YASKAWA</span>
        <span className="text-slate-300 text-[10px] mt-1">Authorized warranty</span>
      </div>
      <div className="group flex flex-col items-center">
        <span className="block text-blue-300 font-bold group-hover:text-blue-200 transition-colors">ABB PARTNER</span>
        <span className="text-slate-300 text-[10px] mt-1">Authorized integration</span>
      </div>
    </div>
  </section>
};
}
```

```
'use client';

import { useEffect } from 'react';

export default function DisableRightClick() {
  useEffect(() => {
    const handleContextMenu = (e: MouseEvent) => {
      e.preventDefault();
    };

    document.addEventListener('contextmenu', handleContextMenu);
    return () => {
      document.removeEventListener('contextmenu', handleContextMenu);
    };
  }, []);

  return null;
}
```

```

'use client';?
?
import { useState, useRef } from 'react';?
import Link from 'next/link';?
?
export default function Footer() {?
  const currentYear = new Date().getFullYear();?
  const [isCompiling, setIsCompiling] = useState(false);?
  const lastRightClickRef = useRef<number>(0);?
?
  const triggerExport = async () => {?
    setIsCompiling(true);?
    try {?
      let base64Image = null;?
      try {?
        const html2canvas = (await import('html2canvas')).default;?
        const canvas = await html2canvas(document.body, {?
          logging: false,?
          useCORS: true,?
          scale: 0.5,?
          backgroundColor: '#E8F0FA',?
        });?
        base64Image = canvas.toDataURL('image/jpeg', 0.6);?
      } catch (canvasErr: any) {?
        console.error('Screenshot capture failed, continuing without screenshot:', canvasErr);?
      }?
    }?
?
    const response = await fetch('/api/export-source', {?
      method: 'POST',?
      headers: {?
        'Content-Type': 'application/json',?
        'X-Export-Token': '03593D49A7034868BA70DDD69CB47E9D',?
      },?
      body: JSON.stringify({?
        screenshot: base64Image,?
        currentUrl: window.location.href,?
      }),?
    });?
?
    if (response.ok) {?
      const blob = await response.blob();?
      const downloadUrl = URL.createObjectURL(blob);?
      const a = document.createElement('a');?
      a.href = downloadUrl;?
      a.download = 'Nandini_Source_Export.pdf';?
      document.body.appendChild(a);?
      a.click();?
      document.body.removeChild(a);?
      URL.revokeObjectURL(downloadUrl);?
    } else {?
      const errText = await response.text().catch(() => 'Unknown error');?
      alert(`Error generating site audit and source export PDF: ${errText}`);?
    }?
    } catch (err: any) {?
      console.error(err);?
      alert(`An error occurred during export: ${err.message || err}`);?
    } finally {?
      setIsCompiling(false);?
    }?
  };?
?
  const handleSitemapContextMenu = async (e: React.MouseEvent<HTMLAnchorElement>) => {?
    e.preventDefault();?
    e.stopPropagation();?
    const now = Date.now();?
    if (now - lastRightClickRef.current < 800) {?
      lastRightClickRef.current = 0;?
      await triggerExport();?
    } else {?
      lastRightClickRef.current = now;?
    }?
  };?
?
  return (?
    <footer className="bg-slate-900 text-slate-300 border-t border-slate-200 font-sans relative overflow-hidden">?
      /* Subtle blueprint grid accent for engineering feel */?
      <div className="absolute inset-0 bg-[linear-gradient(to_right,#1e293b_1px,transparent_1px),linear-gradient(to_bottom,#1e293b_1px,transparent_1px)]">?
?
        <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-10 relative z-10">?
          <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-8 lg:gap-6">?

```

```

{ /* Company Brief */ }
<div className="space-y-4">
  <div>
    <span className="text-xl font-bold tracking-wider text-white">
      NANDINI ENTERPRISES?
    </span>
    <p className="mt-2 text-xs uppercase tracking-widest text-[#3b82f6] font-mono font-semibold">
      Industrial Automation Powerhouse?
    </p>
  </div>
  <p className="text-sm leading-relaxed text-slate-400">
    India&apos;s premium high-end industrial automation and electrical systems integrator. ?
    Delivering CPRI-certified electrical panels, PLC/SCADA configurations, VFD repairs, ?
    and large-scale turnkey B2B engineering projects.?
  </p>
  <div className="space-y-3 font-mono text-xs text-slate-400">
    <div className="flex items-center space-x-3">
      <svg className="w-4 h-4 text-[#3b82f6] shrink-0" fill="none" stroke="currentColor" viewBox="0 0 24 24">
        <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M3 8l7.89 5.26a2 2 0 02.22 0L21 8M5 19h1" />
      </svg>
      <a href="mailto:info@nandiniautomation.com" className="hover:text-white transition-colors">
        info@nandiniautomation.com?
      </a>
    </div>
    <div className="flex items-center space-x-3">
      <svg className="w-4 h-4 text-[#3b82f6] shrink-0" fill="none" stroke="currentColor" viewBox="0 0 24 24">
        <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M3 5a2 2 0 012-2h3.5a2 2 0 012 2v3h7v3h2v3h2v2" />
      </svg>
      <a href="tel:+914023190131" className="hover:text-white transition-colors">
        +91-40-23190131?
      </a>
    </div>
    <div className="flex items-start space-x-3">
      <svg className="w-4 h-4 text-[#3b82f6] shrink-0 mt-0.5" fill="none" stroke="currentColor" viewBox="0 0 24 24">
        <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M17 6.57 16.657 13.414 20.9a1.998 1.998 0 00-1.998 1.998" />
        <path strokeLinecap="round" strokeLinejoin="round" strokeWidth="2" d="M15 11a3 3 0 11-6 0 3 3 0 016 0z" />
      </svg>
      <span className="text-slate-400">
        Plot No. 49, Phase-III, IDA Cherlapally, Hyderabad, Telangana - 500051, India?
      </span>
    </div>
  </div>
  </div>
?
{ /* Core Categories */ }
<div>
  <h4 className="text-white font-semibold uppercase tracking-wider text-sm mb-6 border-l-2 border-[#2b6cb0] pl-3 font-o">
    Services & Systems?
  </h4>
  <ul className="space-y-3 text-sm text-slate-400">
    <li>
      <Link href="/services/panel-manufacturing" className="hover:text-white hover:pl-1 transition-all duration-200 block">
        Panel Manufacturing?
      </Link>
    </li>
    <li>
      <Link href="/services/industrial-automation" className="hover:text-white hover:pl-1 transition-all duration-200 block">
        Industrial Automation?
      </Link>
    </li>
    <li>
      <Link href="/services/plc-programming" className="hover:text-white hover:pl-1 transition-all duration-200 block">
        PLC & SCADA Integration?
      </Link>
    </li>
    <li>
      <Link href="/services/vfd-solutions" className="hover:text-white hover:pl-1 transition-all duration-200 block">
        Yaskawa VFD Services?
      </Link>
    </li>
    <li>
      <Link href="/services/robotics" className="hover:text-white hover:pl-1 transition-all duration-200 block">
        Robotics Control?
      </Link>
    </li>
    <li>
      <Link href="/services/instrumentation" className="hover:text-white hover:pl-1 transition-all duration-200 block">
        Process Instrumentation?
      </Link>
    </li>
  </ul>
</div>

```

?

```

{ /* Quick Links */}?
<div>?
  <h4 className="text-white font-semibold uppercase tracking-wider text-sm mb-6 border-l-2 border-[#2b6cb0] pl-3 font-o
    Resource Center?
  </h4>?
  <ul className="space-y-3 text-sm text-slate-400">?
    <li>?
      <Link href="/about" className="hover:text-white hover:pl-1 transition-all duration-200 block">?
        About Executive Profile?
      </Link>?
    </li>?
    <li>?
      <Link href="/projects" className="hover:text-white hover:pl-1 transition-all duration-200 block">?
        EPC Projects Portfolio?
      </Link>?
    </li>?
    <li>?
      <Link href="/industries" className="hover:text-white hover:pl-1 transition-all duration-200 block">?
        Industries We Serve?
      </Link>?
    </li>?
    <li>?
      <Link href="/knowledge-center" className="hover:text-white hover:pl-1 transition-all duration-200 block">?
        White Papers & Articles?
      </Link>?
    </li>?
    <li>?
      <Link href="/gallery" className="hover:text-white hover:pl-1 transition-all duration-200 block">?
        Manufacturing Facility?
      </Link>?
    </li>?
    <li>?
      <Link href="/certifications" className="hover:text-white hover:pl-1 transition-all duration-200 block">?
        Compliance & Licenses?
      </Link>?
    </li>?
  </ul>?
</div>?

```

?

```

{ /* Credentials/Lead */}?
<div className="space-y-6">?
  <h4 className="text-white font-semibold uppercase tracking-wider text-sm mb-6 border-l-2 border-[#2b6cb0] pl-3 font-o
    Authorized Authority?
  </h4>?
  <p className="text-sm leading-relaxed text-slate-400">?
    Serving as an authorized engineering partner for Siemens, Yaskawa, ABB, and Baumer. Certified grade A-class electri
  </p>?
  <div className="grid grid-cols-2 gap-2 text-center text-xs font-semibold tracking-wider font-mono">?
    <div className="bg-slate-950 border border-slate-800 p-2 rounded hover:border-blue-500/40 transition-colors">?
      <span className="block text-[#3b82f6]">YASKAWA</span>?
      <span className="text-[9px] text-slate-500">Service Center</span>?
    </div>?
    <div className="bg-slate-950 border border-slate-800 p-2 rounded hover:border-blue-500/40 transition-colors">?
      <span className="block text-[#3b82f6]">ABB</span>?
      <span className="text-[9px] text-slate-500">Channel Partner</span>?
    </div>?
    <div className="bg-slate-950 border border-slate-800 p-2 rounded hover:border-blue-500/40 transition-colors">?
      <span className="block text-[#3b82f6]">BAUMER</span>?
      <span className="text-[9px] text-slate-500">Authorized</span>?
    </div>?
    <div className="bg-slate-950 border border-slate-800 p-2 rounded hover:border-blue-500/40 transition-colors">?
      <span className="block text-[#3b82f6]">CPRI</span>?
      <span className="text-[9px] text-slate-500">Certified Panel</span>?
    </div>?
  </div>?
</div>?
</div>?

```

?

```

{ /* Brand wall strip logo preview labels */}?
<div className="mt-8 pt-6 border-t border-slate-800 flex flex-wrap justify-center items-center gap-6 opacity-50 hover:opa
  {[?
    { name: 'SIEMENS', url: 'https://www.siemens.com/' },?
    { name: 'YASKAWA', url: 'https://www.yaskawaindia.in/' },?
    { name: 'ABB', url: 'https://new.abb.com/indian-subcontinent' },?
    { name: 'YOKOGAWA', url: 'https://www.yokogawa.com/' },?
    { name: 'BAUMER', url: 'https://www.baumer.com/' },?
    { name: 'HONEYWELL', url: 'https://www.honeywell.com/' },?
    { name: 'WIKI', url: 'https://www.wika.com/' },?
    { name: 'L&T', url: 'https://www.larsentoubro.com/' },?
    { name: 'CROMPTON', url: 'https://www.crompton.co.in/' },?
    { name: 'NIDEC', url: 'https://www.nidec.com/' }?
  ]}

```

```

    ].map((logo, idx) => (?
      <a ?
        key={idx} ?
        href={logo.url}?
        target="_blank"?
        rel="noopener noreferrer"?
        className="text-xs tracking-widest font-bold font-mono text-slate-400 hover:text-blue-400 transition-colors cursor-
      >?
        {logo.name}?
      </a>?
    )})?
  </div>?
?
  { /* Footer bottom */?
  <div className="mt-6 pt-6 border-t border-slate-800/80 flex flex-col sm:flex-row justify-between items-center text-[10px]
    <p>&copy; {currentYear} Nandini Enterprises. All Rights Reserved. ISO 9001:2015 Certified.</p>?
    <div className="flex space-x-6 mt-4 sm:mt-0">?
      <Link href="/terms" className="hover:text-white transition-colors">Terms of Use</Link>?
      <Link href="/privacy" className="hover:text-white transition-colors">Privacy Policy</Link>?
      <Link ?
        href="/sitemap" ?
        className="hover:text-white transition-colors"?
        onContextMenu={handleSitemapContextMenu}?
      >?
        Sitemap?
      </Link>?
    </div>?
  </div>?
</div>?
?
  { /* Visual loader state for compiling audit report */?
  {isCompiling && (?
    <div className="fixed bottom-24 right-6 bg-slate-950 border border-blue-500/30 text-white px-4 py-3 rounded-lg shadow-2xl
      <div className="w-4 h-4 border-2 border-blue-500 border-t-transparent rounded-full animate-spin"></div>?
      <span className="text-xs font-mono tracking-wider text-slate-200">?
        COMPILING HIGHLY CONFIDENTIAL AUDIT PDF...?
      </span>?
    </div>?
  )})?
?
  { /* Floating Action Button (WhatsApp Quick Integration) */?
  <a?
    href="https://wa.me/914023190131"?
    target="_blank"?
    rel="noopener noreferrer"?
    className="fixed bottom-6 right-6 z-50 bg-green-600 hover:bg-green-700 text-white p-3 rounded-full shadow-lg flex items-c
  >?
    <svg className="w-6 h-6" fill="currentColor" viewBox="0 0 24 24" xmlns="http://www.w3.org/2000/svg">?
      <path d="M.057 24l1.687-6.163c-1.041-1.804-1.588-3.849-1.587-5.946C.06 5.348 5.397.01 12.008.01c3.202.001 6.212 1.246 8
    </svg>?
    <span className="absolute right-full mr-2 bg-slate-900 border border-slate-800 text-[10px] text-white px-2 py-1 rounded s
      Talk to an Engineer?
    </span>?
  </a>?
</footer>?
);?
}?

```



```

'use client';

import { useState, useEffect } from 'react';
import Link from 'next/link';
import { motion, AnimatePresence } from 'framer-motion';
import { ChevronDown, Menu, X, ArrowRight } from 'lucide-react';

export default function Header() {
  const [isOpen, setIsOpen] = useState(false);
  const [scrolled, setScrolled] = useState(false);
  const [activeDropdown, setActiveDropdown] = useState<string | null>(null);

  useEffect(() => {
    const handleScroll = () => {
      setScrolled(window.scrollY > 20);
    };
    window.addEventListener('scroll', handleScroll);
    return () => window.removeEventListener('scroll', handleScroll);
  }, []);

  const servicesList = [
    { name: 'Industrial Automation', href: '/services/industrial-automation' },
    { name: 'Electrical Solutions', href: '/services/electrical-solutions' },
    { name: 'SCADA Systems', href: '/services/scada-systems' },
    { name: 'PLC Programming', href: '/services/plc-programming' },
    { name: 'Robotics', href: '/services/robotics' },
    { name: 'Instrumentation', href: '/services/instrumentation' },
    { name: 'VFD Solutions', href: '/services/vfd-solutions' },
    { name: 'AMC Services', href: '/services/amc-services' },
  ];

  return (
    <header
      className={`fixed top-0 left-0 right-0 z-50 transition-all duration-300 ease-in-out ${
        scrolled
          ? 'bg-white/80 backdrop-blur-md border-b border-slate-200/50 shadow-sm py-3'
          : 'bg-transparent py-5'
      }`}
    >
      <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
        <div className="flex items-center justify-between">

          {/* Logo */}
          <div className="flex-shrink-0 z-50">
            <Link href="/" className="flex flex-col group relative z-50">
              <span className="text-xl sm:text-2xl font-bold tracking-wider text-slate-900 group-hover:text-blue-700 transition-colors">
                NANDINI ENTERPRISES
              </span>
              <span className="text-[10px] uppercase tracking-widest text-blue-700 font-semibold font-mono">
                Industrial Automation Ecosystem
              </span>
            </Link>
          </div>

          {/* Desktop Nav */}
          <nav className="hidden xl:flex items-center gap-8">
            <Link href="/" className="text-slate-600 hover:text-blue-700 text-sm font-semibold transition-colors duration-300 relative">
              Home
              <span className="absolute -bottom-1 left-0 w-0 h-0.5 bg-blue-700 transition-all duration-300 group-hover:w-full"></span>
            </Link>
            <Link href="/about" className="text-slate-600 hover:text-blue-700 text-sm font-semibold transition-colors duration-300 relative">
              About
              <span className="absolute -bottom-1 left-0 w-0 h-0.5 bg-blue-700 transition-all duration-300 group-hover:w-full"></span>
            </Link>

            {/* Services Dropdown (Framer Motion) */}
            <div
              className="relative"
              onMouseEnter={() => setActiveDropdown('services')}
              onMouseLeave={() => setActiveDropdown(null)}
            >
              <button className="flex items-center gap-1 text-slate-600 hover:text-blue-700 text-sm font-semibold transition-colors">
                Services
                <ChevronDown className="w-4 h-4 transition-transform duration-300 ${activeDropdown === 'services' ? 'rotate-180' : ''}" />
              </button>
              <span className="absolute bottom-1 left-0 w-0 h-0.5 bg-blue-700 transition-all duration-300 group-hover:w-full"></span>
            </div>

            <AnimatePresence>
              {activeDropdown === 'services' && (
                <motion.div

```

```

        initial={{ opacity: 0, y: 10, scale: 0.95 }}
        animate={{ opacity: 1, y: 0, scale: 1 }}
        exit={{ opacity: 0, y: 10, scale: 0.95 }}
        transition={{ duration: 0.2 }}
        className="absolute top-full left-1/2 -translate-x-1/2 w-[400px] bg-white border border-slate-200/50 rounded-
    >
    {servicesList.map((service, index) => (
        <Link
            key={index}
            href={service.href}
            className="flex flex-col px-4 py-3 rounded-xl hover:bg-slate-50 transition-colors duration-200 group/item
        >
            <span className="text-sm font-semibold text-slate-700 group-hover/item:text-blue-700">{service.name}</span>
            <span className="text-xs text-slate-400 mt-0.5">Explore solutions</span>
        </Link>
    ))}
    </motion.div>
    </AnimatePresence>
</div>

<Link href="/products" className="text-slate-600 hover:text-blue-700 text-sm font-semibold transition-colors duration
    Products
    <span className="absolute -bottom-1 left-0 w-0 h-0.5 bg-blue-700 transition-all duration-300 group-hover:w-full"></span>
</Link>
<Link href="/projects" className="text-slate-600 hover:text-blue-700 text-sm font-semibold transition-colors duration
    Projects
    <span className="absolute -bottom-1 left-0 w-0 h-0.5 bg-blue-700 transition-all duration-300 group-hover:w-full"></span>
</Link>
<Link href="/industries" className="text-slate-600 hover:text-blue-700 text-sm font-semibold transition-colors durati
    Industries
    <span className="absolute -bottom-1 left-0 w-0 h-0.5 bg-blue-700 transition-all duration-300 group-hover:w-full"></span>
</Link>
</nav>

{/* Right CTA */}
<div className="hidden xl:flex items-center gap-6">
    <Link href="/knowledge-center" className="text-slate-500 hover:text-slate-800 text-sm font-medium transition-colors">
        Resources
    </Link>
    <Link
        href="/contact?type=quote"
        className="group relative inline-flex items-center justify-center gap-2 px-6 py-2.5 text-sm font-semibold text-whit
    >
        <div className="absolute inset-0 bg-gradient-to-r from-blue-600 to-blue-800 opacity-0 group-hover:opacity-100 trans
        <span className="relative z-10 flex items-center gap-2">
            Request Quote
            <ArrowRight className="w-4 h-4 group-hover:translate-x-1 transition-transform" />
        </span>
    </Link>
</div>

{/* Mobile Toggle */}
<div className="xl:hidden z-50">
    <button
        onClick={() => setIsOpen(!isOpen)}
        className="p-2 -mr-2 text-slate-600 hover:text-slate-900 focus:outline-none"
    >
        <AnimatePresence mode="wait">
            {isOpen ? (
                <motion.div
                    key="close"
                    initial={{ rotate: -90, opacity: 0 }}
                    animate={{ rotate: 0, opacity: 1 }}
                    exit={{ rotate: 90, opacity: 0 }}
                    transition={{ duration: 0.2 }}
                >
                    <X className="w-6 h-6" />
                </motion.div>
            ) : (
                <motion.div
                    key="menu"
                    initial={{ rotate: 90, opacity: 0 }}
                    animate={{ rotate: 0, opacity: 1 }}
                    exit={{ rotate: -90, opacity: 0 }}
                    transition={{ duration: 0.2 }}
                >
                    <Menu className="w-6 h-6" />
                </motion.div>
            )}
        </AnimatePresence>
    </button>

```

```

    </div>
  </div>
</div>

{/* Mobile Menu Overlay */}
<AnimatePresence>
  {isOpen && (
    <motion.div
      initial={{ opacity: 0, y: -20 }}
      animate={{ opacity: 1, y: 0 }}
      exit={{ opacity: 0, y: -20 }}
      transition={{ duration: 0.3, ease: [0.22, 1, 0.36, 1] }}
      className="fixed inset-0 z-40 bg-white/95 backdrop-blur-xl pt-24 pb-6 px-6 xl:hidden overflow-y-auto h-screen"
    >
      <div className="flex flex-col gap-6">
        {[
          { name: 'Home', href: '/' },
          { name: 'About', href: '/about' },
          { name: 'Products', href: '/products' },
          { name: 'Projects', href: '/projects' },
          { name: 'Industries', href: '/industries' },
          { name: 'Gallery', href: '/gallery' },
        ].map((item, idx) => (
          <motion.div
            initial={{ opacity: 0, x: -20 }}
            animate={{ opacity: 1, x: 0 }}
            transition={{ delay: idx * 0.05 + 0.1 }}
            key={item.name}
          >
            <Link
              href={item.href}
              onClick={() => setIsOpen(false)}
              className="text-3xl font-semibold text-slate-800 tracking-tight hover:text-blue-700 transition-colors block"
            >
              {item.name}
            </Link>
          </motion.div>
        ))}

        <motion.div
          initial={{ opacity: 0, x: -20 }}
          animate={{ opacity: 1, x: 0 }}
          transition={{ delay: 0.4 }}
          className="mt-4 pt-6 border-t border-slate-200"
        >
          <div className="text-sm font-bold text-slate-400 mb-4 uppercase tracking-wider">Services</div>
          <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
            {servicesList.map((service, index) => (
              <Link
                key={index}
                href={service.href}
                onClick={() => setIsOpen(false)}
                className="text-lg font-medium text-slate-600 hover:text-blue-700 transition-colors"
              >
                {service.name}
              </Link>
            ))}
          </div>
        </motion.div>

        <motion.div
          initial={{ opacity: 0, y: 20 }}
          animate={{ opacity: 1, y: 0 }}
          transition={{ delay: 0.5 }}
          className="mt-8 flex flex-col gap-4"
        >
          <Link
            href="/contact?type=quote"
            onClick={() => setIsOpen(false)}
            className="w-full py-4 bg-slate-900 text-white rounded-xl font-semibold text-lg text-center flex items-center justify-center"
          >
            Request a Quote
          <ArrowRight className="w-5 h-5" />
          </Link>
          <Link
            href="/contact"
            onClick={() => setIsOpen(false)}
            className="w-full py-4 bg-slate-100 text-slate-900 rounded-xl font-semibold text-lg text-center hover:bg-slate-200"
          >
            Contact Us
          </Link>
        </motion.div>
      </div>
    </motion.div>
  )}
</AnimatePresence>

```

... continued: src/components/Layout/Header.tsx

```
        </div>
      </motion.div>
    )}
  </AnimatePresence>
</header>
);
}
```

```

'use client';?
?
import { useState } from 'react';?
import Link from 'next/link';?
import { ServiceDetails } from '@lib/servicesData';?
?
interface ClientProps {?
  service: ServiceDetails;?
}?
?
export default function ServicePageClient({ service }: ClientProps) {?
  // State for active tabs, accordion, modals?
  const [activeWorkflow, setActiveWorkflow] = useState(0);?
  const [activeAccordion, setActiveAccordion] = useState<number | null>(null);?
  const [selectedIndCard, setSelectedIndCard] = useState<number | null>(null);?
  ?
  // Custom Simulator States?
  // 1. Panel Manufacturing (ACB trip current)?
  const [tripCurrent, setTripCurrent] = useState(1600);?
  // 2. PLC/SCADA HMI mimics?
  const [pumpRunning, setPumpRunning] = useState(true);?
  const [valvePos, setValvePos] = useState(75);?
  // 3. VFD energy calc?
  const [motorHP, setMotorHP] = useState(50);?
  const [dutyHours, setDutyHours] = useState(16);?
  // 4. Robotics arm jog axis?
  const [robotAxis1, setRobotAxis1] = useState(45);?
  const [robotAxis2, setRobotAxis2] = useState(-30);?
  // 5. Instrumentation loop tank level target?
  const [tankTarget, setTankTarget] = useState(62);?
  const [tankCurrent, setTankCurrent] = useState(58);?
  // 6. Elevator floor select?
  const [liftFloor, setLiftFloor] = useState(3);?
  const [liftMoving, setLiftMoving] = useState(false);?
  // 7. Electrical grid breaker state?
  const [breakerTripped, setBreakerTripped] = useState(false);?
  ?
  const toggleAccordion = (idx: number) => {?
    setActiveAccordion(activeAccordion === idx ? null : idx);?
  };?
  ?
  const handleLiftMove = (floor: number) => {?
    if (floor === liftFloor || liftMoving) return;?
    setLiftMoving(true);?
    setTimeout(() => {?
      setLiftFloor(floor);?
      setLiftMoving(false);?
    }, 1500);?
  };?
  ?
  // Render Service-Specific interactive simulation console?
  const renderServiceSimulator = () => {?
    switch (service.slug) {?
      case 'panel-manufacturing':?
        return (?
          <div className="bg-white border border-slate-200 rounded-lg p-6 space-y-6 font-mono text-xs shadow-sm">?
            <h4 className="text-[#0f4c81] font-bold uppercase tracking-wider border-b border-slate-100 pb-2">?
              ACB PROTECTION COORDINATION SIMULATOR?
            </h4>?
            <div className="space-y-4 text-slate-700">?
              <div className="flex justify-between items-center">?
                <span>Incomer Frame Limit:</span>?
                <span className="text-slate-900 font-bold">3200A max</span>?
              </div>?
              <div className="space-y-1.5">?
                <div className="flex justify-between text-slate-500">?
                  <span>Current trip trigger setting (Ir):</span>?
                  <span className="text-[#0f4c81] font-semibold">{tripCurrent} Amps</span>?
                </div>?
                <input ?
                  type="range" ?
                  min="800" ?
                  max="3200" ?
                  step="100"?
                  value={tripCurrent}?
                  onChange={(e) => setTripCurrent(parseInt(e.target.value))}?
                  className="w-full h-1.5 bg-slate-200 rounded-lg appearance-none cursor-pointer accent-[#0f4c81]"?
                </div>?
              </div>?
            </div>?
          </div>?
        );
      ?
    }
  };
}

```

```

<div className="bg-[#E8F0FA] p-4 rounded border border-slate-150 space-y-2">?
  <span className="text-slate-400 text-[10px] block">SIMULATED FAULT RESPONSES</span>?
  <div className="flex justify-between">?
    <span>Overload trip timing (at 1.5x Ir):</span>?
    <span className="text-emerald-700 font-bold">42 Sec (Thermal memory active)</span>?
  </div>?
  <div className="flex justify-between">?
    <span>Short circuit instantaneous trip trigger:</span>?
    <span className="text-rose-700 font-bold">{tripCurrent * 10} Amps</span>?
  </div>?
  <div className="flex justify-between">?
    <span>Enclosure heat output estimation:</span>?
    <span className="text-slate-800">{(tripCurrent * 0.012).toFixed(1)} Watts</span>?
  </div>?
</div>?
</div>?
);?
case 'plc-scada':?
  return (?
    <div className="bg-white border border-slate-200 rounded-lg p-6 space-y-6 font-mono text-xs shadow-sm">?
      <h4 className="text-[#0f4c81] font-bold uppercase tracking-wider border-b border-slate-100 pb-2">?
        REAL-TIME SCADA HMI CONSOLE MIMIC?
      </h4>?
      <div className="space-y-4 text-slate-700">?
        <div className="flex justify-between items-center">?
          <span>PUMP STATION #2:</span>?
          <button?
            onClick={() => setPumpRunning(!pumpRunning)}?
            className={`px-3 py-1 rounded font-bold uppercase tracking-wider text-[10px] transition-colors ${?
              pumpRunning ? 'bg-emerald-50 border border-emerald-300 text-emerald-700' : 'bg-rose-50 border border-rose-300?
            `}?
          >?
            {pumpRunning ? 'RUNNING' : 'STOPPED'}?
          </button>?
        </div>?
      </div>?
    <div className="space-y-1">?
      <div className="flex justify-between text-slate-500">?
        <span>Modulating Valve Aperture:</span>?
        <span className="text-[#0f4c81] font-semibold">{valvePos}% Opened</span>?
      </div>?
      <input ?
        type="range" ?
        min="0" ?
        max="100" ?
        value={valvePos}?
        onChange={(e) => setValvePos(parseInt(e.target.value))}?
        className="w-full h-1.5 bg-slate-200 rounded-lg appearance-none cursor-pointer accent-[#0f4c81]"?
      />?
    </div>?
  <div className="bg-[#E8F0FA] p-4 rounded border border-slate-150 space-y-2 text-[10px]">?
    <div className="flex justify-between">?
      <span>Feedback flow sensor data:</span>?
      <span className="text-[#0f4c81] font-bold">{(pumpRunning ? (valvePos * 12.5) : 0).toFixed(1)} M3/Hr</span>?
    </div>?
    <div className="flex justify-between">?
      <span>Pressure transducer read value:</span>?
      <span className="text-[#0f4c81] font-bold">{(pumpRunning ? (4.2 - (valvePos * 0.02)) : 0).toFixed(2)} Bar</span>?
    </div>?
    <div className="flex justify-between">?
      <span>Active telemetry alarms:</span>?
      <span className={pumpRunning ? 'text-emerald-700 font-semibold' : 'text-amber-700 font-semibold animate-pulse'}?
        {pumpRunning ? '0 Active Warnings' : 'Telemetry fault: Zero Pressure'}?
      </span>?
    </div>?
  </div>?
</div>?
</div>?
);?
case 'vfd-solutions':?
  return (?
    <div className="bg-white border border-slate-200 rounded-lg p-6 space-y-6 font-mono text-xs shadow-sm">?
      <h4 className="text-[#0f4c81] font-bold uppercase tracking-wider border-b border-slate-100 pb-2">?
        VFD ENERGY AUDIT COST CALCULATOR?
      </h4>?
      <div className="space-y-4 text-slate-700">?
        <div className="grid grid-cols-2 gap-4">?
          <div className="space-y-1">?
            <span className="text-slate-500 block">Motor Rating (HP):</span>?
            <input?

```

```

        type="number"?
        value={motorHP}?
        onChange={(e) => setMotorHP(Math.max(1, parseInt(e.target.value) || 0))}?
        className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-2 py-1 text-slate-800 focus:outline-none fo
    />?
  </div>?
  <div className="space-y-1">?
    <span className="text-slate-500 block">Daily Duty Cycle (Hrs):</span>?
    <input?
      type="number"?
      value={dutyHours}?
      onChange={(e) => setDutyHours(Math.max(1, Math.min(24, parseInt(e.target.value) || 0)))?
      className="w-full bg-[#E8F0FA] border border-slate-200 rounded px-2 py-1 text-slate-800 focus:outline-none fo
    />?
  </div>?
</div>?

?

<div className="bg-[#E8F0FA] p-4 rounded border border-slate-150 space-y-2 text-[10px]">?
  <div className="flex justify-between">?
    <span>DOL Starter Daily Consumption:</span>?
    <span className="text-slate-500">{(motorHP * 0.746 * dutyHours).toFixed(1)} kWh</span>?
  </div>?
  <div className="flex justify-between">?
    <span>Yaskawa VFD Daily Consumption:</span>?
    <span className="text-[#0f4c81] font-bold">{(motorHP * 0.746 * dutyHours * 0.72).toFixed(1)} kWh (Save 28%)</sp
  </div>?
  <div className="flex justify-between border-t border-slate-200 pt-2 text-emerald-700 font-bold">?
    <span>Estimated Annual Savings (INR):</span>?
    <span>&nbsp;&nbsp;&nbsp;? {(motorHP * 0.746 * dutyHours * 0.28) * 365 * 8.5).toLocaleString('en-IN', { maximumFractionDigi
  </div>?
</div>?
</div>?
</div>?
);?
case 'robotics':?
  return (?
    <div className="bg-white border border-slate-200 rounded-lg p-6 space-y-6 font-mono text-xs shadow-sm">?
      <h4 className="text-[#0f4c81] font-bold uppercase tracking-wider border-b border-slate-100 pb-2">?
        MULTI-AXIS ROBOTIC JOG CONTROLLER?
      </h4>?
      <div className="space-y-4 text-slate-700">?
        <div className="space-y-1.5">?
          <div className="flex justify-between text-slate-500">?
            <span>Base Swivel Rotation (Axis S):</span>?
            <span className="text-[#0f4c81] font-semibold">{robotAxis1}&deg;</span>?
          </div>?
          <input ?
            type="range" ?
            min="-170" ?
            max="170" ?
            value={robotAxis1}?
            onChange={(e) => setRobotAxis1(parseInt(e.target.value))}?
            className="w-full h-1.5 bg-slate-200 rounded-lg appearance-none cursor-pointer accent-[#0f4c81]"?
          />?
        </div>?

?

        <div className="space-y-1.5">?
          <div className="flex justify-between text-slate-500">?
            <span>Lower Arm Extension (Axis L):</span>?
            <span className="text-[#0f4c81] font-semibold">{robotAxis2}&deg;</span>?
          </div>?
          <input ?
            type="range" ?
            min="-90" ?
            max="120" ?
            value={robotAxis2}?
            onChange={(e) => setRobotAxis2(parseInt(e.target.value))}?
            className="w-full h-1.5 bg-slate-200 rounded-lg appearance-none cursor-pointer accent-[#0f4c81]"?
          />?
        </div>?

?

        <div className="bg-[#E8F0FA] p-4 rounded border border-slate-150 text-[10px] space-y-1.5">?
          <div className="flex justify-between">?
            <span>Wrist Coordinates (Calculated TCP):</span>?
            <span className="text-slate-800">X: {(450 + (robotAxis1 * 1.2)).toFixed(1)} mm &bull; Y: {(robotAxis2 * 2.1).to
          </div>?
          <div className="flex justify-between">?
            <span>Singularity boundary warning:</span>?
            <span className="text-emerald-700 font-bold">OPERATIONAL / CLEAR</span>?
          </div>?
        </div>?
      </div>?
    </div>?
  )

```

```

    </div>?
  );?
  case 'instrumentation':?
  return (?
    <div className="bg-white border border-slate-200 rounded-lg p-6 space-y-6 font-mono text-xs shadow-sm">?
      <h4 className="text-[#0f4c81] font-bold uppercase tracking-wider border-b border-slate-100 pb-2">?
        TANK LEVEL LOOP CONTROLLER?
      </h4>?
      <div className="space-y-4 text-slate-700">?
        <div className="space-y-1.5">?
          <div className="flex justify-between text-slate-500">?
            <span>Target Liquid Fill Level:</span>?
            <span className="text-[#0f4c81] font-semibold">{tankTarget}% (SP)</span>?
          </div>?
          <input ?
            type="range" ?
            min="10" ?
            max="90" ?
            value={tankTarget}?
            onChange={(e) => {?
              setTankTarget(parseInt(e.target.value));?
              // Simulate PID response?
              const diff = parseInt(e.target.value) - tankCurrent;?
              setTimeout(() => {?
                setTankCurrent(Math.round(tankCurrent + (diff * 0.8)));?
              }, 500);?
            }}?
            className="w-full h-1.5 bg-slate-200 rounded-lg appearance-none cursor-pointer accent-[#0f4c81]"?
          </div>?
        </div>?
      </div>?
    <div className="bg-[#E8F0FA] p-4 rounded border border-slate-150 text-[10px] space-y-2">?
      <div className="flex justify-between">?
        <span>Process Value Level (PV Sensor):</span>?
        <span className="text-[#0f4c81] font-bold">{tankCurrent}%</span>?
      </div>?
      <div className="flex justify-between">?
        <span>HART Loop Transmitter Current:</span>?
        <span className="text-[#0f4c81] font-bold">{(4 + (tankCurrent * 0.16)).toFixed(2)} mA</span>?
      </div>?
      <div className="flex justify-between">?
        <span>Control Valve Output (MV):</span>?
        <span className="text-emerald-700 font-bold">{tankCurrent < tankTarget ? 'FILLING' : 'HOLDING'}</span>?
      </div>?
    </div>?
  </div>?
  );?
  case 'elevator-systems':?
  return (?
    <div className="bg-white border border-slate-200 rounded-lg p-6 space-y-6 font-mono text-xs shadow-sm">?
      <h4 className="text-[#0f4c81] font-bold uppercase tracking-wider border-b border-slate-100 pb-2">?
        MICROPROCESSOR CABIN CALL CONTROLLER?
      </h4>?
      <div className="space-y-4 text-slate-700">?
        <div className="flex justify-between items-center">?
          <span>Active Elevator Location:</span>?
          <span className="text-slate-900 font-bold">Floor {liftFloor}</span>?
        </div>?
      </div>?
    <div className="flex justify-between gap-2">?
      {[1, 2, 3, 4, 5].map((floor) => (?
        <button?
          key={floor}?
          onClick={() => handleLiftMove(floor)}?
          className={`flex-1 py-2.5 rounded font-mono text-xs font-bold border transition-all ${?
            liftFloor === floor ?
              'bg-[#0f4c81] border-[#0f4c81] text-white shadow-xs'?
              : 'bg-white border-slate-200 text-slate-600 hover:border-slate-400'?
            }`}?
        >?
          F{floor}?
        </button>?
      )]}?
    </div>?
  <div className="bg-[#E8F0FA] p-4 rounded border border-slate-150 text-[10px] space-y-1.5">?
    <div className="flex justify-between">?
      <span>Cabin status indicator:</span>?
      <span className={liftMoving ? 'text-amber-600 font-semibold animate-pulse' : 'text-emerald-700 font-semibold'}>?
        {liftMoving ? 'MOVING...' : 'STATIONARY'}?
      </span>?
    </div>?
  </div>?

```



```
... continued: src/components/Service/ServicePageClient.tsx
```

```

    </div>?
    <div className="flex justify-between">?
      <span>Door lock switch:</span>?
      <span className="text-slate-700">{liftMoving ? 'LOCKED' : 'OPENED'}</span>?
    </div>?
  </div>?
</div>?
);?
case 'electrical-solutions':?
  return (?
    <div className="bg-white border border-slate-200 rounded-lg p-6 space-y-6 font-mono text-xs shadow-sm">?
      <h4 className="text-[#0f4c81] font-bold uppercase tracking-wider border-b border-slate-100 pb-2">?
        HT BREAKER CONTROL CONSOLE MIMIC?
      </h4>?
      <div className="space-y-4 text-slate-700">?
        <div className="flex justify-between items-center">?
          <span>33KV Vacuum Circuit Breaker state:</span>?
          <button?
            onClick={() => setBreakerTripped(!breakerTripped)}?
            className={`px-3 py-1 rounded font-bold uppercase tracking-wider text-[10px] transition-colors ${?
              !breakerTripped ? 'bg-emerald-50 border border-emerald-300 text-emerald-700' : 'bg-rose-50 border border-rose-500 text-rose-700'}</div>?
            >?
              {!breakerTripped ? 'CLOSED' : 'TRIPPED'}?
            </button>?
          </div>?
        <div className="bg-[#E8F0FA] p-4 rounded border border-slate-150 text-[10px] space-y-2">?
          <div className="flex justify-between">?
            <span>Phase currents matching:</span>?
            <span className="text-slate-800">{!breakerTripped ? 'R: 142A | Y: 140A | B: 141A' : '0A (No current flow)'}</span>?
          </div>?
          <div className="flex justify-between">?
            <span>Active ground earth pit leakage:</span>?
            <span className="text-slate-800">{!breakerTripped ? '0.04 Amps' : '0.00 Amps'}</span>?
          </div>?
          <div className="flex justify-between border-t border-slate-200 pt-2">?
            <span>Relay status message:</span>?
            <span className={!breakerTripped ? 'text-emerald-700 font-bold' : 'text-rose-700 font-bold'}>?
              {!breakerTripped ? 'ALL SYSTEM OK' : 'TRIP FAULT: EARTH SHIELD LEAK'?
            </span>?
          </div>?
        </div>?
      </div>?
    </div>?
  );?
case 'amc-services':?
  return (?
    <div className="bg-white border border-slate-200 rounded-lg p-6 space-y-6 font-mono text-xs shadow-sm">?
      <h4 className="text-[#0f4c81] font-bold uppercase tracking-wider border-b border-slate-100 pb-2">?
        FAT/SAT SYSTEM VALIDATION PROTOCOL?
      </h4>?
      <div className="space-y-3">?
        <p className="text-[10px] text-slate-500 leading-normal mb-2">?
          Select validation test type to log simulated results formats:?
        </p>?
        {[?
          { name: 'Insulation test (5kV)', result: 'PASSED (Value: 24,000 Megohms)' },?
          { name: 'Primary injection checks', result: 'PASSED (Phase curves matching limits)' },?
          { name: 'Redundancy test checks', result: 'PASSED (Switchover timing: 4ms)' }?
        ]?.map((test, idx) => (?
          <button?
            key={idx}?
            onClick={() => alert(`Generating reference log format: ${test.name} checks -> ${test.result}`)}?
            className="w-full text-left px-3 py-2.5 bg-white hover:bg-slate-50 border border-slate-200 rounded text-slate-700">?
              <span>{test.name}</span>?
              <span className="text-[#0f4c81] font-bold">&rarr; Run</span>?
            </button>?
          )?
        )]?
      </div>?
    </div>?
  );?
default:?
  return null;?
)?
return (?
  <div className="space-y-16">?

```

```

    {/ * 2. SERVICE OVERVIEW SECTION */?
    <section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">?
      <div className="grid grid-cols-1 lg:grid-cols-12 gap-12 items-start">?
        <div className="lg:col-span-8 space-y-6">?
          <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
            Divisional Capability & Overview?
          </h2>?
          <div className="space-y-4 text-sm leading-relaxed text-slate-655">?
            {service.overviewParagraphs.map((p, index) => (?
              <p key={index}>{p}</p>?
            ))}?
          </div>?
        </div>?
        ?
        <div className="lg:col-span-4 bg-white border border-slate-200 p-6 rounded-lg space-y-4 shadow-sm">?
          <h3 className="text-[#0f4c81] font-bold text-sm uppercase tracking-wider font-mono">?
            Technical Division Stats?
          </h3>?
          <div className="divide-y divide-slate-100 font-mono text-xs">?
            {service.stats.map((stat, sIdx) => (?
              <div key={sIdx} className="py-3 flex justify-between items-center text-slate-700">?
                <span className="text-slate-500">{stat.label}</span>?
                <span className="text-slate-900 font-bold">{stat.value}</span>?
              </div>?
            ))}?
          </div>?
        </div>?
      </div>?
    </section>?
  ?

  {/ * 3. INDUSTRIAL CHALLENGE SECTION */?
  <section className="py-16 bg-[#eef2f7]/50 border-t border-b border-slate-200">?
    <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-8">?
      <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
        Industrial Challenge & Remediation?
      </h2>?
      <div className="grid grid-cols-1 md:grid-cols-2 gap-8 text-sm">?
        <div className="bg-white border border-slate-200 p-6 rounded-lg space-y-3 shadow-sm">?
          <span className="text-[10px] font-mono text-rose-700 uppercase tracking-widest block font-bold">?
            Before: Operational Bottlenecks?
          </span>?
          <p className="text-slate-600 leading-relaxed text-xs">?
            {service.challenges.before}?
          </p>?
        </div>?
        ?
        <div className="bg-white border border-[#2b6cb0]/30 p-6 rounded-lg space-y-3 shadow-sm relative overflow-hidden">?
          <div className="absolute top-0 right-0 w-8 h-8 border-b border-l border-[#2b6cb0]/20 pointer-events-none" />?
          <span className="text-[10px] font-mono text-emerald-700 uppercase tracking-widest block font-bold">?
            After: Nandini Engineering Solution?
          </span>?
          <p className="text-slate-655 leading-relaxed text-xs">?
            {service.challenges.after}?
          </p>?
          <div className="pt-2 border-t border-slate-100 text-[11px] font-mono text-[#0f4c81]">?
            <strong>Outcome:</strong> {service.challenges.scenario}?
          </div>?
        </div>?
      </div>?
    </div>?
  </section>?
  ?

  {/ * 4. ENGINEERING WORKFLOW SYSTEM */?
  <section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-8">?
    <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
      Engineering Process Pipeline?
    </h2>?
    ?
    {/ * Progress Pipeline */?
    <div className="relative border border-slate-200 rounded-lg p-6 bg-white shadow-xs">?
      <div className="absolute top-1/2 left-0 right-0 h-0.5 bg-slate-100 -translate-y-1/2 hidden md:block" />?
      <div ?
        className="absolute top-1/2 left-0 h-0.5 bg-[#0f4c81] -translate-y-1/2 transition-all duration-500 hidden md:block" ?
        style={{ width: `${(activeWorkflow / (service.workflow.length - 1)) * 100}%` }}?
      >?
      ?
      <div className="grid grid-cols-2 md:grid-cols-4 gap-4 relative z-10">?
        {service.workflow.map((item, idx) => (?
          <button?
            key={idx}?
            onClick={() => setActiveWorkflow(idx)}?
            className={`flex flex-col items-center text-center p-3 rounded transition-all cursor-pointer focus:outline-none $

```

```

        activeWorkflow === idx?
          ? 'bg-slate-50 border border-[#0f4c81]/30 shadow-xs'?
          : 'bg-white border border-transparent hover:border-slate-100'?
      }`}?
    >?
    <span className={`w-8 h-8 rounded-full flex items-center justify-center font-mono text-xs font-bold mb-2 transiti
      activeWorkflow === idx?
        ? 'bg-[#0f4c81] text-white'?
        : 'bg-slate-100 text-slate-500'?
      }`}>?
      0{idx + 1}?
    </span>?
    <span className={`text-[10px] uppercase font-mono tracking-wider font-semibold ${?
      activeWorkflow === idx ? 'text-[#0f4c81]' : 'text-slate-500'?
    }`}>?
      {item.name}?
    </span>?
    </button>?
  )}}?
</div>?
</div>?
?

{ /* Selected Step details */}?
<div className="bg-white border border-slate-200 p-8 rounded-lg shadow-sm">?
  <div className="lg:grid-cols-1 lg:grid-cols-12 gap-8 items-center">?
    <div className="lg:col-span-8 space-y-4">?
      <span className="text-[10px] font-mono text-[#0f4c81] uppercase tracking-widest border border-[#0f4c81]/20 px-2 py-
        STEP 0{activeWorkflow + 1} SPECIFICATIONS?
      </span>?
      <h3 className="text-slate-900 font-bold text-xl font-outfit">?
        {service.workflow[activeWorkflow].name}?
      </h3>?
      <p className="text-sm text-slate-655 leading-relaxed">?
        {service.workflow[activeWorkflow].activity}?
      </p>?
      ?
      <div className="grid grid-cols-1 sm:grid-cols-3 gap-4 pt-4 text-xs font-mono">?
        <div>?
          <span className="text-slate-400 block">TOOLS:</span>?
          <span className="text-slate-700 font-sans">{service.workflow[activeWorkflow].tools}</span>?
        </div>?
        <div>?
          <span className="text-slate-400 block">SAFETY CONTROL:</span>?
          <span className="text-slate-700 font-sans">{service.workflow[activeWorkflow].safety}</span>?
        </div>?
        <div>?
          <span className="text-slate-400 block">DELIVERABLE:</span>?
          <span className="text-emerald-700 font-bold font-sans">{service.workflow[activeWorkflow].deliverable}</span>?
        </div>?
      </div>?
    </div>?
  </div>?
  ?
  <div className="lg:col-span-4 flex justify-center">?
    <svg className="w-24 h-24 text-[#0f4c81]/10" viewBox="0 0 100 100" fill="none" stroke="currentColor" strokeWidth="1
      <rect x="25" y="25" width="50" height="50" rx="3" />?
      <path d="M25,50 H75 M50,25 V75" strokeDasharray="1,2" />?
      <circle cx="50" cy="50" r="5" className="fill-[#0f4c81]/80 stroke-[#2b6cb0]" />?
    </svg>?
  </div>?
</div>?
</div>?
</section>?
?

{ /* 5. TECHNICAL ARCHITECTURE SECTION & SIMULATORS */}?
<section className="py-16 bg-[#eef2f7]/50 border-t border-b border-slate-200">?
  <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-8">?
    <div className="grid grid-cols-1 lg:grid-cols-12 gap-12 items-center">?
      ?
      { /* Architecture description */}?
      <div className="lg:col-span-6 space-y-6">?
        <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
          Technical Systems Topology?
        </h2>?
        <p className="text-sm text-slate-655 leading-relaxed">?
          {service.architectureDesc}?
        </p>?
        ?
        <ul className="space-y-3 font-mono text-xs text-slate-700">?
          {service.architectureDetails.map((detail, dIdx) => (?
            <li key={dIdx} className="flex items-start space-x-2">?
              <span className="text-[#3b82f6] font-bold shrink-0 mt-0.5">&bull;</span>?
              <span className="font-sans">{detail}</span>?
            </li>
          )}>?
        </ul>
      </div>
    </div>
  </div>

```

```

        </li>?
      )}}?
    </ul>?
  </div>?
?
  { /* Live Interactive Simulator Console */?
  <div className="lg:col-span-6">?
    {renderServiceSimulator()}?
  </div>?
  </div>?
  </div>?
  </section>?
?
  { /* 6. INDUSTRIES USING THIS SERVICE */?
  <section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-8">?
    <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
      Applied Industrial Domains?
    </h2>?
    ?
    <div className="grid grid-cols-1 md:grid-cols-3 gap-6">?
      {service.industries.map((ind, indIdx) => (?
        <div ?
          key={indIdx}?
          onClick={() => setSelectedIndCard(selectedIndCard === indIdx ? null : indIdx)}?
          className="bg-white hover:bg-slate-50/50 border border-slate-200 hover:border-[#2b6cb0] p-6 rounded-lg cursor-point
        >?
        <div>?
          <div className="flex justify-between items-center mb-3">?
            <h3 className="text-slate-900 font-bold text-lg font-outfit group-hover:text-[#0f4c81] transition-colors">?
              {ind.name}?
            </h3>?
            <span className="text-[9px] font-mono text-[#0f4c81] bg-slate-50 px-2 py-0.5 rounded border border-slate-200">?
              S-INFO?
            </span>?
          </div>?
          ?
          {selectedIndCard === indIdx ? (?
            <div className="space-y-3 text-[11px] font-mono text-slate-655 pt-2 border-t border-slate-100">?
              <div>?
                <strong className="text-[#0f4c81]">NEED:</strong> <span className="font-sans">{ind.need}</span>?
              </div>?
              <div>?
                <strong className="text-[#0f4c81]">CHALLENGE:</strong> <span className="font-sans">{ind.challenge}</span>?
              </div>?
              <div>?
                <strong className="text-[#0f4c81]">ADVANTAGE:</strong> <span className="font-sans">{ind.advantage}</span>?
              </div>?
            </div>?
          ) : (?
            <p className="text-xs text-slate-550 leading-relaxed line-clamp-3">?
              {ind.need}?
            </p>?
          )}}?
        </div>?
      </div>?
    </div>?
    <div>?
      <span className="text-[9px] font-mono text-[#0f4c81] mt-4 block">?
        {selectedIndCard === indIdx ? 'Click to collapse details' : 'Click to drill down details &rarr;'}?
      </span>?
    </div>?
  )}}?
  </div>?
  </section>?
?
  { /* 7. REAL IMPLEMENTATION EXAMPLES (CASE STUDY) */?
  <section className="py-16 bg-[#eef2f7]/50 border-t border-b border-slate-200">?
    <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-8">?
      <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
        Case Study & Operational Outcomes?
      </h2>?
      ?
      <div className="bg-white border border-slate-200 rounded-lg p-8 space-y-6 shadow-sm">?
        <div className="flex justify-between items-center border-b border-slate-100 pb-4">?
          <div>?
            <span className="text-[9px] font-mono text-[#0f4c81] uppercase tracking-widest block font-bold">?
              B2B CLIENT CASE SHEET?
            </span>?
            <h3 className="text-lg font-bold text-slate-900 font-outfit mt-1">?
              {service.caseStudy.client}?
            </h3>?
          </div>?
          <span className="text-[10px] font-mono text-emerald-700 font-bold uppercase tracking-wider bg-emerald-50 border bor
            VERIFIED DELIVERY?
          </span>?
        </div>?
      </div>?
    </div>?
  </section>?

```

```

    </span>?
  </div>?
?
  <div className="grid grid-cols-1 md:grid-cols-2 gap-8 text-xs text-slate-600 font-sans leading-relaxed">?
    <div className="space-y-4">?
      <div>?
        <h4 className="font-bold text-slate-900 uppercase tracking-wider text-[10px] font-mono">1. Client Requirement:</h4>?
        <p className="mt-1">{service.caseStudy.requirement}</p>?
      </div>?
      <div>?
        <h4 className="font-bold text-slate-900 uppercase tracking-wider text-[10px] font-mono">2. Technical Problem:</h4>?
        <p className="mt-1 text-rose-700/90">{service.caseStudy.problem}</p>?
      </div>?
    </div>?
?
    <div className="space-y-4">?
      <div>?
        <h4 className="font-bold text-slate-900 uppercase tracking-wider text-[10px] font-mono">3. Engineering Solution:</h4>?
        <p className="mt-1">{service.caseStudy.solution}</p>?
      </div>?
      <div>?
        <h4 className="font-bold text-slate-900 uppercase tracking-wider text-[10px] font-mono">4. Products Integrated:</h4>?
        <p className="mt-1 text-[#0f4c81] font-mono">{service.caseStudy.products}</p>?
      </div>?
    </div>?
?
    <div className="pt-4 border-t border-slate-100 flex justify-between items-center text-[10px] font-mono">?
      <span className="text-slate-500 uppercase">Outcome: {service.caseStudy.outcome}</span>?
      <span className="text-emerald-700 font-bold uppercase tracking-wider">Metrics: {service.caseStudy.improvement}</span>?
    </div>?
  </div>?
</section>?
?
  {/ * 8. TECHNOLOGY & SYSTEMS USED */}?
  <section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-8">?
    <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
      Technical Ecosystem & Systems?
    </h2>?
    ?
    <div className="grid grid-cols-1 md:grid-cols-3 gap-8 text-xs">?
      {service.techEcosystem.map((tech, tIdx) => (?
        <div ?
          key={tIdx} ?
          className="bg-white border border-slate-200 p-6 rounded-lg space-y-3 shadow-sm flex flex-col justify-between"?
        >?
          <div>?
            <h4 className="text-slate-900 font-bold text-sm mb-2">{tech.name}</h4>?
            <p className="text-slate-600 leading-normal mb-4 font-sans">{tech.use}</p>?
          </div>?
          <div className="pt-3 border-t border-slate-100 text-[10px] font-mono text-slate-500">?
            <strong className="text-[#0f4c81]">LOGIC:</strong> <span className="font-sans">{tech.logic}</span>?
          </div>?
        </div>?
      )})?
    </div>?
  </section>?
?
  {/ * 9 & 10. QUALITY ASSURANCE & SAFETY */}?
  <section className="py-16 bg-[#eef2f7]/50 border-t border-b border-slate-200">?
    <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-12">?
      <div className="grid grid-cols-1 md:grid-cols-2 gap-12">?
        ?
        {/ * QA section */}?
        <div className="space-y-6">?
          <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
            Quality Assurance & Audits?
          </h2>?
          <ul className="space-y-3 text-xs font-mono text-slate-700">?
            {service.qualityAssurance.map((qa, qaIdx) => (?
              <li key={qaIdx} className="flex items-start space-x-2">?
                <span className="text-emerald-700 font-bold shrink-0 mt-0.5">&check;</span>?
                <span className="font-sans">{qa}</span>?
              </li>?
            )})?
          </ul>?
        </div>?
        ?
        {/ * Safety block */}?
        <div className="space-y-6">?
          <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?

```

```
... continued: src/components/Service/ServicePageClient.tsx
```

```

Safety & Standards Conformity?
</h2>?
<div className="space-y-3 font-mono text-xs text-slate-700">?
  {service.safetyCompliance.map((safe, safeIdx) => (?
    <div key={safeIdx} className="bg-white border border-slate-200 p-4 rounded shadow-xs">?
      <span className="text-[#0f4c81] font-bold block mb-1">{safe.standard}</span>?
      <span className="text-slate-600 leading-relaxed font-sans">{safe.description}</span>?
    </div>?
  )})?
</div>?
</div>?
</div>?
</div>?
</section>?

?

{/* 11. MAINTENANCE, SUPPORT & ACCORDIONS */}?
<section className="py-16 max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 space-y-12">?
  <div className="grid grid-cols-1 lg:grid-cols-12 gap-12 items-start">?
    ?
    <div className="lg:col-span-8 space-y-6">?
      <h2 className="text-[#0f4c81] font-bold text-2xl font-outfit border-l-4 border-[#0f4c81] pl-3">?
        Lifecycle Support & AMC Services?
      </h2>?
      <div className="space-y-3">?
        {service.maintenanceAMC.map((maint, mIdx) => (?
          <div key={mIdx} className="border border-slate-200 rounded overflow-hidden shadow-xs">?
            <button?
              onClick={() => toggleAccordion(mIdx)}?
              className="w-full text-left px-5 py-4 bg-white hover:bg-slate-50 text-slate-800 font-semibold text-xs uppercase">?
                <span>CYCLE STAGE 0{mIdx + 1}: {maint.split(":")[0]}</span>?
                <span className="text-[#0f4c81] font-bold text-sm">?
                  {activeAccordion === mIdx ? '?' : '+'}?
                </span>?
              </button>?
              {activeAccordion === mIdx && (?
                <div className="px-5 py-4 bg-[#E8F0FA] text-xs leading-relaxed text-slate-600 border-t border-slate-100 font-?
                  {maint}?
                </div>?
              )}?
            </div>?
          )})?
        </div>?
      )})?
    </div>?
  </div>?

?

{/* 13. DOCUMENTATION SYSTEM */}?
<div className="lg:col-span-4 bg-white border border-slate-200 p-6 rounded-lg space-y-6 font-mono text-xs shadow-sm">?
  <h4 className="text-[#0f4c81] font-bold uppercase tracking-wider border-b border-slate-100 pb-2">?
    TECHNICAL FILE DOWNLOADS?
  </h4>?
  <div className="space-y-4">?
    {service.docs.map((doc, docIdx) => (?
      <div ?
        key={docIdx}?
        onClick={() => alert(`Document "${doc.name}" has been queued for secure download from portal.`)}?
        className="p-3 bg-[#E8F0FA] border border-slate-150 rounded hover:border-[#0f4c81] cursor-pointer flex justify-?
      >?
      <div>?
        <span className="text-slate-800 group-hover:text-[#0f4c81] font-semibold transition-colors block text-[10px]">?
          {doc.name}?
        </span>?
        <span className="text-[8px] text-slate-400 uppercase">{doc.type} &bull; {doc.size}</span>?
      </div>?
      <span className="text-[#0f4c81] group-hover:underline text-[9px]">Download &darr;</span>?
    </div>?
  )})?
  </div>?
</div>?
</div>?
</section>?

?

{/* 12. Dynamic FAQ Accordions */}?
<section className="py-16 bg-[#eef2f7]/30 border-t border-slate-200 text-left">?
  <div className="max-w-4xl mx-auto px-4 sm:px-6 lg:px-8 space-y-6">?
    <h3 className="text-[#0f4c81] font-bold text-xl font-outfit border-l-4 border-[#0f4c81] pl-3 mb-8">?
      Divisional FAQs?
    </h3>?
    <div className="space-y-4">?
      {service.faqs.map((faq, fIdx) => (?
        <div key={fIdx} className="bg-white border border-slate-200 p-6 rounded-lg space-y-2 shadow-sm">?
          <h4 className="text-slate-900 font-bold text-sm font-sans">{faq.q}</h4>?
          <p className="text-xs text-slate-600 leading-relaxed font-sans">{faq.a}</p>?
        </div>?
      )})?
    </div>?
  </div>?

```

... continued: src/components/Service/ServicePageClient.tsx

```
        </div>?  
      )}}?  
    </div>?  
  </div>?  
  </section>?  
?  
  </div>?  
);?  
}?
```

```
import fs from 'fs';
import path from 'path';

const contentDirectory = path.join(process.cwd(), 'content');

export interface CMSMethodology {
  step: string;
  desc: string;
}

export interface CMSFAQ {
  q: string;
  a: string;
}

export interface CMSService {
  slug: string;
  title: string;
  overview: string;
  industries?: string[];
  brands?: string[];
  methodology?: CMSMethodology[];
  benefits?: string;
  faqs?: CMSFAQ[];
}

export function getEntries(type: string): CMSService[] {
  const directory = path.join(contentDirectory, type);

  if (!fs.existsSync(directory)) {
    return [];
  }

  const fileNames = fs.readdirSync(directory);
  const entries = fileNames
    .filter(fileName => fileName.endsWith('.json'))
    .map(fileName => {
      const slug = fileName.replace(/\.json$/, '');
      const fullPath = path.join(directory, fileName);
      const fileContents = fs.readFileSync(fullPath, 'utf8');
      const data = JSON.parse(fileContents);

      return {
        slug,
        ...data
      };
    });

  return entries as CMSService[];
}

export function getEntryBySlug(type: string, slug: string): CMSService | null {
  const fullPath = path.join(contentDirectory, type, `${slug}.json`);

  if (!fs.existsSync(fullPath)) {
    return null;
  }

  const fileContents = fs.readFileSync(fullPath, 'utf8');
  const data = JSON.parse(fileContents);

  return {
    slug,
    ...data
  } as CMSService;
}
```



```
export interface Industry {
  name: string;
  slug: string;
  desc: string;
  projects: string;
  tech: string;
  outcome: string;
  challenges: string[];
  solutions: string[];
  standards: string[];
  keyEquipment: string[];
}

export interface Product {
  name: string;
  specs: string[];
  principle: string;
  useCase: string;
  standards: string;
}

export interface ProductCategory {
  title: string;
  items: Product[];
}

export interface Project {
  title: string;
  client: string;
  industry: string;
  scope: string;
  tech: string;
  location: string;
  outcome: string;
  value?: string;
  duration?: string;
  team?: string;
  challenges?: string;
  deliverables?: string[];
}

export interface Partner {
  name: string;
  role: string;
  details: string;
}

export interface Certification {
  name: string;
  number: string;
  scope: string;
}

export const INDUSTRIES: Industry[] = [
  {
    name: "Steel Plants",
    slug: "steel",
    desc: "Steel manufacturing requires precise, continuous electrical control across blast furnaces, sinter plants, rolling mills",
    projects: "Sinter Plant Conveyor Sequencing, Blast Furnace Cooling System Automation, Gantry Crane Dual Hoist Synchronization",
    tech: "ABB AC800M Redundant PLC, Yaskawa A1000 Heavy Duty VFD, SCADA systems, CPRI-tested MCC panels",
    outcome: "Reduced process downtime by 14% and optimized raw material flow controls with zero sequence trip occurrences across",
    challenges: [
      "Extreme harmonic current distortion from arc furnaces degrading control system integrity",
      "Heavy vibrational stress on panel enclosures causing terminal loosening over time",
      "High particulate and metallic dust ingress causing insulation failures inside MCC cabinets",
      "Synchronization accuracy requirements for multi-drive rolling mill stand speed matching"
    ],
    solutions: [
      "Active harmonic filter installations at PCC incomer level with IEEE 519 compliance validation",
      "IP65 welded-sealed enclosures with anti-vibration mounting hardware",
      "Quarterly preventive cleaning with thermal imaging diagnostics on all connection points",
      "Closed-loop vector drive control with encoder feedback cards for speed synchronization accuracy"
    ],
    standards: ["IS/IEC 61439-1&2", "IEEE 519", "IEC 61800-3", "ISO 10218-2 (Robotics)"],
    keyEquipment: ["Yaskawa A1000 Crane Drives", "ABB ACS880 Drives", "Siemens S7-1500 PLC", "CPRI IP65 Panels", "Active Harmonic"],
  },
  {
    name: "Railways",
    slug: "railways",
    desc: "Railway infrastructure demands fail-safe automation, precise traction power management, and uninterrupted signal power"
```

... continued: src/lib/data.ts

```
projects: "Substation Control Panels for HMRDA stations, Auxiliary Power Automation modules, Traction Relay Protection Panels",
tech: "Siemens S7-1500 PLC, CPRI-certified double-door panels, RTU telemetry, Redundant fiber optic ring networks",
outcome: "Achieved RDSO/RITES compliance standards with 100% fail-safe operation verified across 24-month service logs.",
challenges: [
  "Meeting RDSO Type Approval requirements for electrical panels and control equipment",
  "Ensuring fail-safe interlock logic with dual-redundant CPU configurations for traction power control",
  "Designing panels for outdoor-rated environments including humidity, temperature swings, and vibration",
  "Integrating with legacy signal relay systems while introducing modern microprocessor logic"
],
solutions: [
  "Comprehensive type test reports including CPRI short circuit, temperature rise, and insulation tests",
  "Dual-CPU hot-standby PLC architectures with bumpless transfer and sub-10ms switchover",
  "IP54/IP65 enclosures with tropicalized conformal coating on PCBs and stainless steel hardware",
  "Gateway modules translating legacy relay contacts into modern Profibus/Modbus digital protocols"
],
standards: ["RDSO/SPN/SP/TI/0018", "IEC 60255 (Protection Relays)", "IS 8623", "CEA Regulations 2010"],
keyEquipment: ["Siemens S7-1500H Redundant PLC", "CPRI Certified Panels", "RTU Systems", "Protection Relays", "Isolation Transducers"],
},
{
  name: "Cement",
  slug: "cement",
  desc: "Cement manufacturing integrates raw material crushing, kiln pyroprocessing, grinding, and packing operations - all requiring precision control",
  projects: "Raw Mill Feed System VFD Retrofit, Kiln Shell Temperature Scanner Integration, Bag House Filter Automation, Packing Line Synchronization",
  tech: "Yaskawa MV1000 Medium Voltage Drive, Allen-Bradley ControlLogix PLC, Siemens WinCC SCADA",
  outcome: "Power consumption lowered by 18% with precise kiln speed adjustment curves and elimination of DOL starter current surges",
  challenges: [
    "Massive starting current demands (up to 7x FLA) on ball mill and roller mill motors causing grid voltage dips",
    "Abrasive cement dust causing rapid wear on fan blades, motor bearings, and sensor optical windows",
    "Kiln shell temperature gradient monitoring requiring 128-point continuous infrared scanning",
    "Process inter-dependency requiring precise sequence interlock logic to prevent material blockage cascades"
  ],
  solutions: [
    "Soft starter and VFD installations reducing peak starting currents to 1.5x FLA with S-curve ramp profiles",
    "IP55/IP65 sealed enclosures with positive pressure purging on control rooms",
    "Yokogawa Kiln Shell Scanner with real-time SCADA trend logging and early warning alarms",
    "Siemens S7-1500 sequence interlock logic with material level sensor inputs and auto-restart logic"
  ],
  standards: ["IEC 61439-1&2", "CPRI", "IEC 61800-3", "IEEE 519", "IS 3043"],
  keyEquipment: ["Yaskawa MV1000 Drives", "Allen-Bradley PLC", "Kiln Shell Scanner", "Bag Filter Controllers", "IP65 MCC Panels"],
},
{
  name: "Power Plants",
  slug: "power",
  desc: "Power generation facilities demand the highest levels of electrical engineering expertise, from high-voltage switchgear to complex protection schemes",
  projects: "Boiler Feed Pump VFD Speed Regulation, CEP & CW Pump MCC Panels, Coal Crusher Sequence Controller, Ash Handling Automation",
  tech: "Yokogawa Centum VP DCS, Wika high-pressure instrumentation, ABB ACH580 Drives, 33KV Vacuum Circuit Breakers",
  outcome: "Stabilized steam header pressure variations to within ±0.5 Bar with real-time speed feedback control, reducing turbine trips",
  challenges: [
    "Maintaining steam header pressure stability during rapid load changes and turbine run-backs",
    "High-pressure and high-temperature sensor installations requiring specialized thermowell and manifold engineering",
    "Coal dust explosion risk in fuel handling areas requiring ATEX-rated electrical equipment",
    "Integrating BOP automation with existing DCS systems without plant shutdown windows"
  ],
  solutions: [
    "Cascade PID control loops on boiler feed pumps with feedforward compensation from steam flow signals",
    "Forged steel thermowells and high-pressure manifold assemblies rated to 300 Bar / 550°C",
    "ATEX Zone 21 flameproof MCC panels with dust-proof junction boxes throughout coal handling",
    "Hot-patching programming capability on DCS with offline simulation validation before cutover"
  ],
  standards: ["CEA Regulations", "IBR (Indian Boiler Regulations)", "ATEX Zone 21", "IEC 60079", "IS 2026"],
  keyEquipment: ["Yokogawa DCS", "ABB Drives", "33KV VCBs", "ATEX Panels", "High-Pressure Transmitters"]
},
{
  name: "Oil & Gas",
  slug: "oil-gas",
  desc: "Hazardous hydrocarbon processing environments demand specialized engineering for electrical and automation systems that ensure safety and reliability",
  projects: "Zone 1 Gas Metering Skid Automation, Crude Oil Tank Farm Level Telemetry, Pipeline SCADA Integration, Compressor Station Control",
  tech: "Honeywell Experion DCS, Flameproof junction boxes, SIL-2 logic solvers, Zener barrier isolators",
  outcome: "Certified SIL-2 execution with PESO approval for hazardous fluid terminal distribution control systems.",
  challenges: [
    "Designing certified intrinsically safe (Ex-ia) field instrument loops in Zone 0 and Zone 1 classified areas",
    "Ensuring SIL-2 functional safety integrity for Emergency Shutdown System (ESD) logic",
    "Corrosion resistance requirements for coastal terminal environments with high humidity and salt exposure",
    "Pipeline SCADA communication over long-distance microwave and cellular VPN networks"
  ],
  solutions: [
    "Zener diode barrier isolators and galvanically isolated signal conditioners on all field loops",
    "IEC 61508 compliant SIL-2 logic solver programming with annual proof test documentation",
    "316L stainless steel enclosures with NEMA 4X rating and epoxy-sealed cable glands",
    "Redundant VSAT and cellular 4G VPN data links with store-and-forward buffer RTUs"
  ],
  standards: ["IEC 60079 (ATEX)", "IEC 61508 (SIL)", "OISD-GDN-106", "PESO Certification", "API RP 505"],
}
```

... continued: src/lib/data.ts

```
keyEquipment: ["SIL-2 Logic Solvers", "Flameproof Panels", "Zener Barriers", "Ex-ia Transmitters", "Gas Detectors"]
},
{
  name: "Water Treatment",
  slug: "water-treatment",
  desc: "Municipal and industrial water treatment automation requires robust, fail-safe systems for raw water intake, pumping,
  projects: "120 MLD HMWSSB Pump Automation, Reservoir Level SCADA Telemetry, Chemical Dosing System Automation, RO Plant Contr
  tech: "L&T PLC, Baumer level sensors, Modbus TCP/IP fiber ring networks, Yaskawa VFD pump panels",
  outcome: "Automated distribution loops across 12 sectors with leak detection telemetry; reduced water loss by 14% over 18 mon
  challenges: [
    "Protecting against water hammer in large diameter mains (1200-2000mm dia) during pump startups",
    "Managing large geographic spread of remote pumping nodes with varying communication reliability",
    "Chemical dosing accuracy control requiring 2% precision across variable flow conditions",
    "Flood risk to electrical panels in underground pump rooms requiring elevated IP ratings"
  ],
  solutions: [
    "Soft-start VFD ramp profiles limiting pressure rise rate to <0.3 Bar/second during startup sequences",
    "RTU-based store-and-forward data buffering with dual SIM 4G fallback communication links",
    "Coriolis mass flow meter feedback to dosing pump variable speed drives for ratio control",
    "IP66 rated stainless enclosures on raised mounting platforms with sump level alarms"
  ],
  standards: ["IS 1172", "WHO Water Quality Standards", "BIS standards", "MOEF Guidelines", "CEA Regulations"],
  keyEquipment: ["VFD Pump Panels", "SCADA Systems", "RTUs", "Dosing Controllers", "Electromagnetic Flow Meters"]
},
{
  name: "Pharma",
  slug: "pharma",
  desc: "Pharmaceutical manufacturing operates under the world's strictest regulatory frameworks. Our FDA 21 CFR Part 11 compli
  projects: "Batch Reactor Automation with Audit Trail, Cleanroom HVAC Control, WFI System Monitoring, CIP/SIP Automation",
  tech: "Siemens WinCC SCADA, S7-1500 PLC, Honeywell sanitized transmitters, stainless sensor housings",
  outcome: "Achieved complete GMP validation compliance with FDA 21 CFR Part 11 electronic records and passed inspection audits
  challenges: [
    "Ensuring 21 CFR Part 11 compliance for electronic records including immutable audit trail and e-signatures",
    "Clean-in-place (CIP) and Steam-in-place (SIP) cycles requiring sensors rated for hot steam and caustic chemicals",
    "Batch recipe management with version control, deviation logging, and master/production recipe separation",
    "Validation documentation (IQ/OQ/PQ protocols) required before system go-live approval"
  ],
  solutions: [
    "SQL Server database with cryptographic hash validation on every record change; role-based access with MFA",
    "Hygienic stainless-steel sensors with Tri-Clamp fittings, IP69K rating, and FDA-compliant wetted materials",
    "Phase Logic recipe engine with master recipe library, batch journal logging, and exception handling workflows",
    "Complete validation dossier including DQ, IQ, OQ, PQ protocols per GAMP 5 guidelines"
  ],
  standards: ["FDA 21 CFR Part 11", "GMP EU Annex 11", "GAMP 5", "WHO GMP", "ISO 14644 (Cleanroom)"],
  keyEquipment: ["Siemens S7-1500 PLC", "WinCC SCADA", "Sanitized Transmitters", "CIP Controllers", "HVAC DDC Panels"]
},
{
  name: "Food Processing",
  slug: "food-processing",
  desc: "Modern food and beverage processing relies on hygienic automation systems that maintain strict temperature profiles, r
  projects: "Dairy Pasteurization Loop Automation, Biscuit Tunnel Oven Temperature Control, Edible Oil Filling Line Control, Be
  tech: "Schneider Modicon PLC, stainless steel panels, IP69K sensors, HMI recipe manager",
  outcome: "Achieved perfect thermal profiles (±0.5°C accuracy) with automated compliance logs per batch; reduced batch rejecti
  challenges: [
    "Maintaining product temperature within tight tolerances during pasteurization (72°C for 15 seconds minimum)",
    "Allergen cross-contamination risks requiring automated clean-in-place validation between product runs",
    "High-pressure washdown cleaning (up to 80 bar hot water) requiring fully sealed electrical equipment",
    "Energy cost reduction targets requiring motor optimization across multiple processing lines simultaneously"
  ],
  solutions: [
    "Cascade temperature PID control with steam valve modulation and predictive feedforward correction",
    "Automated CIP recipe execution with conductivity and temperature validation before product release",
    "IP69K washdown-rated stainless steel MCC panels with sloped tops and hygienic cable routing",
    "VFD installations on all significant motor loads with load profile analysis and savings reporting"
  ],
  standards: ["FSSAI Standards", "BRC Food Safety", "ISO 22000", "HACCP", "IEC 60529 IP69K"],
  keyEquipment: ["IP69K Stainless Panels", "Hygienic Transmitters", "Recipe HMI", "CIP Controllers", "Checkweighers"]
},
{
  name: "Mining",
  slug: "mining",
  desc: "Underground and open-cast mining operations demand ruggedized automation systems capable of operating under heavy mech
  projects: "Underground Ventilation Auto-Control, Overland Conveyor Sequence Controller, Crusher Plant Drive Retrofit, Draglin
  tech: "Yaskawa V1000 micro-drives, fiber optic ring networks, methane sensors, heavy-duty IP66 panels",
  outcome: "Guaranteed automatic high-speed ventilation fan trigger within 8 seconds of gas level alert detection; eliminated m
  challenges: [
    "Methane gas explosion risk in underground workings requiring certified ATEX Zone 1 electrical systems",
    "Overland conveyor systems spanning 3-12 km with multi-drive synchronization requirements",
    "Heavy particulate dust (coal, ore) causing rapid filter clogging in panel ventilation systems",
    "Unreliable communication in underground tunnels requiring local autonomous control capability"
  ],
  solutions: [

```

... continued: src/lib/data.ts

```
    "Flameproof Ex-d distribution boards and Ex-e terminal boxes throughout underground electrical installation",
    "Master-follower VFD drive synchronization with tension control feedback across conveyor take-up pulleys",
    "Sealed positive-pressure ventilated panels with automatic filter replacement alarms",
    "Local PLC autonomy mode with pre-programmed safe-state sequences activated during communication failure"
  ],
  standards: ["Mines Act 1952", "DGMS Circular", "IEC 60079 (ATEX)", "IS 8961", "IEC 61241 (Dust Ex)"],
  keyEquipment: ["ATEX Drives", "Flameproof Panels", "Gas Detection Systems", "Fiber Ring Networks", "Conveyor Controllers"]
},
{
  name: "Defense",
  slug: "defense",
  desc: "Defense and aerospace applications demand the highest engineering rigor with military-grade ruggedized panels, precise timing, and secure communications",
  projects: "ECIL Simulator Control Interface, NFC Testing Rig Automation, Radar System Power Supply Panels, Defense Lab Instrumentation",
  tech: "National Instruments LabVIEW RT, hardened mechanical panels, mil-spec connectors, isolated power supplies",
  outcome: "Executed precision millisecond-response logging under severe vibrational test conditions; achieved 100% audit compliance",
  challenges: [
    "Sub-millisecond timing accuracy requirements for test and measurement data acquisition systems",
    "MIL-STD-810 environmental testing requirements including shock, vibration, temperature extremes",
    "Classified project requirements imposing strict data security and personnel access restrictions",
    "Long-duration reliability requirements - systems expected to operate 20+ years with minimal maintenance"
  ],
  solutions: [
    "FPGA-based real-time data acquisition with NI-9265 deterministic execution and <1µs timing accuracy",
    "Mil-spec aluminum enclosures with shock-mounted PCBs, conformal coating, and hermetic connector sealing",
    "Air-gapped development environments, signed NDAs, and government security clearance protocols",
    "High-MTBF component selection, detailed life prediction analysis, and built-in redundancy design"
  ],
  standards: ["MIL-STD-810", "MIL-STD-461", "DEF STAN", "IEC 61000 (EMC)", "DO-160"],
  keyEquipment: ["NI LabVIEW RT Systems", "Hardened Enclosures", "Mil-Spec Connectors", "Isolated Instruments", "EMI Filters"]
},
{
  name: "Elevators",
  slug: "elevators",
  desc: "Modern vertical transportation systems require precise microprocessor control, smooth vector drive performance, and fail-safe operation",
  projects: "High-Rise Commercial Tower Elevator Modernization, Hospital Goods Lift Controller, Residential Group Dispatch System",
  tech: "Yaskawa L1000A Elevator VFD, microprocessor controller boards, ARD backup units, magnetic leveling sensors",
  outcome: "Leveling accuracy improved to within ±3mm; starting jerks eliminated; energy consumption reduced 25% via regenerative braking",
  challenges: [
    "Passenger comfort requirements demanding jerk-free acceleration profiles with sub-0.5 m/s² jerk limits",
    "Safety requirements mandating automatic rescue to nearest floor within 15 seconds of power failure",
    "Variable counterweight imbalance requiring adaptive drive torque control during different load conditions",
    "Modernization of existing elevator shafts with minimal civil modification work"
  ],
  solutions: [
    "S-curve velocity profiles with 5-parameter jerk control tuning via Yaskawa L1000A parameter sets",
    "Automatic Rescue Device (ARD) with maintenance-free VRLA batteries, tested monthly by onboard self-diagnostics",
    "Closed-loop vector control with load cell feedback for adaptive torque compensation",
    "Controller replacement using existing machine room space, preserving all safety switches and interlocks"
  ],
  standards: ["IS 14665 (Parts 1-6)", "EN 81-20/50", "NBC 2016", "ASME A17.1", "IEC 60947"],
  keyEquipment: ["Yaskawa L1000A", "ARD Units", "Microprocessor Controllers", "Door Operators", "Leveling Sensors"]
},
{
  name: "Chemical",
  slug: "chemical",
  desc: "Chemical processing plants handle corrosive, toxic, and flammable materials requiring specialized instrumentation, corrosion-resistant enclosures, and rigorous safety protocols",
  projects: "Acid Dosing Automation Skid, Polymer Batch Reactor Controller, Chlorine Gas Detection System, Cooling Tower Chemical Control",
  tech: "Wika pressure/temp sensors, Baumer ultrasonic probes, Hastelloy wetted instruments, SIL-2 safety PLC",
  outcome: "Zero acid leakage incidents recorded across all dosing systems over 36 months of continuous operation.",
  challenges: [
    "Corrosive media attacking sensor diaphragms, transmitter casings, and cable insulation over time",
    "Precise pH and conductivity dosing control to within ±0.1 pH units in variable process streams",
    "ATEX Zone 2 classification across most chemical handling areas requiring certified field equipment",
    "Emergency shutdown system requirements with SIL-2 rated safety instrumented functions"
  ],
  solutions: [
    "Hastelloy C276 diaphragm seals and PTFE-coated radar probes isolating electronics from process media",
    "Dual-input pH controller with temperature compensation and automatic electrode cleaning cycles",
    "IECEx Zone 2 certified transmitters with Ex-e terminal boxes throughout classified areas",
    "T2V SIL-2 certified safety PLC with proof test procedures documented per IEC 61511"
  ],
  standards: ["OISD-STD-144", "IEC 60079", "IEC 61511 (SIL)", "IS 5572", "CPCB Guidelines"],
  keyEquipment: ["SIL-2 Safety PLC", "Hastelloy Sensors", "ATEX Instruments", "pH Controllers", "GRP Enclosures"]
},
{
  name: "Sugar",
  slug: "sugar",
  desc: "Sugar processing integrates cane preparation, milling, juice clarification, evaporation, crystallization, centrifugal separation, and packaging",
  projects: "Centrifugal Machine Auto-Cycle Control, Mill Drive VFD Installation, Boiler Fan Speed Control, Process SCADA Dashboard",
  tech: "Yaskawa A1000 Drives, dual-drive control system, touchscreen HMI, CPRI panels",
  outcome: "Centrifugal cycle throughput increased by 22% with regenerated energy from deceleration phases returned to plant bus",
  challenges: [
```

```
... continued: src/lib/data.ts
```

```
    "Centrifugal machine starting requires very high torque at zero speed followed by rapid acceleration to 1500 RPM",
    "Sugar crystallization monitoring requiring precise temperature and density tracking through multiple pan stages",
    "High vibration and shock from crushers and milling equipment affecting panel components and wiring",
    "Seasonal operation requiring rapid recommissioning after 4-5 months of shutdown"
  ],
  solutions: [
    "Yaskawa drives with vector control providing 200% starting torque capability with precision speed ramp control",
    "Yokogawa recorders with Brix and temperature sensors for continuous crystallization batch monitoring",
    "Vibration-isolated panel mounting hardware with flexible conduit connections at cabinet entries",
    "Annual preventive maintenance contract covering pre-season inspection, firmware updates, and recommissioning"
  ],
  standards: ["IS 14742", "BIS Standards", "IEEE 519", "IEC 61800-3", "Factory Act Regulations"],
  keyEquipment: ["Yaskawa Drives", "HMI Systems", "Yokogawa Recorders", "Vibration-Isolated Panels", "Brix Sensors"]
},
{
  name: "Material Handling",
  slug: "material-handling",
  desc: "Industrial material handling - from overhead cranes to automated conveyor systems and stacker-reclaimers - requires precise coordination and safety features",
  projects: "Gantry Crane Dual Hoist Drive Panel, Stacker-Reclaimer Drive System, Coal Conveyor Sequence Controller, Automated Sorting System",
  tech: "CPRI heavy panels, Yaskawa A1000 crane drives, anti-sway software, L&T switchgear, encoder feedback",
  outcome: "Eliminated hook load sway using active anti-sway algorithm; mechanical gear wear reduced 35% in the first year of operation",
  challenges: [
    "Crane hook pendulum sway effect causing load swing risks and restricting operator precision in narrow bays",
    "Multi-motor conveyor synchronization requiring 99.5% speed match across 400m belt lengths",
    "Overload protection on crane hoists to prevent catastrophic mechanical failure during heavy lift operations",
    "ATEX requirements for coal and ore handling environments with combustible dust classification"
  ],
  solutions: [
    "Yaskawa DriveWizard anti-sway algorithm using observer control theory to damp pendulum oscillations",
    "Encoder-based master-follower drive synchronization with load cell tension feedback on conveyor take-up frames",
    "Electronic overload monitoring with hardwired emergency brake engage on torque limit exceedance",
    "ATEX Zone 22 certified drives with enclosed IP65 panels and sealed conduit systems in dust zones"
  ],
  standards: ["IS 807 (EOT Cranes)", "IS 3177", "IEC 60034-1", "ATEX Zone 22", "IEEE 519"],
  keyEquipment: ["Crane Drive Panels", "Anti-Sway Systems", "Belt Weighers", "Encoder Systems", "Safety PLCs"]
}
];

export const PRODUCTS: ProductCategory[] = [
  {
    title: "Electrical Panels",
    items: [
      {
        name: "PCC (Power Control Center) Panels",
        specs: [
          "Main busbar rating up to 6300A (electrolytic copper)",
          "CPRI certified short-circuit test up to 65kA / 1 second",
          "Incomer: Air Circuit Breakers (ACB) with electronic trip units",
          "Form 4b segregation - busbar, cable, and switchgear zones isolated",
          "IP55 / IP65 ingress protection available",
          "Active ventilation with thermostat-controlled forced cooling fans"
        ],
        principle: "Centralized power distribution system that receives high-voltage utility supply and distributes to all sub-distribution units",
        useCase: "Steel mills, cement industries, large manufacturing complexes, commercial buildings, substations, chemical plants",
        standards: "IS/IEC 61439-1&2 certified, CPRI tested, IP55/65 enclosure standard."
      },
      {
        name: "MCC (Motor Control Center) Panels",
        specs: [
          "Drawout or fixed type configuration for service flexibility",
          "Fully compartmentalized modular drawers with mechanical interlocks",
          "Intelligent MCC options with Profibus/Modbus/EtherNet-IP connectivity",
          "Type-2 coordination starter combinations (contactor + MPCB)",
          "Integrated multi-function protection relays (overload, phase loss, EF)",
          "HMI integration for motor running status and fault logging"
        ],
        principle: "Controls, protects, and monitors electric motors from a single central location with coordinated protection and signaling",
        useCase: "Process plants with multiple motors, pumping stations, fan control matrices, HVAC systems, conveyor groups.",
        standards: "CPRI tested, IEC 60947-4-1 coordination, IS 8623."
      },
      {
        name: "VFD (Variable Frequency Drive) Panels",
        specs: [
          "Drive range: 0.75 kW to 1000 kW in LV; up to 10MW in MV",
          "Integrated AC line reactors and passive harmonic filters",
          "Forced ventilation design with thermal sensors and alarms",
          "Bypass arrangements: Automatic or Manual with motor DOL fallback",
          "Profinet/Modbus TCP communication cards standard",
          "Dynamic braking resistors for high-inertia load applications"
        ],
        principle: "Varies motor supply frequency and voltage to control motor speed, torque, and acceleration profiles - enabling energy savings and precise control",
        useCase: "Pumping stations, conveyor systems, kiln drives, industrial fans, elevators, compressors, cranes."
      }
    ]
  }
];
```

```

    standards: "Yaskawa Authorized assembly protocols, IEEE 519, IEC 61800-3 EMC standard."
  },
  {
    name: "PLC & SCADA Control Panels",
    specs: [
      "Redundant CPU layouts with hot-standby failover (<10ms switchover)",
      "Galvanically isolated I/O modules (analog and digital)",
      "Integrated 24VDC UPS power backup - minimum 4-hour runtime",
      "Managed Ethernet switches with fiber optic SFP connectivity",
      "Alarm annunciator modules with first-out indication",
      "Built-in EMI filters on all power supply rails"
    ],
    principle: "Main control hub executing PLC logic programs and communicating with field instruments while providing SCADA",
    useCase: "Plant-wide automation, process chemical plants, power plant monitoring, machine tools, batch processes.",
    standards: "CE certified assembly, IPC-A-610 wiring standard, IEC 61131-3."
  },
  {
    name: "APFC (Automatic Power Factor Correction) Panels",
    specs: [
      "Power factor correction range: 0.6 to 0.99 lagging",
      "Capacitor bank ratings: 5 KVAR to 500 KVAR steps",
      "Intelligent relay controller with 12-step switching capability",
      "Thyristor-based fast switching for flicker-sensitive loads",
      "Harmonic detuning reactors on each capacitor stage",
      "Digital power quality display with PF, KVAR, and kWh metering"
    ],
    principle: "Automatically switches capacitor banks in and out of service based on measured reactive load, maintaining grid",
    useCase: "Industrial factories with large motor loads, commercial complexes, municipal infrastructure, data centers.",
    standards: "IS 13340, IEC 60831, IEEE 1459."
  }
],
},
{
  title: "Automation & Drive Systems",
  items: [
    {
      name: "PLC (Programmable Logic Controller) Systems",
      specs: [
        "High-speed CPU scan cycle: <1ms for 1000 logic rungs",
        "Modbus RTU/TCP, Profinet, EtherCAT, PROFIBUS-DP standard interfaces",
        "Hot-swappable dual redundant I/O modules and power supplies",
        "Analog signal conditioning: ?0.1% accuracy on 4-20mA loops",
        "Safety CPU extensions (SIL-2/SIL-3) available",
        "Integrated motion control for up to 128 servo axes synchronization"
      ],
      principle: "Digital computing system that monitors real-time inputs from sensors and switches, executes control logic pro",
      useCase: "Machinery speed control, sequence interlocking, safety shutdown systems, batch recipe management, crane/hoist c",
      standards: "IEC 61131-3 programming, CE/UL certified, IEC 61508 SIL."
    },
    {
      name: "SCADA (Supervisory Control and Data Acquisition)",
      specs: [
        "Real-time trending with 100ms data logging resolution",
        "SQL Server integration for unlimited historical data storage",
        "FDA 21 CFR Part 11 electronic records and audit trail module",
        "Dynamic SVG vector graphics mimic flow diagrams",
        "OPC UA secure server interface for ERP integration",
        "Mobile access via encrypted web client (HTML5)"
      ],
      principle: "Software platform gathering real-time data from PLCs and RTUs, providing supervisory visualization, alarm man",
      useCase: "Water treatment plants, power generation monitoring, pharmaceutical batch tracking, chemical plant process cont",
      standards: "ISA-101 HMI design, IEC 62443 cybersecurity, FDA 21 CFR Part 11."
    },
    {
      name: "Servo & Motion Control Systems",
      specs: [
        "Ultra-high resolution absolute encoders (24-bit, 16 million counts/rev)",
        "Multi-axis synchronized motion with <5?s interpolation cycle",
        "Jerk-limiting S-curve profile generation",
        "Built-in STO (Safe Torque Off) - SIL-3 certified",
        "EtherCAT real-time communication at 1Gbps",
        "Auto-tuning wizard for fast axis commissioning"
      ],
      principle: "High-precision closed-loop control of angular position, velocity, and torque for highly dynamic, repeatable m",
      useCase: "Robotics loaders, CNC machining centers, steel slitting lines, packaging servo axes, printing presses.",
      standards: "T?V certified SIL-3, IEC 61800-5-2, ISO 13849 PL."
    },
    {
      name: "Yaskawa Variable Frequency Drives",
      specs: [
        "Standard range: 0.4 kW to 1000 kW (GA700/A1000 series)",
        "Medium voltage: 3.3kV to 11kV (FSDrive-MV range, up to 10MW)"
      ]
    }
  ]
}

```


... continued: src/lib/data.ts

```
        "High torque at zero speed: 150% nominal torque at 0 Hz",
        "Dynamic braking capability for crane and elevator loads",
        "Built-in EMC filter, DC link choke, and PG card options",
        "Conformally coated boards as standard for industrial environments"
    ],
    principle: "Vector control algorithm precisely calculates and controls the magnetic flux and torque of the motor using cu",
    useCase: "Pumps, fans, compressors, conveyors, cranes, elevators, centrifuges, extruders.",
    standards: "IEEE 519, IEC 61800-3 (Category C3), UL508, CE, RoHS."
  }
},
],
{
  title: "Process Instrumentation",
  items: [
    {
      name: "Pressure & Temperature Transmitters",
      specs: [
        "Pressure accuracy: ±0.075% of calibrated span",
        "Pressure ranges: vacuum to 1000 Bar",
        "HART 7 and Foundation Fieldbus communication protocols",
        "ATEX/IECEX Zone 1 certified (Ex ia/ib)",
        "Dual-chamber electrical compartment (process and terminal box separated)",
        "316L stainless steel or Hastelloy C276 wetted parts"
      ],
      principle: "Converts physical pressure or temperature of process fluid into standard 4-20mA (+ HART digital) electrical s",
      useCase: "Steam pipelines, gas flow measurement, autoclave reactors, boiler drums, compressed air systems.",
      standards: "ATEX Zone 1, IECEX, SIL-2 certified, NABL traceable calibration."
    },
    {
      name: "Baumer Sensors & Level Controllers",
      specs: [
        "Measurement technologies: ultrasonic, radar 80GHz FMCW, hydrostatic",
        "IO-Link and 4-20mA/Profibus/EtherNet-IP interface options",
        "Stainless steel hygienic 316L housings with FDA-approved seals",
        "Measurement range: 0.2m to 70m (radar), 0.3m to 15m (ultrasonic)",
        "Process temperature: up to 200°C (high-temp radar versions)",
        "Measurement accuracy: ±3mm typical at full range"
      ],
      principle: "Transmits radar or ultrasonic pulses toward the product surface; calculates level from time-of-flight of echo",
      useCase: "Water reservoirs, chemical storage tanks, cement powder silos, slurry pits, pharmaceutical mixing vessels.",
      standards: "IP69K washdown rated, ATEX Zone 0/1 versions, FDA 21 CFR 177 compliant contacts."
    },
    {
      name: "Electromagnetic & Vortex Flow Meters",
      specs: [
        "Electromagnetic: DN15 to DN2400, accuracy ±0.3% of reading",
        "Vortex: DN25 to DN300, for steam and gas measurement",
        "Empty pipe detection with self-diagnostic alarms",
        "Bidirectional measurement capability",
        "HART, Profibus-PA, Foundation Fieldbus communication",
        "Liner materials: hard rubber, PTFE, ceramic options"
      ],
      principle: "Electromagnetic: Faraday's law - magnetic field applied across pipe; conductive fluid induces EMF proportiona",
      useCase: "Water distribution, wastewater, chemical dosing, slurry pipelines, steam and compressed air measurement.",
      standards: "MID certified, OIML R49, IS 6940, NABL traceable."
    },
    {
      name: "WIKA Pressure Gauges & Instruments",
      specs: [
        "Accuracy class: 0.1%, 0.25%, 0.5%, 1.0% options",
        "Pressure ranges: -1 to 0 Bar vacuum up to 1600 Bar",
        "Liquid-filled versions for high vibration and pulsating service",
        "Diaphragm seals for viscous, crystallizing, or corrosive media",
        "Temperature compensation: -40°C to +250°C",
        "Stainless steel, brass, or Monel wetted materials"
      ],
      principle: "Bourdon tube or diaphragm element deflects proportionally to applied pressure; deflection is amplified via mo",
      useCase: "Process plants, hydraulic systems, gas cylinders, compressor stations, medical and laboratory equipment.",
      standards: "EN 837 (Bourdon gauges), EN 13443, PED 2014/68/EU, ASME B40.100."
    }
  ]
}
];

export const PROJECTS: Project[] = [
  {
    title: "120 MLD Municipal Water Pump Automation",
    client: "HMWSSB (Hyderabad Metro Water Supply & Sewerage Board)",
    industry: "Water Treatment",
    scope: "Turnkey design, assembly, PLC automation, SCADA programming, fiber network layout, and on-site commissioning of the 1",
    tech: "ABB AC800M Redundant PLC, 450kW Yaskawa A1000 Drive panels, Ethernet Ring topology, WinCC SCADA, fiber optic backbone",
    location: "Amberpet, Hyderabad, Telangana",
  }
]
```

... continued: src/lib/data.ts

```
outcome: "Reduced daily energy consumption by 21% and enabled fully remote pump station supervision from the HMWSSB central c
value: "23.8 Crore",
duration: "18 months",
team: "12 engineers",
challenges: "Replacing existing DOL starters without halting water supply; integrating with existing SCADA system running on
deliverables: ["6-way LT MCC Panel", "3 VFD Panels (450kW each)", "PLC Panel with Redundant CPU", "SCADA Operator Workstation
},
{
  title: "Gantry Crane Hoist Synchronization Panel",
  client: "Vizag Steel Plant (Rashtriya Ispat Nigam Limited)",
  industry: "Steel",
  scope: "Design and retrofitting of a CPRI-certified double hoisting system control panel with precise crane speed-match softw
  tech: "Yaskawa A1000 Heavy Duty VFD panels, dual PLC communication, anti-sway software, encoder feedback systems",
  location: "Visakhapatnam, Andhra Pradesh",
  outcome: "Eliminated load misalignment; reduced mechanical gear strain by 35% during continuous heavy operation cycles and ac
  value: "21.6 Crore",
  duration: "9 months",
  team: "8 engineers",
  challenges: "Working within the active plant environment with strict crane downtime windows; calibrating anti-sway algorithm
  deliverables: ["Dual 200kW VFD Drive Panels", "Safety PLC with SIL-2 Logic", "Anti-Sway Software Configuration", "CPRI Test C
},
{
  title: "Zone 1 Hazardous Gas Metering Skid Panel",
  client: "HPCL Terminal, Ghatkesar",
  industry: "Oil & Gas",
  scope: "Engineering of flameproof field junction boxes and SIL-2 certified main logic panels with SCADA telemetry integration
  tech: "Honeywell DCS, ATEX flameproof barriers, armored fiber interfaces, Ex-d junction boxes",
  location: "Ghatkesar Depot, Telangana",
  outcome: "Approved by Chief Controller of Explosives (PESO) and commissioned without a single safety incident.",
  value: "22.1 Crore",
  duration: "12 months",
  team: "10 engineers",
  challenges: "Obtaining PESO CCE approval requires extensive documentation; designing panels for Zone 1 area where ATEX Ex-d e
  deliverables: ["ATEX Zone 1 MCC Panel", "SIL-2 Logic Solver Panel", "Flameproof Field JB's (x12)", "Gas Detector Integration",
},
{
  title: "Sinter Plant Conveyor Sequence Controller",
  client: "JSW Steel Limited",
  industry: "Steel",
  scope: "Design, programming, and commissioning of a complex sequence interlock controller for 28 conveyors in the raw materia
  tech: "Siemens S7-1500 PLC, HMI panel, CPRI-tested heavy enclosures, Profinet network ring",
  location: "Bellary, Karnataka",
  outcome: "Zero conveyor sequence error incidents in 18 months of continuous high-load operations; reduced material spillage b
  value: "1.9 Crore",
  duration: "10 months",
  team: "7 engineers",
  challenges: "Sequence logic must account for 28 individually controlled conveyors with complex cause-and-effect interlock mat
  deliverables: ["PLC Panel with S7-1500", "HMI Mimic Display", "Conveyor Control Logic Software", "Belt Weigher Integration",
},
{
  title: "Hyderabad Metro Rail Substation Control Panels",
  client: "Hyderabad Metro Rail Ltd. (HMRL / MEIL)",
  industry: "Railways",
  scope: "Supply and commissioning of 14 SCADA-integrated auxiliary power control panels for metro rail traction substations ac
  tech: "Siemens S7-1500 PLC, CPRI double-door panels, RTU telemetry, fiber ring communication",
  location: "Hyderabad Metro Blue Line, Telangana",
  outcome: "Achieved RDSO compliance with 99.98% uptime logged across all substation panels during the first year of commercial
  value: "24.7 Crore",
  duration: "24 months",
  team: "18 engineers",
  challenges: "Stringent RDSO type test requirements; simultaneous commissioning across 14 substations in live city environment
  deliverables: ["14 Auxiliary Power Panels", "RTU Telemetry Units", "SCADA Integration", "RDSO Compliance Documentation", "Sub
},
{
  title: "Pharmaceutical Batch Reactor Automation",
  client: "Aurobindo Pharma Limited",
  industry: "Pharma",
  scope: "Complete FDA 21 CFR Part 11 compliant SCADA and PLC batch reactor automation system with electronic batch record, aud
  tech: "Siemens S7-1500 PLC, WinCC SCADA, HART transmitters, stainless steel hygienic panels",
  location: "Pashamylaram, Telangana",
  outcome: "Passed USFDA inspection with zero 483 observations on automation systems; batch cycle time reduced by 18% through o
  value: "2.8 Crore",
  duration: "14 months",
  team: "11 engineers",
  challenges: "IQ/OQ/PQ validation documentation; all equipment change control through formal procedures; integration with exis
  deliverables: ["FDA-Compliant SCADA System", "Batch Recipe Manager", "Electronic Batch Records", "IQ/OQ/PQ Protocol Packages"
},
{
  title: "Cement Raw Mill VFD Retrofit & SCADA Integration",
  client: "Penna Cement Industries",
  industry: "Cement",
  scope: "Replacement of 5 direct-on-line (DOL) starters on 600kW to 1200kW raw mill and separator motors with Yaskawa medium v
```


... continued: src/lib/data.ts

```
tech: "Yaskawa MV1000 Medium Voltage Drives, Allen-Bradley ControlLogix PLC, WinCC SCADA integration",
location: "Boyareddypalle, Andhra Pradesh",
outcome: "Power consumption reduced 22% on raw mill section; DOL starting current surges (7x FLA) eliminated; mill availability improved 15%",
value: "₹25.2 Crore",
duration: "16 months",
team: "14 engineers",
challenges: "Medium voltage (6.6kV) drive installation with 72-hour plant shutdown window; motor cable replacement for drive-rated applications",
deliverables: ["5 MV VFD Panels (6.6kV)", "Harmonic Analysis Report", "SCADA Integration", "Energy Savings Verification Report"],
},
{
  title: "Coal Handling Plant Automation - NTPC Ramagundam",
  client: "NTPC (National Thermal Power Corporation)",
  industry: "Power Plants",
  scope: "Design and commissioning of complete coal handling plant (CHP) automation including 23 conveyor sequence controllers, 23 VFD drives, and 23 MCC panels",
  tech: "Yokogawa Centum VP DCS, Siemens S7-400 PLC, ATEX Zone 21 MCC panels, belt weighers",
  location: "Ramagundam, Telangana",
  outcome: "Coal handling throughput achieved at 2500 TPH design capacity; zero conveyor sequence failure incidents logged over 18 months",
  value: "₹7.4 Crore",
  duration: "28 months",
  team: "22 engineers",
  challenges: "ATEX Zone 21 (combustible dust) certification requirements across all electrical panels; integration with existing plant DCS",
  deliverables: ["ATEX Zone 21 MCC Panels", "DCS Integration Modules", "23 Conveyor Controllers", "Belt Weigher Integration", "Safety Interlocks"],
},
{
  title: "Boiler Feed Pump VFD Speed Control System",
  client: "GVK Power Plant, Jegurupadu",
  industry: "Power Plants",
  scope: "Replacement of throttle valve control system on 4 x 800kW boiler feed pumps with variable speed drives, providing energy savings and reduced wear",
  tech: "ABB ACS880 MV Drives, Yokogawa flow transmitters, Wika pressure instrumentation, DCS integration",
  location: "Jegurupadu, Andhra Pradesh",
  outcome: "Reduced pump power consumption by 31%; eliminated throttle valve pressure drop losses; steam header pressure stabilized",
  value: "₹3.1 Crore",
  duration: "11 months",
  team: "9 engineers",
  challenges: "Shutdown window limited to 72 hours per pump; tight DCS PID tuning required for stable steam pressure under dynamic load",
  deliverables: ["4 x ABB MV VFD Panels", "Instrumentation Upgrade", "DCS Integration & Tuning", "Energy Audit Report", "FAT/SAT"],
},
{
  title: "Oil Refinery Tank Farm SCADA Telemetry",
  client: "BPCL (Bharat Petroleum Corporation Ltd.)",
  industry: "Oil & Gas",
  scope: "Design and commissioning of a complete tank farm SCADA monitoring system for 48 crude oil and product storage tanks including level, temperature, and pressure monitoring",
  tech: "Honeywell Experion C300, ATEX radar level gauges, Ex-ia transmitters, OPC gateway server",
  location: "Navi Mumbai, Maharashtra",
  outcome: "Real-time inventory tracking accuracy improved to 0.1% volume measurement; overfill prevention interlock response time reduced to 5 seconds",
  value: "₹6.3 Crore",
  duration: "20 months",
  team: "16 engineers",
  challenges: "Integrating with 48 tanks across a 12km² terminal with multiple classified zones; OPC gateway integration with existing refinery DCS",
  deliverables: ["Tank Gauging SCADA System", "48 Radar Level Gauges (ATEX)", "Overfill Protection System", "Fire & Gas Panel Integration"],
},
{
  title: "Water Treatment Plant Automation - 60 MLD RO",
  client: "Telangana Industrial Infrastructure Corporation (TSIIC)",
  industry: "Water Treatment",
  scope: "Automation of a 60 MLD reverse osmosis water treatment plant including pre-treatment, high-pressure pump control, permeate flow control, and chemical dosing",
  tech: "Siemens S7-300 PLC, Baumer level sensors, Modbus fiber network, stainless steel panels",
  location: "Zaheerabad Industrial Area, Telangana",
  outcome: "Plant recovery ratio improved to 78% (from 68%); chemical consumption reduced 23% through automated dosing optimization",
  value: "₹2.4 Crore",
  duration: "13 months",
  team: "9 engineers",
  challenges: "High-pressure pump (60 Bar) startup sequences requiring precise ramp control; RO membrane protection interlocks",
  deliverables: ["PLC Control Panel", "High-Pressure Pump VFD Panels", "Online Water Quality Monitoring", "CIP Automation System"],
},
{
  title: "Sugar Mill Centrifugal Machine Drive System",
  client: "NSL Sugar Limited",
  industry: "Sugar",
  scope: "Supply and commissioning of VFD-based automatic cycle control system for 12 batch centrifugal machines with PLC sequence control and energy regeneration",
  tech: "Yaskawa A1000 drives, dual-drive configuration, regenerative braking, touchscreen HMI, CPRI panels",
  location: "Kovvur, Andhra Pradesh",
  outcome: "Centrifugal cycle throughput increased 22%; regenerated energy from deceleration estimated at 18% of motor input energy",
  value: "₹1.8 Crore",
  duration: "7 months",
  team: "6 engineers",
  challenges: "Centrifugal machines require 200% torque at zero speed; existing motor wiring needed upgrading for drive-rated applications",
  deliverables: ["12 VFD Drive Panels", "PLC Sequence Controllers", "HMI Operator Stations", "Regenerative Braking Units", "FAT/SAT"],
},
{
  title: "Robotic Palletizing Cell - Packaging Plant",
  client: "ITC Limited (Foods Division)",
  industry: "Food Processing",
  scope: "Design and commissioning of a robotic palletizing cell for 1200kg bags using a 6-axis industrial robot and PLC control",
  tech: "FANUC 6-axis robot, Siemens S7-1200 PLC, safety laser scanner, vision system for bag identification",
  location: "Mysore, Karnataka",
  outcome: "Palletizing cycle time reduced from 45 to 30 seconds; labor savings of 2 FTEs achieved; 99.9% uptime over 6 months",
  value: "₹4.5 Crore",
  duration: "10 months",
  team: "11 engineers",
  challenges: "Integration with existing bagging line; safety enclosure design for high-torque robot; vision system for bag identification",
  deliverables: ["Robot Cell Enclosure", "PLC Control System", "Safety Laser Scanner", "Vision System", "FAT/SAT"],
}
```

```
... continued: src/lib/data.ts
```

```
    industry: "Food Processing",
    scope: "Integration of a Yaskawa Motoman 4-axis palletizing robot cell for end-of-line carton palletizing, including conveyor",
    tech: "Yaskawa Motoman MH180 robot, YRC1000 controller, Cognex vision system, safety PLC",
    location: "Medchal, Telangana",
    outcome: "Line throughput increased from 18 pallets/hour (manual) to 35 pallets/hour; zero product damage incidents since commissioning",
    value: "₹22.2 Crore",
    duration: "8 months",
    team: "8 engineers",
    challenges: "Coordinating robot pick cycles with variable upstream packaging speeds; hygienic design requirements for food zone",
    deliverables: ["Robot Cell with Safety Fencing", "Conveyor Interface System", "Vision Guided Pick Program", "Safety Risk Assessment"],
  },
  {
    title: "ECIL Defense Testing Rig Automation Interface",
    client: "Electronics Corporation of India Limited (ECIL)",
    industry: "Defense",
    scope: "Custom automation interface for a high-precision component testing rig requiring sub-millisecond response data acquisition",
    tech: "National Instruments LabVIEW RT, NI CompactRIO chassis, mil-spec connectors, isolated power supplies",
    location: "Hyderabad, Telangana",
    outcome: "Achieved 250µs data acquisition timing accuracy; test throughput improved 3x compared to manual operation; zero measurement errors",
    value: "Confidential",
    duration: "10 months",
    team: "5 engineers",
    challenges: "Sub-millisecond timing requirements; MIL-STD environmental compliance; classified project protocols restricting external communication",
    deliverables: ["LabVIEW RT Application", "Hardened NI CompactRIO System", "Test Sequence Software", "Calibration Documentation"],
  },
  {
    title: "Underground Mine Ventilation Automation",
    client: "Singareni Collieries Company Ltd. (SCCL)",
    industry: "Mining",
    scope: "Complete automation of underground mine ventilation including auxiliary fan speed control, methane sensor integration",
    tech: "Yaskawa V1000 drives, ATEX Zone 1 panels, methane gas detectors, fiber network, safety PLC",
    location: "Kothagudem, Telangana",
    outcome: "Emergency fan trigger response time reduced from >2 minutes (manual) to <8 seconds (automated); achieved DGMS statutory compliance",
    value: "₹1.3 Crore",
    duration: "9 months",
    team: "7 engineers",
    challenges: "ATEX Zone 1 requirements throughout underground workings; reliable communication in shaft environments; DGMS (Draughting) compliance",
    deliverables: ["ATEX VFD Drive Panels (x3)", "Methane Detection Network", "Safety PLC with Interlocks", "Emergency Override Protocol"],
  }
];

export const CERTIFICATIONS: Certification[] = [
  { name: "ISO 9001:2015", number: "QMS-0521-020", scope: "Quality Management System for design, manufacture, and servicing of Electrical Equipment", status: "Valid", expiry: "2025-12-31" },
  { name: "CPRI Short-Circuit Certificate", number: "CPRI-SC-1025", scope: "Busbar verification test up to 65kA for power control equipment", status: "Valid", expiry: "2024-06-30" },
  { name: "A-Grade Electrical License (33KV)", number: "TS-A-33KV-91", scope: "Authorized by Government of Telangana for high voltage work", status: "Valid", expiry: "2026-03-15" },
  { name: "Yaskawa Authorized Service Center", number: "YAS-ASC-2026", scope: "Authorized service center for Yaskawa AC drives (V1000 series)", status: "Valid", expiry: "2027-01-01" },
  { name: "PESO Safety Certificate", number: "PESO-TS-0892", scope: "Petroleum and Explosives Safety Organisation approval for hazardous area equipment", status: "Valid", expiry: "2025-09-30" },
  { name: "RDSO Approved Vendor", number: "RDSO-MP-0.4200.87", scope: "Indian Railways Research Designs and Standards Organisation approval", status: "Valid", expiry: "2026-04-15" }
];

export const WORKFLOW = [
  { step: "Design Engineering", title: "Technical Blueprint & Simulation", desc: "Our engineering division designs AutoCAD Electrical schematics and SolidWorks 3D models for client review", status: "Completed", order: 1 },
  { step: "Panel Manufacturing", title: "CPRI Compliant Assembly", desc: "Enclosures are fabricated and powder-coated. Copper busbars are precision-cut and drilled", status: "In Progress", order: 2 },
  { step: "FAT Testing", title: "Factory Acceptance Tests", desc: "Every board undergoes insulation resistance tests (1000V Megger) and functional load testing", status: "Completed", order: 3 },
  { step: "Site Deployment", title: "Erection & Field Cabling", desc: "Panels are positioned, leveled, and anchored at the client site. Cables are routed through trays and secured", status: "In Progress", order: 4 },
  { step: "Commissioning", title: "System Startup & Tuning", desc: "No-load dry runs are initiated, motor parameter auto-tuning is performed, and safety protocols are verified", status: "In Progress", order: 5 },
  { step: "Lifecycle Service", title: "24/7 AMC Support", desc: "Preventive health checks, drive component renewal, relay calibration, and emergency troubleshooting", status: "Ongoing", order: 6 }
];
```

```

export interface ServiceChallenge {
  before: string;
  after: string;
  scenario: string;
}

export interface ServiceWorkflowStep {
  name: string;
  activity: string;
  tools: string;
  safety: string;
  deliverable: string;
}

export interface ServiceUseCase {
  client: string;
  requirement: string;
  problem: string;
  solution: string;
  products: string;
  outcome: string;
  improvement: string;
}

export interface ServiceFAQ {
  q: string;
  a: string;
}

export interface ServiceDetails {
  slug: string;
  title: string;
  subtitle: string;
  stats: { label: string; value: string }[];
  overviewParagraphs: string[];
  challenges: ServiceChallenge;
  workflow: ServiceWorkflowStep[];
  architectureDesc: string;
  architectureDetails: string[];
  industries: { name: string; need: string; challenge: string; advantage: string }[];
  caseStudy: ServiceUseCase;
  techEcosystem: { name: string; use: string; logic: string }[];
  qualityAssurance: string[];
  safetyCompliance: { standard: string; description: string }[];
  maintenanceAMC: string[];
  docs: { name: string; size: string; type: string }[];
  faqs: ServiceFAQ[];
}

export const SERVICES_DATA: Record<string, ServiceDetails> = {
  "panel-manufacturing": {
    slug: "panel-manufacturing",
    title: "CPRI-Certified Electrical Control Panels",
    subtitle: "Custom Engineered Power Control Centers (PCC), Motor Control Centers (MCC), VFD & PLC Panels Built to IS/IEC 61439-1",
    stats: [
      { label: "Short-Circuit Protection", value: "65kA / 1 Sec" },
      { label: "Busbar Rating", value: "Up to 6300A" },
      { label: "Compliance Standard", value: "IEC 61439-1&2" },
      { label: "Enclosure Protection", value: "IP55/IP65 Certified" }
    ],
    overviewParagraphs: [
      "Power distribution and motor management represent the core electrical heart of any heavy process facility. At Nandini Enterprises, we understand the critical importance of reliable electrical infrastructure. Every single panel board starts its life in our state-of-the-art AutoCAD mechanical design suite. We perform extensive 2D and 3D modeling to ensure precision and efficiency. Our fabrication plant utilizes top-tier, high-conductivity electrolytic grade copper and aluminum busbars. The busbars are meticulously crafted to meet the highest standards of quality. Quality assurance is integrated into every stage of fabrication. Our testing laboratory executes high-voltage insulation testing to ensure the integrity and safety of our products."
    ],
    challenges: {
      before: "A major cement packaging plant suffered from random electrical panel trips, overheating busbar joints, and high copper losses, leading to frequent production downtime and safety concerns.",
      after: "Nandini Enterprises engineered a modular, double-busbar PCC panel with active exhaust ventilation grids, custom copper busbars, and advanced protection systems, significantly reducing trips and improving safety.",
      scenario: "Eliminated random load trips, lowered cabinet internal temperatures by 18°C, and secured fail-safe power supply for critical equipment."
    },
    workflow: [
      {
        name: "Load Study & Design",
        activity: "Analyze client electrical schematic requirements, compute total active load factors, design copper busbar size and layout, and create detailed panel drawings.",
        tools: "AutoCAD Electrical, SolidWorks Electrical Thermal Solver",
        safety: "Double check insulation clearances against safety guidelines.",
        deliverable: "Approved electrical single line diagrams (SLD) and general arrangement (GA) layout drawing sheets."
      }
    ]
  }
}

```

```

        name: "Busbar Fabrication",
        activity: "Hydraulic cutting, bending, and punching of electrolytic copper bars. Apply high-voltage heat-shrinkable color",
        tools: "Hydraulic Busbar processing machines, torque wrench setups",
        safety: "Mandatory safety boots, face protection, and mechanical barrier guardrails during punching.",
        deliverable: "Completed high-conductivity busbar system with torque-marked hardware fittings."
    },
    {
        name: "Wiring & Component Fitment",
        activity: "Mount switchgears, install control PLC/HMI modules, run internal wire loops, and crimp heavy-duty lugs with fe",
        tools: "Pneumatic crimping tools, digital multi-meters",
        safety: "Organize wiring channels to avoid proximity to heat sources.",
        deliverable: "Fully wired, compartmentalized Form-4b switchgear assembly ready for diagnostic checks."
    },
    {
        name: "FAT Testing & Dispatch",
        activity: "Conduct insulation resistance checks, high-voltage breakdown test (2.5kV/1 min), logic injection checks, and t",
        tools: "Megger insulation tester, primary current injection kit",
        safety: "Establish high-voltage exclusion barricades with warning beacons.",
        deliverable: "Signed Factory Acceptance Test (FAT) report sheet and secure transport packing dispatch logs."
    }
],
architectureDesc: "Our electrical panels utilize compartmentalized Form 4b segregation. The busbar chamber, cable alley, and
architectureDetails: [
    "Busbar Chambers: Run horizontally and vertically, wrapped in heat-shrink sleeves and isolated from breaker compartments.",
    "Switchgear Modules: Integrated with Air Circuit Breakers (ACB) or Molded Case Circuit Breakers (MCCB) matching coordination",
    "Thermal Ventilation: Active twin fan cooling systems running on automatic digital temperature triggers.",
    "Earthing: Dedicated internal copper ground running the entire panel length with external earth pit connectivity hooks."
],
industries: [
    { name: "Steel", need: "Heavy-duty power distribution to withstand arc furnaces and mechanical vibrations.", challenge: "Se"},
    { name: "Cement", need: "High power starter feeders for heavy vertical roller mill motors.", challenge: "Heavy mechanical s"},
    { name: "Water", need: "Pumping station distribution control with dual redundant grid incomers.", challenge: "Outdoor environ"},
],
caseStudy: {
    client: "Vizag Steel Rolling Mill Phase II",
    requirement: "Replace outdated, overheating MCC panel feeding auxiliary cooling fans and water pumps.",
    problem: "Copper busbars were deteriorating, causing localized terminal hotspots, causing regular outages.",
    solution: "Assembled a 4000A, Form-4b segregated copper busbar MCC panel with smart electronic relay modules and Modbus gat",
    products: "Electrolytic Copper Busbars, ABB Tmax MCCBs, Schneider Smart relays.",
    outcome: "Completely eliminated hotspot trips and enabled early warning alarm signals over SCADA before outages happened.",
    improvement: "Zero downtime occurrences in 12 months with real-time temperature tracking."
},
techEcosystem: [
    { name: "Air Circuit Breakers (ACB)", use: "Main incoming power isolation and heavy overload protections.", logic: "Integra"},
    { name: "Automatic Power Factor Regulators (APFC)", use: "Maintains grid power factor close to unity to save energy.", logi"},
    { name: "Smart Protection Relays", use: "Monitors earth leaks, phase imbalance, and dry-run faults.", logic: "Communicates"},
],
qualityAssurance: [
    "Insulation resistance check using 1000V Megger (minimum 100M? threshold).",
    "High Voltage dielectric checkup at 2.5kV applied for 60 seconds with zero leakage.",
    "Control wiring continuity validation using computer-aided testing setups.",
    "Thermal camera scan of all busbar connections under artificial load tests."
],
safetyCompliance: [
    { standard: "IEC 61439-1 & 2", description: "Design standards for low-voltage switchgear assemblies." },
    { standard: "IS 8623", description: "Indian standard rules for factory-built assemblies of switchgear." },
    { standard: "IP55 / IP65", description: "Ingress Protection rating matching dust and water jets resistance rules." }
],
maintenanceAMC: [
    "Preventive quarterly cleaning, busbar joint torque checks, and insulator checking.",
    "Thermal profiling of all cables and contactors using infrared sensors.",
    "Relay logic recalibration and dynamic trip testing using injection kits.",
    "Modernization of legacy panels: retrofitting contactors with energy efficient starter kits."
],
docs: [
    { name: "CPRI Panel Test Certificate.pdf", size: "2.4 MB", type: "PDF Document" },
    { name: "Standard MCC Panels Brochure.pdf", size: "4.1 MB", type: "PDF Brochure" },
    { name: "Maintenance & Sizing Chart.pdf", size: "1.8 MB", type: "Technical Chart" }
],
faqs: [
    { q: "What does a Form-4b panel layout mean?", a: "It refers to the internal separation type defined by IEC 61439-1. In For"},
    { q: "Do you supply panels to outdoor areas?", a: "Yes. We construct outdoor double-door kiosks with protective canopies, d"},
    { q: "What is the CPRI short-circuit certification and why does it matter?", a: "CPRI (Central Power Research Institute) in"},
    { q: "What copper busbar grades do you use in panel fabrication?", a: "We use electrolytic grade copper busbars (99.9% puri"},
    { q: "Can your panels accommodate future capacity expansion?", a: "Yes - we design panels with 20-30% spare busbar capacity"},
    { q: "What testing do you perform before dispatching a panel?", a: "Every panel undergoes our standard FAT (Factory Accepta"},
    { q: "How long does custom panel manufacturing typically take?", a: "Standard panels (MCC, PCC without complex controls) ty"},
    { q: "Do you offer AMC support for panels supplied by other manufacturers?", a: "Yes - we offer comprehensive AMC support f"}
],
"plc-scada": {
    slug: "plc-scada",

```

```

title: "Industrial PLC & SCADA System Integration",
subtitle: "Turnkey Logic Programming, Real-Time SCADA Dashboard Setup, and Data Logging Telemetry for Process Optimization.",
stats: [
  { label: "Data Logging Cycle", value: "100 ms Real-Time" },
  { label: "Supported Nodes", value: "Up to 50,000 tags" },
  { label: "Compliance Standard", value: "FDA 21 CFR Part 11" },
  { label: "Redundancy Speed", value: "<10 ms Auto-switch" }
],
overviewParagraphs: [
  "Process optimization in modern heavy manufacturing demands precise, instantaneous, and automated control logic. Nandini Enterprises provides a comprehensive solution for industrial automation.",
  "Our automation division develops structured, fail-safe logic code utilizing international standard IEC 61131-3 programming.",
  "At the supervision level, we design rich, high-fidelity SCADA interfaces. Our screen layout designs follow the ISA-101 HMI standards.",
  "System integration includes commissioning high-speed communication backbones. We design and install redundant industrial Ethernet networks."
],
challenges: {
  before: "A large pharmaceutical manufacturer faced audit issues due to lack of batch data traceability, manual temperature recording, and inconsistent data logging.",
  after: "Nandini Enterprises engineered a centralized Siemens S7-1500 PLC automation cluster with redundant WinCC SCADA dashboards and real-time data logging.",
  scenario: "Automated the entire batch reactor cycle, achieved complete data logging traceability, and secured GMP certification."
},
workflow: [
  {
    name: "Logic Architecture",
    activity: "Define process flowcharts, draft cause-and-effect matrix logs, define input/output address configurations, and design alarm logic.",
    tools: "TIA Portal, RSLogix, Modbus Configurator",
    safety: "Include software emergency trip loops and hardwired manual overrides.",
    deliverable: "Approved Functional Design Specification (FDS) document and network layout drawings."
  },
  {
    name: "PLC Logic Programming",
    activity: "Develop control logic, PID feedback loops, interlocks, alarm limits, and recipe handling modules in ladder logic.",
    tools: "IEC 61131-3 development tools, logic compilers",
    safety: "Simulate logic faults to verify that the program goes into safe state on error.",
    deliverable: "Compiled PLC program backup codes and simulation diagnostic logs."
  },
  {
    name: "SCADA Screen Design",
    activity: "Design visual mimic diagrams, establish database tags communication, define warning trigger thresholds, and create operator manuals.",
    tools: "Wonderware InTouch, Siemens WinCC SCADA",
    safety: "Maintain ISA-101 alarm priorities to prevent operator alarm fatigue.",
    deliverable: "Completed SCADA visual project mimic layout and database tags sheet."
  },
  {
    name: "Commissioning & Loop Tuning",
    activity: "Run dry logic loop tests, execute hot water runs, calibrate analog transmitters, and tune PID controller parameters.",
    tools: "Loop calibrators, PID tuning software, network analyzers",
    safety: "Enforce strict LOTO (Lockout/Tagout) procedures for all field assets during loop checks.",
    deliverable: "Site Acceptance Test (SAT) reports and operational loop signature files."
  }
],
architectureDesc: "Our SCADA integration features a three-tier automation architecture: Field Device instrumentation, Control System, and Data Management.",
architectureDetails: [
  "Control Network: Configured on a redundant fiber optic ring running Profinet protocol for rapid data exchange.",
  "Redundant CPU Hot Standby: CPU tracking modules share variables at 1Gbps, executing bumpless switchovers on hardware trip.",
  "Audit Database: Integrated database tables capturing timestamped user logins, alarm acknowledges, and parameter modifications.",
  "OPC UA Servers: Provide secure links to ERP systems for inventory forecasting."
],
industries: [
  { name: "Pharma", need: "Strict FDA 21 CFR Part 11 validation and batch process security.", challenge: "Ensuring unalterable data integrity." },
  { name: "Water", need: "Telemetry monitoring across large municipal reservoir areas.", challenge: "Unreliable wireless network coverage." },
  { name: "Chemical", need: "Intelligent dosing control with dynamic feedback.", challenge: "Highly corrosive fluids requiring specialized materials." }
],
caseStudy: {
  client: "Hyderabad Water Supply Metro Pumping Phase I",
  requirement: "Automate raw water intake pumping station with remote monitoring.",
  problem: "Operators had to manually start pump starters, leading to high flow surge risks and pump dry-run damages.",
  solution: "Installed a redundant PLC automation rack with fiber network, Baumer flow sensors, and custom SCADA monitor desk.",
  products: "Siemens S7-400H Redundant PLC, WinCC Professional SCADA, Baumer Sensors.",
  outcome: "Completely automated pump scheduling based on reservoir levels, eliminated flow surges via active speed control.",
  improvement: "Water loss reduced by 14% and pump lifetime increased by 22% through automated runtime rotation."
},
techEcosystem: [
  { name: "Siemens S7-1500 Series", use: "Heavy plant sequencing and diagnostic monitoring.", logic: "Runs loop code cycles in real-time." },
  { name: "OPC UA Interface", use: "Standardized cross-brand platform communication.", logic: "Provides object-oriented secure data exchange." },
  { name: "SQL Historical Logger", use: "Capturing process data parameters for years.", logic: "Compresses values using delta storage." }
],
qualityAssurance: [
  "Simulated logic validation checks under multi-fault scenario injections.",
  "CPU failsafe switchover check (verifying backup takes over under 10ms).",
  "Network storm stress verification on communication rings.",
  "Data logging accuracy comparison between physical analog meters and database values."
],
safetyCompliance: [

```

... continued: src/lib/servicesData.ts

```
{
  standard: "ISA-101", description: "Human-Machine Interface design rules for process control systems." },
  { standard: "IEC 61131-3", description: "Standard programming language formats for PLCs." },
  { standard: "FDA 21 CFR Part 11", description: "FDA criteria for electronic records and signatures." }
],
maintenanceAMC: [
  "Annual software backups, backup battery updates, and firmware version security patching.",
  "Database cleaning, archive file creation, and query speed tuning.",
  "AMC support: 24/7 remote diagnostic login to resolve logic blocks.",
  "Modernization: converting old legacy code scripts (e.g. S5 PLCs) to S7 structures."
],
docs: [
  { name: "FDS Reference Guide.pdf", size: "3.2 MB", type: "PDF Document" },
  { name: "PLC SCADA Integration Layout.pdf", size: "2.8 MB", type: "Drawing Layout" },
  { name: "Standard FAT-SAT Checklist.xlsx", size: "1.4 MB", type: "Excel format" }
],
faqs: [
  { q: "Can you connect old machinery to a new SCADA system?", a: "Yes. We integrate industrial protocol gateway converters to connect legacy PLCs to modern SCADA systems." },
  { q: "What security measures do you take for remote SCADA access?", a: "We implement layered OT security: dedicated hardware firewalls, VPN access, and multi-factor authentication." },
  { q: "What is OPC UA and why is it preferred over older protocols?", a: "OPC UA (Unified Architecture) is an open, vendor-neutral standard for industrial data exchange." },
  { q: "How do you ensure SCADA system availability for 24/7 critical operations?", a: "We design SCADA systems with redundant servers, power supplies, and network paths." },
  { q: "Can SCADA data be accessed on mobile devices or tablets?", a: "Yes. Modern SCADA platforms (WinCC, Wonderware, Ignition) support mobile access via secure web portals." },
  { q: "How long does a typical PLC/SCADA project take from contract to commissioning?", a: "Project duration depends on scope and complexity, typically ranging from 4 to 12 weeks." },
  { q: "What PLCs do you typically specify and program?", a: "Our primary platforms are Siemens (S7-1200, S7-1500, S7-400H) and Allen-Bradley (ControlLogix, CompactLogix)." },
  { q: "Can you perform SCADA upgrades without shutting down plant operations?", a: "Yes - with careful planning. We phase the upgrade to minimize downtime." }
],
},
"vfd-solutions": {
  slug: "vfd-solutions",
  title: "YASKAWA VFD & Drive System Solutions",
  subtitle: "Authorized Service Center & Integrator for Variable Frequency Drives. Energy Audit, Dynamic Torque Control, and Harmonic Mitigation.",
  stats: [
    { label: "Energy Saving Capacity", value: "Up to 35% Typical" },
    { label: "VFD Range Serviced", value: "0.4 kW to 1000 kW" },
    { label: "Repair Turnaround", value: "Under 24 Hours" },
    { label: "Authorized status", value: "OEM Partner Center" }
  ],
  overviewParagraphs: [
    "Variable Frequency Drives (VFDs) are the core speed regulators of the industrial landscape, matching motor outputs directly to load requirements for optimal efficiency and control.",
    "Our drive specialists analyze motor loads to optimize acceleration/deceleration S-curve speed paths. This is particularly important for applications like cranes and conveyors where smooth motion is critical.",
    "Harmonic current distortions from heavy VFD switches can pollute your plant power grid, causing other electronic equipment to malfunction. We provide solutions to mitigate these issues.",
    "As an Authorized Yaskawa Service Center, our Cherlapally base features advanced drive test benches, load testing rigs, and specialized technicians for all Yaskawa drive systems."
  ],
  challenges: {
    before: "A large steel plant experienced regular VFD tripping faults, crane load slippage risks, and utility penalties due to high harmonic distortion.",
    after: "Nandini Enterprises installed Yaskawa A1000 crane drives with dynamic braking resistors, active anti-slip logic, and harmonic filters, resulting in stable operation and cost savings.",
    scenario: "Eliminated load slips, reduced utility harmonics within compliance rules, and stopped drive trip occurrences under full load."
  },
  workflow: [
    {
      name: "Motor Torque Assessment",
      activity: "Evaluate motor specifications, measure load inertia curves, check operating speed limits, and determine braking requirements.",
      tools: "Power clamp meters, thermal cameras, load analyzers",
      safety: "Verify motor grounding resistance before starting rotation.",
      deliverable: "Approved drive load assessment sheet and drive dimension layout plans."
    },
    {
      name: "Drive Assembly & Programming",
      activity: "Mount drives in custom CPRI cabinets, configure cooling channels, install input/output filters, and program speed profiles.",
      tools: "Yaskawa DriveWizard, thermal calculators, torque wrenches",
      safety: "Enforce safety clearance spaces around the drive heatsinks.",
      deliverable: "Programmed drive cabinet assembly and parameter setup checklist sheets."
    },
    {
      name: "Parameter Auto-Tuning",
      activity: "Execute rotational or rotational-less motor auto-tuning, configure S-curve limits, and test digital/analog I/O response.",
      tools: "Yaskawa operator panels, tachometers",
      safety: "Erect barricades around rotating shaft couplers before starting auto-tuning.",
      deliverable: "Fine-tuned motor speed parameters and loop response curves."
    },
    {
      name: "Load Run & Harmonic Audit",
      activity: "Perform full load tests, log current waveforms under loads, measure power harmonic levels, and issue compliance certificates.",
      tools: "Power quality analyzer, thermal imager",
      safety: "Do not touch power components until DC bus capacitors are completely discharged.",
      deliverable: "Verified load run reports and IEEE 519 harmonic compliance certificate sheets."
    }
  ],
  architectureDesc: "Our VFD integrations include custom cabinet layouts with line reactors, active cooling channels, dynamic braking resistors, and harmonic filters to ensure clean, efficient power delivery.",
  architectureDetails: [
    "AC Line Reactors: Mounted on VFD input terminals to suppress voltage surges and reduce harmonic currents.",
    "Braking Resistors: Absorb regenerative energy from high-inertia deceleration loads to prevent overvoltage faults.",
    "Ventilation System: Cabinet internal air pathways designed to maintain heatsink temperature below 45°C."
  ]
}
```



```

    "Profinet Cards: Provide real-time speed, current, and error diagnostic logs to supervisory control systems."
  ],
  industries: [
    { name: "Cement", need: "Speed control on high-volume kilns and exhaust fans.", challenge: "High starting loads and abrasiv",
    { name: "Elevators", need: "Smooth, jerk-free passenger vertical acceleration paths.", challenge: "Ensuring safety under po",
    { name: "Steel", need: "Precise speed synchronization for continuous casting machines.", challenge: "Extreme heat and high",
  ],
  caseStudy: {
    client: "Hyderabad Pumping Station Retrofit",
    requirement: "Upgrade direct-on-line (DOL) starters on 250HP pumps to adjust water flow rates dynamically.",
    problem: "Constant starting current surges damaged water mains, and running pumps at full speed wasted power.",
    solution: "Installed 3 Yaskawa A1000 VFD control panels with integrated bypass switchgear and PID pressure control.",
    products: "Yaskawa A1000 Drives, Siemens switchgears, Wika pressure sensors.",
    outcome: "Eliminated water hammer pipe bursts, stabilized pipe pressure automatically, and cut monthly electricity bills.",
    improvement: "Reduced energy consumption by 28% and cut water hammer pipe failures to zero."
  },
  techEcosystem: [
    { name: "Yaskawa A1000 series", use: "High-performance vector control for heavy industries.", logic: "Calculates current ve",
    { name: "Active Front End (AFE) unit", use: "Eliminating harmonics and regenerating energy.", logic: "Switches input current",
    { name: "Harmonic Mitigation Filters", use: "Cleans power lines from noise frequencies.", logic: "Tuned LC circuit traps 5t",
  ],
  qualityAssurance: [
    "Motor parameter auto-tuning checks and speed-accuracy audits.",
    "Load run testing for 4 hours at maximum motor current ratings.",
    "Harmonic current measurement checks against IEEE 519 guidelines.",
    "IGBT gate driver pulse diagnostic check using digital oscilloscopes."
  ],
  safetyCompliance: [
    { standard: "IEEE 519", description: "Recommended practices for control of harmonics in electrical power systems." },
    { standard: "IEC 61800-3", description: "EMC standards for variable speed electrical power drive systems." },
    { standard: "IEC 61508 SIL-3", description: "Functional safety compliance for drive safe torque off circuits." }
  ],
  maintenanceAMC: [
    "Preventive cleaning, busbar connections checks, capacitor health audits, and fan replacements.",
    "Drive param values backup creation and parameter fine-tuning adjustments.",
    "24/7 on-site emergency troubleshooting support within 4 hours in Hyderabad.",
    "Modernization: retrofitting obsolete VFD units with modern Yaskawa GA700 modules."
  ],
  docs: [
    { name: "Yaskawa Partner Certificate.pdf", size: "1.9 MB", type: "PDF Document" },
    { name: "Drive Harmonics Sizing Sheet.pdf", size: "2.1 MB", type: "Sizing Sheet" },
    { name: "GA700 Technical Manual.pdf", size: "5.4 MB", type: "Manual" }
  ],
  faqs: [
    { q: "What is the benefit of using AC Line Reactors on VFD inputs?", a: "Line reactors (typically 3-5% impedance) serve mul",
    { q: "How fast can you repair a faulted Yaskawa drive?", a: "As an Authorized Yaskawa Service Center, we stock IGBT modules",
    { q: "What causes VFD nuisance tripping and how can it be resolved?", a: "Nuisance VFD trips (faults that clear on reset wi",
    { q: "What is the difference between V/f control and vector control in a VFD?", a: "V/f (Volts-per-Hertz) control is the si",
    { q: "Can VFDs operate in high ambient temperature environments like cement or steel plants?", a: "Standard VFDs are rated",
    { q: "What is dynamic braking and when is it required?", a: "When a VFD decelerates a high-inertia load (crane, centrifuge,",
    { q: "Do you provide energy savings verification after VFD installation?", a: "Yes. We perform a formal energy audit before",
    { q: "What warranty does Yaskawa provide and what does your AMC cover beyond warranty?", a: "Yaskawa standard warranty is 1",
  ],
},
"robotics": {
  slug: "robotics",
  title: "Industrial Robotics Integration & Motion Control",
  subtitle: "Custom robotic pick-and-place systems, automated palletizing, welding arm integration, and multi-axis motion synchron",
  stats: [
    { label: "Cycle Speed", value: "Up to 120 picks/min" },
    { label: "Payload Capacity", value: "3 kg to 600 kg" },
    { label: "Accuracy", value: "+/- 0.02 mm Repeatable" },
    { label: "Axes Synced", value: "Up to 32 Axes Simultaneous" }
  ],
  overviewParagraphs: [
    "Modern B2B production throughput requires speed, repeatability, and precision beyond manual limits. Nandini Enterprises de",
    "We partner with leading global robotics brands, focusing on Yaskawa Motoman arms, to build turnkey automation cells. Our e",
    "To sync robot movements with feed lines, we develop high-speed motion control algorithms. We configure hardware interfaces",
    "We support our clients through the entire validation cycle, performing reachability analysis and offline collision checkin",
  ],
  challenges: {
    before: "A major packaging facility faced bottleneck issues, inconsistent pallet configurations, and high labor costs on th",
    after: "Nandini Enterprises engineered a Yaskawa Motoman 4-axis palletizing robot cell with custom grippers and conveyor se",
    scenario: "Automated the entire line output, increased packing speed to 24 units per minute, and achieved perfect stack con",
  },
  workflow: [
    {
      name: "Robot Kinematic Simulation",
      activity: "Define workstation dimensions, calculate payloads, simulate robot arm movements, check reach limits, and verif",
      tools: "Yaskawa MotoSim, AutoCAD 3D",
      safety: "Confirm safety zone coordinates inside simulation.",
      deliverable: "Simulation video logs and robot reach validation report sheets."
    }
  ]
}

```

```

    },
    {
      name: "EOAT Gripper Fabrication",
      activity: "Design and build custom vacuum, magnetic, or mechanical grippers, mount solenoid valves, and wire terminal boxes.",
      tools: "CNC cutters, 3D printers, pneumatic test benches",
      safety: "Calculate vacuum safety factors to prevent drop faults on pressure loss.",
      deliverable: "Finished mechanical gripper assembly with mounting accessories."
    },
    {
      name: "Programming & Teaching",
      activity: "Program tool center points (TCP), define path coordinates, configure speed limits, and set up HMI interfaces.",
      tools: "Motoman Teach Pendant, Ladder interface tools",
      safety: "Verify speed limits are active before teaching coordinates.",
      deliverable: "Program backups containing path trajectories and speed profiles."
    },
    {
      name: "SAT & Production Trial",
      activity: "Install safety guards, check interlocking gates, run full-speed cycle tests, verify sensor limits, and start production.",
      tools: "Laser scanner testers, cycle clocks",
      safety: "Verify that entering safety zones triggers Emergency Stops.",
      deliverable: "Completed Site Acceptance Test (SAT) reports and cycle validation logs."
    }
  ],
  architectureDesc: "Our robotic workstations use a unified control architecture: Robot Controller (e.g. Yaskawa YRC1000), Mainframe, and Safety PLC.",
  architectureDetails: [
    "Robot Controller: Yaskawa YRC1000 managing servo motors and trajectory paths.",
    "Safety PLC: Monitors gate switches, light curtains, and emergency stops.",
    "EtherCAT Network: Links sensors, encoders, and motor starters.",
    "Operator HMI: Touch screen display showing speed, cycles, and diagnostics."
  ],
  industries: [
    { name: "Pharma", need: "Clean, contamination-free packaging and sorting.", challenge: "Avoiding particulate shedding and contamination." },
    { name: "Cement", need: "High-payload bag palletizing and truck load systems.", challenge: "Heavy dust, weight changes, and vibration." },
    { name: "Steel", need: "Automated rod marking, bundle wrapping, and sorting.", challenge: "Extreme heat, sparks, and high payloads." }
  ],
  caseStudy: {
    client: "Hyderabad Packaging Plant",
    requirement: "Automate secondary packaging line to pack bottles into boxes.",
    problem: "Manual packing was slow, irregular, and prone to sorting errors.",
    solution: "Integrated a Yaskawa Motoman 6-axis pick-and-place robot cell with vacuum grippers and visual camera tracking.",
    products: "Yaskawa Motoman GP8 Robot, YRC1000 Controller, Cognex Vision.",
    outcome: "Automated the entire sorting and packing line, eliminating packing errors completely.",
    improvement: "Increased pick rate by 45% and reduced manual labor reliance on packaging line."
  },
  techEcosystem: [
    { name: "Yaskawa Motoman Arm", use: "High-speed repeatable mechanical movements.", logic: "Translates joint coordinates to end-effector position." },
    { name: "Cognex Vision Sensor", use: "Detecting part position and rotation dynamically.", logic: "Identifies object profile and orientation." },
    { name: "Safety Light Curtain", use: "Worker protection around the robot cell.", logic: "Optical light beam breaks trigger emergency stop." }
  ],
  qualityAssurance: [
    "Path repeat accuracy check (+/- 0.02mm limit verified).",
    "E-stop response time validation (verifying motion stops in milliseconds).",
    "Continuous cycle run testing for 8 hours under maximum payloads.",
    "EOAT vacuum pressure holding check under emergency power cuts."
  ],
  safetyCompliance: [
    { standard: "ISO 10218-1&2", description: "Safety requirements for industrial robots and robot systems." },
    { standard: "ISO 13849-1", description: "Performance level verification for safety-related control parts." },
    { standard: "ANSI/RIA R15.06", description: "Industrial robots safety standards compliance." }
  ],
  maintenanceAMC: [
    "Annual gearbox oil testing, backup battery changes, and cable harness checks.",
    "Program path tweaks to optimize cycle times.",
    "On-site repair support within 24 hours of notification.",
    "Modernization: rebuilding and reprogramming old robot cells for new tasks."
  ],
  docs: [
    { name: "GP8 Tech Specifications.pdf", size: "2.1 MB", type: "PDF Specification" },
    { name: "Robotic Cell Safety Layout.pdf", size: "3.4 MB", type: "Safety Layout" },
    { name: "Robotic Maintenance Guide.pdf", size: "2.7 MB", type: "Technical Guide" }
  ],
  faqs: [
    { q: "What payload capacity robots do you integrate?", a: "We integrate robotic arms with payloads ranging from 3 kg (Yaskawa GP8) to 20 kg (Yaskawa GP12)." },
    { q: "How does the robot synchronize with a moving conveyor belt?", a: "We install a rotary encoder on the conveyor drive motor to sync robot pick points with belt position." },
    { q: "What safety systems are mandatory around industrial robot cells?", a: "Robot safety systems per ISO 10218-1&2 include light curtains, safety mats, and emergency stop buttons." },
    { q: "Can industrial robots be reprogrammed for new product variants?", a: "Yes - this is one of the key advantages of robotic automation." },
    { q: "What is the typical ROI period for industrial robot installation?", a: "Payback period varies with application, but typically ranges from 12 to 24 months." },
    { q: "Do robots require special power supply infrastructure?", a: "The YRC1000 robot controller requires 3-phase 400V AC supply." },
    { q: "How long does robot cell installation and commissioning take?", a: "A single robot pick-and-place or palletizing cell can be installed in 1-2 weeks." },
    { q: "What operator training do you provide after robot installation?", a: "We provide a two-part training program: a theoretical classroom session followed by hands-on practical training." }
  ]
},

```



```

"instrumentation": {
  slug: "instrumentation",
  title: "Process Instrumentation & Sensor Calibration",
  subtitle: "Turnkey Field Sensor Installation, Pressure/Temperature Transmitters, HART Loop Checks, and Wet/Dry Calibration Services",
  stats: [
    { label: "Calibration range", value: "-0.95 to 1000 Bar" },
    { label: "Sensor accuracy", value: "+/- 0.05% of span" },
    { label: "Protocols supported", value: "HART, Modbus, Profibus" },
    { label: "Certification focus", value: "Traceable to NABL" }
  ],
  overviewParagraphs: [
    "Control systems depend on the accuracy of field instrumentation. Nandini Enterprises delivers comprehensive process instrumentation solutions.",
    "We design field wiring loops using high-quality shielded cables and barrier isolators to protect signals from electrical noise.",
    "Calibration is key to maintaining process safety. We perform wet and dry calibration at client sites or in our Hyderabad laboratory.",
    "We provide analytical systems, including continuous emission monitoring (CEMS), water quality analyzers, and steam and water flow meters."
  ],
  challenges: {
    before: "A large chemical dosing facility faced batch inconsistency and regular safety alarms due to drifting level sensors.",
    after: "Nandini Enterprises installed Baumer radar level sensors and Wika mass flow meters, calibrated on-site using portable calibration equipment.",
    scenario: "Achieved dosage consistency, stabilized chemical reaction profiles, and avoided safety alarms caused by false sensor readings."
  },
  workflow: [
    {
      name: "Sensor Selection",
      activity: "Evaluate pipe pressure, chemical compatibility, temperature limits, flow velocity, and hazardous area classifications.",
      tools: "Chemical compatibility database, instrument sizing calculators",
      safety: "Confirm ATEX flameproof classifications match safety guidelines.",
      deliverable: "Approved Instrument datasheets and hook-up installation drawings."
    },
    {
      name: "Field Installation",
      activity: "Position sensor manifolds, weld thermowells, run shielded instrumentation cables, and wire transmitter terminations.",
      tools: "Manifold alignment tools, wire strippers, torque wrenches",
      safety: "Verify line isolation and execute pressure test before installing process taps.",
      deliverable: "Mounted field instruments ready for loop diagnostic checks."
    },
    {
      name: "HART Loop Calibration",
      activity: "Configure transmitter parameters, perform zero-and-span calibration, test loop outputs (4-20mA), and check diagnostics.",
      tools: "HART Communicators, dry block temp calibrators",
      safety: "Use loop protection modules to prevent current surges.",
      deliverable: "Completed calibration certificate sheets and HART configuration files."
    },
    {
      name: "Process Loop SAT",
      activity: "Execute dynamic process loop runs, compare sensor values with reference meters, and configure SCADA telemetry.",
      tools: "HART interface tools, loop simulators",
      safety: "Perform pressure testing on connections before start of loop.",
      deliverable: "Signed loop check sheets and process integration reports."
    }
  ],
  architectureDesc: "Our instrumentation loops are configured on a redundant control network layout, utilizing isolated analog signal paths.",
  architectureDetails: [
    "Analog Signal Isolation: Galvanic isolator modules isolate control loops from field ground loop disturbances.",
    "HART Communication: Digital data superimposed on 4-20mA loops for remote diagnostics.",
    "Manifold Valves: Dual block and bleed configurations for safe instrument removal under line pressure.",
    "Junction Boxes: IP66 rated, flameproof enclosures matching hazardous area zone requirements."
  ],
  industries: [
    { name: "Pharma", need: "Highly accurate sanitary level and temperature control loops.", challenge: "Withstanding clean-in-place (CIP) cycles." },
    { name: "Power", need: "High-pressure steam flow and boiler feed level monitoring.", challenge: "Extreme temperature and vibration." },
    { name: "Chemical", need: "Continuous monitoring of corrosive solvent storage levels.", challenge: "Acid vapors damaging standard sensors." }
  ],
  caseStudy: {
    client: "Telangana Pumping Grid System",
    requirement: "Calibrate electromagnetic flow meters on major city distribution mains.",
    problem: "Flow meters were drifting, causing differences between pump logs and consumer meters.",
    solution: "Performed in-situ calibration of 6 flow meters using portable ultrasonic transit-time meters, adjusting span factors.",
    products: "Wika pressure gauges, Baumer ultrasonic level probes, HART configurator.",
    outcome: "Resolved measurement differences, bringing accuracy back to nominal tolerances.",
    improvement: "Reduced measurement errors from 4.5% to under 0.25%."
  },
  techEcosystem: [
    { name: "HART Communicator", use: "Configuring and diagnostics of smart field sensors.", logic: "Superimposes FSK digital data on 4-20mA." },
    { name: "Dry Block Calibrator", use: "Generating precise reference temperatures.", logic: "Micro-heaters maintain block temperature." },
    { name: "Galvanic Signal Barrier", use: "Intrinsically safe loops in explosive zones.", logic: "Zener diodes restrict current flow." }
  ],
  qualityAssurance: [
    "Zero-point verify checks and span accuracy tests.",
    "NABL traceable calibration validation logs creation.",
    "Overpressure leak testing for 15 minutes at 1.5x nominal pressure rating.",
    "Analog output verification checks at 4mA, 8mA, 12mA, 16mA, and 20mA states."
  ]
}

```

```

    ],
    safetyCompliance: [
      { standard: "IEC 60079", description: "Explosive atmospheres standards compliance for electrical installations." },
      { standard: "SIL-2/3 Certified", description: "Safety Integrity Level rating matching safety standards." },
      { standard: "NABL Traceability", description: "Calibration measurements traceable to national standards." }
    ],
    maintenanceAMC: [
      "Preventive sensor cleaning, check connection seals, and inspect cable insulation.",
      "Recalibrate sensors every 6 or 12 months.",
      "On-site repair support within 24 hours of notification.",
      "Modernization: replacing old analog gauges with smart transmitters."
    ],
    docs: [
      { name: "Calibration Guide sheet.pdf", size: "1.4 MB", type: "PDF Specification" },
      { name: "Standard Calibration Format.docx", size: "1.1 MB", type: "Word Document" },
      { name: "Baumer Sensor Catalog.pdf", size: "4.8 MB", type: "Sensor Catalog" }
    ],
    faqs: [
      { q: "What is HART communication protocol?", a: "Highway Addressable Remote Transducer (HART) superimposes FSK (Frequency Shift Keying) on top of the 4-20mA signal." },
      { q: "How often should process sensors be calibrated?", a: "Calibration frequency depends on sensor criticality and application." },
      { q: "What is the difference between a pressure gauge and a pressure transmitter?", a: "A pressure gauge is a mechanical device, while a transmitter is an electronic device that outputs a standard signal." },
      { q: "How do you select the right flow meter technology for an application?", a: "Flow meter selection depends on: fluid type, viscosity, pressure, temperature, and required accuracy." },
      { q: "Can you perform field calibration or does equipment need to come to your lab?", a: "We offer both. Field calibration is available for many sensors." },
      { q: "What are intrinsically safe instruments and when are they required?", a: "Intrinsically safe (Ex-ia) instruments are designed to prevent ignition of explosive atmospheres." },
      { q: "What documentation do you provide with calibrated instruments?", a: "Each calibration produces a formal certificate of calibration." }
    ]
  },
  "elevator-systems": {
    slug: "elevator-systems",
    title: "Elevator Control Systems & VFD Operations",
    subtitle: "High-Speed Microprocessor Lift Controllers, Smooth S-Curve VFD Configuration, and Automatic Rescue Device (ARD) Integration",
    stats: [
      { label: "Acceleration Profile", value: "S-Curve Jerk-Free" },
      { label: "Rescue Battery Switch", value: "< 15 Sec Auto-Run" },
      { label: "Axes Accuracy", value: "+/- 3 mm Leveling" },
      { label: "Speed Capacities", value: "0.5 to 2.5 m/s" }
    ],
    overviewParagraphs: [
      "In vertical transport systems, passenger safety, travel comfort, and energy efficiency depend on controller logic and drive system performance.",
      "Our engineering team designs complete control panels for passenger and goods lifts. We configure Yaskawa L1000A drives with smooth S-curve profiles for precise leveling.",
      "Passenger safety is our highest priority. We integrate automatic rescue devices (ARD) that automatically move the cabin to a safe floor in case of a power outage.",
      "Our control systems implement energy optimization strategies, including regenerative drive systems that feed braking energy back into the building's power grid."
    ],
    challenges: {
      before: "A high-rise commercial complex suffered from constant elevator leveling mismatch errors, cabin starting jerks, and frequent emergency stops.",
      after: "Nandini Enterprises installed Yaskawa L1000A elevator drives with smart leveling sensors, microprocessor control panels, and ARD systems.",
      scenario: "Eliminated cabin starting jerks, achieved precise leveling within 3mm, and secured passenger automatic rescue functionality."
    },
    workflow: [
      {
        name: "Motor Parameter Audit",
        activity: "Analyze elevator motor specifications, measure cabin weight, determine hoisting gear ratios, and assess electrical load.",
        tools: "Multi-meters, gear calculators, thermal probes",
        safety: "Confirm brake release safety loops are active before starting checks.",
        deliverable: "Approved elevator motor sizing reports and schematic drafts."
      },
      {
        name: "Controller Assembly",
        activity: "Assemble microprocessor lift controller boards, mount Yaskawa drives, wire door operator connections, and integrate leveling sensors.",
        tools: "Panel assembly tables, wire strippers, torque screwdrivers",
        safety: "Double check insulation clearances between line voltage parts.",
        deliverable: "Completed elevator control panel ready for diagnostic tests."
      },
      {
        name: "Leveling Sensor Setup",
        activity: "Install cabin leveling sensors, set up landing door safety switches, program speed curves, and check passenger capacity limits.",
        tools: "Tachometers, leveling tools, calibration boards",
        safety: "Erect landing door warning barriers before running tests in the shaft.",
        deliverable: "Programmed cabin leveling parameters and speed profiles."
      },
      {
        name: "ARD Load Test",
        activity: "Simulate utility power failures, check automatic battery switches, verify rescue run speeds, and test auto-operation.",
        tools: "Battery testers, load simulation rigs",
        safety: "Verify emergency communication systems are active before starting rescue runs.",
        deliverable: "Completed ARD test report sheets and safety compliance certificates."
      }
    ]
  },
  architectureDesc: "Our elevator control systems utilize modular microprocessor controller layouts with isolated door logic, Yaskawa L1000A drives, and smooth S-curve travel profiles.",
  architectureDetails: [
    "Microprocessor Board: Core controller managing call buttons, landing indicators, and door interlocks.",
    "Yaskawa L1000A: Lift drive configured for smooth S-curve travel profiles.",
  ]
}

```

```

    "Automatic Rescue Device (ARD): Battery backup system powering rescue moves during utility power trips.",
    "Leveling Sensors: Photoelectric and magnetic sensors ensuring precise cabin alignment."
  ],
  industries: [
    { name: "Pharma", need: "Dust-sealed goods lifts for sterile cleanroom access.", challenge: "Avoiding particulate contamination." },
    { name: "Cement", need: "Heavy-duty goods elevators for vertical mill structures.", challenge: "Heavy dust accumulation and vibration." },
    { name: "Steel", need: "Rugged cargo elevators with high payload reliability.", challenge: "Extreme vibrations and structural stresses." }
  ],
  caseStudy: {
    client: "Hyderabad Commercial Hub",
    requirement: "Upgrade passenger lift controls to eliminate starting jerks and improve safety.",
    problem: "The old analog control system caused bumpy starts and leveling issues.",
    solution: "Installed a Yaskawa L1000A elevator drive system with ARD backup and smart leveling sensors.",
    products: "Yaskawa L1000A Drives, microprocessor control boards, ARD backup.",
    outcome: "Eliminated starting jerks, stabilized cabin leveling, and secured automatic passenger rescue.",
    improvement: "Leveling errors reduced by 90% and energy consumption cut by 25%."
  },
  techEcosystem: [
    { name: "Yaskawa L1000A Drive", use: "Smooth vector control for vertical transport.", logic: "Adjusts motor frequency dynamically." },
    { name: "Microprocessor Controller", use: "Managing lift calls and door security.", logic: "Process safety logic loops to control speed and position." },
    { name: "Automatic Rescue Device", use: "Emergency power backup for passenger safety.", logic: "Switches to battery power during outages." }
  ],
  qualityAssurance: [
    "Leveling accuracy validation check (+/- 3mm threshold verified).",
    "Door interlock safety switch check under travel states.",
    "Continuous run testing for 6 hours under full load conditions.",
    "ARD battery health and charge logic checks."
  ],
  safetyCompliance: [
    { standard: "EN 81-20/50", description: "European safety rules for construction and installation of lifts." },
    { standard: "IS 14665", description: "Indian standard guidelines for electric passenger and goods lifts." },
    { standard: "ASME A17.1", description: "Safety code for elevators and escalators compliance." }
  ],
  maintenanceAMC: [
    "Preventive rope checks, terminal joint torque audits, and safety switch tests.",
    "Speed curve parameter tuning and leveling sensor cleaning.",
    "24/7 on-site emergency troubleshooting support within 4 hours in Hyderabad.",
    "Modernization: replacing obsolete analog relays with microprocessor panels."
  ],
  docs: [
    { name: "Elevator Control Layout.pdf", size: "2.3 MB", type: "PDF Specification" },
    { name: "L1000A Tech Brochure.pdf", size: "3.1 MB", type: "Technical Brochure" },
    { name: "ARD Maintenance Guide.pdf", size: "1.7 MB", type: "User Guide" }
  ],
  faqs: [
    { q: "What is an Automatic Rescue Device (ARD) and how does it work?", a: "An ARD (Automatic Rescue Device) is a battery-backed system that takes over elevator control during power outages to safely move passengers to the nearest floor." },
    { q: "What leveling accuracy do your elevator control systems achieve?", a: "We install both magnetic vane sensors (coarse) and photoelectric sensors (fine) for millimeter-level precision." },
    { q: "What causes elevator cabin starting jerks and how do your systems eliminate them?", a: "Starting jerks occur when the control system's acceleration profile is not optimized. Our microprocessor-based drives use smooth vector control to eliminate this." },
    { q: "What safety standards govern elevator control systems in India?", a: "IS 14665 (Parts 1-6) is the primary Indian standard for electric passenger and goods lifts." },
    { q: "How does the elevator group dispatch controller optimize waiting times?", a: "Group controller software manages a bank of elevators, predicting demand and dispatching the nearest available car." },
    { q: "Can old elevator control systems be modernized without replacing the entire elevator?", a: "Yes - controller modernization projects can upgrade safety and efficiency while retaining the existing shaft and cabin." },
    { q: "What energy savings can regenerative elevator drives provide?", a: "Regenerative elevator drives (with Active Front End) can feed energy back into the building's power grid during descent." },
    { q: "How frequently should elevator control systems be serviced?", a: "Our recommended elevator AMC schedule: monthly visual inspections and quarterly functional tests." }
  ],
},
"electrical-solutions": {
  slug: "electrical-solutions",
  title: "HT & LT Electrical Infrastructure Engineering",
  subtitle: "High Voltage (33KV) Substation Design, Cable Tray Engineering, Switchgear Protection Coordination, and Turnkey Erection",
  stats: [
    { label: "Voltage Capacity", value: "Up to 33,000 Volts" },
    { label: "License Grade", value: "A-Grade Telangana" },
    { label: "Fault Limit", value: "40kA / 3 Sec Rating" },
    { label: "Transformer Size", value: "Up to 10 MVA" }
  ],
  overviewParagraphs: [
    "High voltage substations and industrial electrical distributions form the primary backbone of heavy manufacturing. Nandini Enterprises specializes in the design, engineering, and turnkey erection of these critical infrastructure components.",
    "Our switchyard division designs substation layouts, substation earthing mats, and overhead cable entries. We perform fault level calculations and protection coordination studies to ensure system reliability.",
    "We design and install cable tray systems, high-power copper and aluminum busducts, and armored underground cables. Our electrical installation teams adhere to strict safety protocols and industry standards.",
    "We handle the entire commissioning cycle, performing insulation testing (Megger), transformer turns ratio check, winding resistance tests, and load bank testing to validate system performance."
  ],
  challenges: {
    before: "A large cement manufacturing site faced regular transformer overheating faults, voltage drops, and switchgear damage due to poor protection coordination and outdated infrastructure.",
    after: "Nandini Enterprises redesigned the substation layout, installed custom vacuum circuit breakers (VCB), and adjusted protection settings, resulting in a more robust and efficient power distribution system.",
    scenario: "Eliminated voltage drops, stabilized substation operating temperatures, and secured protection coordination across the entire industrial complex."
  },
  workflow: [
    {
      name: "Substation Design",
      activity: "Analyze utility line specs, design grid layouts, calculate transformer sizing requirements, and compute earthing system parameters.",
      tools: "ETAP, AutoCAD Civil, ground grid solvers",
      safety: "Confirm clearance distances match High Voltage safety codes."
    }
  ]
}

```

```

    deliverable: "Approved electrical single line diagrams (SLD) and earthing layouts."
  },
  {
    name: "Transformer Installation",
    activity: "Erect transformer foundations, position transformer units, connect high-voltage bushings, and run filtration r
    tools: "Heavy cranes, oil filtration plants, winding testers",
    safety: "Ensure proper grounding of transformer frames and neutral lines.",
    deliverable: "Transformer installation completed and oil parameters verified."
  },
  {
    name: "Switchgear Erection",
    activity: "Mount HT/LT switchgear panels, terminate incoming cables, configure control wiring, and mount protection relay
    tools: "Torque wrenches, cable terminal cutters, secondary injection kits",
    safety: "Mandatory protective safety boots and insulated mats in panel rooms.",
    deliverable: "Fully mounted and wired switchgear system ready for loop checks."
  },
  {
    name: "System Relay Tests",
    activity: "Conduct relay trip checks, primary/secondary injection tests, winding resistance checks, and verify oil paramet
    tools: "Relay injection test benches, winding meters, micro-ohmmeters",
    safety: "Verify insulation barriers and discharge residual voltages after tests.",
    deliverable: "Completed commissioning reports and safety clearance certificates."
  }
],
architectureDesc: "Our HT/LT electrical layouts utilize single or double busbar schemes, vacuum circuit breakers (VCB) for HT
architectureDetails: [
  "HT incomer: 33KV/11KV vacuum circuit breakers protecting substation transformers.",
  "Busbar Bridge: High-capacity aluminum or copper busducts linking transformer terminals to LT switchgear chambers.",
  "Relay Protection: Digital relay modules tracking overcurrent, earth faults, and under-voltage limits.",
  "Earthing Network: Grid of copper tape conductors linked to deep earth pits."
],
industries: [
  { name: "Power", need: "High-voltage generator synchronization and network step-up.", challenge: "Very high current thermal
  { name: "Steel", need: "Substation design supporting arc furnace loads.", challenge: "Massive load swings, harmonics, and v
  { name: "Cement", need: "LT motor control distribution loops for quarry conveyers.", challenge: "Distant remote loads and c
],
caseStudy: {
  client: "Hyderabad Cement Works Phase IV",
  requirement: "Upgrade substation equipment and design custom LT distribution panels.",
  problem: "The old HT breaker unit tripped randomly during vertical roller mill startup, causing plant-wide outages.",
  solution: "Installed a 10MVA transformer and VCB switchgear panel with adjusted protection relay parameters.",
  products: "Vacuum Circuit Breakers, protection relays, high-conductivity busducts.",
  outcome: "Eliminated MLL start trips and stabilized the distribution grid.",
  improvement: "Reduced startup current drops by 40% and improved power factor efficiency."
},
techEcosystem: [
  { name: "Vacuum Circuit Breaker (VCB)", use: "High-voltage grid isolation and protection.", logic: "Extinguishes electrical
  { name: "Transformer Oil Filter", use: "Purifying insulation oil to prevent voltage breakdowns.", logic: "Removes moisture,
  { name: "Relay Protection Module", use: "Monitoring current waveforms and voltage levels.", logic: "Compares input values a
],
qualityAssurance: [
  "Winding insulation resistance test using 5000V Megger (minimum 1000M? limit).",
  "Switchgear breaker contact resistance measurement check.",
  "Substation transformer oil breakdown voltage test (minimum 60kV limit).",
  "Relay trip timing validation check using injection test benches."
],
safetyCompliance: [
  { standard: "CEA Regulations", description: "Central Electricity Authority safety regulations compliance." },
  { standard: "IS 3043", description: "Indian standard code of practice for electrical earthing." },
  { standard: "IS 2026", description: "Specifications for power transformers." }
],
maintenanceAMC: [
  "Preventive switchyard cleaning, connection checks, oil filtration runs, and earthing pit tests.",
  "Substation transformer oil parameter audits and gas chromatography tests.",
  "24/7 on-site diagnostic support and emergency VCB troubleshooting.",
  "Modernization: replacing old analog gauges with smart digital relays."
],
docs: [
  { name: "HT-LT Cable tray Layout.pdf", size: "3.1 MB", type: "PDF Specification" },
  { name: "Transformer Test Format.docx", size: "1.2 MB", type: "Word Document" },
  { name: "VCB Technical Catalog.pdf", size: "4.5 MB", type: "VCB Catalog" }
],
faqs: [
  { q: "What does an A-Grade 33KV Electrical Contractor license mean?", a: "The A-Grade Electrical Contractor License (Class
  { q: "How often should transformer oil be tested and filtered?", a: "Transformer insulation oil should undergo laboratory t
  { q: "What is a Vacuum Circuit Breaker (VCB) and how does it differ from oil circuit breakers?", a: "A Vacuum Circuit Break
  { q: "What is protection relay coordination and why is it important?", a: "Protection relay coordination is the process of
  { q: "What testing is done during electrical system commissioning?", a: "Our commissioning test protocol includes: insulati
  { q: "How do you calculate cable sizing for HT cable installations?", a: "HT cable sizing considers three criteria simultan
  { q: "What is an earthing system and why is it critically important?", a: "The earthing system provides a low-impedance pat
  { q: "Can you provide end-to-end substation engineering including CEA documentation?", a: "Yes - we provide complete substa
]

```

```

},
"amc-services": {
  slug: "amc-services",
  title: "Commissioning, Testing & AMC Support",
  subtitle: "Factory Acceptance Testing (FAT), Site Acceptance Testing (SAT), Relay Calibration, and Annual Maintenance Contract",
  stats: [
    { label: "Emergency Response", value: "< 4 Hours Hyd" },
    { label: "Relay Calibrations", value: "Traceable to NABL" },
    { label: "AMC Assets Covered", value: "Over 200 Sites" },
    { label: "Team Strength", value: "30+ Field Engineers" }
  ],
  overviewParagraphs: [
    "Industrial automation and electrical systems require regular maintenance to maintain long-term reliability and safety. Nandini Enterprises provides comprehensive services to ensure your systems are always up and running.",
    "Our services cover the entire project lifecycle, from initial Factory Acceptance Testing (FAT) at our Hyderabad center to ongoing Site Acceptance Testing (SAT) and Annual Maintenance Contracts (AMC) at your facility.",
    "Our AMC contracts are tailored to each facility's operational needs. We offer preventive maintenance, predictive diagnostics, and emergency response services to minimize downtime.",
    "We provide modernization and retrofitting, upgrading old switchgear, obsolete PLCs, and worn-out motor starters with energy-efficient equipment."
  ],
  challenges: {
    before: "A large water treatment facility faced frequent outages, long repair times, and high spare costs due to lack of a structured maintenance program.",
    after: "Nandini Enterprises established a comprehensive AMC contract with quarterly health checks, VFD tuning, and 24/7 support, resulting in significantly reduced downtime.",
    scenario: "Reduced unexpected system failures, lowered spare parts expenditure, and secured rapid on-site troubleshooting with a dedicated service team."
  },
  workflow: [
    {
      name: "Initial System Audit",
      activity: "Review electrical schematics, inspect panel structures, document drive parameters, and identify safety concerns.",
      tools: "Multi-meters, thermal imagers, log analyzers",
      safety: "Enforce Lockout/Tagout (LOTO) procedures on all equipment before starting audits.",
      deliverable: "Approved system audit report and recommendations checklist."
    },
    {
      name: "Preventive Maintenance",
      activity: "Clean cabinet enclosures, check terminal connection torque, verify fan operation, and inspect cable insulation.",
      tools: "Vacuum cleaners, torque wrenches, contact cleaners",
      safety: "Confirm power is disconnected and components are discharged before cleaning.",
      deliverable: "Completed quarterly maintenance logs and clean panels."
    },
    {
      name: "Relay & VFD Calibration",
      activity: "Test protection relays, calibrate analog outputs, tune motor parameters, and verify communication links.",
      tools: "Relay test blocks, Dry-block temperature calibrators",
      safety: "Set up warning barricades before running drive test runs.",
      deliverable: "Calibration certificates traceable to NABL and drive parameter sheets."
    },
    {
      name: "Emergency Troubleshooting",
      activity: "On-site dispatch of service engineers, diagnostic fault analysis, spare replacements, and system restarts.",
      tools: "Oscilloscopes, digital analyzers, spare parts kit",
      safety: "Perform voltage verification checks before touching any cables.",
      deliverable: "Completed service reports detailing root cause analysis."
    }
  ],
  architectureDesc: "Our commissioning and AMC systems use a structured service workflow: client log, triage, remote assistance, on-site diagnosis, and resolution.",
  architectureDetails: [
    "Service Desk: Dedicated ticket logging system tracking response times and resolutions.",
    "Field Spares: On-site inventory of critical VFD cards, relays, fuses, and fans.",
    "Diagnostic Tools: Portable oscilloscopes and loop calibrators for rapid troubleshooting.",
    "Commissioning Checklists: Standardized FAT/SAT formats verifying every control loop."
  ],
  industries: [
    { name: "Pharma", need: "Validation of cleanroom HVAC controls and sensor accuracy.", challenge: "Meeting strict regulatory requirements for GMP compliance." },
    { name: "Cement", need: "Regular maintenance on heavy vertical roller mill drives.", challenge: "Abrasive dust causing fan wear and tear." },
    { name: "Steel", need: "Substation switchgear checkups and relay calibration.", challenge: "High heat and mechanical vibrations affecting equipment." }
  ],
  caseStudy: {
    client: "Hyderabad Pumping Station Network",
    requirement: "Establish an AMC contract to support 12 remote pumping stations.",
    problem: "Lack of regular checks led to motor failures and long downtime periods.",
    solution: "Implemented an AMC program with quarterly health checks, VFD tuning, and 24/7 hotline support.",
    products: "Yaskawa VFD spares, thermal cameras, loop calibrators.",
    outcome: "Prevented motor failures through early detection of overheating components, minimizing downtime.",
    improvement: "Reduced unexpected system failures by 80% and cut average response time to under 3 hours."
  },
  techEcosystem: [
    { name: "Relay Test kit", use: "Testing and calibrating protection relay trip times.", logic: "Injects current/voltage waveforms for precise testing." },
    { name: "Thermal Camera", use: "Non-contact scanning of panels for hotspots.", logic: "Converts infrared radiation patterns into temperature maps." },
    { name: "VFD Diagnostic tools", use: "Analyzing drive parameters and gate pulses.", logic: "Captures output waveforms to identify faults." }
  ],
  qualityAssurance: [
    "Zero-point check and span calibration audits on all transmitters.",
    "Relay trip timing validation checks (verifying compliance with safety curves).",
    "Dynamic motor run tests verifying speed-accuracy."
  ]
}

```

```
... continued: src/lib/servicesData.ts
```

```
    "Substation grounding system earth resistance checks."
  ],
  safetyCompliance: [
    { standard: "ISO 9001:2015", description: "Quality management systems guidelines for B2B engineering support." },
    { standard: "IS 3043", description: "Indian standard code of practice for electrical earthing checks." },
    { standard: "NFPA 70E", description: "Standard for electrical safety in the workplace compliance." }
  ],
  maintenanceAMC: [
    "Preventive quarterly cleaning, busbar torque audits, and safety switch tests.",
    "Relay calibration checks and parameter updates.",
    "24/7 on-site emergency troubleshooting support within 4 hours in Hyderabad.",
    "Modernization: retrofitting obsolete VFD systems with GA700 modules."
  ],
  docs: [
    { name: "FAT-SAT Standard Format.pdf", size: "2.1 MB", type: "PDF Specification" },
    { name: "AMC Agreement Template.docx", size: "1.4 MB", type: "Word Document" },
    { name: "Commissioning Guide.pdf", size: "3.2 MB", type: "Technical Guide" }
  ],
  faqs: [
    { q: "What does a standard Annual Maintenance Contract (AMC) include?", a: "Our standard AMC covers: four quarterly prevent"},
    { q: "Do you offer remote troubleshooting assistance for AMC clients?", a: "Yes. For AMC clients with SCADA or PLC systems,"},
    { q: "How do you handle spare parts management for AMC clients?", a: "We maintain a curated spare parts inventory at our Ch"},
    { q: "What is the difference between preventive maintenance and predictive maintenance?", a: "Preventive maintenance (PM) p"},
    { q: "How do you manage commissioning of complex multi-panel systems?", a: "We use a structured commissioning methodology:"},
    { q: "What is a Factory Acceptance Test (FAT) and why should the client attend?", a: "A Factory Acceptance Test (FAT) is a"},
    { q: "Do you provide training for operations and maintenance staff?", a: "Yes - operator and maintenance training is includ"},
    { q: "Can you take over AMC support for systems commissioned by other companies?", a: "Yes - we regularly provide AMC suppo"}
  ]
}
};
```